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August 12, 2024

FEDERAL STANDARD

TESTING METHOD OF LUBRICANTS, LIQUID FUELS, AND RELATED PRODUCTS

AMSC N/A FSC 91GP

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Foreword

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INTRODUCTION

TESTING METHOD OF LUBRICANTS, LIQUID FUELS, AND RELATED PRODUCTS

1. SCOPE

1.1 This test method standard is composed of standard test methods for sampling, inspecting and testing lubricants, liquid fuels and related products. It is issued pursuant to the Federal Property and Administration Services Act of 1949, as amended and its application to the purchase of commodities referred to herein is mandatory on all Federal agencies.

1.2 The test methods of this standard, when cited in a specification form an integral part of that specification. Such a citation, although referenced to the basic number of a method is intended to apply to the latest version of the method as revised (see paragraph 2) and as amended by any subsequent Change Notices to this standard.

CAUTION

Change Notices are not cumulative.
Therefore, all notices must be consulted to ensure that the latest requirements are observed

1.3 Some methods are identical to those published by the ASTM International. These methods are listed in the indexes together with their corresponding ASTM method numbers and the details of such test methods are omitted from this standard.

2. NUMBERING SYSTEM

2.1 The test methods are identified by numerical designations, each consisting of a basic number and in some instances a revision number (decimal)

a. The basic number of a method is its official designation and should be used without revision number whenever the method is referenced. The number always refers to the same method of determination and the same accuracy, and the same end results regardless of revision status.

b. Revision numbers appear as decimal addition to the basic numbers. These are assigned to basic numbers when changes are made in the method to clarify it or to give additional details that will increase the reproducibility of the test results.

3. NOTES

3.1 Purchase of Specification and Standards. Federal and military specifications and standards may be obtained under general provisions in the index of specifications and standards. They can be found in the websites also. The index may be obtained from https://quicksearch.dla.mil/qsSearch.aspx.

3.2 Purchase of ASTM test methods. ASTM test methods may be purchased from ASTM International, https://www.astm.org/

3.3 Patent notice. When Government drawings, specifications or other data are used for any

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purpose other than in connection with a definitely related government operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the government may have furnished, formulated or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacturer, use or sell any patented invention that may in any way be related thereto.

3.4 Class I and II ozone depleting substances (ODS) shall not be used in FED-STD-791 or any referenced procedures.

TABLE I. Numerical index of test method.

Method No	Method Title	Replacement test	Test cancelled by
103.6	Color Of Gasoline	ASTM D2392	Army
202.1	Cloud Intensity At Low Temperature	None	Army
203.1	Pour Stability At Low Temperature	None	Army
204.1	Diluted Pour Point	None	Navy
313.4	Penetration Of Lubricating Greases After Prolonged Working		
321.3	Oil Separation From Lubricating Greases (Static Technique)	ASTM D6184	Navy
335.3	Gear Wear	None	Army
339.6	Performance Of Diesel Engine Lubricating Oils Under Severe	None	Navy
347.3	Performance Of Diesel Engine Lubricating Oils Under Severe Operating Condition	None	Navy
350.3	Evaporation Loss Of Lubricating Greases And Oils (High Temperature)		
352.1	Effect Of Evaporation On Flammability	None	Army
353.1	Evaporation	None	Army
354.1	Performance Of Lubricating Oils In A Two Cycle Diesel Engine Under Cyclic, Turbo-Supercharged Conditions	None	Army
355	Performance Of Lubricating Oils In A Two Cycle Diesel Engine Under Cyclic, Turbo-Supercharged Conditions	ASTM D5862	Army
361.5	Stroking Properties Of Hydraulic Brake Fluids		
500.1	Induction System Deposit (ISD) Tendencies Of Motor Gasoline	ASTM D6241	Army
550.1	Emulsification Tendencies Of Petroleum Fuels By Multiple Contact Extractions	None	Army
1110.2	Quenching Speed	None	Army
1151.3	Explosive Vapors In Boiler Fuel Oil		

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TABLE I. Numerical index of test method – Continued.

Method No	Method Title	Replacement test	Test cancelled by
1203.0	Determination of Particulate Matter in Aviation and Distillate Fuels by Automatic Particle Counting		
1303.2	Flock Point (Refrigerant Compressor Oil)	None	Navy
2508.1	Thermal Stability Of Lubricating And Hydraulic Fluids	None	Army
3004.6	Trace Sediment in Lubricating Oils	ASTM D2273	Army
3005.5	Dirt Content of Grease		
3006.3	Contamination In Engine Oil	None	Navy
3007.2	Water Displacement and Water Stability		
3009.3	Solid Particle Contamination in Hydraulic Fluids	None	Army
3010.2	Solid Particle Contamination in Aircraft Turbine Engine Lubricants (Gravimetric Procedure)		
3011.1	Particulate Contamination in Aerospace Hydraulic Fluid Using Particle Counter	None	Air Force
3012.1	Determination Of Particulate Matter in Aerospace Hydraulic Fluids		
3013.2	Determination Of Particulate Contamination in Synthetic Turbine Engine Lubricants		
3044.7	Trace Sediment in Hydraulic Fluds		
3110.1	Fatty Oil in Cutting Oils by Infrared Spectrophotometry	None	Army
3180.2	Sulfur In Cutting Fluids (Active and Total)	ASTM D1662	Army
3201.7	Emulsion (Petroleum and Petroleum –Like Products)	ASTM D1401	Army
3205.3	Emulsion (Soluble Cutting Oils)	None	Army
3213.2	Foaming Characteristics Of Aircraft Turbine Lubricating Oils (Static Foam Test)	ASTM D892	Navy
3214.1	Foaming Characteristics of Aircraft Turbine Lubricating Oils (Dynamic Foam Test)	None	Air Force
3289.1	Total Solids in Water For Aircraft Injection System	None	Air Force
3403.3	Compatibility Of Turbine Lubricating Oils		
3407.2	Oxidation And Thermal Stability of Aircraft Engine Lubricating Oils (CLR Engine)	None	Navy
3410.2	High Temperature Deposits and Oils Degradation Characteristics Of Aviation Turbine Oil	None	
3411.2	Thermal Stability and Corrosivity Of Aircraft Turbine Engine Lubricants	None	
3430.2	Compatibility Characteristics of Universal Gear Lubricants	None	Air Force

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TABLE I. Numerical index of test method – Continued.

Method No	Method Title	Replacement test	Test cancelled by
3432.2	Compatibility Of Elastomers with Aircraft Turbine Lubricants		
3433.1	Compatibility Of Synthetic Aircraft Turbine Lubricants with Silicone Rubber		
3440.1	Storage Volubility Characteristics of Universal Gear Lubricants	ASTM D7603	Army
3450	Bearing Deposition of Aviation Turbine Engine Lubricants	FTM 3410.2	Air Force
3451.4	Stability Of Lubricating Oils (Work Factor)	None	Navy
3452.2	Bearing Compatibility of Turbine Oils	None	Navy
3456.2	Channeling Characteristics of Lubricants	None	Army
3457.3	Hydrolytic Stability		
3458.2	Low Temperature Stability Test for Oil		
3459.1	Low Temperature Stability	FTM 3458	Army
3463.3	Stability Of Grease in Hot Water (Water Immersion)		
3465.1	Storage Stability Test or Fluids and Lubricants	None	Air Force
3467.2	Storage Stability of Lubricating Grease		
3470.1	Homogeneity And Miscibility of Oils	ASTM D6922	Army
3472	Sonic Shear Stability of Polymer-Containing Oils	ASTM D2603	Navy
3480.3	Volatility Of Blended Oils		
3500.2	Monobasic Acid Components of Synthetic Ester Lubricants by Gas Chromatography		
3603.6	Swelling Of Synthetic Rubbers	ASTM D4289	Army
3604.3	Swelling Of Synthetic Rubber by Aircraft Turbine Lubricants		
3710.1	Molybdenum Disulfide Purity	None	Army
3720.3	Molybdenum Disulfide Content of Lubricating Grease		
3722.3	Molybdenum Disulfide Content of Non-Soap Thickened Lubricating Greases		
3816.1	Film Thickness of Dry Solid Film Lubricants	ASTM D1005	Army
4001.3	Corrosion Protection by Coating: Salt-Spray (Fog) Test	ASTM B117	Army
5003.3	Deposit- Forming Tendencies of Aircraft Turbine Lubricants		
5101.7	Neutrality (Qualitative)	ASTM D1093	Navy
5102.1	Acid And Base Number by Extraction (Color-Indicator Titration)	None	Navy
5304.5	Corrosiveness Of Greases or Semi-Solid Products At 25°C	None	Army
5305.2	Corrosiveness Of Lubricants At 232° C (450° F)		

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TABLE I. Numerical index of test method – Continued.

Method No	Method Title	Replacement test	Test cancelled by
5306.5	Corrosiveness Of Emulsifiable Cutting Fluids	None	Army
5307.2	Corrosiveness And Oxidation Stability of Aircraft Turbine	ASTM D4636	Air Force
5308.7	Corrosiveness And Oxidation Stability of Light Oils	ASTM D4636	Army
5309.6	Corrosiveness Of Grease (Copper Strip, 100 °C)		
5321.3	Corrosion Of Lead by Lubricating Oils		
5322.2	Corrosiveness Of Oil on Bimetallic Couple	ASTM D6547	Army
5327	Fuel System Icing Inhibitors in Hydrocarbon Fuels (Iodometric Method)	None	Air Force
5329.3	Corrosion Protection (Humidity Cabinet)		
5330	Fuel System Icing Inhibitor in Hydrocarbon Fuel (Calorimetric Method)	None	Air Force
5331.2	Corrosion Protection of Steel Against Sulfurous Acid-Salt Spray By Solid Film Lubricants		
5333.1	Fuel System Icing Inhibitor by Freezing Point Depressing Method	None	Air Force
5340.2	Fuel System Icing Inhibitor in Hydrocarbon Fuels (Refractometer Method)	ASTM D5006	Air Force
5341	Fuel System Icing Inhibitor in Hydrocarbon Fuels (Redox Method)	None	Air Force
5342	Fuel System Icing Inhibitor in Hydrocarbon Fuels (Hand Refractometer Method)	ASTM D5006	Navy
5350.1	Silting Index of Hydrocarbon Fuels	None	Navy
5414.5	Resistance Of Grease to Fuel		
5415.2	Resistance Of Grease to Water and A 1:1 Water Ethanol Solution		
6050.1	Nitrate- Type Ignition Improvers in Diesel Fuel	None	Army
6052.2	High Temperature, High Pressure Spray Ignition		
6053.2	Manifold Ignition Test		
6505.2	Load Carrying Capacity of Universal Gear Lubricants by The Timken Machine	ASTM D2782	Navy
6508.3	Load Carrying Capacity of Lubricating Oils (U.S. Navy Ryder Gear Machine)		
6509.2	Gear-Fatigue Characteristics of Aircraft Gas Turbine Lubricants At 204°C	None	Navy
6511.2	Load Carrying Capacity of Lubricating Oils At 204° C	None	Navy
6516.2	Oscillation Test of Grease in Helicopter Bearings	None	Navy
6517	High Temperature Gear Load –Carrying Capacity of Lubricating Oils	None	Air Force

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TABLE I. Numerical index of test method – Continued.

Method No	Method Title	Replacement test	Test cancelled by
6520.1	Extreme Pressure Properties of Lubricants (Four Ball Testers)	ASTM D2783	Navy
7501.1	Low Temperature Torque Test Method for Lubricating Greases		
7502.1	Solvent Cleaning Power by a Soil Test Method		
7503.0	High-Temperature Rolling Contact Fatigue Life Testing (RCF) Of Lubricating Oils and Bearing Materials (Ball-On-Rod RCF Tester)		
7504.0	Efficiency Of Axle Gear Lubricants		
7505.0	Load Carrying Capacity of Lubricating Oils Using a Modified FZG Gear Test Rig		
7506.1	Gear Lubricant Compatibility for Stryker Vehicle Limited Slip Differential Clutches		
7507.0	Isentropic Bulk Modulus Of Aviation Turbine Fuels and Diesel Fuels		
7508.1	Method For Detailed Hydrocarbon Analysis of Middle Distillate Fuels by Two-Dimensional Gas Chromatography		
7509.0	Method For Determination of Filtration Time and Total Solids		

TABLE II. Numerical index of cancelled and superseded test method.

Title of Cancelled Test Method	Cancelled Method Number	Substitute Method
Material Handling Safety Precaution	1000	None
Color of Gasoline	103.6	ASTM D2392
Cloud Intensity at Low Temperature	202.1	None
Pour Stability at Low Temperature	203.1	None
Diluted Pour Point	204.1	None
Oil Separation from Lubricating Greaser (Static Technique)	321.3	ASTM D6184
Gear Wear	335.3	None
Performance of Diesel Engine Lubricating Oil Under Severe Operating Condition	339.6	
Performance of Aviation Piston Engine Oil in CLR Engine (Low Temperature Dispersancy and Detergency)	347.3	None
Effect Of Evaporation of Flammability	352.1	None
Evaporation	353.1	None

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TABLE II. Numerical index of cancelled and superseded test method – Continued.

Title of Cancelled Test Method	Cancelled Method Number	Substitute Method
Performance of Arctic Lubricating Oil in a two-cycle diesel Engine under cyclic, turbo supercharge condition	354.1	None
Performance of Engine Lubricating Oils in a Two Cycle Diesel Engine under cyclic Turbo supercharge condition	355	ASTM D5862
Induction System Deposit (ISD) Tendencies of Motor Gasoline	500.1	ASTM D6241
Emulsification Tendencies of Petroleum Fuels by Multiple Contact Extractions	550.1	None
Quenching Speed	1110.2	None
Flock Point (Refrigerant Compressor Oil)	1303.2	None
Thermal Stability of Lubricating and Hydraulic Fluid	2508.1	None
Trace Sediment in Lubricating Oils	3004.6	ASTM D2273
Contamination Of Engine Oil	3006.3	None
Solid Particle Contamination in Hydraulic Fluids	3009.3	None
Particulate Contamination in Aerospace Hydraulic Fluids Using the HIAC Particulate Counter	3011.1	None
Fatty Oil in Cutting Oils by Infrared Spectrophotometry	3110.1	None
Sulfur In Cutting Fluids (Active and Total)	3180.2	ASTM D1662
Emulsion (Petroleum and Petroleum –Like Products)	3201.7	ASTM D1401
Emulsion (Soluble Cutting Oils)	3205.4	None
Foaming Characteristics of Aircraft Turbine Lubricating Oils (Dynamic Foam Test)	3214.1	None
Total Solids in Water for Aircraft Injection System	3289.1	None
Oxidation And Thermal Stability of Aircraft Turbine Engine Lubricating Oils (CLR Engine)	3407.2	None
Compatibility Characteristics of Universal Gear Lubricants	3430.2	None
Storage Solubility Characteristics of Universal Gear Lubricants	3440.1	ASTM D7603
Bearing Deposition of Aviation Turbine Engine Lubricants	3450	FTM 3410.2
Stability Of Lubricating Oils (Work Factor)	3451.4	None
Bearing Compatibility of Turbine Oils	3452.2	None
Channeling Characteristics of Lubricants	3456.2	None
Low Temperature Stability	3459.1	FTM 3458.2
Storage Stability Test or Fluids and Lubricants	3465.1	None
Homogeneity And Miscibility of Oils	3470.1	ASTM D6922
Sonic Shear Stability of Polymer-Containing Oil	3472	ASTM D2603
Swelling Of Synthetic Rubbers	3603.5	ASTM D4289
Molybdenum Disulfide Purity	3710.1	
Film Thickness of Dry Solid Film Lubricants	3816.1	ASTM D1005

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TABLE II. Numerical index of cancelled and superseded test method – Continued.

Title of Cancelled Test Method	Cancelled Method Number	Substitute Method
Corrosion Protection by Coating: Salt-Spray (Fog) Test	4001.3	ASTM B117
Neutrality (Qualitative)	5101.7	ASTM D1093
Acid And Base Number by Extraction (Color Indicator)	5102.1	None
Corrosiveness Of Greases or Semi-Solid Products At 25°C	5304.5	None
Corrosiveness Of Emulsifiable Cutting Fluids	5306.5	None
Corrosiveness And Oxidation Stability of Aircraft Turbine Lubricants	5307.2	ASTM D4636
Corrosiveness And Oxidation Stability of Light Oils	5308.7	ASTM D4636
Corrosiveness Of Oil on A Bimetallic Couple	5322.2	ASTM D6547
Fuel System Icing Inhibitor in Hydrocarbon Fuels (Iodometric Method)	5327	None
Fuel System Icing Inhibitor in Hydrocarbon Fuels (Calorimetric Method	5330	None
Fuel System Icing Inhibitor by Freezing Point Depression Method	5333.1	None
Fuel System Icing Inhibitor in Hydrocarbon Fuels (Refractometer Method)	5340.2	ASTM D5006
Fuel System Icing Inhibitor in Hydrocarbon Fuels (Redox Method)	5341	None
Fuel System Icing Inhibitor in Hydrocarbon Fuels (Hand Refractometer Method)	5342	ASTM D5006
Silting Index of Hydrocarbon Fuels	5350.1	None
Nitrate- Type Ignition Improvers in Diesel Fuel	6050.1	None
Load Carrying Capacity of Universal Gear Lubricants by The Timken Machine	6505.2	ASTM D2782
Gear Fatigue Characteristics of Aircraft Gas Turbine	6509.2	None
Load Carrying Capacity of Lubricating Oils At 204º	6511.2	None
Oscillating Test of Grease in Helicopter Bearings	6516.2	None
High Temperature Gear Load – Carrying Capacity of Lubricating Oils	6517	None
Extreme Pressure Properties of Lubricants (Four Ball Tester)	6520.1	ASTM D2783

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TABLE III. Numerical index of ASTM Equivalents for Federal standard test methods.

ASTM Test Method	FED-STD-791 Test Method
B117	4001.3
D1005	3816.1
D1093	5101.7
D1401	3201.7
D1662	3180.2
D2273	3004.6
D2392	103.6
D2603	3472
D2782	6505.2
D2783	6520.1
D4636	5307.2 and 5308.7
D5006	5327, 5330, 5340.2 and 5342
D5862	355.3
D6184	321.3
D6241	500.1
D6547	5322.2
D6922	3470.1
D7603	3440.1
D7619	1203.0

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TABLE IV. Identifying symbols and addresses of activities responsible for preparation and maintenance of test methods.

Symbol	Service	Activity
NAWCAD	Navy	Department of the Navy Naval Air Systems Command Fuels and Lubricants Team Air-4.4.122229 Elmer Road, Bldg. 2360 Patuxent River, MD 20670-1534
AFPET/PTP	Air Force	Air Force Petroleum Office, Product Support Directorate 2430 C St., Bld. 70, Area B Wright-Patterson AFB, OH 45433
AR	Army	Department of the Army US Army Armament Research and Development Center Dover, New Jersey 07801-5001
SH	Navy	Department of the Navy Naval Sea Systems Command Washington D.C. 20362
AFWAL/POSF	Air Force	Department of the Air Force Air Force Wright Aeronautical Laboratories POSF
AFRL/RQTM	Air Force	Air Force Research Laboratory Turbine Engine Division Engine Mechanical Systems Branch 1790 Loop Road North, Bld. 490, Area B Wright- Patterson AFB, OH 45433-7103
AT	Army	US Army DEVCOM GVSC Fuel and Lubricants Branch Detroit Arsenal, MI 48397

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Method 313.4
September 30, 2019

313.4 – PENETRATION OF LUBRICATING GREASES AFTER PROLONGED WORKING

1. SCOPE

1.1 This method is used for determining the consistency of lubricating greases that have been subjected to severe mechanical working.

2. SUMMARY

2.1 The method consists of working the grease mechanically under specified conditions for 100,000 double strokes (approximately 28 h), then checking the worked penetration in accordance with ASTM D217.

3. SAMPLE SIZE

3.1 Sufficient grease to be tested to pack grease worker.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Grease worker, as specified in the "Cone Penetration of Lubricating Grease - ASTM D217, with the following differences:

a. Machine grease-working, 60 double strokes per minute.
b. Plunger plate, 270 holes 0.16 mm (1/16 inch) diameter, arranged as in ASTM D217 Figure 1: Penetrometer.

5. MATERIALS

5.1 Grease to be tested.

6. PROCEDURE

6.1 Pack sample in grease worker and assemble and vent grease worker as described in "Procedure for Worked Penetration of ASTM D217.

6.2 Mount grease worker in machine and work grease for 100,000 double strokes while maintaining the ambient temperature at 25°C ± 1°C (77°F ± 2°F).

6.3 Remove the worker from the machine, making certain that plunger is pushed all the way down.

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6.4 Open the vent cock and insert the thermometer as in "Procedure for Worked Penetration of ASTM D217.

6.5 Place grease worker in the water bath, bring temperature of sample to 25°C ± 1°C (77°F ± 2°F) and determine worked penetration in accordance with ASTM D217.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Report penetration after mechanical working as determined in paragraph 6.5.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	4	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	4	Army - AT

Method Prepared By:
Navy - NADC - 1986

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Method 350.3
September 30, 2019

350.3 – EVAPORATION LOSS OF LUBRICATING GREASES AND OILS (HIGH TEMPERATURE)

1. SCOPE

1.1 This method describes the test procedure and apparatus for the determination of evaporation loss of lubricating greases and oils for applications where evaporation loss is a factor. Evaporation loss data can be obtained at any temperature in the range 99°C to 538°C (210°F to 1,000°F).

2. SUMMARY

2.1 The method consists of heating the oils or greases in containers vented with 2 L/min of air in order to determine the percentage evaporation loss of the lubricant.

3. SAMPLE SIZE

3.1 Approximately 20 grams of the grease or oil to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Evaporation cell and attachments conforming to the dimensional tolerances indicated in FIGURE 6 and FIGURE 7. Other structural details are given below.

4.1.1 The body and cover of the cell shall be constructed of stainless steel and air-heating coil also of stainless steel.

4.1.2 The sample cups (recommended maximum weight 200 grams each), hood, eduction tube, and orifice shall be constructed of stainless steel. A suitable material is an alloy steel conforming to grade S, type 304, of ASTM A204/A204M. To facilitate removal and separation of the cup and hood for inserting the sample and weighing, the sample cup shall be threaded to the hood and this in turn to the eduction tube of the cover. The cover of the cell shall be made air- tight.

4.2. Air supply system, capable of supplying to the cell the required flow of air free of mechanically entrained particles. A 61.0 cm (24 inch) length of 12.7 cm (5 inch) diameter pipe, reduced to 2.54 cm (1 inch) diameter at each end, packed with glass wool at each end, and filled with a suitable drying agent has been found satisfactory for filtering and drying. Means must be provided for filling the drying and filtering tube.

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4.3 Heating blocks of cast aluminum with evaporation cells and air coils cast within. The heating block shall be heavily insulated and equipped with sufficient electric heating elements to maintain the test temperature. There shall be one thermocouple provided for each heating element. FIGURE 1, FIGURE 2, and FIGURE 3 show a satisfactory heating block.

4.4 Temperature controller. A suitable temperature controller capable of maintaining temperatures between 99°C to 538°C ± 0.5°C. A suitable temperature recorder shall also be provided.

4.5 Flowmeter. A rotameter calibrated to deliver air at a rate of 2.58 g/min ± 0.02 g/min between 15.6°C to 29.4°C (60° to 85°F) (2 L/min at standard temperature and pressure). It shall be furnished with a needle valve and may be mounted as shown in FIGURE 7.

4.6 Evaporation tester stand. A suitable stand for supporting the test apparatus is shown in FIGURE 5.

5. MATERIALS

5.1 VM&P Naphtha.

5.2 Petroleum Ether

5.3 Tissue, Facial.

6. PROCEDURE

6.1 Greases

6.1.1 Clean the grease-sample cup and hood thoroughly, rinsing at least three times using VM&P Naphtha followed by several rinses with petroleum ether. Dry thoroughly with forced air.

6.1.2 Weigh the clean grease-sample cup and hood to the nearest milligram. Remove the hood and fill the cup with sample, taking care to avoid occlusion of air. Smooth the surface level with the rim of the cup with a straight edged spatula. Remove with a clean cloth any grease which may remain on the rim or threads of the cup. Thread the hood tightly onto the cup without disturbing the smoothed grease surface. Weigh the assembly and record the net weight of the sample to the nearest milligram.

6.1.3 With cover in place, but without the hood and sample cup attached, allow the evaporation cell to acquire the temperature of the block ± 2°C (1°F), at which the test is to be made. Allow the block to maintain the temperature of the test at least 2 hours before beginning the test. During this period, allow clean air to flow through the cell at the prescribed rate, 2.58g/min ± 0.02 g/min (2 L/min at standard temperature and pressure), as indicated by the rotameter. Then remove the cover, thread the weighed hood and sample cup into place, and replace the cover. Tighten the three knurled cover- tightening screws securely to prevent air leakage under the cover. Pass clean air through the cell for 22 h ± 5 min.

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6.1.4 At the end of the 22-hour period, remove the assembled sample cup and hood from the cell, and allow to cool to room temperature. Determine the net weight of the sample to the nearest milligram.

6.2 Oils.

6.2.1 Clean the oil-sample cup and hood as described in 6.1.1.

6.2.2 Weigh the clean oil-sample cup and hood to the nearest milligram. Transfer, by means of a pipette, 10.00 ± 0.05 grams of sample to the cup. Assemble the cup and hood, being careful not to splash oil on the underside of the hood. Weigh the assembly and record the net sample weight to the nearest milligram.

6.2.3 Evaporate the sample as described in 6.1.3 and 6.1.4.

7. CALCULATION

7.1 Calculate the evaporation loss of the sample as follows:

$$Evaporation\ loss, percent\ by\ weight\ = \frac{S\ -\ W}{S} * 100$$

where:
S = initial weight of sample, in grams.
W = weight of sample, in grams, after the test.

8. REPORTING

8.1 Report the evaporation loss percentage.

9. PRECISION

9.1 Results should not differ from the mean by more than the following amounts:

Repeatability: 2.5% of mean
Reproducibility: Not established

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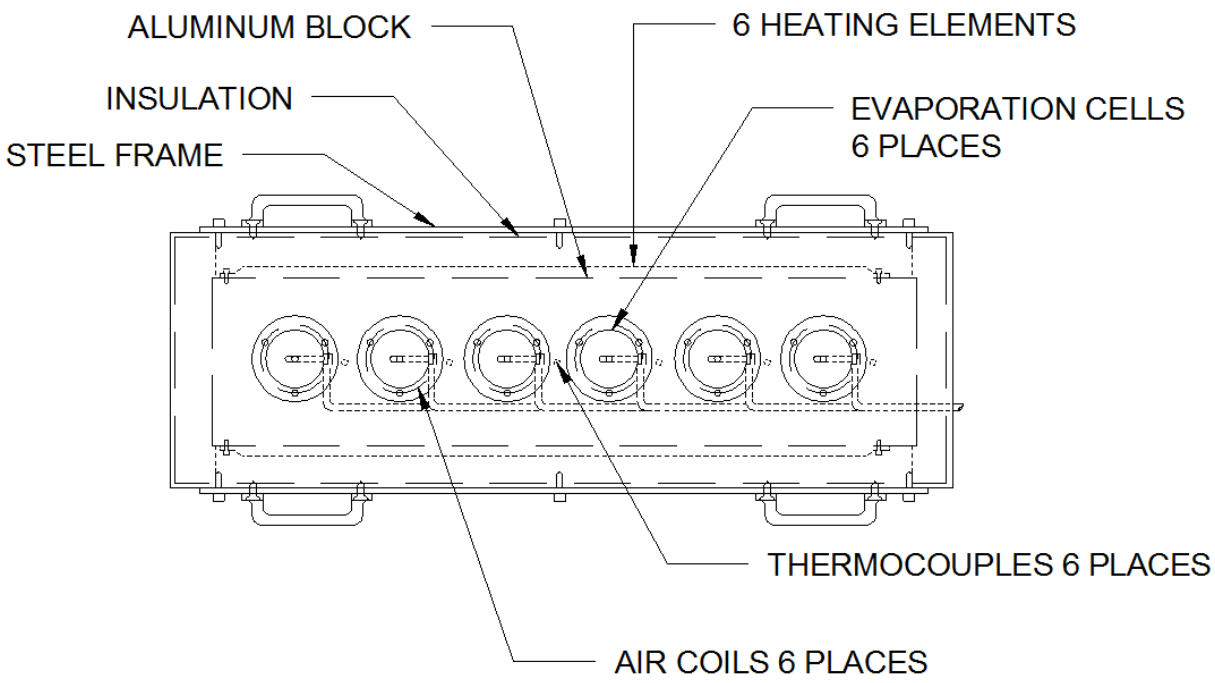


FIGURE 1. Top view high temperature evaporation tester.

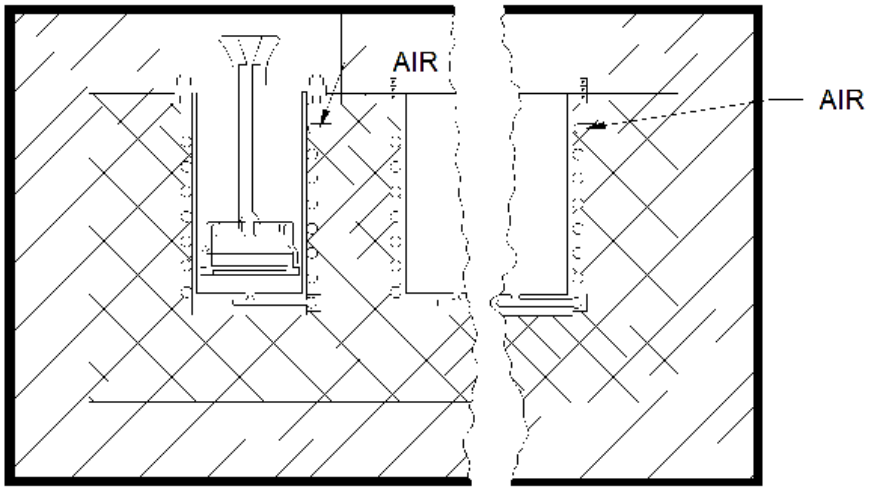


FIGURE 2. Side view high temperature evaporation tester.

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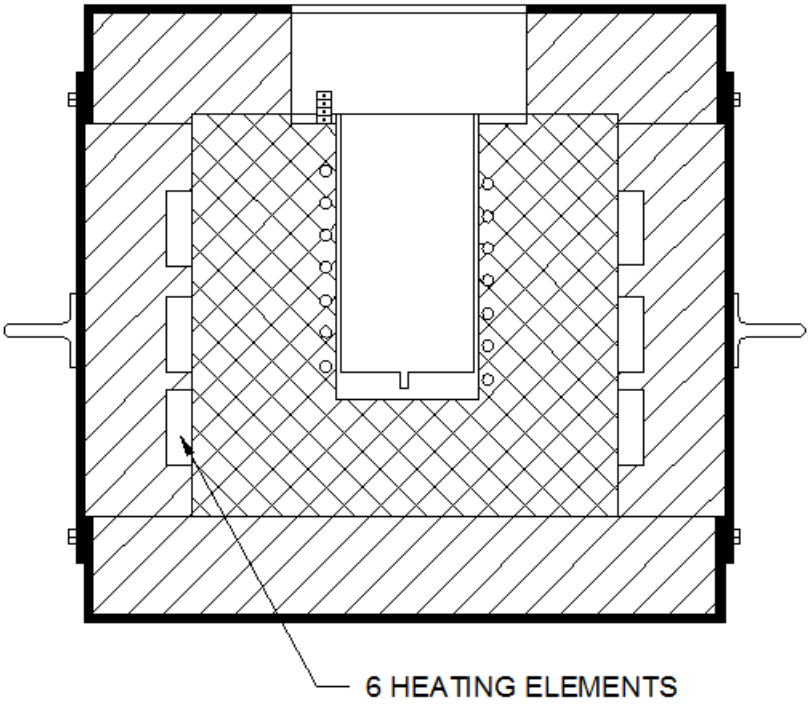


FIGURE 3. Cut end view high temperature evaporation tester.

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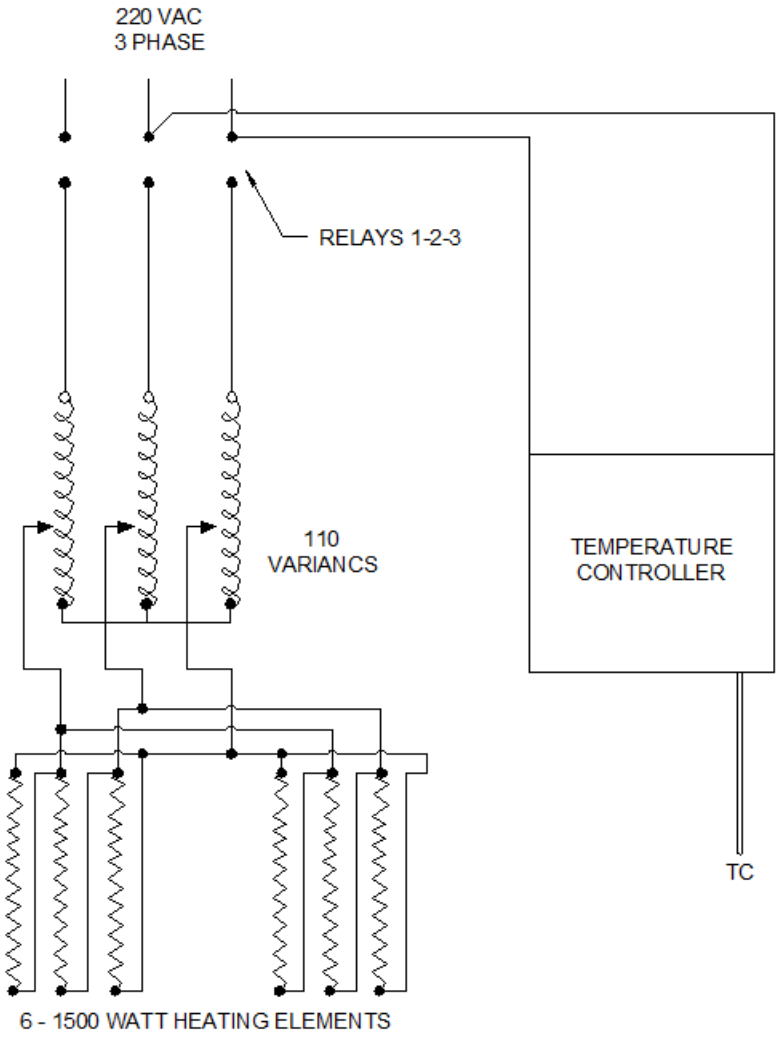


FIGURE 4. High temperature evaporation tester.

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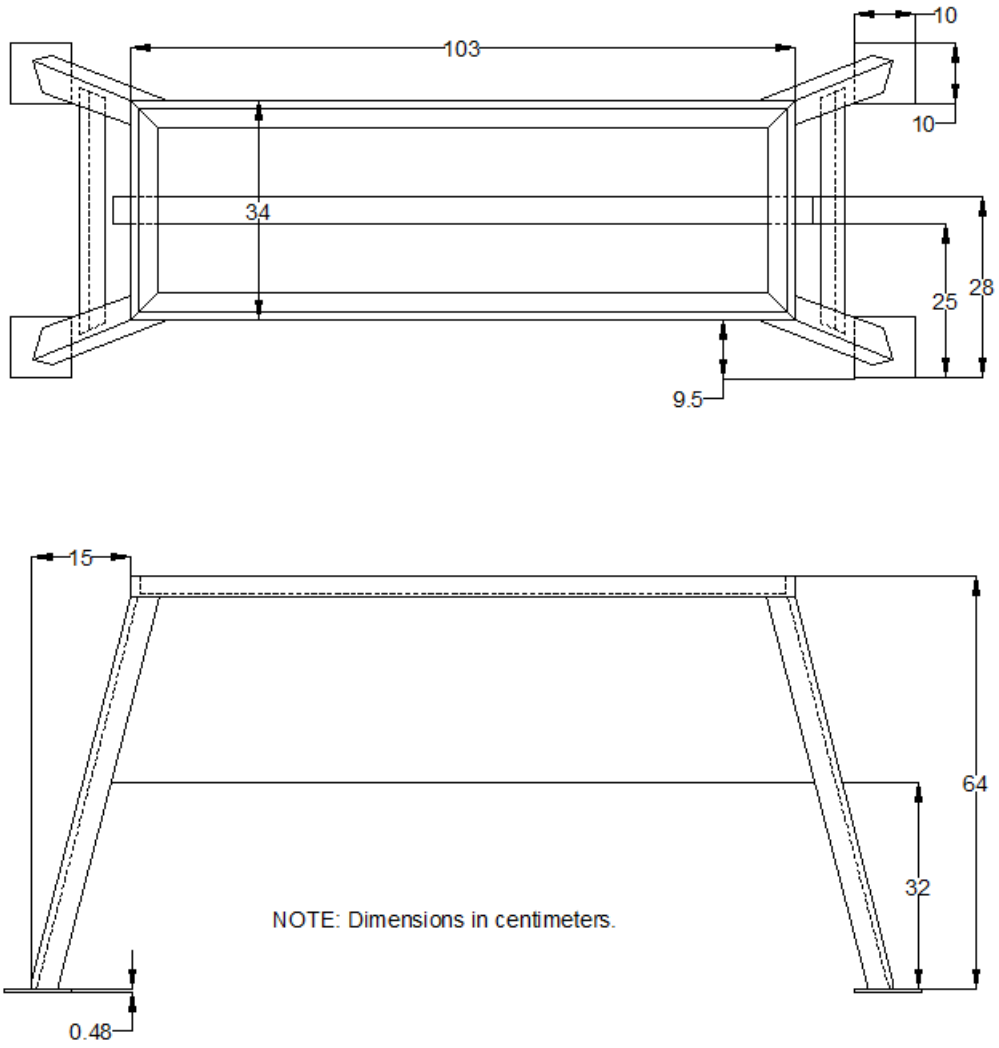
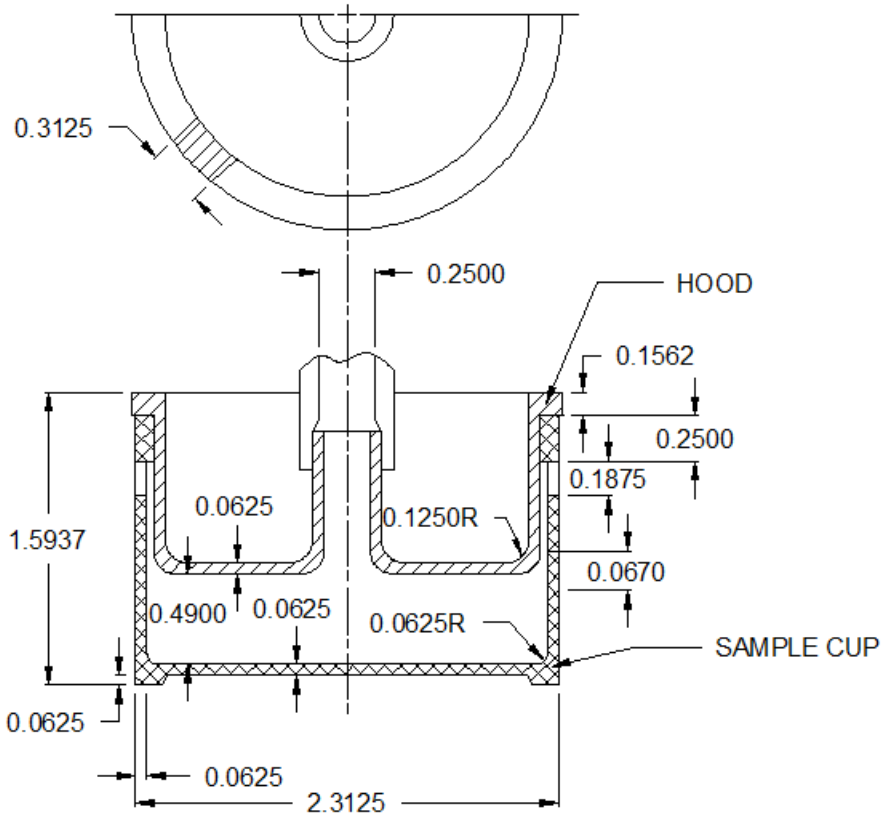


FIGURE 5. Evaporation tester stand.

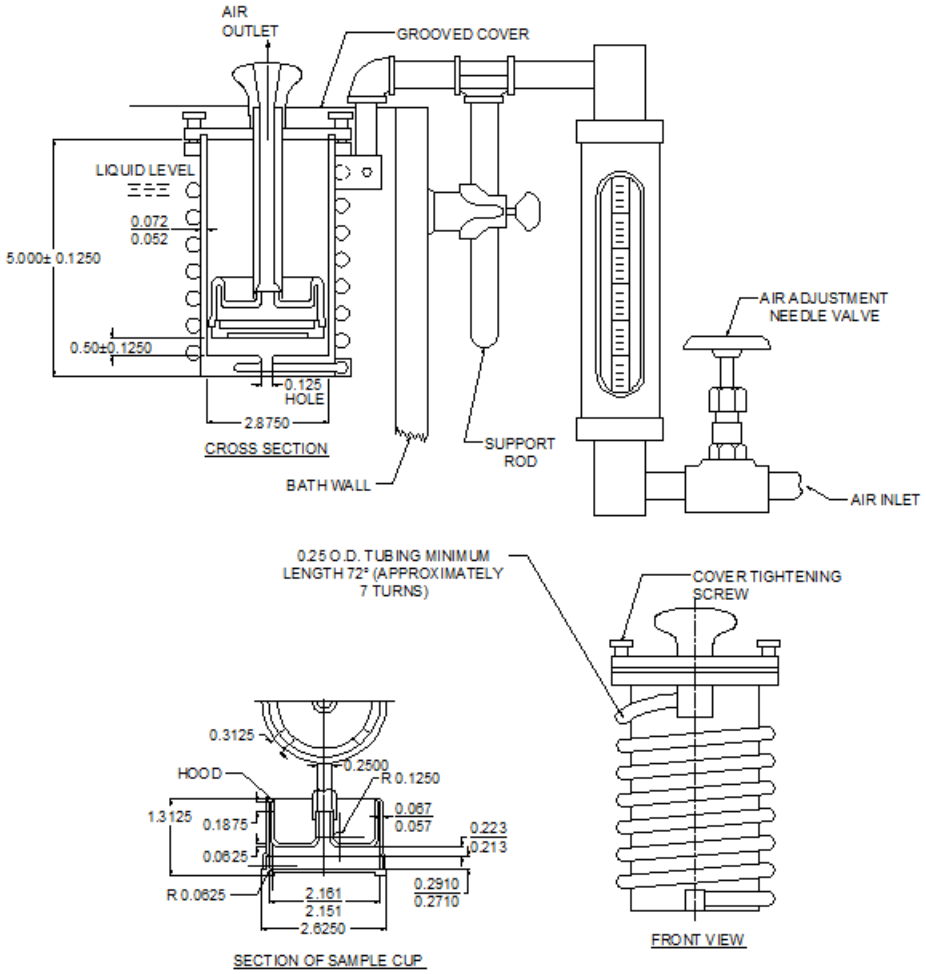
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NOTE:
1. Dimensions in inches.
2. All dimensions±.0156 unless otherwise specified.

FIGURE 6. Oil sample cup.

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NOTE:
1. Dimensions in inches.
2. All dimensions ± .0156 unless otherwise specified.

FIGURE 7. Oil sample cup.

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT
30 Sep 2019	Replaced 1,1,1-Trichloroethane with VM&P Naphtha and petroleum ether	Replacement chemicals are less toxic. They are used by AF PTP laboratory.	3	Army – AT Air Force – AFPET/PTP

Method Prepared By:
Navy - NADC – 1986

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Method 361.5
September 30, 2019

361.5 – STROKING PROPERTIES OF HYDRAULIC BRAKE FLUIDS

1. SCOPE

1.1 This method describes a procedure for evaluating the stroking properties of brake fluids used in the hydraulic brake systems of automotive vehicles. This method, with slight modifications, may be used to evaluate hydraulic brake rubber cups against a standard brake fluid.

2. SUMMARY

2.1 The method utilizes standard SAE International hydraulic brake parts mounted in a manner similar to that of an automobile brake system. The test brake fluid is subjected to various stroking rates, pressures, and temperatures designed to evaluate the lubrication properties of the brake fluid. This method complements the chemical and physical tests required by the brake fluid specifications.

2.2 It is essential to have meaningful reference data before a significant interpretation of test results can be made. Reference data must include comparable test information on a brake fluid of known service performance characteristics.

3. SAMPLE SIZE

3.1 Approximately 3.8 L (1 gal) of the brake fluid to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 SAE Standard.

J 1703 - Motor Vehicle Brake Fluid

4.2 ASTM Standard.

D2240 - Rubber Property-Durometer Hardness

4.3 Apparatus.

4.3.1 Wheel assemblies. Four left front wheel brake assemblies.

4.3.1.2 Hub and brake drum assemblies. Four left front wheel hub and brake drum assemblies.

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4.3.1.3 Helper springs. One helper spring per wheel assembly.

4.3.2 Brake pressure actuating mechanism. A suitable actuating mechanism for applying pressure to the master cylinder push rod to simulate, as closely as possible, the braking operation in a vehicle.

4.3.2.1 Type of actuating mechanism. The actuating mechanism shall be operated by suitable air pressure or hydraulic means. Mechanical actuation is not recommended.

4.3.2.2 Pressure application. The amount of pressure applied by the actuating mechanism shall be adjustable. The mechanism shall be capable of applying sufficient thrust to the master cylinder to create a minimum pressure of 6.9 gage MPa (1000 psig) in the simulated brake system.

4.3.3 Stroking rate and rate-of-load application. The actuating mechanism shall be designed to permit adjustable stroking rates ranging from 500 to 1,000 strokes per hour as desired. The ratio of percent time-of-load application to non-load application shall also be adjustable, from a minimum ratio 25/75 percent to a maximum of 60/40 percent per stroke cycle. (A "60/40 percent ratio per stroke cycle indicates that pressure is applied during 60% of the duration of the stroke cycle, while the pressure is released, or not applied, during the remaining 40% of the cycle.)

4.3.4 Pressure gages. Two recording pressure gages of 0 to 6.9 gage MPa (0 to 1000 psig) capacity with connecting tubing. One of the gages shall have a 24-hour cycle, while the other gage shall have a maximum cycle of 6 minutes. Each gage shall be provided with a shut off valve for removing air from the connecting tubing.

4.3.5 Revolution counter. A mechanical or electrical revolution counter with recording capacity for 500,000 strokes.

4.3.6 Tubing. Approximately 4.6 m (15 ft) of 0.63 cm (1/4 inch) O.D. copper or steel tubing with inverted flare nuts, connectors, adaptors, and fittings.

4.3.7 Stand. A stand or frame for mounting the four left front wheel assemblies as shown in FIGURE 1.

4.3.8 Cabinet or oven. A large cabinet or oven, of sufficient capacity to house the stand with the four mounted wheel assemblies, together with the master cylinder and necessary connections.

4.3.9 Temperature control. The cabinet or oven shall be insulated and shall be equipped with suitable means of heating and cooling to provide the specified test temperatures. The temperatures shall be thermostatically controlled within 2.8°C (5°F) of the specified test temperature.

4.3.10 Durometer, Type A-shore durometer. The description of the durometer, calibration, and application in testing rubber hardness may be found in ASTM D2240.

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5. MATERIALS

5.1 Ethyl alcohol.

5.2 Crocus cloth. Crocus cloth shall conform to the requirements of ANSI B74.18.

5.3 Wheel cylinder assemblies. Four cast iron housing straight bore hydraulic brake wheel cylinders having a diameter of approximately 2.8 cm (1-1/8 inch). SAE part number RM-14a.

5.4 Master cylinder assembly. One cast iron housing hydraulic brake master cylinder having a diameter of approximately 2.8 cm (1-1/8 in.) and fitted with an uncoated steel standpipe. SAE part number RM-15a.

5.5 Rubber cups. All wheel cylinder and master cylinder cups used in the test brake system shall be made of SBR type elastomer. SAE part numbers for wheel cylinders and master cylinders are RM-3, RM-4 and RM-5.

5.6 Wheel cylinder pistons. Unanodized aluminum pistons (SAE RM-12) made from SAE AA 2024 aluminum alloy are to be used.

5.7 Master cylinder piston. Use a SAE RM-13 piston made from SAE CA 360 copper-base alloy (half hard).

NOTE: All standard SAE parts are available from the SAE International, 400 Commonwealth Drive, Warrendale, PA 15096.

6. PROCEDURE.

6.1 Preparation of the test apparatus.

6.1.1 Inspection and cleaning of parts.

6.1.1.1 Wheel cylinder assemblies. The wheel cylinder assemblies shall be completely disassembled. All rubber parts, with the exception of the cylinder boots, shall be discarded. The expanders in the assembly shall also be discarded. All metal parts shall be cleaned with 95% ethyl alcohol to remove preservative oils, grease, and dirt, then dried in a stream of dry air, and examined for cuts, scores, galling, or pitting. Any part having any of these flaws shall be replaced. Light stains on the cylinder walls shall be removed by rubbing with crocus cloth and 95% ethyl alcohol. Any cylinder which cannot be freed of stains by this procedure shall be discarded.

6.1.1.2 Master cylinder assemblies. The master cylinder assemblies shall be completely disassembled. All rubber parts, including the piston washer if it forms an integral part of the primary cup, shall be discarded. The metal parts shall be cleaned, air-dried, examined for flaws, and light stains removed from the cylinder walls in the manner specified for the wheel cylinder assemblies (see 6.1.1.1). Any deficient parts shall be replaced. The cylinder's relief and supply

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parts shall be inspected for burrs and wire-edge. If present, they may be removed with a burring tool or a jeweler's file.

6.1.1.3 Rubber parts. The rubber parts to be used in the test shall be cleaned with a lint-free cloth and 95% ethyl alcohol to remove all dirt and grease, then dried in a stream of dry air and examined for cuts, molding flaws, or blisters. Any deficient part shall be discarded.

6.1.1.4 Tubing. The tubing used as brake fluid lines shall be flushed with 95% ethyl alcohol, and dried by blowing dry air through it, prior to connecting to the test apparatus.

6.2 Initial measurements and tolerance.

6.2.1 Cylinders. The internal diameter of the master cylinder and of each of the wheel cylinders shall be measured with a micrometer, to the nearest 0.0025 cm (0.001 inch). Four readings, approximately 45° apart, shall be taken on each cylinder, and their average considered as the diameter of the cylinder. If the largest and smallest diametrical readings differ from each other by more than 0.0075 cm (0.003 inch), the cylinder in question shall be discarded.

6.2.2 Pistons. The external diameter of the master cylinder piston and of each of the wheel cylinder pistons shall be measured with a micrometer, to the nearest 0.0025 cm (0.001 inch). Two readings, at right angles to each other, shall be taken on each piston, and their average considered as the diameter of the piston. If the two readings differ from each other by more than 0.0025 cm (0.001 inch) the piston in question shall be discarded.

6.2.2.1 Clearance of piston in cylinder. Each piston shall be checked for clearance within the cylinder for which it is intended. If the clearance is less than 0.0075 cm (0.003 inch) or more than 0.0125 cm (0.005 inch), the cylinder or the piston shall be replaced by another similar part.

6.2.3 Rubber cups.

6.2.3.1 Base and lip diameters of the cups. The internal base and lip diameters of each rubber cup shall be determined with a micrometer, to the nearest 0.0025 cm (0.001 inch). The cups shall be placed on a flat surface for 24 hours before measuring. In measuring the diameters of the cups, care should be taken that the micrometer does not extend more than 0.08 cm (1/32 inch) beyond the edges of the cup. If difficulty is encountered in measuring the diameter of the master cylinder secondary cup, the cup may be placed on a mandrel having a slight taper. Two readings at right angles to each other, shall be taken on the base and lip of each cup, and their average considered as the base and lip diameters of the cup. If the two readings on the base or the two readings on the lip differ by more than 0.0025 cm (0.001 inch) respectively, the cup in question shall be discarded.

6.2.3.2 Hardness of rubber. The durometer hardness of each cup shall be measured at four different points near the center of each cup. The average of the four readings shall be considered as the hardness of the rubber cup. For the sake of uniformity the durometer readings should be made as follows: The proper rubber anvil is inserted in the fixture, and the rubber cup placed on the anvil. The back edge of the durometer is placed on the shelf of the fixture, with the bottom

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surface of the durometer kept parallel to the surface of the base of the cup. The durometer is grasped between the thumb and the third and fourth fingers of one hand, with the index finger placed on the top of the depressor. The indentor point is lowered gently until it rests on the surface of the cup. With a minimum amount of rocking, pressure is then applied so that the pressure plate makes firm contact with the surface of the cup. The durometer reading is taken as soon as the dial needle comes to rest. If the needle does not come to rest, then the reading is taken at that moment when a definite change in the speed of the needle is observed, with the reading estimated to the nearest whole point of the dial scale.

6.2.3.2.1 Wheel cylinder cups. The average durometer hardness of the wheel cylinder cups shall be 60 ± 3 points. Nonconforming cups shall be discarded.

6.2.3.2.2 Master cylinder cups. The average durometer hardness of the master cylinder cups shall be 63 ± 3 points. Nonconforming cups shall be discarded.

6.3 Assembling, adjusting, and checking the apparatus.

6.3.1 Assembly of brake parts. After the required measurements have been taken and recorded, the wheel cylinders and the master cylinder shall be assembled for test. All parts with the exception of the cylinder housings and the rubber boots, shall be dipped in the hydraulic fluid under test when they are assembled. The assembled cylinders shall be checked for ease of operation, and then installed in their respective places in the simulated brake systems. Helper springs shall be installed in holes beneath the shoe flanges, pushed up and hooked over the flanges (see FIGURE 1).

6.3.2 Shoe clearance adjustment. The brake shall be adjusted to have a clearance of 0.102 cm ± 0.0075 cm (0.040 inch ± 0.003 inch) between the shoe and the drum, at the toe.

6.3.3 Filling and bleeding the brake system. The assembled simulated brake system shall be filled with the fluid under test by means of a pressure pot, or by hand. If filled by hand the master cylinder shall be stroked several times to completely fill the system. All the wheel cylinders and the tubing connected to the pressure gages shall be bled to remove the entrapped air from the system.

6.3.4 Checking the system for leaks. After the system has been bled, a pressure of 6.9 gage MPa (1,000 psig) shall be applied to the system. The system shall be slowly stroked for 10 minutes, after which an inspection shall be made for leaks. The pressure shall be released, and the brake shoe clearance rechecked (6.3.2).

6.3.5 Fluid level. A constant level of test fluid shall be maintained at the master cylinder throughout the test. Accurate record shall be kept of the number of mL of the fluid added to the system during the test. It may be necessary to install special anti-splash plugs on the master cylinder reservoir to avoid loss of the fluid due to splashing.

6.4 Test procedure.

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6.4.1 Procedure A. Stroking properties at high temperatures.

6.4.1.1 Conditions of test.

6.4.1.1.1 Test temperature. Unless otherwise specified the test temperature shall be 120°C ± 3°C (248°F ± 5°F). The test shall be started at room temperature (23°C ± 5°C). The rise in temperature from room temperature to the test temperature shall take place in not more than 8 h from the start of the test.

6.4.1.1.2 Pressure. The initial stroking pressure in the simulated brake system shall be adjusted at room temperature (23°C ± 5°C), to 3.4 gage MPa (500 psig) and recorded. The pressure stroke pattern shall be adjusted at 60/40 percent stroke ratio 17 strokes/minutes, with the aid of the short-cycle pressure recorder, so as to produce a relatively smooth curve (see FIGURE 2). Accurate record shall be kept of any pressure variations during the test. The pressure stroke pattern shall be compared with the initial pattern at intervals of 50,000 strokes during the test.

6.4.1.2 Brake inch or burnishing run. The stroking rate shall be adjusted at 1,000 ± 50 strokes per hour, using the 60/40 percent stroke ratio and the stroking pressure of 3.4 MPa gage (500 psig) applied at room temperature (23°C ± 5°C). While at room temperature, the system shall be stroked 16,000 ± 500 strokes. At the end of this stroking period the system shall be checked for fluid level. The cylinder shall be filled to the given level, if necessary, and the test resumed.

6.4.1.3 Duration of test. The test shall be run until 94,000 strokes have been completed. The number of strokes given in the burnishing run (see 6.4.1.2) and in coming to test temperature (see 6.4.1.1.1) shall be included as part of total 94,000 strokes. The number of strokes required, if other than 94,000 shall be specified in the material specification.

6.4.2 Procedure B. Stroking properties at sub-zero temperatures.

6.4.2.1 Conditions of test.

6.4.2.1.1 Test temperature. The test temperature shall be -40°C ± 3°C (-40°F ± 5°F) or -55°C ± 3°C (-67°F ± 5°F), as specified. The test shall be started at room temperature (23°C ± 5°C). The drop in temperature from room temperature to the specified test temperature shall take place in 8 to 24 hours, following the start of the test.

6.4.2.1.2 Pressure. The initial stroking pressure in the simulated brake system shall be adjusted at room temperature (23°C ± 5°C) to 3.4 gage MPa (500 psig) and recorded. The pressure stroke pattern shall be adjusted at 60/40 percent stroke ratio, with the air of the short cycle pressure recorder so as to produce a relatively smooth curve (see FIGURE 2). Accurate record shall be kept of any pressure variations during the test.

6.4.2.2 Break in or burnishing run. The stroking rate shall be adjusted at 1,000 ± 50 strokes/h, using the stroking pressure of 3.4 gage MPa (500 ± 10 psig) applied at room temperature (23°C ± 5°C). While at room temperature the system shall be stroked 16,000 ± 500 strokes, at the end of

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which the system shall be checked for leaks, and proper brake shoe clearance and the master cylinder checked for fluid level. The cylinder shall be filled to the given level, if necessary. The stroking rate shall then be adjusted to 720 ± 50 strokes/h, using a 40/60 percent stroke ratio. The pressure-stroke pattern shall be adjusted, with the aid of the short-cycle pressure recorder, to give a relatively smooth curve (see FIGURE 2). Accurate record of observed pressure variations shall be compared with the initial pattern at approximately every 20,000 strokes. The stroking cycle, if other than that listed in this paragraph, shall be specified in the product specification.

6.4.2.3 Duration of test. The test shall be run until 94,000 strokes have been completed. The number of strokes given in the burnishing run (see 6.4.2.2) and in coming to the specified test temperature shall be included as part of the total 94,000 strokes. The number of strokes required, if other than 94,000, shall be specified in the product specification.

6.5 Observations.

6.5.1 Observations during test.

6.5.1.1 Pressure change. Observations at 12-hour intervals shall be made of the stroking pressure in the simulated system, to determine changes from the initial pressure.

6.5.1.2 Functioning of wheel cylinder pistons. The functioning of the wheel cylinder pistons shall be observed. Improper functioning of the pistons, such as freezing of the piston in the cylinder, or failure of the brake shoe to be released from the brake drum, shall be recorded. However, very slight drag of the brake shoe on the drum, when the drum is rotated by hand, shall not be considered as improper functioning of the cylinder pistons. Mechanical defects of the brake mechanism may cause excessive pressure rise or improper functioning of the wheel cylinder pistons. If inspection of the malfunction shows this to be the case, the test shall be repeated using new parts.

6.5.1.3 Leakage (or evaporation). The leakage (or evaporation) loss of the hydraulic brake fluid at 24,000-stroke intervals shall be determined by the mL of the fluid required to maintain the constant fluid level in the master cylinder reservoir. The level of the fluid should be checked daily.

6.6 Observations and measurements at end of test.

6.6.1 Leakage. The equipment shall be allowed to come to room temperature within 16 h following completion of the required number of strokes. The brake wheel cylinder shall be examined for evidence of excessive leakage. The assembly shall then be stroked an additional 100 strokes at 3.4 gage MPa (500 psig), and the cylinders again examined for evidence of excessive leakage.

6.6.2 Disassembly of cylinders. The wheel cylinders and the master cylinder shall be removed from the simulated system. The inlet ports of the cylinders shall be plugged or capped immediately upon removal to retain the fluid remaining therein. Each cylinder shall then be carefully disassembled. The fluid remaining in the cylinder shall be drained into a small

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screwcap jar, and saved for examination for evidence of sludging, jelling, sedimentation, and grittiness.

6.6.3 Inspection. A visual inspection of the disassembled cylinder shall be made, and a record made of the type and quantity of dirt; gum deposition; rusty, scored, and galled areas on the surface of the pistons and on the cylinder walls; chipping, scoring, and general condition of the rubber cups. All parts shall then be cleaned with 95% ethyl alcohol, and dried in a stream of air. The metal parts shall then be re-examined for evidence of corrosive action, such as pitting, etching, and severe stain.

6.6.4 Measurements.

6.6.4.1 Cylinders. The internal diameter of each wheel cylinder and of the master cylinder shall be measured as specified in 6.2.1.

6.6.4.2 Pistons. The external diameter of the master cylinder piston and of each wheel cylinder piston shall be measured as specified in 6.2.2.

6.6.4.3 Rubber cups. Measurements of the base and lip diameters of the rubber cups shall be made as specified in 6.2.3.1. Durometer hardness of the cups shall be determined as specified in 6.2.3.2. The diameter measurements and the hardness determinations shall be made within 1 hour following removal from the cylinder assembly.

7. CALCULATIONS

7.1 Change in cylinder and piston diameter:

$$C = C1 - C2$$

where:

C = change in cylinder diameter.
C_1 = original cylinder bore diameter. C_2 = final cylinder bore diameter.

$$P = P1 - P2$$

where:

P = change in piston diameter.
P_1 = original piston diameter.
P_2 = final piston diameter.

7.2 Change in base and lip diameter of rubber cups.

$$B = B2 - B1$$

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where:

B = change in rubber cup base diameter.
B1 = original base diameter.
B2 = final base diameter.

$$L = L_2 - L_1$$

where:

L = change in rubber cup lip diameter.
L_1 = original lip diameter.
L_2 = final lip diameter.

7.3 Lip diameter interference set.

$$LDIS = \frac{L_1 - L_2}{L_1 - C_1} * 100$$

where:

LDIS = lip diameter interference set percentage.
L_1 = original lip diameter of rubber cup.
L_2 = final lip diameter of rubber cup.
C_1 = original cylinder bore diameter.

7.4 Change hardness of rubber cup.

$$H = H_2 - H_1$$

where:

H = change in durometer hardness.
H_1 = original hardness value.
H_2 = final hardness value.

7.5 Change in fluid pressure.

$$P = P_2 - P_1$$

where:

P = change in system pressure.
P_1 = original pressure value.
P_2 = final pressure value.

7.6 Change in fluid volume

$$V = V_2 - V_1$$

where:

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V = change in fluid volume.
V_1 = original volume.
V_2 = final volume.

8. REPORTING

8.1 A summary of the test conditions and the inspection results shall be made at the completion of the stroking test. Reports shall be made on the following:

a. Degree (slight, moderate, or excessive) of corrosion, scoring, and galling of metal parts.
b. Change in the initial diameter of the cylinders and pistons (see 7.1).
c. Change in the base and lip diameters of the rubber cups (see 7.2).
d. Percentage of lip diameter interference set (see 7.3).
e. Change in the durometer hardness of rubber cups (see 7.4).
f. Degree (slight, moderate, or excessive) of swelling, tackiness, scoring, scuffing, or blistering of the rubber cups.
g. Pressure change in the system at end of each successive 12-hour interval (see 7.5).
h. Volume loss of fluid due to leakage or evaporation, at the end of each successive 24000-stroke interval during the test (see 7.6).
i. Volume loss of the fluid at the end of the test (see 6.2.1 and 7.6).
j. Degree (slight, moderate, or excessive) of sludging, jelling, sedimentation, or grittiness of the fluid after testing (see 6.2.2).
k. Observed improper functioning of the wheel cylinder pistons.

9. PRECISION

9.1 Duplicate test runs should show the same general trend for a given hydraulic brake fluid, particularly in the matter of leakage during and after test, excessive increase in the stroking pressure, and improper functioning of the wheel cylinder pistons.

9.2 The test shall be repeated using new parts if mechanical difficulties, which cannot be attributed to the fluid proper, are encountered during the test.

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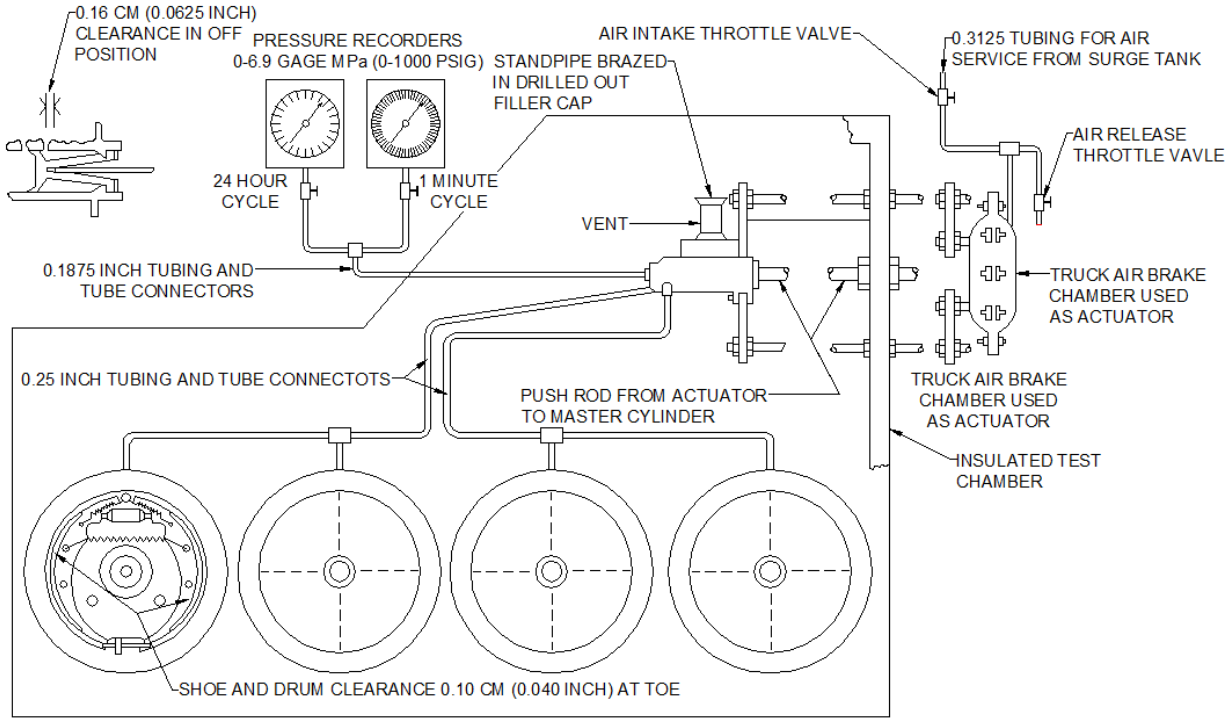


FIGURE 1. Diagrammatic sketch of suitable apparatus.

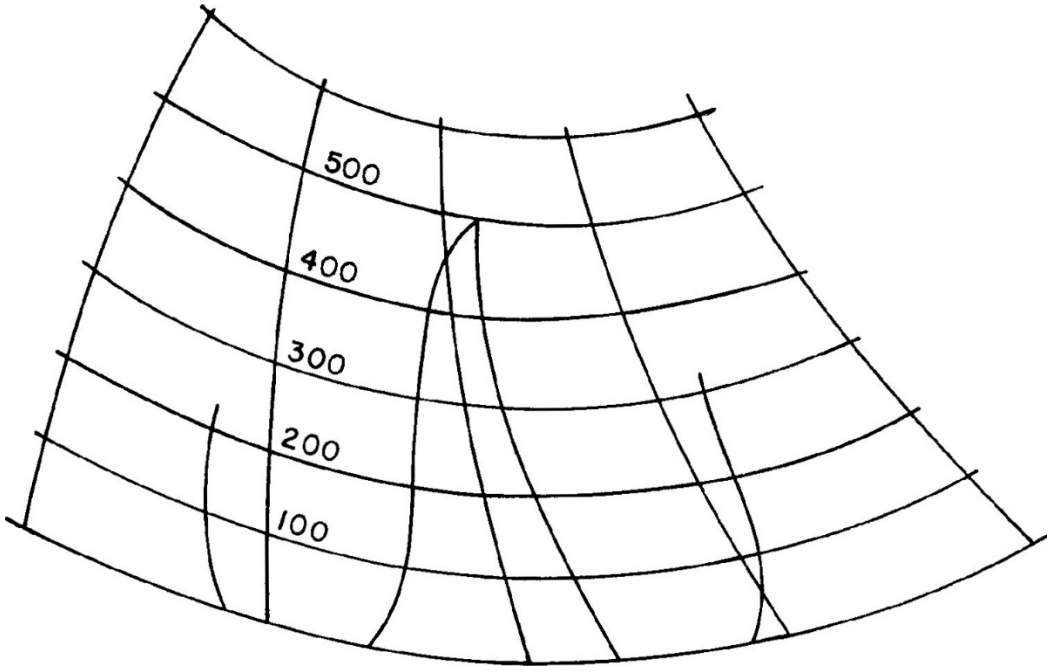


FIGURE 2. Pressure-stroke pattern 60/40 cam, 17 strokes per minute, chart speed- 1 rpm.

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30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	5	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	5	Army - AT

Method Prepared By:
Army - ME - 1986

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Method 1151.3
September 30, 1986

1151.3 – EXPLOSIVE VAPORS IN BOILER FUEL OIL

1. SCOPE

1.1 This method is used for determining the approximate percent explosiveness of gas and vapor mixture given off by fuel oil.

2. SUMMARY

2.1 The test consists of measuring the explosiveness of fuel-oil vapors produced under specified conditions, using a hot-wire type combustible-gas indicator that has been calibrated against a hydrocarbon gas of known explosiveness.

3. SAMPLE SIZE

3.1 Approximately 1 L (1 qt) of oil representative of the lot to be tested.

CAUTION: The one sample is to have been immediately stoppered (allowing a minimum of air space) upon removal from the lot.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Indicator, combustible-gas, hot-wire (Explosimeter Model No. 2, Mine Safety Appliances Co., Pittsburgh, Pa., or equivalent).

4.2 Flask, volumetric, 2 L, with 2-hole rubber stopper.

4.3 Graduated cylinder, 100 mL.

4.4 Tank, water (deep enough to accommodate 2 L volumetric flask and 100 mL graduated cylinder.).

4.5 Tube, drying, containing anhydrous calcium chloride.

4.6 Bulb, aspirator.

4.7 Vapor generating apparatus (see FIGURE 1), consisting of:

a. Bottles (2), glass, 1 qt, wide mouth.
b. Stopper, rubber, 2-hole (for 1 qt bottle).
c. Stopper, rubber, 3-hole (for 1 qt bottle).
d. Thermometer, 40° to 82°C (100° to 180°F) (ASTM E1).

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e. Tube, outlet, copper, approximately 3/16 inch O.D., sealed at one end with approximately 1/8 inch hole, 1/2 in from sealed end.
f. Tubing, glass and rubber, as required.
g. Clamp, screw (for rubber tubing). Bath heating, water, 66° to 69°C (150° to 155°F).

4.9 Compressed-air supply.

5. MATERIALS

5.1 Gas, hydrocarbon, with known lower limit of explosiveness (in volume percentage of gas concentration).

6. PROCEDURE

6.1 Prepare the 2-hole rubber stopper (for the volumetric flask) by inserting a short length of glass tubing in one hole and the sealed end of the copper outlet tube, used in FIGURE 1, in the other hole. Position the outlet tube so that the stopper seals the side opening of the tube; seal the outside end of the glass tube with a short length of rubber tubing and a screw clamp.

6.2 Prepare an air-gas mixture of 50% explosiveness (for use in calibrating the combustible-gas indicator), as follows:

a. Compute the volume of hydrocarbon gas (see. 5.2) required to produce a 2 L air-gas mixture of 50-percent explosiveness.
b. Place approximately 500 mL of water in the 2 L volumetric flask, and invert in the water tank.
c. Measure the required amount of hydrocarbon gas, using an inverted water-filled 100-mL graduated cylinder supported in the water tank so it will contain the computed volume of gas when the water levels in the graduate and the tank are the same.
d. Transfer the gas to the volumetric flask by carefully raising the lip of the graduate beneath the neck of the submerged flask.
e. Add air to the volumetric flask until the water levels (inside and outside) are at the graduation mark.
f. Insert the prepared 2-hole stopper (see 6.1) into the volumetric flask taking care to maintain the water seal of the gas mixture. Remove the flask from the tank and invert several times.

6.3 Calibrate the combustible-gas indicator by passing the air-gas mixture through the indicator and noting the reading as follows:

a. Fill one of the quart bottles about half full of water, and insert its 2 hole stopper (and tubing) as indicated for bottle A of FIGURE 1
b. Connect the water-filled bottle to the rubber tubing on the volumetric flask.
c. Connect the copper outlet tube to the drying tube, and the drying tube to the combustible-gas indicator. Adjust the indicator needle to zero.

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d. Reposition the copper outlet tube so that its side opening is exposed to the air-gas mixture, and apply low-pressure air to the water-filled bottle.
e. Adjust the screw clamp to produce a uniform flow of water of approximately 500 mL in 20 seconds, and record reading at which the indicator becomes stable.

6.4 Prepare the sample as follows:

a. Fill one of the quart bottles (bottle B) half full of test sample.
b. Insert the 3-hole stopper with its inlet tube, outlet tube, and thermometer. Position the outlet tube so that its side opening is covered by the stopper. Seal the inlet tube with a short length of rubber tubing and a screw clamp. Position the thermometer so that its bulb is approximately centered in the test sample.
c. Immerse the bottle (containing the sample) in the heating bath so that the level of the sample is below the level of the bath, and allow the sample to reach approximately 52°C (125°F). During the immersion (approximately 5 min) shake occasionally to ensure even heating.

6.5 Test the explosiveness of the gas and vapor mixture of the sample as follows:

a. When the temperature of the sample reaches 52°C (125°F), remove the bottle and shake vigorously for a total of 5 minutes, with periodic immersions of the bottle in the water bath to maintain the fuel oil temperature at 52°C (125°F).
b. Fill bottle A approximately one-half full of water at about 52°C (125°F) and as soon as bottle B has been shaken for 5 minutes, connect the two bottles as shown in FIGURE 1.
c. Connect the outlet tube to the combustible-gas indicator, zero the indicator needle, and push the copper outlet tube down to expose its opening to the vapors in the bottle.
d. Apply low-pressure air to bottle A. Then while observing the indicator needle, slowly open and adjust the screw clamp to permit the water to flow into the sample-filled bottle at a uniform rate over a period of approximately 20 seconds. When the reading becomes constant, stop the flow, and record the reading.

CAUTION: Do not allow fuel oil to flow into the combustible-gas indicator.

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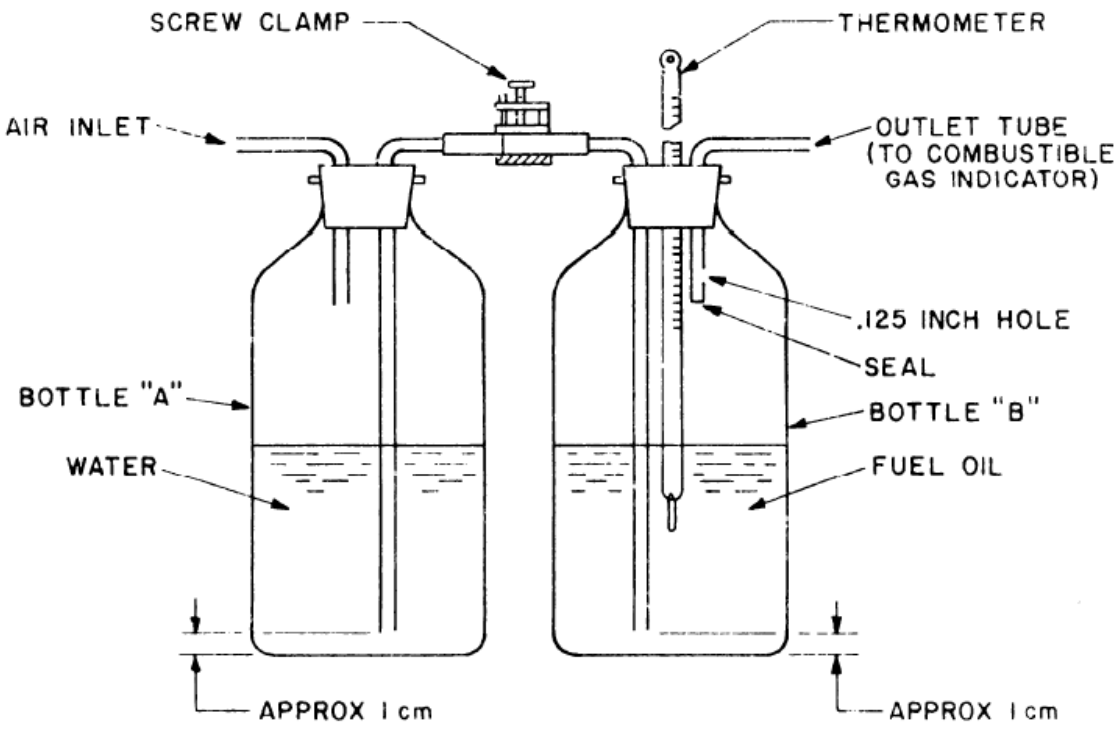


FIGURE 1. Vapor generating apparatus.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Report the sample as having passed if its combustible-gas indicator reading does not exceed the reading for the hydrocarbon gas.

9. PRECISION

9.1 Precision data have not been developed for this method.

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Method Prepared By:
Navy - SH - 1986

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Method 1203.0
December 18, 2025

1203.0 – DETERMINATION OF PARTICULATE MATTER IN AVIATION AND DISTILLATE FUELS BY AUTOMATIC PARTICLE COUNTING

1. SCOPE

1.1 This method describes the determination of particulate matter and water droplets in aviation turbine, aviation gasoline, diesel, and biodiesel blend fuels for ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c) particle counting channels via on-line and container sampling methods with an automatic particle counter system operating on the light extinction principle and calibrated to ISO 11171 *Hydraulic fluid power – Calibration of automatic particle counters for liquids*. Results are expressed by ISO Codes as specified by ISO 4406, *Hydraulic fluid power — Fluids — Method for coding the level of contamination by solid particles*. In this method, the ISO code is written in the form of w/x/y/z, where w, x, y, and z are the scale numbers for particle sizes ≥4 µm(c), ≥6µm(c), ≥14 µm(c), and ≥30 µm(c), respectively, as defined in ISO 4406. This document when cited in product specifications becomes a mandatory test requirement.

2. SUMMARY

2.1 There are two procedures provided in this automatic particle counting method, an on-line sampling method and a container sampling method.

2.2 In the on-line sampling method, the automatic particle counter sampling line is connected directly to the sample port of a flowing fuel stream, the fuel is sampled through automatic particle counter to determine the number of particles and size distribution by the light extinction/light obscuration principle.

2.3 In the container sampling method, an aliquot from a sample is collected in a clean sample container suitable for particle count analysis. The sample container is tumbled and sampled into an automatic particle counter to determine the number of particles and size distribution by the light extinction/light obscuration principle.

3. SAMPLE SIZES

3.1 The sample size for the on-line methodology is measured by the automatic particle counter and consists of a 30 mL flush of the optical measurement cell and plumbing followed by a 10 mL test sample, followed by a second 30 mL flush and second 10 mL test sample. A third 30 mL flush may be performed followed by a third 10 mL test sample may be required.

3.2 Minimum sample size for the container sampling method analysis is 125 mL.

4. APPARATUS

4.1 An automatic particle counter as defined in ISO 11171 operating on the laser light

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obscuration principle calibrated to ISO 11171 with a minimum of four channels, to include ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c), and a liquid light extinction/light obscuration sensor suitable for counting particles in the desired range particle size distribution.

NOTE: The sole sources of the apparatus qualified to this method at this time are the SETA-AvCount3 and AvCount Lite particle counters, available from Stanhope-Seta, London Street, Chertsey, Surrey KT16 8AP UK, and the JF-PC Automatic Particle Counter from D-2 Incorporated, 6 Otis Park Drive, Bourne, MA 02532. Alternate apparatus shall demonstrate equivalence through the procedure established in ASTM D6708, *Standard Practice for Statistical Assessment and Improvement of Expected Agreement Between Two Test Methods that Purport to Measure the Same Property of a Material*, with no bias-correction being considered to further improve the agreement between results between manufacturers.

4.2 *Filter Apparatus,* general purpose for filtering heptane or other solvents.

4.2.1 *Filters,* cellulose, glass fiber, or polycarbonate 0.45 µm.

4.2.2 *Sample container*, Epoxy-lined steel is the preferred container configuration for holding fuel sample to be tested. See ASTM D4306 *Standard Practice for Aviation Fuel Sample Containers for Tests Affected by Trace Contamination* for further discussion.

4.2.3 *Waste container*, for collecting tested sample.

5. MATERIALS

5.1 Solvent dispenser – Suitable dispensers fitted with a ≤0.45 µm Mixed Cellulose Ester (MCE) membrane filter (Millipore HAWP 047 or equivalent) at the outlet. If solvent has been pre-filtered using a ≤ 0.45 µm size filter, then an inline filter is not required.

5.2 Particle-clean solvent – Heptane (reagent grade) filtered through a ≤ 0.45 µm MCE (Millipore HAWP 047 or equivalent) membrane filter prior to use.

5.3 Calibration fluids – fluid conforming to NIST standard reference material 2806x for particle sizes 30 µm and smaller, where x is the letter used by NIST to designate the batch number of the certified primary calibration suspension.

5.4 Verification fluid – fluid identified by the instrument manufacturer to verify instrument performance between required calibrations.

5.5 Clean jet fuel – any jet fuel with an ISO cleanliness code of 15/13/9/4 or lower, per ISO 4406 as amended in paragraph 8.1.

5.6 Dust-free laboratory wipes.

6. CALIBRATION AND VERIFICATION:

6.1 The instrument shall be calibrated with NIST traceable calibration standards in

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accordance with ISO 11171.

6.2 The automatic particle counter shall be calibrated according to ISO 11171 a minimum of every 60 months, upon failure to meet the requirements of a verification sample, after any repairs that have a potential of affecting the calibration, or any time there is a reason to question the accuracy of test results.

6.3 Verify the correct operation of the automatic particle counter at least every 12 months by using the verification fluid. The result obtained shall be equal to or less than $R/\sqrt{2}$ of the measurement plus the uncertainty of the verification fluid, from the certified ≥4 µm(c) value of the verification fluid, where R is the reproducibility of the test; see Table I. If the result obtained is not within this value, ensure the sample preparation is in accordance with the manufacturer's instructions, check the verification fluid's validity date, and run a further test using the filtered heptane to confirm that the inlet tube and cell assembly are free from contaminants; then repeat the verification. If the result is still not within the allowed tolerance, contact the manufacturer.

7. PROCEDURE

7.1 On-line sampling procedure

7.1.1 Ensure the automatic particle counter is set up.

7.1.2 Connect the sample and waste hoses to the particle counter and the sample test port and waste container/port per manufacturer's instructions.

7.1.3 Start the test sequence in accordance with manufacturer's instructions.

7.1.4 The flow rate of the test sample shall be fixed and not adjustable by the operator, and shall be the same for testing, verification, and calibration.

7.1.5 The pump draws 30 mL of the test sample through the sample inlet and directly into the automatic particle counter to flush and clean. The first 10 mL test sample is then drawn into the particle counter for measurement. Record the measurements which shall be provided in counts per mL.

7.1.6 Immediately, starting within approximately 15 seconds of the completion of the first test measurement, repeat the 30 mL flush and a second 10 mL test measurement. Record the first and second test measurements in counts per mL.

7.1.7 Record the particle count measurements as the average of the counts obtained from 7.1.5 and 7.1.6 for the ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c) channel bands per mL.

7.1.8 If time permits, analyze the ≥4 µm(c) results from 7.1.5 and 7.1.6. Average the two ≥4 µm(c) results. If the difference between the two results is less than 10% of their average or less than 200 particles per mL, the result of the test is the average of the particle count for ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c) channel bands per mL. If the difference is equal to or

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greater than 10% of their average and greater than or equal to 200 particles per mL then repeat the 30 mL flush and complete a third 10 mL test sample measurement.

7.1.8.1 Record the particle count measurements as the average of the counts obtained from test sample 2 (see 7.1.6) and test sample 3 (see 7.1.8) for the ≥4 μm(c), ≥6 μm(c), ≥14 μm(c), and ≥30 μm(c) channel bands per mL.

7.1.9 Follow the manufacturer's instructions regarding procedures when switching off the automatic particle counter. Flush the automatic particle counter with filtered solvent or clean jet fuel as needed (e.g., following contaminated samples where there may be a possibility for carry-over contamination), and check for cleanliness according to paragraph.

7.2 Container sampling analysis

NOTE: Samples containing visible particulate matter and/or water shall not be analyzed.

7.2.1 Unless otherwise specified, take a sample of at least 125 mL in accordance with ASTM D4057 *Standard Practice for Manual Sampling of Petroleum and Petroleum Products*, ASTM D5854 *Standard Practice for Mixing and Handling of Liquid Samples of Petroleum and Petroleum Products*, or other comparable sampling practices.

7.2.2 Before commencing a test, tumble the test specimen for a minimum of 1 minute and then allow approximately 60 seconds before immediately starting the test. If the test is not started within approximately 90 seconds after completion of tumbling, gently re-tumble the test specimen for an additional minute.

7.2.3 Consistent disbursement particulates in the fuel sample can be achieved by either: (a) tumbling the sample container by hand end over end for at approximately 60 revolutions per minute, or (b) inverting the test specimen container back and forth approximately 60 times per minute.

7.2.4 It is permissible to test the sample directly from the sample container if the sample container, container closure and sample volume allow the test specimen to be drawn by the automatic particle counter, and the sample has been tumbled as described above.
NOTE: Testing directly from the sample container reduces the possibility of introducing particle contamination into the test specimen. However, the remaining sample could then be unsuitable for carrying out other types of test methods due to possible cross contamination.

7.2.5 Clean the outside of the sample hose before each test sequence by washing the outside in particle-clean solvent (see 5.2) and drying with dust-free laboratory wipes. Connect the clean sample hose to the particle counter and the waste hose per manufacturer's instructions.

7.2.6 Ensure the cleaned test sample hose is sufficiently below the level of the fuel to allow enough fuel to be used in the test sequences. Ensure the hose end is not touching the sides of the sample container and is not on the bottom of the sample container.

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7.2.7 Start the test sequence in accordance with manufacturer's instructions.

7.2.8 The flow rate of the test sample shall be fixed and not adjustable by the operator, and shall be the same for testing, verification, and calibration.

7.2.9 The pump draws 30 mL of the test sample through the sample hose and directly into the automatic particle counter to flush and clean. The first 10 mL test sample is then drawn into the optical measurement cell for measurement. Record the measurements that shall be provided in counts per mL.

7.2.10 Immediately, starting within approximately 15 seconds of the completion of the first test measurement, repeat the 30 mL flush and a second 10 mL test sample measurement. Record the measurements which shall be per mL.

7.2.11 Record the particle count measurements as the average of the counts obtained from 7.2.9 and 7.2.10 for the ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c) channel bands per mL.

7.2.12 If time permits analyze the ≥4 µm(c) results from 7.2.9 and 7.2.10. Average the two ≥4 µm(c) results. If the difference between the two results is less than 10% of their average or less than 200 particles per mL, the result of the test is the average of the particle count for ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c) channel bands per mL obtained. If the difference is equal to or greater than 10% of their average and greater than or equal to 200 particles per mL, then repeat the 30 mL flush and complete a third 10 mL test sample measurement.

7.2.12.1 Record the particle count measurements as the average of the counts obtained from second 10 mL test sample measurement (see 7.2.10) and the third 10 mL test sample measurement (see 7.2.12) for the ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c) channel bands per mL.

7.2.13 Follow the manufacturer's instructions regarding procedures when switching off the automatic particle counter. Flush the automatic particle counter with filtered solvent or clean jet fuel as needed (e.g., following contaminated samples where there may be a possibility for carry-over contamination).

8. REPORTING

8.1 Report ISO code. In this test method, the ISO code is written in the form of w/x/y/z, where w, x, y, and z are the scale numbers for particle sizes ≥4 µm(c), ≥6µm(c), ≥14 µm(c), and ≥30 µm(c) channels, respectively, as defined in ISO 4406.

NOTE: ISO 4406 specifies 3 scale numbers however for reporting purposes this method uses a 4th scale number to indicate the possible presence of free water at the ≥30 µm(c) channel.

8.2 Report particle size cumulative count for ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c); per mL, per user defined requirements.

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9. PRECISION

9.1 General – The on-line precision values given in Table II were derived from a 2024 interlaboratory study (ILS) that used seven instruments and individual operators at a single location to test 11 samples in duplicate and in random order. The container sample precision values given in Table I were derived from a 2024 ILS that used seven instruments and individual operators at a single location to test 19 samples in duplicate and in random order.

NOTE: ILS research report, ASTM D7619, *Standard Test Method for Sizing and Counting Particles in Light and Middle Distillate Fuels, by Automatic Particle Counter,* Equivalence of the D-2 Incorporated JF-PC-N, is available from the Defense Technical Information Center (DTIC). The Accession Document Number is AD1340295.

9.1.1 As the precision was determined from results obtained at a single location, the Reproducibility may not be comparable when results obtained at different times and locations are compared, due to sampling, shipping, storage, and environmental conditions.

9.1.2 The precision was obtained by statistical examination of the laboratory study test results according to ASTM D6300 *Standard Practice for Determination of Precision and Bias Data for Use in Test Methods for Petroleum Products, Liquid Fuels, and Lubricants*.

9.1.3 A single site ILS was carried out as automatic particle counters are unable to differentiate between particulate matter and free water droplets. Free water droplets are fuel solubility and temperature dependent therefore the amount of free water droplets in fuel can change during transport due to temperature and handling.

9.1.4 Repeatability – The difference between successive test results obtained by the same operator with the same apparatus under constant operating conditions on nominally identical test material would, in the normal and correct operation of the test method, exceed the value below only in one case in 20. Table II shows the repeatability and applicable ranges for particle sizes (cumulative count) ≥4 µm(c), ≥6 µm(c), and ≥14 µm(c) for on-line samples. Table I shows the repeatability and applicable ranges for particle sizes (cumulative count) ≥4 µm(c), ≥6 µm(c), ≥14 µm(c) and ≥30 µm(c) for container samples.

9.1.5 Reproducibility, Single Site – The difference between two test results independently obtained by different operators using different apparatuses on nominally identical test material at the same location would, in the normal and correct operation of the test method, exceed the value below only in one case in 20. Table II shows the reproducibility and applicable ranges for particle sizes (cumulative count) ≥4 µm(c), ≥6 µm(c) and ≥14 µm(c) for on-line samples. Table I shows the reproducibility and applicable ranges for particle sizes (cumulative count) ≥4 µm(c), ≥6 µm(c), ≥14 µm(c), and ≥30 µm(c) for bottle samples.

NOTE: On-line samples analyzed for the interlaboratory study used to generate the repeatability and reproducibility of this method did not contain sufficient ≥30 µm(c) particles to calculate precision of this method. ≥30 µm(c) particles are still required to be reported.

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TABLE I. Container sample precision and range per 1 mL.

Parameter size band	Range of results	Repeatability, r	Reproducibility, R
≥4 µm(c)	360-14999	$0.5513*x^{0.7809}$	$0.4525*x^{0.9168}$
≥6 µm(c)	115-31321	$0.4771*x^{0.8451}$	$1.0649*x^{0.8451}$
≥14 µm(c)	20-2871	$0.1498*x^{1.0868}$	$0.3924*x^{1.0868}$
≥30 µm(c)	20-1592	$1.1746*x^{0.8928}$	$1.9265*x^{0.8583}$

Where: x = the average of the 2 results being compared.

TABLE II. On-line precision and range per 1 mL.

Parameter size band	Range of results	Repeatability, r	Reproducibility, R
≥4 µm(c)	44-3499	$1.1792*x^{0.6721}$	$3.4659*x^{0.6721}$
≥4 µm(c)	3500-25489	$0.1123*x^{0.9616}$	$3.4659*x^{0.6721}$
≥6 µm(c)	8-24820	$0.3498*x^{0.8498}$	$1.0796*x^{0.8498}$
≥14 µm(c)	1-2232	$0.5061*(x+0.0001)^{0.8854}$	$1.104*(x+0.0001)^{0.8854}$

Method Prepared By:
Army - AT - 2025

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Method 3005.5
September 30, 2019

3005.5 – DIRT CONTENT OF GREASE

1. SCOPE

1.1 This method is used for determining the size and concentration of foreign particles in lubricating greases.

2. SUMMARY

2.1 The test consists of applying a known quantity of grease to a microscope slide, and examining the slide with a microscope to determine the size and number of particles present.

3. SAMPLE SIZE

3.1 Approximately l mL of grease to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Microscope, mechanical stage, 60 diameter magnification, with eyepiece micrometer (not filar) calibrated against a suitably ruled, stage micrometer.

4.2 Slide, microscope, approximately 2.5 cm ± 5.0 cm with cover glass.

4.3 Template (see FIGURE 1), steel (shim stock), 0.01 cm thick, approximately 2.5 to 5.0 cm, slotted from one end for a distance of approximately 2.0 cm with a slot width of approximately 1.0 mm. The template thickness and slot width are accurately measured so that the end area of the slot can be calculated. (A hemacytometer may be used in place of the template). See Note 1.

NOTE 1: Calculation of test results will be simplified if the template is ground to such thickness, or the slot width is adjusted so that the end area of the slot is 0.01 cm^2.

4.4 Vacuum chamber, 1.0 cm of mercury (absolute), or lower (to accommodate microscope slide).

4.5 Spatulas (2)

5. MATERIALS

5.1 The grease to be tested

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6. PROCEDURE

6.1 Compute the cross sectional area of the end of the slot (see 7.1).

6.2 Clean template, microscope slide, cover glass, and spatulas.

6.3 Prepare the sample for examination as follows:

a. Open a fresh container of grease to be tested, and with a clean spatula, scrape off and discard the surface layer.
b. Hold the template by the un-slotted end, and place it against the clean microscope slide. Using another clean spatula fill the slot with the test grease. Make sure the corners are filled and the surface of the grease is level with, or slightly above the sides of the slot. Then press the side of the slot firmly against the slide.
c. Place the slide and template in the vacuum chamber for 10 to 15 minutes to remove bubbles or voids from the grease in the slot. (See Note 2).

NOTE 2: Vacuum exposure may be omitted if the grease is free from entrained air.

d. Slide the clean cover glass onto the template to shear off surplus grease. Then press the cover glass firmly against the template until it is parallel to the microscope slide.

6.4 Mount the assembled slide on the stage of the microscope. Adjust the illumination so that light passes straight up through the grease in the slot, as well as straight across the surface of the grease.

6.5 Determine the size and number of opaque particles.

a. Adjust the stage and micrometer eyepiece so that the micrometer scale coincides with one side of the slot, and so that one end of the scale coincides with one corner of the slot.
b. Move the stage slowly, so that the slot passes across the field of view. Count each opaque inclusion having a size of 25 µm or more. Particle size is determined by measuring along the largest dimension of the particle.
c. Record the count in three columns headed: "25 to 75 µm", "75 to 125 µm", and "over 125 µm".
d. When the micrometer scale has reached the opposite side of the slot, move the stage along the axis of the slot for a distance equal to the length of the scale. (Use a minor irregularity on the wall of the slot, or a specific particle in the grease, as a point of reference.) Scan this section completely across the slot.
e. Continue scanning successive sections in the slot until approximately three-quarters of the area (1 to 1.5 cm) of the slot has been inspected.
f. Record the number of sections scanned, and the total number of particles according to size.

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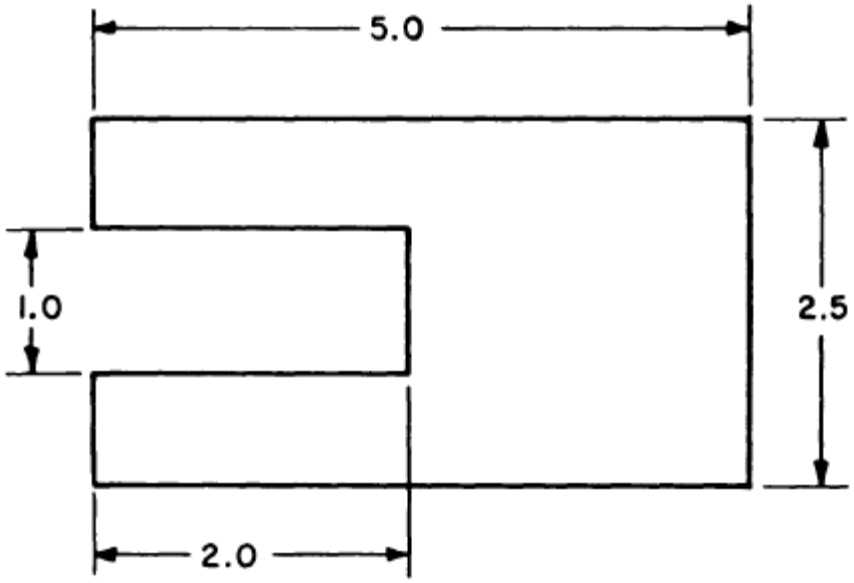


FIGURE 1. Template.

7. CALCULATIONS

7.1 Calculation of slot cross sectional area.

$$T = D - W$$

where:
T = end area of template slot.
D = depth or thickness of template (cm).
W = width of slot (cm).

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7.2 Compute the number of particles per cubic centimeter of grease, using the following formulas:

25 µm or larger:

$$\frac{(A + B + C)}{TSN}$$

75 µm or larger:

$$\frac{(B + C)}{TSN}$$

Over 125 µm:

$$\frac{C}{TSN}$$

where:
A = total number of particles 25 to 75 µm.
B = total number of particles 75 to 125 µm.
C = total number of particles 125 µm or larger.
N = number of sections of grease scanned.
S = length of micrometer scale, cm.
T = end area of template slot, sq cm.

NOTE 3: If dirt in excess of specification limits is not found in the first run, the grease shall meet the requirements of the test.

7.3 If dirt in excess of the specification limits is found in first run, two more runs shall be conducted. If dirt in excess of specification limits is found in one or both of these runs, the grease shall fail the test.

8. REPORTING

8.1 Report the number of particles in the following ranges:

25-75 µm
75-125 µm
125 or larger µm

9. PRECISION

9.1 Precision data have not been developed for this method.

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	5	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	5	Army - AT

Method Prepared By:
Navy - NADC - 1986

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Method 3007.3
September 30, 2019

3007.3 – WATER DISPLACEMENT AND WATER STABILITY

1. SCOPE

1.1 This method is used to determine the water displacing ability of corrosion preventive oils and their stability in the presence of water.

2. SUMMARY

2.1 Steel test panels are immersed in distilled water and then either the test oil or a water emulsion of the test oil. After exposure in a static humidity chamber the panels are examined for the presence of rust.

3. SAMPLE SIZE

3.1 Approximately 60 mL of the test oil.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Standards

4.1.1 ASTM D1748, Rust Protection by Metal Preservative In The Humidity Cabinet.

4.2 Static humidity chamber (a desiccator containing some distilled water).

4.3 Oven, forced draft, 54°C ± 1°C (130°F ± 2°F).

4.4 Desiccator, containing an indicating desiccant.

4.5 Panel hook, stainless steel (for handling panels during processing).

4.6 Erlenmeyer flasks, 125 mL.

4.7 Beakers, tall-form, 500 mL.

4.8 Petri dishes, 10 cm in diameter and 2 cm in height.

4.9 Controlled atmosphere (to provide 25°C ± 3°C [77°F ± 5°F]) and a maximum of 50 percent relative humidity.

4.10 Shaded fluorescent light. (A 15W balance illuminator type that will permit the panel to be viewed from all angles at a distance of 3 inch).

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5. MATERIALS

5.1 Test panels, FS 1009 steel, 5.1 cm by 7.6 cm by 0.16 cm (2.0 inch by 3.0 inch by l/16 inch), fabricated from open hearth "killed-type low carbon, No. 4 soft temper cold rolled sheet or strip conforming to ASTM A109/A109M. Cold drawn bar stock is not satisfactory.

5.3 Heptane.

5.4 Isopropyl alcohol.

5.5 Distilled water.

5.6 Silica sand, white dry, sharp. The size shall be such that it meets the following sieve requirements ASTM E11:

a. 100% must pass through a No. 10 sieve.
b. Minimum of 90% must pass through a No. 20 sieve.
c. Maximum of 10% permitted to pass through a No. 50 sieve.

5.7 Surgical gauze.

5.8 Absorbent tissue.

5.9 Emulsion, water and oil prepared as follows:

a. Place 50 mL of the test sample and 5 mL of distilled water in a 125 mL Erlenmeyer flask.
b. Shake mixture thoroughly and store overnight in an oven a 54°C ± 1°C (130°F ± 2°F).
c. Remove from oven and cool at 25°C ± 3°C (77°F ± 5°F) for testing.
d. Shake the mixture immediately prior to placing in the petri dish.

6. PROCEDURE

6.1 Clean the steel panels as follows:

a. Round the edges and ream out the suspension holes in accordance with ASTM D1748.
b. Wipe the surfaces with solvent soaked rags and scrub with surgical gauze swabs in a beaker of hot naphtha.
c. Rinse in a beaker of hot methyl alcohol.
d. Cool the panels in a desiccator until further processing.
e. Test panels must be handled with a panel hook at all times, avoiding contact with any type of contaminated surface.
f. Maintain the solvents at a temperature high enough to keep the temperature of the panels above the dew point during the cleaning operation.

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Sandblast the panels as follows:

a. Sandblast the edges and lightly sandblast the back of the panels with silica sand.
b. Sandblast the unnumbered side, or test surface, of the panels to a fresh, uniformly abraded surface.
c. Immediately after sandblasting, place the panels in a beaker of anhydrous methyl alcohol.
d. Heat the methyl alcohol so that the solvent will evaporate from the panels immediately upon withdrawal from the solvent.
e. Remove remaining residue by holding the panels in a rack at 20° from the vertical and spraying downward with naphtha.
f. Spray the test surface, then the back of the panel and the test surface again.
g. Rinse the panels in hot naphtha and hot methyl alcohol and store in a desiccator until cool.
h. Panels are to be used the same day as prepared.

NOTE 1: CAUTION. Methyl alcohol and naphtha are both toxic and flammable. Do not permit them to come in contact with the skin or breathe their fumes. Use only in a well ventilated area. Keep all flames away from these materials.

6.3 Three test panels are processed with the original oil as follows:

a. Dip the panels into distilled water, momentarily, and drain in a vertical position for not more than 5 seconds, with the bottom edge in contact with absorbent tissue.
b. Immerse the panels horizontally, and without agitation, for 15 seconds in a petri dish containing 50 mL of the test sample maintained at 25°C ± 3°C (7.70°F ± 5°F) and 50% maximum relative humidity.
c. Drain the panels momentarily and place in the static humidity chamber for 1 hour at 25°C ± 3°C (77°F ± 5°F).
d. At the completion of the static humidity exposure, remove the methanol.
e. Examine the test surfaces for the presence of rust, mottling, or other abnormal surface stains. The outer 0.32 cm (1/8 inch) edge of the panels and 0.32 cm (1/8 inch) around the holes is excluded from the evaluation.

6.4 Three test panels are processed with the oil-water emulsion in accordance with the procedure outlined in paragraph 6.3 and similarly examined.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Follow the specification requirements.

9. PRECISION

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9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Updated materials list: added heptane, removed naphtha, changed methyl alcohol to Isopropyl alcohol	Updates more accurately reflect actual lab practices.	3	Army - AT
30 Apr 2026	Replaced QQ-S-698 with A109/A109M -	QQ-S-698 was cancelled	3	Army - AT

Method Prepared By:
Army - AR – 1986

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Method 3010.2
September 30, 2019

3010.2 – SOLID PARTICLE CONTAMINATION IN AIRCRAFT TURBINE ENGINE LUBRICANTS (GRAVIMETRIC PROCEDURE)

1. SCOPE

1.1 This method describes a gravimetric procedure for the determination of solid particle contamination in packaged aircraft turbine lubricants and related fluids.

2. SUMMARY

2.1 A known volume of fluid is filtered through a pre-weighed membrane filter. The increase in filter weight is determined after washing and drying and reported as total contaminant. The test filter is ashed and the inorganic material remaining is reported as ash.

3. SAMPLE SIZE

3.1 Approximately 1 L (1 qt) of turbine engine lubricant. Less volume can be used but the total volume filtered should be measured.

3.2 The sample must be thoroughly representative of the material to be tested. This requires vigorous agitation of the fluid before sampling. The importance of sampling cannot be too strongly emphasized. All suspected sediment must be homogeneously suspended in the oil. When packaged liter (quart) samples are available the entire liter (quart) should be used.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Membrane filters, white, plain, 4.7 cm diameter, pore sizes 1.2 and 0.22 µm.

4.2 Filter holder, 4.7 cm.

4.3 Analytical balance, sensitivity ± 0.1 mg.

4.4 Petri dishes, 5.0 cm in diameter.

4.5 Oven, non-circulatory air, capable of maintaining a temperature of 90 °C ± 2 °C (194 °F ± 3.6 °F)

4.6 Forceps, flat-bladed with unserrated tips.

4.7 Vacuum system.

4.8 Muffle furnace capable of maintaining a temperature of 810 °C ± 5 °C (1490 °F ± 9 °F).

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4.9 Crucibles, 30 mL capacity, squat form.

4.10 Desiccator, with desiccant.

5. MATERIALS

5.1 Isopropyl alcohol, reagent grade.

5.2 Petroleum ether, reagent grade, boiling point range 30° to 60°C.

5.3 Detergent, liquid, water soluble

6. PROCEDURE

6.1 Preparation of apparatus and containers.

6.1.1 All components of the filtration apparatus and containers must be washed with hot tap water containing detergents.

6.1.2 Rinse with hot tap water.

6.1.3 Rinse with distilled water.

6.1.4 Rinse with filtered isopropyl alcohol.

6.1.5 Rinse with filtered petroleum ether.

6.1.6 Place clean plastic film or clean aluminum foil over containers and filtration apparatus.

NOTE 1: Rinse a sufficient number of times to completely remove the previous solvent.

6.2 Preparation of reagents.

6.2.1 Filter the petroleum ether, and isopropyl alcohol, through a 0.22µm membrane filter. Discard the first 50 mL of each and filter enough solvents to complete testing of samples to be filtered. Containers used for filtered reagents must be cleaned initially as described in Section 6.1.

NOTE 2: Ground the filter mechanism when filtering flammable solvents to prevent buildup of static electricity.

6.3 All equipment used for handling samples must be cleaned as described in Section 6.1. Precaution must be taken to minimize apparatus contamination from all sources including airborne particles.

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6.3.1 Duplicate tests should be performed whenever more than 1 L (1 qt) of lubricant from the same lot or batch is available.

6.4 Preparation of test filters.

6.4.1 Identify each filter by marking the rim with a ballpoint pen, or by marking the petri dishes that will contain them. Place each filter separately in petri dish and dry it for 1/2 h in an oven regulated to 90 °C ± 2 °C (194 °F ± 3.6 °F). The petri dishes are then placed in a desiccator and moved to the balance room in the vicinity of a balance on which weighing's are to be made. Allow 1 hour for the closed desiccator and contents to come to equilibrium with ambient temperature. Use forceps to place the test membrane on the balance pan by handling the edge only.

6.4.2 An air ionizer must be kept permanently on the base of the balance adjacent to the pan to eliminate weighing errors due to electrostatic charge effects. Make all weighing's to the nearest 0.1 mg.

6.5 Laboratory filtration and contamination determination.

6.5.1 Thoroughly clean outside of sample container in the region to be opened by washing with detergent and water and rinsing with tap water and filtered isopropyl alcohol. Shake the sample of lubricant for 15 minutes. Cautiously open can so as to prevent the introduction of metal chips into the sample.

6.5.2 If using less than 1 qt of fluid, measure the amount of sample by volume to be filtered prior to pouring the sample through the filter membrane. This measurement will be used in the sediment calculation.

6.5.3 To keep airborne contaminants on the membrane surface to a minimum, pour some of the sample into the filter funnel before applying vacuum. Filter the entire contents of the sample and thoroughly wash down the inside of the sample can with filtered petroleum ether, passing the washings through the filter.

6.5.4 Wash down the inside of the filter funnel. This will complete the transfer of the contaminant to the filter membrane. With the vacuum applied, carefully remove the funnel. Wash the periphery of the membrane by directing a gentle stream of petroleum ether from the edge toward the center, taking care not to wash any contaminant from the surface of the membrane.

6.5.5 Carefully remove the test membrane from the filter base, using clean flat forceps, and place it in a covered petri dish. Dry and reweigh as described in Section 6.4, taking care not to wash any contaminant from the surface of the membrane.

NOTE 3: For greater accuracy, control membranes may be used as described in ASTM D2276 (Particulate Contaminant in Aviation Turbine Fuels).

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6.6 Determination of non-combustible contamination or ash.

6.6.1 Clean crucibles with hot detergent, wash and rinse with tap and distilled waters and dry. Place crucibles in a muffle furnace maintained at 810°C (1500°F) for 1/2 h. Remove and place crucibles in a desiccator. Allow closed desiccator and contents to cool and come to equilibrium with the atmosphere in the vicinity of the balance for 2 h. Handle crucibles with clean tongs. Weigh crucibles to the nearest 0.1 mg. Repeat this procedure until constant weight is obtained.

6.6.2 Handling the membrane filter only with clean forceps, lay it, contaminant side upwards, in the center of the crucible. Wet it with filtered isopropyl alcohol and push it gently into the crucible.

6.6.3 Cover the crucible containing the filter membrane with a porcelain lid and place it in a cold muffle furnace. Start the furnace and allow to come to 810 °C ± 5 °C (1490 °F ± 9 °F). Hold at this temperature for 1/2 hour. Remove and place crucibles in a desiccator. Allow closed desiccator and contents to cool and come to equilibrium with the atmosphere in the vicinity of the balance for 2 hours. Handle crucibles with clean tongs. Weigh crucibles to the nearest 0.1 mg. Repeat this procedure until constant weight is obtained. Increase in weight is inorganic material or ash.

NOTE: An alternate method is to place the crucible in a triangle and put a porcelain cover over it. Place a small flame under the crucible to one side. When dry, char the filter membrane, being careful not to ignite. Increase the burner flame until all organic matter is distilled from the filter. Place the crucible in a preheated muffle furnace 810 °C ± 5 °C (1490 °F ± 9 °F) for 1/2 h. Weigh as in 6.6.3.

7. CALCULATIONS

$$\textit{Contaminant, mg per L of oil} = \frac{A-B}{C}$$

where:
A = final weight of the membrane filter with the material collected in filtration, mg.
B = initial weight of the test membrane, mg.
C = volume of sample in L filtered through the test membrane.

$$\textit{Ash, mg per L of oil} = \frac{D-B}{C}$$

where:
D = final weight of the crucible and noncombustible material, ash, mg.
B = initial weight of the crucible, mg.
C = volume of sample in L filtered through the test membrane.

8. REPORTING

8.1 Contamination, mg per L of oil.

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8.2 Ash, mg per L of oil.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Added tolerances to oven temperature	Need a range for control	2	NAWCAD
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT
18 May 2021	Removed extra numbers from solvents in materials section	The additional numbers did not add information	2	NAWCAD

Method prepared by
Navy - NAWCAD - 2019

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3012.2 – DETERMINATION OF PARTICULATE MATTER IN AEROSPACE HYDRAULIC FLUIDS

1. SCOPE

1.1 This method describes the determination of particulate matter in aerospace hydraulic fluids with an automatic particle counter system able to be calibrated to ISO 11171. This document when cited in product specifications becomes a mandatory test requirement.

2. SUMMARY

2.1 An aliquot from a sample is collected in a clean bottle suitable for particle count analysis. The sample bottle is agitated and either placed into the automatic particle counter (APC) or directly connected to the sample inlet for the APC to determine the number of particles and size distribution.

3. SAMPLE SIZES

3.1 Minimum sample size for analysis is 120 mL (4 oz).

4. APPARATUS

4.1 Automatic particle counter with a minimum of five channels and a liquid sensor suitable for counting particles in the desired individual or range of sizes. The APC shall have a built-in pressure/vacuum chamber that can subject samples to pressures up to 4 bar (60 psi) minimum or have via external accessory a means of removing air bubbles from the sample via sonification. The APC shall also include an automatic bottle sampler that automatically delivers a predetermined volume of fluid at a controlled flow rate (sensor dependent).

5. MATERIALS

5.1 APC Sample bottles - Cylindrical flat-bottomed glass bottles fitted with a polypropylene threaded cap forming a seal with the bottle without the use of any insert, approximately 250 mL capacity or APC manufacturer's recommended size.

5.2 Solvent dispenser – Suitable dispensers fitted with a ≤0.45 µm Mixed Cellulose Ester (MCE) membrane filter (Millipore HAWP 047 or equivalent) at the outlet. If solvent has been pre-filtered using a ≤ 0.45 µm size filter, then an inline filter is not required.

5.3 Particle-clean solvent – Hexane or manufacturer specified solvent (Certified ACS) filtered through a ≤ 0.45 µm MCE (Millipore HAWP 047 or equivalent) membrane filter prior to use.

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5.4 Detergent – Micro-90 cleaning solution (available through Fisher Scientific, Sigma-Aldrich, and Cole- Parmer) or equivalent.

6. CALIBRATION

6.1 The instrument shall be calibrated with NIST traceable calibration standards in accordance with ISO 11171 *Hydraulic fluid power – Calibration of automatic particle counters for liquids*.

6.2 The APC shall be calibrated annually, after any repairs that have a potential of affecting the calibration, or at any time there is a reason to question the accuracy of test results.

7. PROCEDURE

7.1 SAMPLE BOTTLE PREPARATION

7.1.1 Sample bottle cleaning procedure:

7.1.1.1 Wash sample bottles and caps thoroughly with warm tap water containing detergent (reference paragraph 5.4).

7.1.1.2 Rinse with warm tap water.

7.1.1.3 Rinse with either distilled or deionized water.

7.1.1.4 Oven dry at 90 °C ± 5°C.

7.1.1.,5 Allow to cool at room temperature in a suitable contamination-free environment.

7.1.1.,6 Rinse with filtered solvent.

7.1.1.7 Drip dry.

7.1.1.8 Place cap on bottles and store at room temperature, in a suitable contamination-free environment.

7.1.2 Sample bottle cleanliness verification:

7.1.2.1 Select a representative sample bottle from a given batch of sample bottles cleaned per paragraph 7.1.1 .

7.1.2.2 Determine the cleanliness level of the selected clean sample bottle using ISO 3722 *Hydraulic fluid power – Fluid sample containers – Qualifying and controlling cleaning methods Clause 4 Cleanliness Determination Procedure*.

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7.1.2.2.1 Filtered solvent will be used to verify sample bottle cleanliness.

7.1.2.2.2 Use cumulative particle count method (vs. differential) on the APC.

7.1.2.3 The reported sample bottle cleanliness level shall have no more than 100 particles greater than 6 microns (c) in size per mL of sample bottle.

7.2 SAMPLE PREPARATION

7.2.1 Hermetically sealed containers.

7.2.1.1 Thoroughly clean the outside of the container, particularly the top where it will be punctured, with filtered solvent.

7.2.1.2 Shake the container vigorously for one minute to ensure adequate distribution of particles in the fluid.

7.2.1.3 Puncture the cleaned surface with a tool cleaned and rinsed with filtered solvent, making an opening that is no larger than necessary.

7.2.1.4 Transfer a mid-stream sample of at least 120 mL (or manufacturer's recommended volume) to an APC sample bottle and cap it.

7.2.2 Non-hermetically sealed containers.

7.2.2.1 Thoroughly clean the outside with filtered hexanes paying particular attention to the cap region.

7.2.2.2 Shake container vigorously for one minute to ensure adequate distribution of particles in the fluid.

7.2.2.3 Transfer a mid-stream sample of at least 120 mL (or manufacturer's recommended volume) to an APC sample bottle and cap it.

7.3 SET-UP PARAMETERS

7.3.1 For differential particle count, the particle counter will have the following set-up:

7.3.1.1 Flow rate – 60 mL ± 2 mL/min (or that recommended by manufacturer).

7.3.1.2 Vacuum rate –53.5 kPa (15.8 in Hg) minimum if using a build-in pressure/vacuum chamber for air bubble removal.

7.3.1.3 Sample volume – Minimum of two equal volumes of at least 20 mL each, run consecutively or minimum number of equal volumes based on APC manufacturer defined sample size requirement run consecutively to equal a minimum of 40 mL.

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7.3.1.4 Set particle counter to differential (vs. cumulative) mode.

7.3.2 For cumulative particle count, the particle counter will have the following set up:

7.3.2.1 Flow rate – 60 mL ± 2 mL/min (or that recommended by manufacturer).

7.3.2.2 Vacuum rate - 53.5 kPa (15.8 in Hg) minimum if using a build-in pressure/vacuum chamber for air bubble removal.

7.3.2.3 Sample Volume - Minimum of two equal volumes of at least 20 mL each, run consecutively or minimum number of equal volumes based on APC manufacturer defined sample size requirement run consecutively to equal a minimum of 40 mL.

7.3.2.4 Set particle counter to cumulative (vs. differential) mode.

7.3.3 Ensure that the particle counter is turned on at least 30 minutes prior to start the analysis or follow manufacturer's recommendation.

7.3.4 Prior to daily analysis, run a sample of filtered solvent to verify particle cleanliness level of the APC. For differential particle count, the particle counter must meet SAE AS4059 *Contamination Classification for Hydraulic Fluids* Contamination Level 00 (see TABLE I) limits. For cumulative particle count, the particle counter must meet SAE AS4059 Contamination Level 00 (see TABLE II).

TABLE I. Maximum differential particle contamination limits for the solvent.

Particle Size Ranges, µm [1]		Number of Particles / 100 mL (max)
SAE AS4059 [2]	SAE AS4059 [3]	
6 - 14 (c)	5 - 15	125
14 - 21 (c)	16 - 25	22
21 - 38 (c)	26 - 50	4
38 - 70 (c)	51 - 100	1
>70 (c)	>100	0

NOTES:

1 The symbol (c) is used to designate the APC was calibrated to ISO 11171.

2 Particle size ranges established based on the particle calibration sizes used in ISO 11171 (accomplished using ISO Medium Test Dust (MTD)).

3 Particle size ranges established based on the particle calibration sizes used in ISO 4402 (accomplished using Air Cleaner Fine Test Dust/Arizona Road Dust (ACTFD)).

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TABLE II. Maximum cumulative particle contamination limits for the solvent.

Particle Size, µm [1]		Number of Particles / 100 mL (max)
SAE AS4059 [2]	SAE AS4059 [3]	
>4 (c)	>1	390
>6 (c)	>5	152
>14 (c)	>15	27
>21 (c)	>25	5
>38 (c)	>50	1
>70 (c)	>100	0

NOTES:

1 The symbol (c) is used to designate the APC was calibrated to ISO 11171.

2 Particle size ranges established based on the particle calibration sizes used in ISO 11171 (accomplished using ISO Medium Test Dust (MTD)).

3 Particle size ranges established based on the particle calibration sizes used in ISO 4402 (accomplished using Air Cleaner Fine Test Dust/Arizona Road Dust (ACTFD)).

7.4 SAMPLE ANALYSIS

NOTE: Samples containing visible particulate matter and/or water shall not be analyzed.

7.4.1 Shake the sample bottle vigorously for one minute. Testing must be initiated within two minutes if using an APC with a build-in pressure/vacuum chamber. If treating the sample to a sonification bath for air bubble removal, testing must be initiated within two minutes of completion of the sonification treatment. Re-shake APC sample bottle vigorously for one minute if required.

7.4.1.1 If using an APC with a build-in pressure/vacuum chamber:

7.4.1.1.1 Place the APC sample bottle in the sample assembly and apply a vacuum of at least 53.5 kPa (15.8 in Hg) for 40 seconds to remove air bubbles.

7.4.1.1.2 Turn-off the vacuum source and immediately start particle count.

7.4.1.2 If using a sonification instrument before introducing the sample to the APC:

7.4.1.2.1 Place the APC sample bottle in a sonification bath and sonicate for two minutes.

7.4.1.2.2 Remove the APC sample bottle from the sonification bath and connect to the sample inlet side of the APC and immediately start particle count.

7.4.2 Flush the APC with a small volume of sample.

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7.4.3 Record count result per calibrated channel.

7.4.4 Flush the APC with filtered solvent as needed (e.g., following contaminated samples where there may be a possibility for carry-over contamination), and check for cleanliness according to paragraph 7.3.4.

8. CALCULATIONS

8.1 Average the particle count of the runs (discarding any statistically suspect data) per volume used per channel. If the particle count averaging is done automatically by the APC based on the manufacturers recommended number of particle count runs, verify the results of each run to ensure no statistically suspect data was used in determining the reported particle count.

9. REPORTING

9.1 Results will always be reported based on differential particle count unless another reporting format is requested by the sample submitter.

9.2 For differential particle count, report the number of particles obtained per 100 mL in the desired size ranges.

9.3 For cumulative particle count, report the number of particles obtained per 100 mL in the desired sizes.

9.4 To report results as an ISO Code, use reported results obtained following guidance in paragraphs 7.3 and 9.2 for cumulative particle count. Using the cumulative particle counts for the >4 µm (c), >6µm (c), and >14 µm (c) particle sizes, refer to Table 1 of ISO 4406, *Allocation of scale numbers,* to determine the corresponding Scale number. The reporting format for the ISO Code shall be IAW ISO 4406 and reported as X/Y/Z where X, Y, and Z are Scale numbers from Table 1 of ISO 4406 for particle counts in the >4 µm (c), >6µm (c), and >14 µm (c) ranges respectively.

10. PRECISION

10.1 Precision data has not been developed for this method.

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Complete Revision	ACFTD no longer available	1	Air Force - PTPS
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	1	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	1	Army - AT
17 Jun 2024	Complete Revision	Expand method to allow for alternative particle counters that may be used to perform the test method. Expand test method results reporting options to include differential, cumulative, or ISO Code options.	2	Air Force - PTPS

Method Prepared By:
PTPS – 2024

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Method 3013.2
September 30, 2019

3013.2 – DETERMINATION OF PARTICULATE CONTAMINATION IN SYNTHETIC TURBINE ENGINE LUBRICANTS

1. SCOPE

1.1 This method describes a gravimetric procedure for the determination of particulate contamination in synthetic turbine engine lubricating oils.

2. SUMMARY

2.1 A known volume of oil is filtered through a pre-weighed membrane filter. The increase in filter weight is determined after washing and drying and is considered particulate contaminant. Particulate contaminant per liter of oil is calculated from the volume of oil filtered and the weight of particulate contaminant.

3. SAMPLE

3.1 Approximately 1 L of the oil to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Membrane filters, 4.7 cm diameter, absolute pore size of 3 µm and 0.45 µm. Filters must be compatible with the required reagents and with polyolester lubricants at elevated temperatures. Millipore MF filters or equivalent shall be used.

4.2 Filter holder, 4.7 cm diameter.

4.3 Analytical balance, sensitivity ± 0.1 mg.

4.4 Petri dishes, 10.0 cm diameter with covers.

4.5 Oven, non-circulatory air, capable of maintaining temperature of 90°C.

4.6 Forceps, flat-bladed without serrated tips.

4.7 Vacuum system, 62.5 cm Hg or better.

4.8 Laboratory wash bottle.

5. MATERIALS

5.1 Isopropyl alcohol, reagent grade [1, 3, 0].

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5.2 Petroleum ether, 30°C to 60°C boiling range, reagent grade [1, 4, 0].

6. PROCEDURE

6.1 Sampling

6.1.1 The sample must be representative of the material to be tested. When samples represent packaged quart containers, the entire contents must be filtered. When samples represent packaged 8-oz containers, the entire contents of at least two, but no more than four, of the 8 oz containers must be filtered. When the material is packaged in containers larger than 1 qt, a representative 1 L (1 qt) sample must be taken in a clean container for the filtration test. The entire contents of these samples must be filtered.

6.2 Preparation of apparatus and containers.

6.2.1 Wash all components of the filtering apparatus with hot water and detergent.

6.2.2 Rinse with hot tap water.

6.2.3 Rinse with distilled water.

6.2.4 Rinse with filtered isopropyl alcohol.

6.2.5 Rinse with filtered petroleum ether.

6.2.6 Place clean plastic wrap or aluminum foil over the filtering apparatus when not in use.

6.3 Preparation of reagents.

6.3.1 Filter the isopropyl alcohol and petroleum ether through a 0.45 µm membrane filter. Discard the first 50 mL of each solvent filtered. Filtering equipment and storage containers for the solvents must be cleaned as described in 6.2.

NOTE: Ground the filter mechanism to prevent build-up of static electricity.

6.4 Preparation of test filters.

6.4.1 Place a 3 µm membrane filter in a petri dish and dry for 30 minutes in an oven regulated to 90°C with petri dish lid ajar. Remove petri dish and with lid closed, place in vicinity of balance on which weighing's are to be made. Allow filters to cool for 30 minutes and weigh to the nearest 0.1 mg. Use forceps to handle the test membranes. An air ionizer must be utilized during all weighing's to eliminate errors due to electrostatic charge effects.

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6.5 Filtering of test sample.

6.5.1 Inspect visually the top of the sample container for dust and dirt. If present, wash top of container using laboratory wash bottle containing filtered petroleum ether.

6.5.2 Shake sample mechanically for 5 minutes prior to test. Puncture small opening in top of container (vent in case of screw cap containers) and heat to a maximum of 60°C to facilitate filtering. Lubricant formulations which filter readily (30 minutes or less per L [qt]) need not be heated.

6.5.3 Cautiously open containers so as to prevent the introduction of metal particles into the sample. Pour approximately 50 mL of sample into the assembled filtering apparatus containing the pre-weighed filter and apply vacuum. Filter entire contents of the sample container and record filtering time in minutes. All of the sample must be filtered through one filter. Thoroughly wash down the inside of the sample container with filtered petroleum ether, passing the washings through the filter.

6.5.4 Wash down the inside of the filter funnel using laboratory wash bottle containing filtered petroleum ether. With vacuum still applied, carefully remove the funnel. Wash the periphery of the membrane by directing a gentle stream of filtered petroleum ether from the edge toward the center, taking care not to wash any contaminant from the surface of the membrane.

6.5.5 Carefully remove the test membrane from the filter base and place it in a petri dish. Dry and reweigh as described in 6.4.1.

7. CALCULATIONS

7.1 Calculate particulate contaminant per liter of oil from the initial and final weight of the test filter and the volume of sample filtered.

$$Particulate\ contaminant, \frac{mg}{liter} of\ test = \frac{A - B}{C}$$

where:
A = final weight of test filter in mg.
B = initial weight of test membrane in mg.
C = volume of sample measure in liters passed through the filter.

8. REPORTING

8.1 Report the contaminant in mg/L and filtering time in minutes.

9. PRECISION

9.1 Precision data have not been developed for this method.

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT

Method Prepared By:
Air Force – AFRL/RQTM - 2019

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Method 3044.7
January 2023

3044.7 – STANDARD TEST METHOD FOR TRACE SEDIMENT IN LUBRICATING OILS

1. SCOPE

1.1 This test method covers the determination of trace amounts (less than 0.05 % by volume) of sediment in lubricating oils. Since oil-soluble material precipitated by the specified solvent is not intended as part of the measured sediment, the test method is not applicable in cases where precipitated oil-soluble components will appreciably contribute to the sediment readings.

1.2 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.3 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

2. SUMMARY

2.1 This test measures the trace level amount of sediment that is naphtha-insoluble and can be separated by centrifuging. Excessive amounts of sediment in oil could lead to system malfunction in critical applications.

3. SAMPLE SIZE

3.1 The minimum sample size is 100 mL.

4. STANDARDS AND APPARATUS

4.1 CAUTION – New operators should review the operating manual thoroughly before use of this equipment.

4.2 Centrifuge, meeting all the safety requirements for normal use and capable of whirling two or more filled centrifuge tubes at a speed which can be controlled to give a relative centrifugal force (rcf) between 600 and 700 at the tip of the tubes. The revolving head, trunnion rings, and trunnion cups, including the rubber cushion, shall be soundly constructed to withstand the maximum centrifugal force capable of being delivered by the power source. The trunnion cups and cushions shall firmly support the tubes when the centrifuge is in motion. The centrifuge shall be enclosed by a metal shield or case strong enough to eliminate danger if any breakage occurs. Calculate the speed of the rotating head as follows:

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$$\text{rpm} = 1337\sqrt{\text{rcf}/d} \qquad (1)$$

where:
rcf = relative centrifugal force, and
d = diameter of swing, in millimetres, measured between tips of opposite tubes when in rotating position.

The relationship between the diameter swing, relative centrifugal force, and revolutions per minute is given in TABLE I.

TABLE I. Rotation speeds for centrifuges of various diameters.

Diameter of Swing, mm[A]	RPM at 600 rcf	RPM at 700 rcf
483	1490	1610
508	1450	1570
533	1420	1530
559	1390	1500

A: Measured in mm between tips of opposite tubes when rotating position.

4.3 Centrifuge Tube, cone-shaped, conforming to the dimensions given in FIGURE 1, and made of thoroughly annealed glass. The graduations, numbered as shown in FIGURE 1, shall be clear and distinct, and the mouth shall be constructed in a shape suitable for closure with a cork. Scale-error tolerances and smallest graduations between various calibration marks are given in TABLE II. Calibrated centrifuge tubes shall be purchased from the manufacturer.

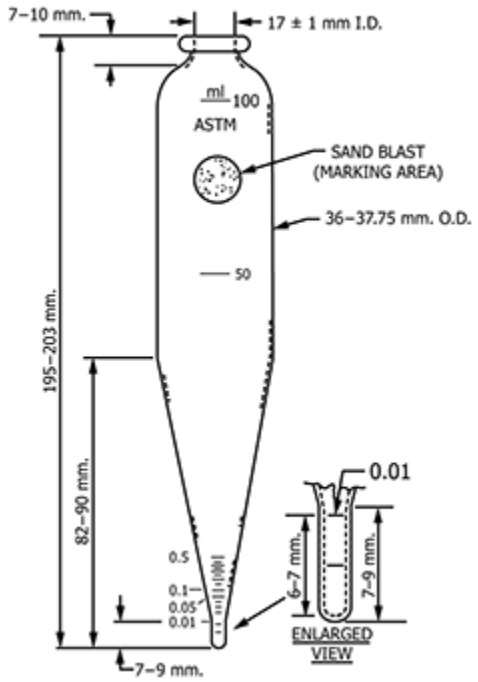


FIGURE 1. Trace sediment tube.

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TABLE II. Trace sediment tube calibration tolerances.

Range, mL	Smallest Scale Division, mL	Scale Error, max, mL
0 to 0.01	0.005	±0.001 at 0.01
0.01 to 0.05	0.01	±0.005
0.05 to 0.15	0.05	±0.01
0.15 to 0.30	0.05	±0.02
0.30 to 0.50	0.05	±0.03
0.50 to 50	none	±1.0
50 to 100	none	±1.0

5. MATERIALS

5.1 Hexanes, reagent grade, minimum purity. (Warning—Extremely flammable. Harmful if inhaled.)

5.1.1 Reagent grade minimum purity hexanes are sometimes referred to or sold by other names such as precipitation naphtha, petroleum naphtha, petroleum ether, ligroine, petroleum benzin, or industrial naphthas.

5.1.2 Before use, the hexanes should be free of any extraneous material that might affect the final test readings. For this purpose, it should either be filtered through a membrane filter or centrifuged several times and decanted just prior to its use.

6. PROCEDURE

6.1 Measure 50 mL ± 1 mL of hexanes (see 5.1) into each of two clean, dry centrifuge tubes at room temperature. (Warning—Extremely flammable.) Then fill each tube to the 100 mL mark with the oil sample and close tightly with a softened cork covered with a thin pliable plastic film that is resistant to petroleum products (not a rubber stopper). Shake sample well to ensure complete mixing, then invert each tube at least 20 times allowing the liquid to drain thoroughly from the tip of the tube at each inversion. If the liquid refuses to drain upon inversion, gently tap the inverted tube against the palm of the hand to jolt the liquid out of the tip. Place the tubes in a water bath at 32 °C to 35 °C for 5 min ± 1 min. Momentarily remove the corks to relieve any pressure, and invert each tube again at least 20 times exactly as before. The success of this method depends to a large degree upon having a thoroughly homogeneous mixture which will drain completely as possible from the tip when the tube is inverted.

6.2 Balance the two centrifuge tubes or pairs of tubes with their respective trunnion cups and place them on opposite sides of the centrifuge head. Then whirl them for 10 min ± 1 min at a rate sufficient to produce a relative centrifuge force (rcf) between 600 and 700 at the tips of the whirling tubes (see 4.2). At the end of the 10 min whirling period, decant the mixture carefully, allowing the sediment to remain in the tubes. Measure another 50 mL ± 1 mL of the hexanes into each of the two tubes, then add the oil sample until the total volume reaches the 100 mL mark. Then stopper the tubes, repeatedly invert, heat, and again invert as described in 6.1. Whirl the tubes in the centrifuge for 10 min ± 1 min as before, repeating the 10 min whirling operation until the volume of sediment in each tube remains constant for three consecutive readings. In general,

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not more than four whirlings will be required for oils having low sediment values. Record the final reading for sediment in each tube.

7. CALCULATIONS

7.1 If sediment is present in one or both tubes, as recorded in 6.2, average the final readings for sediment in the two tubes containing the sample to obtain the average volume of sediment per 100 mL sample.

8. REPORTING

8.1 If no sediment is present in either tube, as determined and recorded in 6.2, report the sample result as 0.

8.2 If the average result determined in 7.1 is >0 % and <0.01 %, report the result to the nearest 0.001 %.

8.3 If the average result determined in 7.1 is within the range of 0.01 % to 0.05 %, report the result to the nearest 0.01 %.

9. PRECISION

9.1 The precision of this test method as determined by statistical examination of interlaboratory results is as follows:

9.1.1 Repeatability- The difference between two test results, obtained by the same operator with the same apparatus under constant operating conditions on identical test material, would in the long run, in the normal and correct operation of the test method, exceed the following values only in one case in twenty:

Sediment, percent volume	Repeatability
0.000 to 0.002	0.001
0.003 to 0.005	0.001
0.006 to 0.01	0.002

9.1.2 Reproducibility- The difference between two single and independent results obtained by different operators working in different laboratories on identical test material would, in the long run, in the normal and correct operation of the test method, exceed the following values only in one case in twenty:

Sediment, percent volume	Reproducibility
0.000 to 0.002	0.001
0.003 to 0.005	0.002
0.006 to 0.01	0.003

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
1 FEB 2023	Method reinstated	This method was previously transferred to ASTM after 3044.6. It was re-designated ASTM 2273 until 2022 when it was cancelled. Army acquired the copyright and re-instated the method under its previous method designation.	7	Army - AT

Method prepared by:
Army- AT - 2023

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Method 3213.2
September 30, 2019

3213.2 – FOAMING CHARACTERISTICS OF AIRCRAFT TURBINE LUBRICATING OILS (STATIC FOAM TEST)

1. SCOPE

1.1 This method is used to determine the foaming characteristics of aircraft turbine lubricating oils under specified temperature and air flow conditions.

2. SUMMARY

2.1 This static foam test is basically the ASTM D892 foam test modified to be substantially more severe. Ratings of synthetic lubricant foaming tendencies using this method have been consistent with existing operational turbine engine experience. The modifications primarily involve the use of a 500 mL rather than the 1000 mL graduated cylinder and a greatly increased air flow rate.

3. SAMPLE

3.1 Approximately 300 mL of the material to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Foam Test Apparatus; Refer to ASTM D892 for Test Bath requirements. The apparatus consists of an air inlet device, 2.54 cm diameter fused crystalline alumina grain spherical gas diffusion stone, and a liquid bath equipped with heater, stirrer and temperature controller. The bath must be capable of maintaining a bath temperature of 80°C ± 1°C throughout a test cycle. Maximum stone pore diameter shall not be greater than 80 µm. Stone permeability at a water pressure of 250 mm shall be 3000 to 6000 mL of air/min. Refer to ASTM D892 for maximum pore diameter and permeability test procedures.

4.2 Cylinder graduated; 500 mL (Standard glass laboratory ware with pouring spout). Corning No. 3022 or equivalent.

4.3 Stopper, rubber; fitted for air tube.

4.4 Flowmeter; capable of delivering 1000 ± 20 mL air/min as measured at 20 °C (68 °F) and 101 gage kPa (760 mm Hg).

4.5 Timer; 1 s graduations, accurate to l s or better.

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4.6 Thermometer; 1°C increments, accuracy of 1°C or better. An ASTM 12C (-20°C to +102 C) thermometer is suitable. Digital contact thermometers are suitable with a range of at least 20°C to 90°C, with accuracy of 1°C or better.

4.7 Oven, drying; standard laboratory equipment.

5. MATERIALS

5.1 Solvents, dry cleaning; such as Stoddard Solvent, petroleum distillate [0,2,0], VM&P naptha, or similar.

5.2 Toluene conforming to ASTM D892 specifications [2,3,0].

5.3. Petroleum ether (Boiling Range 30°C to 60°C), ACS [1,4,0].

5.4 Detergent, liquid, water soluble

"WARNING: Toluene is toxic and flammable. Keep away from open flames. Avoid breathing vapors and contact with skin. Use in a laboratory hood or with adequate ventilation."

6. PREPARATION OF APPARATUS

6.1 The graduated cylinder, air inlet tube and diffuser stone must be cleaned thoroughly after each test. Between tests, these items must be stored in a clean, dust-free area.

6.1.1 The cylinder shall be cleaned by rinsing with dry cleaning solvent or petroleum distillate, then another rinse with toluene followed by washing with detergent and water. Rinse thoroughly with distilled water and dry in oven. Alternatively, cleaning methods listed in ASTM D892 may be used.

6.1.2 Clean stone and air tube by rinsing with dry cleaning solvent. Then immerse stone in toluene and, using vacuum and air pressure, flush the solvent back and forth at least 5 times. Finally, rinse stone with petroleum ether and dry using clean air. Wipe the outside of the air tube with a clean cloth moistened with toluene followed by a clean dry cloth. Alternatively, cleaning methods listed in ASTM D892 may be used.

6.2 PROCEDURE

6.2.1 Measure 200 mL of lubricant into clean cylinder. Immerse the cylinder in bath maintained at 80°C ± 1°C (bath liquid above 500 mL cylinder mark). Install air tube and stone so that the stone just touches the bottom of the cylinder and is approximately at the center of circular cross section. Allow 15 min for lubricant to reach thermal equilibrium. Make all air supply connections. Adjust air flow to 1000 ± 20 mL/min.

6.2.2 Record the maximum foam volume generated during the 30 min test period. Foam volume is defined as the foam-air interface (mL) minus the foam-liquid interface. If no foam-liquid

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interface is discernible, the foam-air interface measured in mL shall be defined as the foam volume. At the end of the test period, disconnect the air supply. Record the time required for the foam to collapse to no more than a single ring of bubbles at the lubricant surface periphery, or record volume of foam remaining after 5 min, whichever occurs first. Disassemble and clean apparatus.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Report the following:

Maximum Foam Volume, mL _____
Foam Collapse Time, s _____
Foam Volume Remaining after 5 min, mL _____

9. PRECISION

9.1 Repeatability. Limited test data indicates the following test repeatability:

TABLE I. Repeatability requirements for foam volume.

Foam volume	Repeatability
50 mL	10 mL
150 mL	20 mL
250 mL	25 mL

9.2 Reproducibility. This has not been established.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Removed source information. Referred to ASTM D892.	Source information outdated.	2	Air Force – AFRL/RQTM
30 Sep 2019	Added information for Digital Contact Thermometers	Gives alternate to liquid thermometers	2	Air Force – AFRL/RQTM

Method prepared by:
Air Force - AFRL/RQTM - 2019

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Method 3403.3
September 30, 2019

3403.3 – COMPATIBILITY OF TURBINE LUBRICATING OILS

1. SCOPE

1.1 This method is used for determining the compatibility of aircraft turbine lubricants with specific referee lubricants.

2. SUMMARY

2.1 This method consists essentially of preparing mixtures of the sample lubricant and referee lubricant in three concentrations, maintaining the mixtures at a fixed temperature for 7 days, and then centrifuging them for ten-minute intervals until the volumes of sediment becomes constant.

3. SAMPLE SIZE

3.1 Approximately 300 mL of the aircraft turbine lubricant to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Oven, gravity-convection, 105°C ± 1°C (221°F ± 2°F) (to accommodate three 250-mL flasks).

4.2 Erlenmeyer flasks, (3), Pyrex, 250-mL with glass stoppers.

4.3 Centrifuge tubes, cone-shaped, 100 mL (see ASTM D2273).

4.4 Centrifuge, capable of spinning two or more centrifuge tubes, and applying a relative centrifugal force of 600 to 700 at the tips of the tubes.

4.5 Bath, hot water, 93°C ± 3°C (200° ± 5°F) (for simultaneous heating of centrifuge tubes to be spun).

5. MATERIALS

5.1 Referee lubricant (in accordance with lubricant specification), 300 mL.

5.2 n-Heptane.

5.3 Cleaning solution. Any suitable laboratory glass cleaning fluid may be used providing it removes all evidence of staining and deposits from previous tests.

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6. PROCEDURE

6.1 Clean all flasks and centrifuge tubes as follows:

a. Thoroughly rinse with n-Heptane in hood, and allow to air dry in hood.
b. Immerse in glass cleaning solution for at least 2 h.
c. Rinse first with tap water, then with distilled water, and dry in an oven 105°C (220°F).

6.2 Prepare three 200-mL mixtures, in 250-mL Erlenmeyer flasks, using the following proportions:

a. Sample, 180 mL; referee lubricant, 20 mL.
b. Sample, 100 mL; referee lubricant, 100 mL.
c. Sample, 20 mL; referee lubricant, 180 mL.

6.3 Stopper with a cork stopper and identify each flask.

6.4 Manually shake each flask vigorously for 1 minutes.

6.5 Loosen the stoppers and heat the flasks in an oven for 168 h at 105°C ± 3°C (221°F ± 5°F).

6.6 After the 168-hour period, remove the flasks, and allow them to cool to room temperature.

6.7 Manually shake each flask (one at a time) vigorously 1 minutes, and immediately fill two clean centrifuge tubes to the 100 mL mark with its contents. Identify the contents of each tube.

NOTE 2: Do not fill more pairs of tubes than can be spun by the centrifuge at one time. Allow the remaining filled flasks to stand until the first set of tubes have been completely centrifuged.

6.8 Process the centrifuge tubes as follows:

a. Heat the tubes for 5 minutes in a hot-water bath at 93°C ± 3°C (200°F ± 5°F).
b. Mount the tubes opposite each other in the centrifuge, being careful to locate the tubes to achieve proper balance of the centrifuge.
c. Turn on the centrifuge and increase the speed until the relative centrifugal force at the tips of the tubes is 600 to 700. Run the centrifuge for 10 minutes.
d. After the ten-minute period, record the amount of sediment in each tube to the nearest 0.001 mL.
e. Repeat the 10 minutes centrifuging and sediment readings until the amount of sediment remains constant for three consecutive readings.

7. CALCULATIONS

7.1 Compute the average sediment volume to the nearest 0.001 mL of each tube pair.

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8. REPORTING

8.1 Report the average volume of sediment in each pair of centrifuge tubes.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Solvent substitution	ODC removal	3	Air Force – AFRL/RQTM
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT

Method prepared by:
Air Force - AFRL/RQTM - 2019

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Method 3410.2
September 30, 2019

3410.2 – HIGH TEMPERATURE DEPOSIT AND OIL DEGRADATION CHARACTERISTICS OF AVIATION TURBINE OILS

1. SCOPE

1.1 This method describes procedures used to evaluate the thermal stability characteristics of turbojet lubricants under varying levels of severity.

2. SUMMARY

2.1 Sample of the lubricant is subjected to one of the severity levels (type 1, 1-1/2, 2, or 3) for a controlled number of hours. The number of test hours can either be 100 hours for a standard or corrosion inhibitor class oil, 200 hours for high thermal stability or enhanced ester class oil, or 48 hours for a type 3 (MIL-PRF-7808) oil. At periodic intervals during the test and at the end of test, samples of the oil are examined for both physical and chemical changes when compared to the sample of new oil. A visual inspection is made at the end of test of the accumulated deposits on the bearing and bearing housing. A weighted numerical rating system is used for rating individual areas as well as for obtaining an overall rating. Additional data on relative sludge forming tendencies on oils are obtained by weighing a 100-mesh filter element and a 40-mesh filter element.

3. SAMPLE SIZE

3.1 Approximately 8000 mL of the aircraft turbine lubricant to be tested.

4. REFERENCES, STANDARDS AND APPARATUS

4.1 The apparatus used in this method is the ERDCO® high temperature bearing head mounted on a suitable drive test stand, similar to the ERDCO® universal test stand. It is not mandatory to use the "universal drive to power the bearing. The support oil system and the test oil system are separate units and of special design for the ERDCO® high temperature bearing head.

4.1.1 ERDCO® high temperature bearing head. The bearing head is divided internally into two main sections. The front or test section with its separate test oil system houses the unshielded 100-mm straight roller test bearing. The temperature of the test bearing is controlled by supplying heat to the outer race of the bearing. The outer race is secured in a bearing mount assembly containing wound Calrod® heating elements packed in aluminum oxide for even heat distribution. The rear or support section of the bearing head houses the externally loaded, main shaft support bearing, together with a separate oil system. The test and support sections of the modified bearing head are separated by a screw thread seal to prevent mixing of the test and support oil. Details covering the bearing head and screw thread seal assembly are shown in

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FIGURE 1.

4.1.1.1 Test oil jet. A standard No. 60 drill size jet located at 12 o'clock on a 0.13335-m (5.250 inch) diameter should be used. The size of the jet should be checked with a No. 60 drill and a visual check of axial straightness made with the drill inserted in the jet. Corrections for orifice size and axial straightness should be made by silver soldering the orifice end of the jet tube and redrilling.

4.1.2 Test bearing. An unshielded 100 mm straight roller bearing is mounted on the free end of the shaft in the test oil compartment of the bearing head. The test bearing is loaded radially by the slight downward displacement of the free end of the shaft caused by a straight downward force from the loading piston acting on the outer race of the main support ball bearing shown in FIGURE 1. The test bearing is lubricated by a single jet located at 12 o'clock which supplies essentially a solid stream of oil midway between the I.D. of the outer race and the O.D. of the retainer on the top or unloaded position of the test bearing.

4.1.3 Test bearing mount. The bearing mount is made with an outer race removal feature which permits the removal of the bearing outer race for cleaning and inspection without disturbing the aluminum oxide packing. Three spot-type thermocouples are located 120° apart (radially around the I.D. of the bearing mount and indicate the skin temperature on the periphery of the outer race, which is pressed into contact with these thermocouple junctions when it is installed in the bearing mount. One of these three thermocouple leads is connected to a differentiating type temperature controller (proportional controller) or to a suitable autotransformer. "ON-OFF" type control is not recommended.

4.1.3.1 Three thermocouple junctions are silver-soldered to the outer-race housing. The soldering operation should leave enough excess metal to provide good contact with the outer race when it is pressed into the mount. Each of the three thermocouple leads are routed out directly through the bearing housing or end cover at 120° intervals.

4.1.3.2 Thermocouple packing. Teflon or an equivalent should be used for all thermocouple leads where they pass through the housing or end cover fitting. This also applies to the two Calrod® elements where they pass out through the end cover.

4.1.4 Test oil system. A schematic of the test oil system is shown in FIGURE 1. The test oil cycle starts at the tank where three 1500 W, series connected heater elements (refer to 4.1.7) maintain the desired temperature by means of a differentiating type temperature controller or a suitable autotransformer. "ON-OFF" type control is not recommended. The test oil is then pumped from the tank to the bearing head where it passes through a No. 60 drill size orifice and lubricates the test bearing. The test oil is drained from the bearing head at three locations. The front drain handles splash oil which does not pass through the test bearing, whereas the rear drain collects only oil that has gone through the bearing. The middle drain is looped into a 360° configuration to slow oil flow and lessen air entrainment in the oil. The three drains go into a sump from which the test oil is pumped back into the top of the tank.

4.1.4.1 The test oil system provides for the installation of a fine 100-mesh filter screen

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(stainless steel, ERDCO®P/N 1RB 2006 S.S. for Type 1-1/2 test conditions) in the pressure line only just ahead of the jet to collect the sludge particles circulated in the test oil system. Pressure taps are located before and after the fine strainer in order to monitor the degree of screen plugging.

4.1.4.2 Heat loss from the test oil system has been minimized by placing a 0.6 cm (1/4 inch) insulating box around the test oil tank. For screening tests at 149°C (300°F) oil inlet temperature, no other insulation is usually used. However, at 177°C (350 °F) and 204°C (400°F) oil inlet temperature, the test oil lines are wrapped, if necessary, with 2.5 cm (1 inch) wide insulating tape. In addition, insulating covers are placed over the test oil pressure pump housing. The amount of insulation required is fixed for any individual system by the oil inlet temperature requirements for each test. For all types of operation, the test oil tank is vented to atmosphere, although plant exhaust suction systems are permissible as long as suction pressures do not exceed 12.7 cm (5.0 inch) of water. Normally, the test tank vapor space is slightly pressurized (0 to 0.76 cm [0 to 0.3 in] Hg) due to back pressure from vapor condensers in the vent line.

4.1.4.2.1 Only stainless steel tubing or pipe lines and fittings (including sump below head) should be used in the test oil system. No insulation is required on scavenge lines but is required on pressure lines and pump housings, as necessary to maintain the test oil temperature conditions; however, the test oil in the thermocouple tee should be well insulated at all times. The tee should be located not more than 7.6 cm (3 inch) from end cover and should be large enough to provide an oil annulus around the thermocouple. The tip of the thermocouple should be located in the center of the tee. A 40-mesh filter screen (stainless steel, ERDCO®P/N 1RB 2007 S.S. for type 1-1/2 test conditions) should be used in the test oil scavenge filter housing which should be located between the bearing head and scavenge pump.

4.1.4.3 Test oil tank.

4.1.4.3.1 Overall configuration. The overall configuration and general construction details of the test oil tank should conform to the detailed ERDCO® Part No. 01RB2160. The tank and baffle is constructed from AMS 5504C material; (AISI 410) 12.5% chromium steel sheet and welding rod. Baffle plate height should be checked and adjusted if necessary to ensure that the top of the plate is 12.1 cm (4-3/4 inch) from the bottom of the tank. The top of the stirrer propeller is to be even with the top of the baffle plate as determined by a straight edge.

4.1.4.3.2 Cover gasket material. A Teflon sheet gasket or equivalent cut to match the outside flange dimensions should be used between the cover and mounting flange around the top of the tank.

4.1.4.3.3 Stirrer and stirrer mounting. A Lightning Model L 1/30 hp motor having a no-load (in air) shaft speed of 1800 rpm should be used in conjunction with a 5.1 cm (2.0 inch) diameter, three-blade propeller.

4.1.5 Air system. Saturated air is metered to the test oil compartment in the bearing head. The air supply for the test bearing compartment should be shop air connected in the following sequence: air supply to pressure regulator to flow control valve to rotameter to water saturator to

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water separator tank to end cover fitting on bearing head. The air directed through the end cover (at about 11 o'clock or 1 o'clock on the same diameter as the test oil jet) into the bearing compartment is used primarily to control the pressure drop across the seal so as to ensure against oil transfer from the support section into the test oil section. A 1/4- H 1/8-inch tubing to pipe fitting should be used for the metered air supply to the rig.

4.1.5.1 Air to the test head is metered through a rotameter. Prior to each run, the rotameter is calibrated against a wet test gas meter for a flow of 9.90 L/min (0.35 CFM). The procedure is to disconnect the air supply line at the fitting where the air enters the head and then connect the supply line to a wet test gas meter which in turn discharges to atmosphere. A mercury manometer is located immediately upstream from the rotameter. The settings of the manometer and rotameter are observed and recorded for a wet test gas meter reading of 9.90 L/min (0.35 CFM). The air line is then reconnected to the bearing head and the previously observed manometer and rotameter settings are maintained throughout the test.

4.1.5.2 Seal (test support). The screw-type seal (P/N IRBB-6204) should be vented to the test cell at both top and the bottom of the seal housing connections.

4.1.5.3 Connections for seal differential pressure measurement. The lines to the manometer used to measure the differential pressure across the seal should be connected to top, front of the bearing head, and the rear vertical surface of the bearing housing between 11 and 1 o'clock 2.5 cm (1 inch) above the junction of the rear case with the main housing. After startup there should be no venting of the manometer lines. In this connection, the pipe plug in the top of the gear box, if used, should be removed and replaced with a small valve to ensure atmospheric pressure in the gear box and hence in the support system.

4.1.6 Fenwal thermostatic control switch (Cat. No. 17802-510 or equivalent) (-18°C to 315°C [0°F to 600°F] normally closed). This switch should have a stainless steel sheath and be located in end of tank opposite stirrer motor 10.8 cm (4-1/4 inch) from bottom and on the center line of the tank.

4.1.7 Heater unit. A Chromalox tubular heater unit (MT0345) having a rating of 4500 W, rated at 115-120 V should be series connected to a 220 V supply and controlled by means of a differentiating type temperature controller or a suitable auto-transformer. Three thermocouples installed 31.8 cm (12-1/2 inch) from terminal end of tank should be used. All thermocouple lead wire should be of the Ceramo, a metal-sheathed type. Test oil heater thermocouples are to be attached to the tubular heater by clamps, welded shoes, or spot weld.

4.2 Graduated cylinder, borosilicate, 1000 mL.

5. MATERIALS

5.1 Lubricating oil, qualified under MIL-PRF-23699.

5.2 Dry cleaning solvent, (A-A-59601D)

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5.3 Oakite LRS (REL-4576)

5.4 No. 400 emery paper.

5.5 No. 30 grit aluminum oxide.

5.6 Bi-solvent; isopropyl alcohol: acetone - 1:1 [1, 3, 0].

5.7 Heptane

6. PROCEDURE

6.1 Cleaning procedure.

6.1.1 Test oil tank. The test oil tank should be removed from the stand at the end of each test. After the deposits have been rated and photographed the tank assembly should be completely disassembled. The interior of the tank and the immersed part of the heater should each be thoroughly cleaned by solvents (10% Oakite LRS/water mixture followed by bi-solvent), hand scrubbing, and or mild abrasive blast to remove all traces of deposits from previous test. An electrical continuity check should be made on the immersion heaters before the tank is reinstalled on the stand. Instances of apparent electrical "shorts in the immersion heater circuit can often be cured by additional air or oven drying.

6.1.2 Setup for type 3 test conditions. This section only applies to the type 3 bearing rig conditions which are run for the MIL-PRF-7808 oil. This is the only type of bearing rig test where metal specimens are added to the test oil sump.

6.1.2.1 Metal Specimen Orientation. Metal specimens, 0.635 cm (1/4 inch) inside diameter by 1.905 cm (3/4 inch) outside diameter by 0.081 cm (0.032 inch) thick, can be placed in the test-oil sump in order to provide information concerning possible metal attack by the lubricant. The metal specimens should be mounted on a rod and separated by spacers of approximately 0.32cm (1/8 inch) thickness and 0.79 cm (5/16 inch) outside diameter. The exact mounting order of specimens will be noted in the applicable specification. The metal specimens are mounted approximately 5.08 cm (2 inches) below the level of the cold oil level. Typical metal specimens are aluminum, silver, bronze alloy, M-50, Waspaloy®, and titanium. The applicable specification will note what metal specimens and rod/spacer materials should be used.

6.1.2.2 Metal specimen cleaning procedure. Clean and pre polish the specimens by appropriate means of choice. If the specimens are being reused from a previous test, no pits, etchings, or signs of corrosion should be visible at this point. Finish with No. 400 grit silicon carbide paper, removing all marks that may have been left by previous polishing. The specimens should not be handled with bare fingers from this point; handle them only with a clean cloth or paper, or with tongs or forceps. Cotton swab wash the specimens with solvents, first with n-heptane and then with acetone, final rinse in clean acetone, air dry and weigh to the nearest 0.1 mg. If there is to be any delay before weighing, store under n-heptane until ready.

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6.1.2.3 Metal specimen analysis. After test, the metal corrosion specimens are disassembled from the holder and solvent-rinsed to remove residual test oil. (The specimens are stored in a desiccator if final processing is delayed). The individual specimens are *n*-heptane swabbed using a series of cotton swabs or pads until clean pads are noted. The coupons are then rinsed in clean *n*-heptane and acetone, air-dried, and weighed to the nearest 0.1 mg. Significant specimen weight change is reported in mg/cm^2 ignoring edge areas in the calculation of exposed surface area. A significant weight change is typically defined as a weight variation of ± 0.20 mg/cm^2 or more.

6.1.3 Test oil pressure and scavenge pumps. The test oil pressure and scavenge pumps are to be completely disassembled and cleaned by solvents by ultrasonic cleaning with a 10% Oakite/water mixture followed by bi-solvent to remove all deposits from the previous test.

6.1.4 Oil out sump. The oil sump located directly below the bearing head should be disassembled and cleaned with bi-solvent followed by ultrasonic cleaning in a 10% Oakite/water mixture followed by a bi-solvent, if necessary. Rinse in bi-solvent and air dry.

6.1.5 Test oil lines. All of the straight run test oil lines and fittings (elbows, tees) should be removed from the test oil system and hand reamed with wire brushes having bristles of steel or other metal except copper in a ultrasonic bath of cleaning solvent (10% Oakite/water mixture followed by bi-solvent). Oil lines with bends greater than 10° should be new at the start of each test.

6.1.6 Oil filter assemblies. All oil filter housings in the test and support oil systems should be removed and wire brushed in cleaning solvent. A soft fiber brush is usually adequate in scrubbing the filter elements after a short soaking in a solvent bath.

6.1.7 Gage lines. The oil lines connected to gages on the instrument console require cleaning with air after each test.

6.1.8 Bearing head. The bearing head should be disassembled as far back as the seal and all of the removable parts, except the test bearing, are to be thoroughly cleaned by scouring with No. 400 emery paper, or glass beads (vacuum-blasting), and solvent to remove all traces of deposits from the previous test. In cases of extreme dirtiness, the entire seal assembly should be removed, and the parts solvent cleaned using No. 400 emery paper if necessary.

6.1.9 Test bearing. For qualification tests a new 100 mm test bearing should be used for each endurance run. The new bearing should be cleaned in dry cleaning solvent for 10 minutes, spun by hand, and finally dipped in test oil prior to installing.

6.2 Bearing head assembly.

6.2.1 Screw thread seal. In installing the seal (P/N IRBB 6204) the right hand threads on the O.D. of the seal must be toward the front (test compartment) end of the rig. Reversing the seal position will cause the test and support oil to be whirled toward the center of the seal by the rotating action of the shaft and enhance the possibility of mixing the two oils.

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6.2.2 Test bearing heater mount. The bearing outer race should be lightly pressed into the bearing heater mount with the serial number facing the front of the rig. The recommended fit on the bearing outer race is -0.000508 cm to + 0.00178 cm (-0.0002 inch to + 0.0007 inch). On repeated usage, the I.D. of the bearing mount may become worn and result in a loose fit which might tend to cock the outer race. It may also result in erroneous temperature readings of the bearing outer race. When this happens, the I.D. of the mount should be chrome plated and brought back to proper dimension or that section of the mount replaced.

NOTE: The heater mount must be disassembled and repacked with No. 30 grit aluminum oxide after each test.

6.2.2.1 Reassembly of heater mount. Prior to assembly of the heater mount, the heater coils should be centered and checked to ensure that no point of the coils comes within 0.318 cm (1/8 inch) of the inside surfaces of the front and rear mount plates. A method of accomplishing this is to assemble the unit without retainer ring; this allows visual inspection of the location of the heater- coils within the assembly. Areas where the coils come within 0.318 cm (1/8 inch) of either front or rear inner faces are marked. During final assembly, transite spacers 0.32 H 0.64 H 1.3 cm (1/8 H 1/4 x 1/2 inch) are placed at these marked points. In reassembling the heater mount, care should be exercised to ensure that the front and rear vertical faces are parallel to each there and square with the I.D. of the mount. If these conditions are not maintained, the concentricity of the bearing outer race in relation to the axis of the shaft may be affected. Before installing the heater mount in the rig, the bearing outer race thermocouples should be checked for continuity.

6.2.3 Test bearing. Before installing the test bearing on the hub, it should be examined visually for any obvious mechanical defects. If the bearing appears satisfactory, the inner race and cage assembly should be pressed onto the hub with the serial number facing the front of the rig. The recommended fit of the bearing inner race on the hub is 0.0020 cm to 0.0036 cm (0.0008 inch to 0.0014 inch) tight. After repeated usage the O.D. of the bearing hub may become worn and result in a loose bearing fit. When this happens, the hub should be chrome plated back to proper dimension or replaced. After the bearing has been installed in the bearing head, turn the shaft by hand to make sure it turns freely. Bearing roughness can usually be detected in this manner. When properly installed the rollers should be evenly centered in the outer race. If the rollers are more than 0.32 cm (1/8 inch) off center on the outer race, check for missing shaft assembly parts or an improperly installed bearing outer race. It is important that the lock nut on the test bearing retainer be tight since it imparts the axial pinch on all of the shaft assembly parts back to the load bearing and in addition helps to prevent the bearing inner race from turning on the hub during operation.

6.2.4 End cover. Before installing the end cover on the rig, check the test oil jet tube alignment to make sure it is aimed midway between the test bearing outer race and the cage. A gasket, made from material to withstand high temperatures, should be installed under the end cover.

6.2.5 Bench check of seal operation. After assembly of the bearing head, the screw thread seal operation should be checked under a 2240-N (500-lb) radial load to the test bearing by

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applying 352 gauge kPa (51 psig) shop air pressure to the loading cylinder and hand rotating the shaft to detect rubbing. If any rubbing occurs, the bearing head should be disassembled and a slight amount of material removed from either the screw threads or the seal plate, depending on the location and severity of the rubbing. The material is removed from the seal plate if the rubbing is localized, and the screw thread is ground if the rubbing is severe and occurs over a considerable area. On a new build, the seal plate may be coated with Prussian blue to determine the area of contact. On later rebuilds, the rubbing can usually be detected without the use of the dye. The seal should be checked for rubbing after every build.

6.2.6 Miscellaneous joint compounds. No pipe dopes, thread compounds, or gasket sealants should be used in the assembly of the rig or test oil system. Teflon tape (no adhesive) is an effective sealant when applied to pipe threads.

6.3 Rig installation and stand checks. After assembly, the bearing head is installed on the stand. The test oil tank is filled with 8 L (2 gal) of the oil to be tested. This oil is heated to 121°C (250°F) in the tank and then circulated for 60 minutes through the test oil system, including the bearing head. The oil is then drained from the tank and the bottom sections of the two oil filter housings. Fresh test oil is then added to the tank until the fluid level is 1.3 cm (1/2 inch) above the baffle plate with the test oil pressure and scavenge lines filled in preparation for the test.

6.3.1 A qualified DOD-PRF-85734 lubricating oil is used in the support system and is not normally changed after each run. Drain periods on the support oil are governed by the amount of deterioration as determined by laboratory tests and the degree of the filter plugging in the support oil system.

6.4 Critical test items. Before starting a test, care should be taken that critical items such as dimensional tolerances and heater electrical connections conform to those listed in TABLE I.

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TABLE I. Critical test items.

	Minimum, cm	Maximum, cm	Minimum, inch	Maximum, inch
1. Dimensional tolerances				
a. Test bearing outer race fit in heater mount	-0.00051	0	-0.0002	0
b. Test bearing inner race fit in heater mount	-0.0020		-0.0008	
c. Axial run-out of front vertical face of test bearing outer race assembled in rig	0.000		0.000	
d. Screw type seal	0.046		0.018	
2. Electrical connections				
a. Test bearing heater. This is a single wrapped 4700-W unit which should be connected to 220-V power source.				
b. Test oil tank immersion heater. This unit is made up of three 1,500-W heaters which should be series-connected to a 220-V power source.				

TABLE II. Operating conditions for different severity levels of ERDCO® high temperature bearing test.

Condition	Type			
	1	1-1/2	2	3
Test oil-in temperature, °C	149° ± 2.5°	177° ± 2.5°	204° ± 2.5°	
Test tank bulk oil temperature, °C	171° ± 2.5°	199° ± 2.5°	227° ± 2.5°	177± 2.5°
Test bearing temperature -max., °C	260° ± 2.5°	260° ± 2.5°	260° ± 2.5°	260° ± 2.5°
Air flow to end cover, L/min	9.9 ± 1.4	9.9 ± 1.4	9.9 ± 1.4	9.9 ± 1.4
Test oil-in temperature, °F	300° ± 5°	350° ± 5°	400° ± 5°	
Test tank bulk oil temperature, °F	340° ± 5°	390° ± 5°	440° ± 5°	440° ± 5°
Test bearing temperature -max., °F	500° ± 5°	500° ± 5°	500° ± 5°	500° ± 5°
Air flow to end cover, CFM	0.35 ± 0.05	0.35 ± 0.05	0.35 ± 0.05	0.35 ± 0.05

NOTE: *The type 3 method is the only test condition that utilizes metal coupons added to the test oil sump.

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6.5 Operating conditions.

6.5.1 Throughout the entire test the variables for different test severities (types 1, 1-1/2, 2, and 3) as shown in TABLE II should be controlled within the limits specified.

6.5.2 Initial test oil jet calibration. To calibrate the test oil jet, the test tank should be heated until bulk oil temperature is about 149°C (300°F). Turn on air to the end cover. Then the test oil and support oil pumps are turned on, and the rig speed increased to 10,000 rpm. The rig speed should be brought up immediately after oil circulation is started as the screw thread seal depends on relative motion for its sealing action. The previous setting on the test oil pressure pump or a minimum of 69 gage kPa (10 psig) oil pressure at the jet should be used to ensure against possible oil starvation during this initial startup. The pressure drop across the seal is maintained at 0.8 cm to 1.5 cm (0.3 inch to 0.6 inch) of test oil by making fine adjustments to the gear box vent valve. Under these conditions of air flow, the pressure in the test compartment should be 0.25 cm - 0.51 cm (0.1 inch - 0.2 inch) of test oil above rig room pressure conditions. As soon as a 121°C (250°F) oil temperature at the jet has stabilized, a minimum of two separate flow checks of one-minute duration each should be made at the three-way valve position on the pressure side of the scavenge pump. The oil pressure at the jet should be adjusted using the variable speed drive unit on the test oil pump until the jet oil flow is 600 mL/minutes. Once the flow rate of 600 mL/min is established for an oil, the flow at this condition is maintained constant during the test.

6.5.3 After the oil jet calibration step, the conditions in TABLE III are allowed to stabilize:

TABLE III. Test conditions.

Oil-in temperature	121°C	250°F
Oil flow rate	600 ± 30 mL/min	600 ± 30 mL/min
Bearing speed	10,000 ± 100 rpm	10,000 ± 100 rpm
Radial load	352 gage kPa (2220 N)	51 psig (500 lb)
Support oil-in temperature	71°C - 82°C	160°F - 180°F
Support oil pressure	552-689 gage kPa	80-100 psig
Gear box oil pressure	172-207 gage kPa	25-30 psig
Air flow to end cover	9.9 ± 1.4 L/min	0.35 ± 0.05 cfm
Pressure drop across seal	0.8-1.5 cm of test oil (test compartment higher)	0.3-0.6 in. of test oil (test compartment higher)

6.5.3.1 After the foregoing conditions stabilize, the run is continued until the test bearing temperature from the highest reading thermocouple levels off (about one hour). This temperature is then noted and used as an indication of the mechanical condition of the bearing. Previous tests on bearing stabilization at a 121°C (250°F) oil inch temperature to the jet indicate that a temperature range of 149°C to 177°C (300°F to 350°F) is normal for a properly functioning 100-mm ERDCO® test bearing. Higher stabilization temperatures indicate the test bearing condition is questionable and the bearing should probably be replaced before continuing with the test.

6.5.4 Once the test bearing has reached a satisfactory stabilized temperature, the temperature

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setting on the temperature controller on the test tank should be increased to raise bulk oil temperatures to about 28°C (50°F) higher than the desired temperature of the oil at the jet 149°C (300°F) for the test of type 1 lubricants, 177°C (350°F) for the type 1-1/2 lubricants, and 204°C (400°F) for the test of type 2 lubricants). In order to reduce the length of warmup time required to reach the desired test oil jet temperature, the Calrod heater should be turned on simultaneously with the change in test tank setting to bring the bearing outer race temperature (max) to 260°C (500°F). At this point, the three thermocouples attached to the bearing outer race should be checked to make sure that the thermocouple at the 11 o'clock position is attached to the Temperature Controller. Improper hookup of the bearing thermocouples will result in poor control of test bearing temperatures.

6.5.5 When the desired test conditions shown in TABLE IV are reached, the test oil flow rate is again checked twice and adjusted if necessary to obtain 600 mL/min flow at the three-way valve position in the scavenge line. Then the timer for the endurance test is begun. These data are recorded on a data acquisition system continuously during the course of the test.

6.5.6 Test cycle. The endurance test is run so that there is at least a seven-hour shutdown period out of every 24 hours of operation. Except for emergency stops, all shutdowns should be made in accordance with the detailed procedure as outlined in 6.6.4 warmup time following shutdown periods is not considered part of the endurance test time.

6.5.7 Used oil analysis. The following laboratory tests are run on the new and used oil samples:

TABLE IV. Required testing for used oil analysis.

Test	Method	50 mL samples
Viscosity at: 40°C (104°F), cSt Neutralization No.	 ASTM D445 SAE-ARP5088	 X X

6.5.8 Test oil flow rate checks. During operation the test oil flow rate should be checked at the three-way valve in the scavenge line twice every start-up. For convenience, a 1000-mL, borosilicate glass graduated cylinder should be used to collect the test oil during the flow check. The volume of oil collected in a one- or two-minute period should check the original volume flow in mL/min within 5.0%. In order to prevent too large a drop in test oil level in the tank during flow checks, a maximum continuous flow period should be 2 minutes. The hot test oil collected in the glass graduate should be dumped back into the test tank after each flow check. With the friction type variable speed drive unit on the test oil pressure pump, it is not possible to obtain a non-slip drive on the pump after oil pressures reach about 276 gage kPa (40 psig). At this point the test oil will usually have become so viscous it can no longer be effectively pumped and the test should be considered ended.

6.5.9 At the completion of the endurance test, the test bearing is restabilized at 121°C (250°F) oil in temperature to determine the relative change in stabilized temperatures compared

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to the results obtained at the start of the test. These data can be used to determine the change in frictional characteristics of the bearing resulting from metal wear or excessive oil deposits.

6.5.10 The rig is then shut down in accordance with the shutdown procedure described in 6.6.4.

6.5.11 When the system cools to room temperature, the tank and test oil system should be drained into a suitable container so that the volume of test oil remaining can be determined. After the volume of test oil has been measured and recorded, 3.8 L (1 gal) of the test oil should be poured into a clean container and saved for laboratory analyses.

6.5.12 Rig photographs. As soon as possible, the rig should be disassembled. Although color is preferred, black-and-white photographs, measuring at least 7.6 H l0.2 cm (3 H 4 inch) or slides should then be taken of the following parts:

a. Bearing compartment with cover removed - front view.
b. Heater mount - rear side.
c. Cover plate - rear side.
d. Seal plate in housing - heater and bearing removed.
e. Test bearing - inner race, rollers and cage assembly.
f. Test bearing - outer race, inside diameter.
g. Test oil tank - baffle plate.
h. Test oil tank - baffle removed.

6.5.13 After the photographs are completed, a more detailed visual inspection is made of these parts to determine the average depth, coverage, consistency, and type of oil deposits, plus the extent of abnormal metal wear on the test bearing cage, roller race. Rate deposits according to 6.7.

6.6 Procedures

6.6.1 Flushing procedure.

a. Install a fine mesh (100 mesh) filter at the inlet of the spray nozzle and a coarse mesh (40 mesh) filter at the inlet of the scavenge pump. It is not necessary to pre-weigh the filters.
b. The same type of oil is used to flush the system as used in the test. Service the test oil tank with 8000 mL of test oil.
c. Ensure the tank is vented by means of an exhauster/blower arrangement and that the vent is open.
d. Ensure that the test oil supply valve at the inlet side of the pressure pump is closed.
e. The watt-hour meter reading for the test oil heater should be recorded before the flush is begun.
f. At the control panel turn on the following switches in the order shown:
    i) Control panel power.
    ii) Support test oil cart power.

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iii) Stirrer.
iv) Timer.
v) Test oil heater.
vi) Strip chart recorder.
vii) Flowmeter timer.
vii) Doric.

g. Provide full power (220 V) to the test oil heater.
h. When the temperature of the test oil in the tank exceeds 121°C (250°F) turn on the pressure and scavenge pumps. Immediately open the test oil supply valve (on the inlet side of the pressure pump) to allow oil to draw to the pressure pump. The pump speed should be maintained such that the pressure is a minimum of 138 gage kPa (20 psig) and the flow is 600 ± 30 mL/minute.
i. Adjust the test oil heater as needed to maintain a temperature of 121°C (250°F) at the test oil jet.
j. While the oil is circulating draw two samples of approximately 700 mL each and check for debris. If the samples are clean, return them to the system. If the samples contain debris, drain the oil and begin the flush again.
k. Circulate the oil for 60 minutes and then turn off the tank heater.
l. When the test oil tank temperature has cooled to 38°C (100°F), close the oil supply valve on the inlet side of the pressure pump.
m. Turn off the following switches:
i) Pumps
ii) Timer.
iii) Stirrer.
iv) Recorder.
v) Flowmeter.
vi) Doric.
vii) Support test oil cart power.
viii) Control panel power.
n. Drain the test oil tank. Take one sample of the oil (100 mL). Mark the samples and the remaining oil:
Flush oil
B.R. Test
Test #
Oil Code
Date
o. After removing the two flush filters, clean and store them.

6.6.2 Preparation for test.

a. Start the data acquisition system.
b. Install two pre-weighed filters. A fine mesh (100 mesh) filter is installed on the spray nozzle inlet and a coarse mesh (40 mesh) filter is installed on the inlet line of the scavenge pump.
c. Close the oil supply valve. Check the oil level in the test oil and support oil tanks. The test oil tank should be charged with test oil at room temperature until the fluid level is

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1.3 cm (1/2 inch) above the baffle plate with the pressure and scavenge lines filled. This level should be marked and will establish the "FILL" mark or the minimum operating level during test operation. The expansion of the test oil upon heating to test conditions will establish the "FULL" mark. The support oil tank should contain approximately 23 to 30 L (6 to 8 gal) of a MIL-PRF-23699, MIL-PRF-7808, or DOD-PRF-85734 lubricating oil.

d. Ensure that the test oil tank vent is open and that any vent line drain is closed.

e. Check the main air supply pressure, adjusting it if necessary to maintain a minimum of 207 gage kPa (30 psig).

f. Open water valves to dynamic and support oil cooler.

g. Check dynamic water pressure: it should be 310 gage kPa (45 psig).

h. Adjust the air pressure for the load cylinder to 207 gage kPa (30 psig).

i. Set the load valve to the zero load position (clockwise direction).

j. Add distilled water to the vapor bottle until the desired level is reached.

k. Open air bleeds on both sides of the differential manometer for the screw thread oil seals which should be filled with test oil.

l. Record the watt hour meter readings from the test oil tank heater and bearing heater meters. Turn on the electric power (breaker boxes) to the test oil tank heater, support oil tank heater, test bearing heater, and the bearing drive motor.

m. Turn off automatic mode switch.

n. Turn on the following switches on the control panel:

i) Control panel power.
ii) Data Acquisition System.
iii) Support test oil cart power.
iv) Stirrer motor.
v) Timer (adjust as necessary).
vi) Test oil heater.
vii) Doric.
vii) Water heater.
ix) Load by-pass.

o. When the test oil temperature reaches 121°C (250°F) turn on the support oil heater. The temperature controllers should be adjusted to obtain 138°C (280°F) and 82°C (175°F) in the test oil and support oil tanks, respectively. In the test oil tank, the maximum heater skin temperature should not exceed 29°C (85°F) above the test oil tank target temperature during warmup. Higher temperatures indicate improper electrical hookup of excessively high supply voltage which should be corrected before continuing the test program.

p. When the bulk oil temperature in the test tank reaches 138°C (280°F) turn on the air supply to the end cover and adjust to approximately 9.9 L/min (0.35 ft^3/min). Air pressure should be held at 34.4 gage kPa (5.0 psig).

q. Start the test oil pressure and scavenge pumps. Ensure that the flow is 69 gage kPa (10 psig) minimum at the test oil jet. The test oil flowrate is 600 ± 30 mL/minutes.

r. Check the oil supply valve between the test oil tank and the variable speed pressure pump; it should be wide open.

s. Start the support oil pumps; there should be at least 414 gage kPa (60 psig) manifold pressure. At the screw thread seal there should be at least 0.8 cm (0.3 inch) of oil

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positive pressure relative to the support oil side (manometer leg to test compartment lower).

t. Set and maintain the support oil pressure at 552 to 689 gage kPa (80 to 100 psig) by means of a regulating valve.

u. Allow oil temperature at the support scavenge line to reach at 38.8°C least (100 °F).

v. Start the bearing drive motor; the bearing speed is preset at 10,000 rpm.

w. Turn off the load by-pass switch.

x. Apply a load to the test bearing. Increase the load on the bearing gradually with the load controls. The load cylinder pressure should be held at 352 gage kPa (51 psig).

y. Continue running until the following conditions have been stabilized:

Test oil in temperature	121°C ± 2.5°C	(250°F ± 5°F)
Test tank bulk oil temperature	138°C ± 2.5°C	(280°F ± 5°F)
Air flow, end cover	9.9 ± 1.4 L/min	(0.35 ± 0.05 ft^3/min)

z. Make at least two separate flow checks of one-minute duration each at the three-way valve position on the pressure side of the scavenge pump. The measured volume flow rate should be 600 ± 30 mL/minutes. The hot test oil collected during the flow check should be poured immediately back into the test oil tank through the fill port and the port recapped.

aa. Continue running for one hour at these conditions with the bearing heater off. If during or at the end of this period, the maximum bearing temperature has exceeded 177°C (350°F), shut down the rig, install a new test bearing, and repeat items 1 through 26. If the maximum bearing temperature has not exceeded 177°C (350°F) during this period, proceed with the next step.

ab. Turn on the test bearing heater and adjust indicator flag to the target temperature on the temperature controller.

ac. Check the three test bearing outer-race thermocouples to make sure that the 11 o'clock thermocouple is connected to the temperature controller. Maintain 260°C (500°F) on the 11 o'clock position thermocouple.

ad. Adjust the test oil tank thermal switch so that the target temperature for the type of test being run is met within 30 minutes (minimum) and 60 minutes (maximum). Maximum heater skin temperature should not exceed 29°C (85°F) above test oil target temperature during the warmup period. The nozzle temperature should be maintained at 177°C ± 2.5°C (350°F ± 5°F) for type 1-1/2 test.

TABLE V. Target temperatures by test type.

Test type	1	1-1/2	2
Target temperature	171° ± 2.5°C (340° ± 5°F)	199° ± 2.5°C (390° ± 5°F)	227° ± 2.5°C (440° ± 5°F)
Oil-in temperature	149° ± 2.5°C (300° ± 5°F)	177° ± 2.5°C (350° ± 5°F)	204° ± 2.5°C (400° ± 5°F)

ae. Continue running until the conditions of TABLE V for the type of test requested have stabilized.

af. Take at least two separate flow checks of one-minute duration at each of the three-way valve position on the pressure side of the scavenge pump. Flow rate should be 600 ±

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30 mL/minutes. The hot test oil collected during the flow check should be poured immediately back into the test oil tank through the fill port and the port recapped.

ag. Continue endurance testing. At the completion of the endurance test, the test bearing is to be restabilized at 121°C (250°F) oil inch temperature to determine the relative change in stabilized temperatures compared to the results obtained at the start of the test. 6.6.3 Daily startup sequence.

i) Update data acquisition system.

ii) Check oil level through visual sight gauge. Add enough oil so that the bottom of the test oil meniscus is even with the mark that was made in step 3 of procedure 6.6.2. Open vent drain valve and remove all oil from the vent line.

iii) Repeat steps 4 to 23 for startup procedure.

iv) Turn on test bearing heater and adjust as necessary to reach 260°C ± 2.5°C (500°F ± 5°F) on the highest reading thermocouple.

v) Continue running until the conditions have stabilized for the type of test being run as shown in TABLE V.

vi) Make at least two separate flow checks of one-minute duration each at the three-way valve position on the pressure side of the scavenge pump. The measured volume flow rate should be 600 ± 30 mL/minutes.

6.6.3 Daily running procedure.

a. Monitor the data acquisition system for maintenance of test temperatures within correct operating ranges. Monitor gauges and flowmeters as well.

b. Take a sample of test oil (100 mL) from the three-way valve in the scavenge line or with an automated system every 10 h (including warmup time along with endurance time). The first test sample should be fresh test oil before the oil is introduced into the system. If the 10-hour sampling period happens to fall during a shutdown period, take the samples just before shutdown or after first reaching type 1, 1-1/2, 2, or 3 test conditions from next startup of rig. Mark each sample:

B.R Test Test #
Oil Code
Date
Hour of Sample

c. Makeup oil should be added immediately after each 100 mL sample is taken by adding sufficient quantity of unused test oil to the tank to bring the oil level in the visual sight gage to the full mark.

d. The test oil filter changes should be made after shutting down on the first day. If the pressure drop across the 100-mesh filter exceeds 21 gage kPa (3 psig) during operation, the rig should be shut down in accordance with the shutdown procedure under 6.6.5, and a new weighed filter reinstalled. The dirty 100-mesh filter element and the 40-mesh filter element are to be weighed and the amount of the weight gain recorded. Prior to weighing, the 100- and 40-mesh filter screens should be allowed to drain for 1 hour at room temperature.

e. Actual running time should conform to the following conditions:

i) 17-hour maximum test time between shutdowns.

ii) Minimum shutdown period of 7 h.

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iii) Minimum daily running time of 6 h.

6.6.4 Shutdown procedure.

a. Shut off bearing and tank heaters.
b. Open air regulated loading valve on console and reduce loading pressure to 172 gage kPa (25 psig).
c. Continue running at these conditions until test bearing temperature (maximum) drops to 204°C (400°F).
d. Reduce load completely and shut off drive motor.
e. When the output shaft stops turning, shut off test pumps, support pumps and heat, test tank stirrer, main electrical and water switches, and all air valves.

6.6.5 post-stabilization run.

a. This procedure is done in place of the last shutdown.
b. Repeat steps 1 through 3 of the shutdown procedure.
c. Reduce load on bearing completely.
d. Allow the test oil tank temperature and the oil in temperature to reach 138°C (280°F) and 121°C (250°F) respectively. Maintain these temperatures.
e. When the test temperatures stabilize, run for one hour and monitor the maximum temperature of the test bearing.
f. Record the highest bearing temperature.
g. Shut off bearing drive motor.
h. Repeat step 5 of the shutdown procedure.

7. CALCULATION

7.1 Demerit system for rating deposits in ERDCO® bearing machine.

7.1.1 Outline of method.

7.1.1.1 Cleanliness demerits. Bearing head cleanliness shall be reported in this demerit system by the assignment of values of 0 to 20 to the inspection results, as follows:

0 shall designate a new or thoroughly clean condition.
20 shall represent the worst condition that could be expected.

7.1.1.2 TABLE VI shows the numerical demerits to be assigned to different types and degrees of deposits:

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TABLE VI. Deposit rating demerit scale.

Type of deposit	Degree	Demerit values
Flaked carbon	Heavy	20
	Moderate	19
	Light	18
Blistered carbon	Heavy	17
	Moderate	16
	Light	15
Crinkled	Heavy	14
	Moderate	13
	Light	12
Smooth carbon	Heavy	11
	Moderate	10
	Light	9
Sludge	Heavy	8
	Moderate	7
	Light	6
Varnish	Heavy	5
	Moderate	3
	Light	1
Clean		0

NOTE: All demerits shall be reported to the nearest whole number

7.1.1.3 TABLE VII defines deposit types and degrees.

TABLE VII. Description of bearing deposit rating terms.

Type of deposit	Degree	Description
Flaked Carbon		Carbonaceous coating that cannot be removed by wiping with a rag. Broken blisters, peeling.
	Heavy	Flakes are closely packed with a majority of the coating raised off the surface.
	Moderate	Flakes are moderately scattered with minimal visually observable thickness.
	Light	Flakes are widely scattered with minimal visually observable thickness.

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TABLE VII. Description of bearing deposit rating terms – Continued.

Type of deposit	Degree	Description
Blistered carbon		Dull, matte finish type coating with multiple raised areas that are scattered bubbles or blisters.
	Heavy	Measureable thickness greater than 1/64 inches with a majority raised from the surface.
	Moderate	Visually observable thickness except for the raised areas.
	Light	No visually observable thickness except in the raised areas.
Crinkled carbon		Dull, matte finish type coating with multiple raised areas that can be intersecting and black in color.
	Heavy	Measureable thickness greater than 1/64 inches with a majority raised from the surface.
	Moderate	Visually observable thickness except for the raised areas.
	Light	No visual observable thickness except in the raised areas.
Smooth carbon		Dull, matte finish type coating that is black in color and cannot be wiped off.
	Heavy	Measureable thickness greater than 1/64 inches that can be easily scraped off.
	Moderate	Visually observable thickness that cannot be easily scraped off.
	Light	No measureable thickness and can easily be scraped off.
Sludge		Shiny, oily emulsion of carbon and oil. Can be wiped off with a rag.
	Heavy	Thickness greater than 1/64 inches.
	Moderate	Visual observable thickness.
	Light	Just covering.
Varnish		Varnish or lacquer-like coating shiny.
	Heavy	Shiny non-transparent with a measureable .thickness.
	Moderate	Non-transparent with no measureable thickness.
	Light	Transparent with no measureable thickness.
Clean		Clean - no deposit.

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7.1.2 Computation of overall rating of the ERDCO® high temperature bearing head.

7.1.2.1 An area demerit shall be determined in accordance with the area covered by the deposits as follows:

$$\frac{Percent\ area\ covered}{10} = Area\ demerit$$

7.1.2.2 A rating shall be obtained by multiplying the area demerit by the demerit value assigned factor assigned in TABLE VI. The rating for each item shall be modified by multiplying by the weight factor by TABLE VIII which will yield a demerit rating.

7.1.2.2.1 Bearing rating and demerits are determined from the following categories:

Rollers	1. face
	2. rear
	3. front
Cage	4. between rollers
	5. rear
	6. front
Outer ring	7. path
	8. rear
	9. front
	Sum of Ratings x 5 (Weight Factor)
	Demerit for Bearing
Inner ring	10. rear
	11. front

7.1.2.3 The total bearing head deposit rating shall be the sum of the demerits for each item of TABLE VIII.

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TABLE VIII. Bearing head deposit weight factor rating.

No.	Item		Weight factor	Demerit
1.	End cover	Multiply rating by	1	=
2.	Spacer and nut	Multiply rating by	2	=
3.	Heater-front	Multiply rating by	3	=
4.	Heater-rear	Multiply rating by	3	=
5.	Seal plate	Multiply rating by	1	=
6.	Bearing	Multiply rating by	5	=

7.1.2.4 The overall rating of the ERDCO® high temperature bearing head shall be the total bearing head deposit rating divided by 6.

7.1.2.5 An example of the calculations described is shown in TABLE IX.

TABLE IX. Example of detailed rating.

Item	Description	Area demerit 7.1.2.1	Demerit scale 7.1.1.2	Rating 7.1.2.2	Factor 7.1.2.2 TABLE VI	Demerit
End cover	Clean	1	0	0	1	0
Space and Nut	10% L. sludge	1	6	6		
	90% M.	9	3	27	2	66
	Varnish			= 33		
Heater - front	25% Clean	2.5	0	0		
	25% L. Varnish	2.5	1	2.5		
	25%L. Crinkled carbon	2.5	12	30	3	195
	25% M.	2.5	13	32.5		
	Crinkled carbon			= 65		

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TABLE IX. Example of detailed rating – Continued.

Item	Description	Area demerit 7.1.2.1	Demerit scale 7.1.1.2	Rating 7.1.2.2	Factor 7.1.2.2 TABLE VI	Demerit
Heater - Rear	10% L varnish	1	1	1		
	10% M varnish	1	3	3		
	10% H varnish	1	5	5	3	195
	20% M sludge	2	7	14		
	50% L smooth carbon	5	9	45		
				=68		
Seal - plate	5% M varnish	0.5	3	1.5		
	95% clean	9.5	0	0	1	1.5
				=1.5		
Bearing:						
a. Rollers:						
(1) Face	100% M	10	3	30		
(2) Rear	varnish	5	1	5		
(3) Front	50% L. varnish	5	3	15		
	50% M varnish			=50		
b. Cage:						
(4) Between rollers		10	0	0		
(5) Rear	100%clean					
	50%clean	5	6	30		
(6) Front	50%L sludge	5	0	0		
	50%clean	5	6	30		
	50% L sludge	5		=60		
c. Outer ring:	75% clean	7.5	0	0		
(7) Path	25% M varnish	2.5	3	7.5		

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TABLE IX. Example of detailed rating – Continued.

Item	Description	Area demerit 7.1.2.1	Demerit scale 7.1.1.2	Rating 7.1.2.2	Factor 7.1.2.2 TABLE VI	Demerit
(8) Rear	30%M varnish	3	3	9		
	40%H varnish					
	30% L smooth					
	carbon					
(9) Front	20% clean	2	0	0		
	40%M varnish	4	3	12		
	40%H varnish	4	5	20		
				=95.5		
d. Inner ring:						
(10) Rear	50% L varnish	5	1	5		
	50% M sludge	5	7	35		
(11) Front	50% L varnish	5	1	5		
	50% L crinkled carbon	5	12	60		
				=105		
Bearing (total)				310.5/11		141
				=28.25		= 607.5
Overall ratings						
(Average of items)				607.6/6		
1, 2, 3, 4, and 6)				=101.2		

8. REPORTING

8.1 As required by the specification.

9. PRECISION

9.1 Precision data have not been developed for this method.

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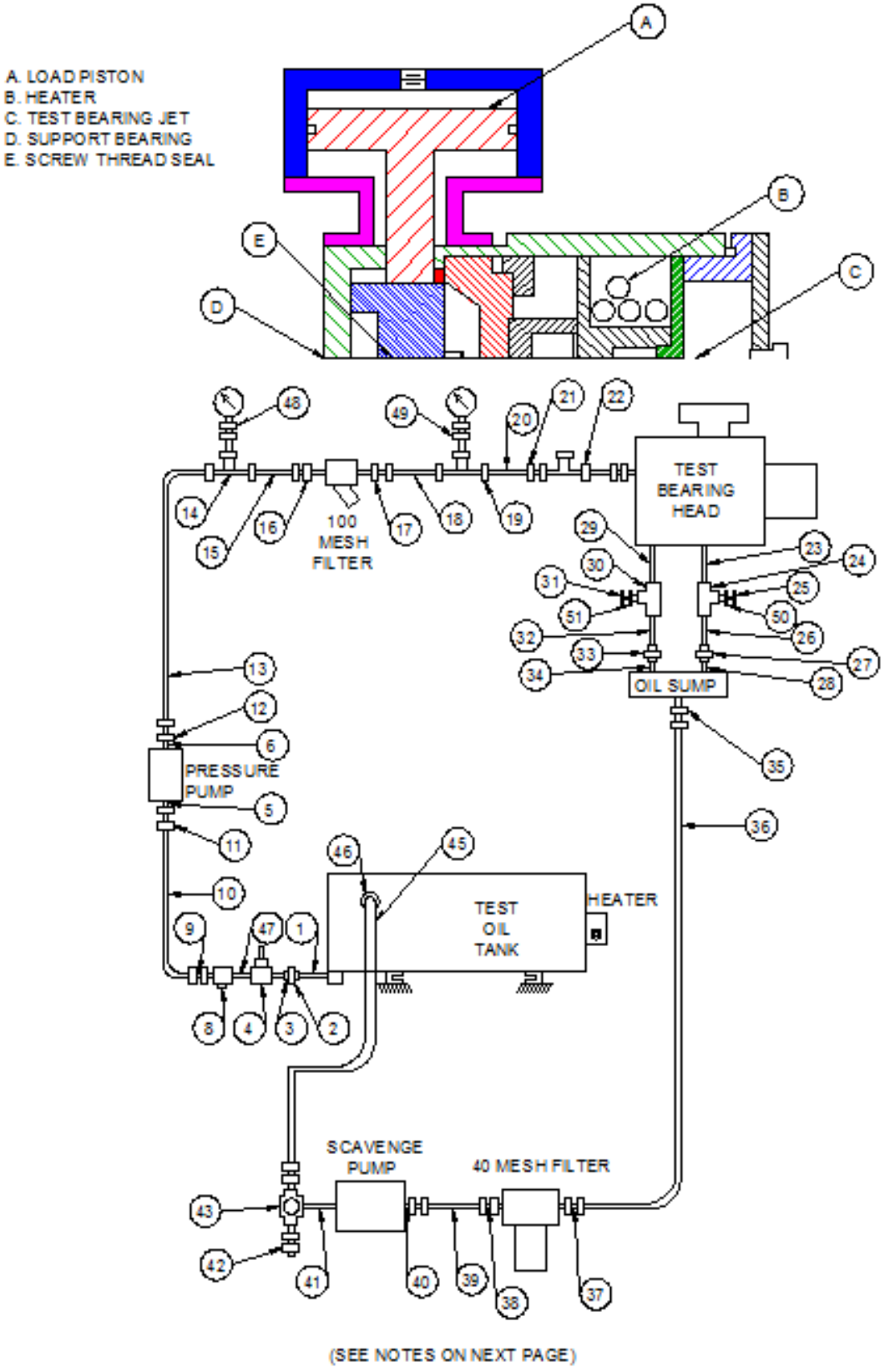


FIGURE 1. Schematic of test oil system.

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Notes on FIGURE 1:

1. 3/8 pipe nipple - 1-1/2 long
2. 3/8 pipe union
3. 3/8 pipe nipple - 1-1/2 long
4. 3/8 valve (iron)
5. 3/4 by 3/8 reducer
6. 3/4 by 3/8 reducer
7. Omitted
8. 3/8 pipe tee
9. 3/8 tube H 3/8 NPT-male
10. 3/8 tubing
11. 3/8 tube H 3/8 NPT-male
12. 3/8 tube H 3/8
13. 1/2 tube
14. 1/4 tube tee
15. 1/4 tube
16. 1/4 tube H 3/8 NPT-male
17. 1/4 tube H 3/8 NPT-male
18. 1/4 tube
19. 1/4 tube tee
20. 1/4 tube
21. 1/4 tube H 1/8 NPT-female
22. 3/16 tube H 1/8 NPT-male tee (branch T/C modified)
23. 3/8 pipe nipple - 1-1/2 long
24. 3/8 pipe tee
25. 3/16 tube H 1/8 NPT-male (T/C modified)
26. 3/8 pipe nipple - 1-1/2 long
27. 3/8 pipe union
28. 3/8 pipe nipple - 1-1/2 long
29. 3/8 pipe nipple - 1-1/2 long
30. 3/8 pipe tee
31. 3/16 tube H 1/8 NPT-male (T/C modified)
32. 3/8 pipe nipple - 1-1/2 long
33. 3/8 pipe union
34. 3/8 pipe union - 1-1/2 long
35. 1/2 tube H 1/2 NPT-male
36. 1/2 tubing
37. 1/2 tube H 3/8 NPT-male
38. 1/2 tube H 3/8 NPT-male
39. 1/2 tube1. 3/8 pipe nipple - 1-1/2 long
40. 1/2 tube H 1/2 NPT-male
41. 1/2 pipe npl H 3 long
42. 1/2 tube H 1/2 NPT-male
43. 1/2 three way valve (iron)
44. 1/2 tube H 1/2 NPT-male
45. 1/2 tube
46. 1/2 tube H 3/8 NPT-male
47. 3/8 pipe NPT - 1-1/2 long
48. 1/4 tube bulkhead connector (steel)
49. 1/4 tube bulkhead connector
50. 3/8 by 1/8 reducing bushing
51. 3/8 by 1/8 reducing bushing

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Updated solvents to add in Oakite LRS	Added new solvent since last revision	2	NAWCAD
30 Sep 2019	Updated rating table criteria	Old table was inaccurate	2	NAWCAD
30 Sep 2019	Updated the calculation section to include the actual calculation	Seemed appropriate	2	NAWCAD
30 Sep 2019	Removed references to a strip chart and added in DAQ system	Updated	2	NAWCAD
30 Sep 2019	Added in section for type 3 testing to accommodate Air Force method	The Air Force cancelled their method that is now captured in this method	2	NAWCAD
30 Sep 2019	Control temperature changed from second hottest point to the 11 O'clock position	Reflects how the method is actually run	2	NAWCAD
30 Sep 2019	Updated test table for each oil sample	Reflects accurate methods that are used	2	NAWCAD
30 Sep 2019	Included new section 6.1.2 to address setup a new condition added to the method	The new condition addresses the Air Force requirements and allows them to cancel their older method	2	NAWCAD
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT
17 May 2021	Editorial changes to fix numbering.	Editorial corrections	2	NAWCAD

Method Prepared By:
-Navy - NAWCAD – 2019

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Method 3411.2
September 30, 2019

3411.2 – THERMAL STABILITY AND CORROSIVITY OF AIRCRAFT TURBINE ENGINE LUBRICANTS

1. SCOPE

1.1 This method describes a procedure for determining the thermal stability and corrosiveness of aircraft turbine lubricants in an environment free of air and moisture. The catalytic effects of mild steel on the decomposition of the fluids and the corrosion of the steel by the thermal degradation products of the lubricant are measured.

2. SUMMARY

2.1 A weighed mild steel specimen is placed in a glass thermal stability test cell with a measured amount of test lubricant. Air and moisture are removed from the test cell by heating and shaking under vacuum prior to sealing the tube. The sealed test cell is then heated at a given temperature for a specified period of time. The stability of the lubricant is assessed by determining the change in viscosity, total acid number and by the appearance of the residual lubricant at the end of the heating period. The steel specimen is weighed and examined under a microscope to determine changes due to corrosion.

3. SAMPLE SIZE

3.1 Approximately 10 mL for each steel corrosion test.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Test cell – A borosilicate glass test cell having the dimensions given in FIGURE 1 is recommended as the sample container.

4.2 Heating bath, 9.5 L (10 qt) capacity, (manufactured by Sta-Warm Electric, Ravena, Ohio or equivalent). An aluminum block or sand bath may be used in place of the heating bath if so desired. The resistance wire-heated aluminum block must have a suitable number of openings of the proper diameter (approximately 2.0 cm) and depth (approximately 21 cm) so that the sealed test cell is completely immersed in the heater. The temperature of the aluminum block or heating bath shall be capable of being controlled within ± 1°C (± 2°F) at the test temperature. The temperature gradient between oil sample and the aluminum block or bath shall be kept within 1°C (± 2°F).

4.3 Test cell holder – When a heating bath is used, a heavy-wall steel sample tube holder of the proper size to contain the test cell shall be provided for safety reason.

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4.4 Becton, Dickinson and Company Yale Luer-Lock (20YL), or equal 20 mL hypodermic syringe with a No. 14 square-tip needle approximately 20 cm (8 inch) long.

4.5. Vacuum pump, capable of reducing the pressure to at least 10 µm of mercury or less.

4.6 McLeod gauge or other suitable pressure measuring device.

4.7 Water bath, capable of maintaining a temperature of 80°C ± 3°C (175°F ± 5°F) and sufficient size to contain the test cell.

4.8 Laboratory safety shield, constructed of either safety glass or wire reinforced plastic.

4.9 Balance, sensitivity 0.1 mg.

4.10 Microscope, 20X magnification.

4.11 Forceps, stainless steel.

5. MATERIALS

5.1 Toluene, reagent-grade.

5.2 Acetone, reagent-grade.

5.3 Nitric-sulfuric acid mixture, approximately 1:10.

5.4 Abrasive paper, silicon carbide, 240 grit.

5.5 Steel specimens of approximately 0.15, 0.7, and 7.5 cm made from C1008/C1010 cold rolled half hard with #2 finish and #4 round edge.

5.6 Detergent, liquid, water soluble

6. PROCEDURE

6.1 Definitions

6.1.1 Thermal stability is the resistance to permanent changes in properties of the lubricant caused solely by heat.

6.1.2 Corrosion is any surface damage of the steel specimen resulting in a weight change.

6.2 Preparation of apparatus.

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6.2.1 Cutting test cells - The cells are cut in two, either on an emery cutoff wheel or by circumscribing the tube with a hot wire about 17 cm from the bottom. Test cells can also be precut conforming to the thermal stability test cell.

NOTE 1: Circumscribing with a hot wire is preferred because it is cleaner and safer.

6.2.2 Cleaning test cell - Clean the divided test cell by scrubbing with a suitable detergent, rinsing in tap water, heating in hot nitric-sulfuric acid mixture, rinsing with distilled water and drying in an oven at 105°C (225°F) for l h. The cells can be stored in a desiccator until used.

6.2.3 Preheat and adjust aluminum block or heating bath temperature to 274°C ± 1°C (525°F ± 2°F).

6.2.4 Preparation of steel specimens.

6.2.4.1 Cut samples approximately 0.15, 0.7, and 7.5 cm. Precut samples can be procured and are acceptable for this test method.

6.2.4.2 Remove pits and other imperfections on a fine grinding wheel.

6.2.4.3 Polish with 240 grit silicon carbide paper to remove all surface blemishes from both sides and all four edges.

NOTE 2: In all succeeding steps, handle the steel specimens with filter paper or forceps.

6.2.4.4 Clean all metal dust from the samples by rubbing vigorously with lint-free tissue.

6.2.4.5 Wash sample strips in warm toluene, then in warm acetone and allow to air dry.

6.2.4.6 Weigh each specimen to within 0.1 mg.

6.3 Test procedure.

6.3.1 Duplicate tests are conducted.

6.3.2 A freshly prepared steel specimen is placed in the lower half of the cleaned cell, which is then fused to the top half.

NOTE 3 Proper annealing of the fused glass cell will prevent rupture and leaks during testing.

6.3.3 Introduce 6 mL of the oil samples below the capillary end of the test cell with the aid of the hypodermic syringe. With this sample volume the metal strip is only partially immersed in the oil leaving approximately 1.3 cm (1/2 inch) exposed.

6.3.4 Connect the fused test cell containing the oil sample and metal specimen to the vacuum pump. This can be done with vacuum tubing connected through a Kjeldahl connecting bulb, T_s joint 24/40, and a No. 1 rubber stopper drilled to hold the thermal stability tube.

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6.3.5 Mount the test cell, while still connected to the operating vacuum pump, in a water bath maintained at 80° ± 3°C (175° ± 5°F). Degas the cell and its contents while shaking gently for 20 minutes. A Thomas-Boerner oscillating platform shaking apparatus may be used. Mounting the tubes about 16 cm above the platform. A manifold can be used to evacuate more than one cell at a time. If frothing persists, continue the heating and shaking for another 10 minutes.

NOTE 4: Certain ester based as well as other type synthetic lubricants are hydrolytically unstable at the test temperature of 274°C (525°F). Therefore, the procedure described in 6.3.5 to remove all water before sealing the cell is necessary. The pressure in the cell, as measured with a McLeod gauge, or other suitable measuring device, shall be no more than 10 µm before sealing.

6.3.6 While the test cell is still under vacuum it is sealed and drawn off at the capillary approximately 3 cm above where it joins the cell body. (See Note 3).

6.3.7 Weigh the test cell to the nearest mg.

6.3.8 Insert the glass test cell in the heater [preheated to 274°C ± 1°C (525° ± 2°F)] for period of 96 hours. If a heating bath is used, insert the test cell in the steel holder and place the assembly in the bath.

NOTE 5: Safety shields must always be used as a precaution when placing the test cells in the heater, removing the cells from the heater, and opening the cells. Tremendous pressures are generated during the test period.

6.3.9 At the conclusion of the heating period, remove the test cell and allow to cool to ambient temperature. Wipe down the outside of the cell with suitable solvent.

6.3.10 Visually observe the samples for evidence of insolubles, phase separation, interactions between metal and oil, or changes in color of the oil. A significant difference in final color between duplicate tests will necessitate retesting since color differences indicate improper preparation of one or both of the test cells.

6.3.11 Reweigh the test cell. Any significant loss in weight indicates leakage of the test cell and the test must be repeated.

6.3.12 As a safety measure the cells must be cooled before cutting them open because of the pressure which develops in testing. Any convenient cooling method may be used such as dry ice - acetone, liquid nitrogen, cooling chambers, etc.

6.3.13 Remove the oil sample and steel specimen by carefully breaking the test cell. This may be accomplished by scribing a mark around the test cell near the top and applying a hot wire to the scratch mark. (See Note 5).

6.3.14 Remove the steel specimen with forceps, wash with a stream of toluene or an equivalent solvent to remove the oil residue from a wash bottle to remove oil then dip in two successive warm toluene washes and a final dip in warm acetone. Wipe with lint-free tissue and

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air dry. Weigh to the nearest 0.1 mg. Using a 20X microscope, carefully examine each steel specimen and note any evidence of pitting or etching either above or below the liquid level.

6.3.15 Inspect the tube for insolubles and gum deposits.

6.3.16 Determine the viscosity at 40°C (104°F) of both the original lubricant and the lubricant after exposure to heat in accordance with ASTM D445.

6.3.17 Determine the neutralization number on both the original and heated lubricant in accordance with ASTM D664.

7. CALCULATIONS

7.1 $$Percent\ viscosity\ change = \frac{V_2 - V_1}{V_1} * 100$$

where:
V_1 - viscosity, Initial
V_2 - viscosity after test

7.2 $$Total\ Acid\ Number\ change = N_2 - N_1$$

where:
N_1 - Total Acid Number, Initial
N_2 - Total Acid Number after test

7.3 $$Change\ in\ weight\ of\ the\ steel\ specimen\ in\ mg/cm2 = \frac{B-A}{12.96}$$

where:
A - Metal weight after test in mg
B - Metal weight before test in mg

8. REPORTING

8.1 The report shall include the following:

a. Visual appearance of test cell, insolubles and gum deposits.
b. Visual appearance of lubricant after test.
c. Visual appearance of steel specimen above and below the liquid level.
d. Change in the neutralization number of the lubricant sample.
e. Percentage change in viscosity at 40°C (104°F).
f. Change in weight of the steel specimen in mg/cm^2.

9. PRECISION

9.1 Precision data have not been developed for this method.

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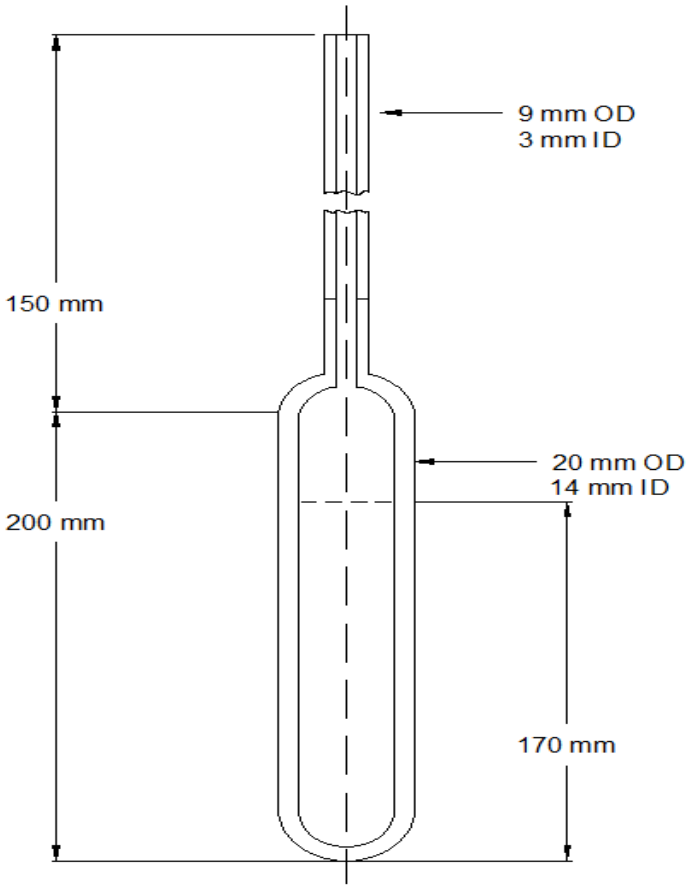


FIGURE 1. Thermal stability-corrosion test cell.

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30 Sep 2019	Added sand bath to section 4.2	This apparatus is what our lab uses	2	NAWCAD
30 Sep 2019	Added precut samples can be used	Samples are readily available	2	NAWCAD
30 Sep 2019	Changed the temperature for the viscosity measurement from 100 F to 104 F	40 C is the standard temperature run for viscosity now	2	NAWCAD
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT
20 May 2021	Removed extra numbers behind solvents in the materials section	Additional numbers did not add extra information	2	NAWCAD
30 April 2026	Section 5.5: changed Steel specimens made from QQ-S-698 mild steel to steel specimens made from C1008/C1010 cold rolled half hard with #2 finish and #4 round edge.	QQ-S-698 was cancelled	2	NAWCAD

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Navy – NAWCAD- 2019

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Method 3432.2
September 30, 2019

3432.2 – COMPATIBILITY OF ELASTOMERS WITH AIRCRAFT TURBINE LUBRICANTS

1. SCOPE

1.1 This method is used for determining effects of aircraft turbine lubricants on elastomers.

2. SUMMARY

2.1 It consists of determining the swelling effect, change in elongation, change in tensile strength and change in hardness of standard fluorocarbon and fluorosilicone elastomer specimens after being immersed in the sample for 72 hours, at temperatures of 175°C (347°F) for fluorocarbon specimens and 150°C (302°F) for fluorosilicone specimens.

3. SAMPLE SIZE

3.1 Approximately 1000 mL of the lubricant to be tested.

4. APPARATUS

4.1 Bath; aluminum block or oil bath capable of maintaining sample temperature within 1°C (1.8°F) of required test temperature. Aluminum block baths will have 5.0 mm ± 1 cm in thickness of insulation. Oil baths will have a cover with openings for immersion of test tubes.

4.2 Test tubes, heat-resistant, 30 cm length and 3.9 cm outside diameter.

4.3 Hanger, specimen, made of stainless steel or nichrome wire.

4.4 Corks, standard size number 19.

4.5 ASTM Die C as described in ASTM D412.

4.6 Bench Marker as described in ASTM D412.

4.7 Micrometer as described in ASTM D412.

4.8 Testing Machine for elongation and tensile strength measurements as described in ASTM D412.

4.9 Grips as described in ASTM D412.

4.10 Durometer, Type A Shore as described in ASTM D2240.

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4.11 Stamp Pad and Ink as described in ASTM D412.

4.12 Punch, Leather, 0.159 cm (1/16 inch) diameter.

4.13 Balance, sensitivity 0.1 mg, suitable for weighing in both air and water.

4.14 Tubing, Pyrex, 0.8 cm O.D.

4.15 Beaker, 400 mL.

5. MATERIALS

5.1 Standard test rubber sheets, type F-A, conforming to SAE AMS 3217/4.

5.2 Standard test rubber sheets, type FS, conforming to SAE AMS 3217/5.

5.3 Cleaning solution. Any suitable laboratory glass cleaning fluid may be used providing it removes all evidence of staining and deposits from previous tests.

5.4 Dry cleaning solvent (such as Stoddard Solvent or similar).

5.5 Ethyl alcohol, ACS grade [0, 3, 0].

5.6 Acetone, ACS grade [1, 3, 0].

5.7 Distilled water.

5.8 Filter paper.

5.9 Detergent, liquid, water soluble

6. PROCEDURE

6.1 Clean all glassware as follows:

a. Rinse in dry cleaning solvent, and allow to air dry (see note 1).

NOTE 1: CAUTION. Dry cleaning solvent is both toxic and flammable. Do not breathe its fumes nor allow it to come in contact with the skin. Keep flames away from the dry cleaning solvent.

b. Soak in glass cleaning solution at a temperature of 71°C (160°F) for 1 hour (minimum time), if glassware contains oil deposits from a previous test.
c. Wash glassware with laboratory detergent. Rinse thoroughly with tap water and then with distilled water.
d. Dry in an oven at 100°C to 105°C (212° to 221°F) and store in dust free location.

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6.2 Clean suspension device and hooks thoroughly with dry cleaning solvent, wipe with a clean dry cloth, air dry and store in a dust free location.

6.3 Inspect the sheets of standard stock elastomer for flaws. Sections of an elastomeric sheet having a flaw will not be used for specimens. Cut four specimens from sheets of standard elastomer stock, with all sheets being from a single production batch, for each lubricant to be tested. Cut the specimens using ASTM Die C as described in method ASTM D412. Punch a 0.159 cm (1/16 inch) hole in each end of the cut elastomer specimens using a leather punch. The hole shall be centered and 0.64 cm (1/4 inch) from the end of the specimen. Weigh each test specimen in air and in water in accordance with method ASTM D471 (and record weights). Using a type A Shore Durometer obtain four Durometer readings (two on each large end) in accordance with method ASTM D2240. Determine the initial hardness of each specimen by averaging the four readings.

6.4 Preheat and adjust immersion bath to maintain lubricant sample test temperature of 175°C ± 1°C (347°F ± 1.8°F) for F-A elastomers specimens and 150°C ± 1°C (302°F ± 1.8°F) for FS elastomer specimens. A suggested method of controlling these temperatures is to place a similar test tube containing 140 mL of lubricant in the bath at the same depth as the test lubricant. A thermocouple is then placed in this tube at a depth equal to the depth of the small section of the elastomeric specimens when suspended in the test lubricant tubes. The test tube containing the elastomeric specimens and test lubricant is placed into the bath to a depth which places the level of the test lubricant 4.1 cm (1-5/8 inch) above the top of the heating medium and not from the top of the insulation of bath covering.

6.5 Immerse the elastomeric specimens in the heated test lubricant by means of a specimen hanger made from stainless steel or nichrome wire. The hanger shall have two hooks at one end permitting one elastomeric specimen to be hung from each hook. The two specimens shall hang parallel and 1.3 cm (1/2 inch) apart. Attach the wire specimen hanger to a standard size number 19 cork and fit with two lengths of 0.8 cm Pyrex glass tubing (chimneys). Insert one chimney, 7.6 cm (3 inch) in length, through the cork such that the chimney extends 1.3 cm (1/2 inch) above the top of the cork. Insert the second chimney 12.7 cm (5 inch) in length, through the cork such that the chimney extends 8.9 cm (3 1/2 inch) above the top of the cork. Place the cork fitted with the chimneys, specimen hanger and specimens into the pre-heated test tube containing the test oil. Adjust the length of the specimen hanger so that when the cork is placed tightly into the test tube, the specimens will hang with equal distance of test lubricant above and below the specimens using 140 mL of test lubricant. Hang only two elastomeric specimens in one tube and adjust cork such that specimens do not contact the side of the test tube.

6.6 Remove the elastomeric specimens from the test tube after 72 hours with the time starting when tubes are first inserted into the bath. Place the elastomeric specimens in a beaker of fresh test lubricant which is at room temperature and allow to cool for a minimum of 30 minutes. Remove the specimens from the cooling oil and place them into a beaker of acetone for a minimum of 14 seconds and a maximum of 20 seconds. Dry the specimens with filter paper by blotting. Determine volume swell, hardness, elongation and tensile strength of the elastomeric specimens.

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6.7 Determine a base line for elongation and tensile strength measurements for each batch of standard elastomer stock. Cut eight specimens in accordance with 6.3 with no more than two specimens being cut from any one sheet. Determine the elongation and tensile strength of each untreated specimen in accordance with method ASTM D412 using a 2.5 cm (1 inch) bench marker. Use the average of these values in determining elongation change and tensile strength change of the oil treated specimens.

6.8 Weigh each oil treated specimen in air and in water in accordance with method ASTM D472 and calculate percent volume swell for each specimen. Calculate the percent volume swell of the test lubricant as the average of four individual volume swell determinations.

6.9 Determine hardness readings on the specimens as described in 6.3. Calculate hardness change for the test lubricant as the difference between the initial hardness and the final hardness.

6.10 Determine elongation and tensile strength of the four specimens in accordance with method ASTM D412 using a 2.54 cm (1 inch) bench marker. Average the four elongations measurements. Calculate the percent change in elongation, see 7.1.

6.11 Determine the tensile strength of the four specimens in accordance with method ASTM D412 and average the four readings. Calculate the percentage change in tensile strength, see 7.2.

7. CALCULATIONS

$$Elongation, \% \, change \; = \; \frac{Final \; Elongation \; - \; Initial \; Elongation}{Initial \; Elongation} * 100$$

7.1 Calculate the percent change in elongation as follows:

7.2 Calculate the percent change in tensile strength as follows:

$$Tensile \; strength, \% \, change \; = \; \frac{Final \; Tensile \; strength \; - \; Initial \; Tensile \; strength}{Initial \; Tensle \; strength} * 100$$

8. REPORTING

8.1 Report the percentage change in the elongation and tensile strength.

9. PRECISION

9.1 Precision data have not been developed for this method.

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Rubber Sheet Specifications updated	Specifications had been superseded	2	Air Force – AFRL/RQTM
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT

Method prepared by:
Air Force – AFRL/RQTM – 2019

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Method 3433.1
September 30, 2019

3433.1 – COMPATIBILITY OF SYNTHETIC AIRCRAFT TURBINE LUBRICANTS WITH SILICONE RUBBER

1. SCOPE

1.1 This method is used for determining the compatibility of synthetic aircraft turbine lubricants with a standard silicone rubber composition representative of commercial seal materials. It determines the extent of deterioration of a silicone rubber standard by the lubricating fluids.

1.2 Advanced synthetic lubricant fluids for aircraft turbine engines with modified additive systems must be compatible with materials of construction used in aircraft engines particularly with silicone rubber seals. This test was devised to reveal rubber seal-lubricating fluid compatibility problem areas.

2. SUMMARY

2.1 This method provides a procedure for exposing silicone rubber test specimens to the influence of synthetic lubricants under definite conditions of temperature and time for purpose of measuring the resulting deterioration by noting the changes in physical properties before and after immersion in the test liquid.

3. SAMPLE SIZE

3.1 Approximately 500 mL of the lubricant to be tested.

4. APPARATUS

4.1 Oven, gravity convection, 121°C ± 1°C (250°F ± 2°F).

4.2 Beakers, tall-form, heat resistant, 600 mL.

4.3 Suspension device, glass, for holding test rubber specimens

4.4 Hook, stainless steel or nichrome wire, for suspending test rubber sheets from balance.

4.5 Balance, sensitivity, ± 0.1 mg.

4.6 Dies and cutters - ASTM die A or equivalent so that width of rubber specimen suspended between jaws of tensile tester is 1.27 cm (1/2 inch).

4.7 Durometer - Type A Shore as described in ASTM D2240.

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4.8 Micrometer - To measure thickness of rubber as described in ASTM D412.

4.9 Testing Machine for elongation and tensile strength measurements as described in ASTM D412.

4.10 Borosilicate tubing - thermocouple holder closed at one end, 18 cm length - 0.3 cm inside diameter and 0.5 cm outside diameter.

4.11 Thermocouple and recorder - to record temperature of test oils.

5. MATERIALS

5.1 Standard silicone test rubber sheets, may be obtained from Propulsion Lubricants Team, 22229 Elmer Road, Building 2360, Unit 4, Patuxent River, MD 20670-1534. Test sheets should be 15x15 cm (6x6 inch) by 0.19 cm thick (0.075 inch).

5.2 Cleaning solution, glass (concentrated sulfuric acid saturated with potassium or sodium dichromate).

5.3 Methyl alcohol, ACS, particle free.

5.4 Toluene, ACS.

5.5 Acetone, ACS, particle free.

6. PROCEDURE

6.1 Definitions

6.1.1 Tensile stress - the applied force per unit of original cross sectional area of specimen.

6.2 Test specimen preparation

6.2.1 Test specimens are to be cut using ASTM die A or equivalent.

6.2.2 Straight specimens may be prepared where it is not practical to cut ASTM dumbbell specimens. The minimum length must be 10.2 cm (4 inch). The width should be 1.27 cm (1/2 inch) between the jaws which are 2.54 cm (1 inch) apart.

6.3 Test procedure

6.3.1 Volume swell, change in hardness and change in tensile strength are all determined relative to original properties, from the post-test condition of the same specimens. The original tensile strength and hardness shall be determined, using duplicate specimens cut from the same sheet from which the test specimens were obtained. The duplicate specimens should be cut from

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an area of the sheet which is immediately adjacent to the area from which the test specimens were cut. The original rubber specimen volume is determined from the test specimen prior to immersion in the oil sample.

6.4 Preparation

6.4.1 Clean all glassware.

6.4.1.1 Rinse in solvent, rinse with acetone and air dry.

6.4.1.2 Soak in glass cleaning solution for 2 hours (minimum).

6.4.1.3 Rinse thoroughly with tap water, and then with distilled water.

6.4.1.4 Dry in oven at 100°C (212°F) and store in dust-free location.

6.4.2 Preheat and adjust oven to maintain a temperature of 121°C ± 1°C (250°F ± 2°F).

6.4.3 Prepare standard test silicone rubber specimens.

6.4.3.1 Do not buff.

6.4.3.2 Cut three pieces (per 6.2) from the sheet. Pieces must be cut with the grain of the rubber as indicated by two grain marks at the top of the slab.

6.4.3.3 Punch a 0.3175 cm (1/8 inch) diameter hole at the center of the 10.16 cm (4 inch) side, about 1.905 cm (3/4 inch) from the top edge.

6.4.3.4 Mark each piece for identification by cutting small pieces from one or more corners.

6.4.3.5 Clean the suspension device and hooks thoroughly with toluene, air dry, and store in a dust free location.

6.4.3.6 Attach the wire suspension device to each piece.

6.4.4 Determine original volume of silicone test specimens.

6.4.4.1 Weigh each rubber test specimen and its hook in air to the nearest 0.1 mg by suspending it from the balance by its hook. Record the weights.

6.4.4.2 Reweigh (to the nearest 0.1 mg) each rubber test specimen while suspended in distilled water at room temperature, being careful to remove all air bubbles from the sheet and prevent the sheet from touching the water container. Record the weights.

6.4.4.3 Determine (and record) the displacement weight of each piece by subtracting the weight in water from the weight in air.

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6.5 Testing.

6.5.1 Fill a 600 mL beaker with 500 mL of the oil sample.

6.5.2 Attach the rubber specimen to the suspension device, and submerge the rubber test specimen in the sample so that the top edges are no less than 0.635 cm (1/4 inch) below the surface. Make certain that the sheets do not touch each other or contact the sides or bottom of the beaker.

6.5.3 Place the beaker in the oven at 121°C ± 1°C (250°F ± 2°F) for 96 h, or the specified conditions for the appropriate oil specification.

6.5.4 Insert a thermocouple in a glass thermocouple holder and place assembly into oil. The temperature of each oil should be recorded during the test.

NOTE 1: Test liquids must not be reused.

6.5.5 At the end of test time, remove the beaker from the oven and allow it to cool at room temperature for 30 ± 5 minutes.

6.6 Posttest determination of physical properties.

6.6.1 Remove the rubber test specimens from the sample and wash in a fresh beaker of methyl alcohol, then rub gently with the fingers to remove all oil from the surface and rinse with a jet spray of methyl alcohol. Caution: Wear protective gloves.

6.6.2 Allow the rubber pieces to air dry for several minutes.

6.6.3 Volume swell.

6.6.3.1 Measure the displacement weight of the rubber test specimens per 6.4.4. Calculate the percent change in volume swell as described in 7.1.

6.6.4 Hardness.

6.6.4.1 The hardness of the rubber specimen shall be determined in accordance with ASTM Method D2240, Type A durometer.

6.6.4.2 The thickness of the rubber specimen for the hardness test should be at least 0.3 cm. Therefore, a single-ply thickness of silicone rubber may be used as a cushion.

6.6.4.3 After swell testing, hardness readings can be conducted on the end of each test specimen outside of the area to be used for tensile testing.

6.6.4.4 Apply the pressure foot to the specimen. Read the scale 15 s after the pressure foot is in firm contact with the specimen.

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6.6.4.5 The hardness value of the test specimen shall be the average value obtained for the three test specimens.

6.6.5 Tensile strength.

6.6.5.1 Determine the loss in tensile strength by use of Method D412, Standard Method of Tensile Testing of Vulcanized Rubber.

6.6.5.2 Three measurements shall be made for thickness; one at the center and one at each end of the specimen. The median of the three measurements shall be used as the thickness in calculating the cross sectional area.

6.6.5.3 When rubber specimens of standard thickness are used as per section 5.2 of this test method and the difference between the maximum and minimum thickness does not exceed 0.0762 cm (0.003 inch), it is not necessary to calculate the cross-sectional area for each piece.

6.6.5.4 Place a specimen in the grips of the testing machine, using care to adjust it symmetrically in order that the tension will be distributed uniformly over the cross section. If tension is greater on one side of the specimen than the other, maximum strength of the rubber will not be developed. Start the machine and record the stress at the time of the rupture. The rate of elongation should be 20 ± 2 in/min. Calculate the percent change in tensile strength as described in 7.3.

7. CALCULATIONS

7.1 Calculate the percentage change in volume (V) of each rubber sample as follows:

$$V = \frac{(A - B)100}{B}$$

where:
V = change in volume, percent.
A = displacement weight at conclusion of test, mg.
B = displacement weight before immersion in sample, mg.

7.2 Calculate the percentage change in hardness (H) of each rubber sample as follows:

$$H = \frac{(C - D)\ 100}{C}$$

where:
H = change in hardness, percent.
C = hardness before test.
D = hardness after test.

7.3 Calculate the change in tensile strength.

7.3.1 Observed tensile strength.

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F/E = tensile strength in mPa (psi).
F = observed force at break in N (lb).
E = Cross-sectional area of the unstretched specimen in cm^2 (in^2).

7.3.2 Change in tensile strength, percent

$$T = \frac{(J - K)\ 100}{J}$$

where:
T = change in tensile strength, percent.
J = tensile strength before test.
K = tensile strength after test.

NOTE 2: See section 6.6.5.3

8. REPORTING

8.1 The report shall include the following:

8.1.1 Results calculated in accordance with paragraph 7, Calculations.

8.1.2 All observed and recorded data on which the calculations are based.

8.1.3 Date of test.

8.1.4 Temperature of test.

8.1.5 Type and dimensions of specimens used.

8.1.6 Immersion fluid used.

8.1.7 Exposure period.

8.1.8 Condition of exposed specimen from visual and manual examination.

8.1.9 Test apparatus used for hardness.

8.1.10 Type of tensile test apparatus.

9. PRECISION

Precision data have not been developed for this method.

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Removed reference to U.S. Air Force Specification bulletin 539	Bulletin has been cancelled and superseded	1	NAWCAD
30 Sep 2019	Updated responsible organization location from Trenton, NJ to Patuxent River, MD	We moved from NJ to MD about 20 years ago	1	NAWCAD
30 Sep 2019	Removed the solvent Naphtha	No longer used	1	NAWCAD
30 Sep 2019	Added an elongation rate to 6.6.5.4.	Helps make the method more precise	1	NAWCAD
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	1	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	1	Army - AT
20 May 2021	Removed extra odd numbers after each solvent in the materials section.	Additional numbers did not add additional information.	1	NAWCAD

Method Prepared By:
Navy – NAWCAD-2019

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Method 3457.3
September 30, 2019

3457.3 – HYDROLYTIC STABILITY

1. SCOPE

1.1 This method is used for determining the resistance of a finished oil to reaction when in contact with water.

2. SUMMARY

2.1 The test consists of tumbling under specified conditions, a mixture of the oil and water in a bottle containing a copper strip, and then testing for changes in the oil, water, and copper.

3. SAMPLE SIZE

3.1 Approximately 125 grams of the oil to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Standards.

4.1.1 ASTM D445 – Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity).

4.1.2 ASTM D974 – Standard Test Method for Acid and Base Number by Color-Indicator Titration.

4.1.3 ASTM D1500 – Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale).

4.2 Oven, convection, air; controllable to ± 0.5°C (± 1°F) in temperature range of 66°C and 93°C (150°F and 200°F).

4.3 Fluid container (carbonated beverage bottle), 210 mL (7 oz), capable of being sealed with a cork-lined steel cap.

4.4 Tumbling mechanism (for rotating the fluid container end-over-end at 5 rpm, inside the oven).

4.5 Brush, short-bristled, stiff (typewriter cleaning brush, or equal).

4.6 Gooch filter crucible.

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4.7 Centrifuge, 1500 rpm.

4.8 Centrifuge tubes, cone-shaped, 100 mL.

4.9 Microscope, 20x magnification.

4.10 Balance, sensitivity ± 0.2 mg.

5. MATERIALS

5.1 Emery paper, 00 or finer.

5.2 Litmus paper.

5.3 Petroleum ether (O-E-751) [1, 4, 0].

5.4 VM&P naphtha.

5.5 *n*-Hexane (technical grade) [1, 3, 0].

5.6 Sodium sulfate, anhydrous (ACS) [u, u, u].

5.7 Copper strip (ASTM B187/B187M, ASTM B272), 16 to 22 gage (B&S), 1.3 by 5.1 cm.

6. PROCEDURE

6.1 Determine the acid number of the oil by ASTM D974.

6.2 Determine the kinematic viscosity of the oil by ASTM D445.

6.3 Heat the oven to the temperature required by the specification.

6.4 Weigh (to within 0.5 g) 75 grams of the sample and 25 grams of distilled water and place in the clean fluid container (see 4.3).

6.5 Using the filter paper to prevent fingerprints during handling, polish all surfaces of the copper strip with the fine emery paper.

6.6 Wash the copper strip in warm petroleum ether or *n*-hexane, dry, and weigh to within 0.2 mg, and immediately place the strip in the container.

NOTE 1: CAUTION. Petroleum ether and *n*-hexane are both flammable. Use these materials in a well ventilated area. Do not breathe their fumes nor permit them to come in contact with the skin. Keep all flames away from these materials.

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6.7 Cap the bottle (with inert seal in place), and mount it in the tumbling mechanism in the oven.

6.8 Allow the mechanism to rotate the container end-over-end at 5 rpm for the time required by the specification.

6.9 After the container has rotated for the required time, filter the liquid contents through a tared Gooch filter crucible lined with a thin layer of asbestos. Save the filtrate. (The operator should wear heavy asbestos gloves and a face shield while handling the hot beverage bottle). Rinse the copper strip and beverage bottle with 50 mL of distilled water and 50 mL of petroleum ether and filter through the Gooch filter crucible. Save only the water washings and add them to the filtrate.

6.10 Process any insolubles that remain in the crucible, as follows:

a. Wash the precipitate with 50 mL of distilled water, and add the washings to the filtrate.
b. Wash the precipitate with 50 mL of petroleum ether, and dry the crucible at 66°C to 93°C (150°F to 200°F).
c. Determine and record the increase in weight of the crucible to within 0.2 mg.

6.11 Process the filtrate as follows:

a. Weigh a clean centrifuge tube to within 0.2 mg.
b. Transfer the filtrate to the tube, and centrifuge the tube for 15 minutes at 1500 rpm.
c. After centrifuging, decant the oil and water layers to suitable separate containers.

6.12 Process the oil layer from the centrifuge tube as follows:

a. Wash the oil with 25 mL portions of distilled water until the water is neutral to litmus. Save the water from this washing.
b. Examine the oil for insolubles, and remove any that still exist, using a combination of filtration, centrifuging, and solvent extraction as necessary.
c. When all insolubles have been removed, dry the oil with anhydrous sodium sulfate.
d. Determine the acid number of the oil by ASTM D974, and calculate the change from the original acid number.
e. Determine the kinematic viscosity of the oil by ASTM D445, and calculate the change from the original viscosity.
f. Determine the color of the oil by ASTM D1500.

6.13 Process the water layer from the centrifuge tube and the washings as follows:

a. Combine the water from the centrifuge tube with the water from the washings.
b. Determine the acid number of this water (based upon the original weight of the oil sample) by ASTM D974.

6.14 Determine the weight of any insolubles remaining in the centrifuge tube as follows:

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a. Add 50 mL of new test oil to the tube, and centrifuge until the oil is clear.
b. Carefully decant the oil, and dry the tube in the oven to remove all traces of water.
c. Again wash the contents of the tube with new test oil, centrifuge until clear, and decant the oil.
d. Wash the residue with petroleum ether (to remove the remaining oil); centrifuge, decant, and dry the tube in the oven.
e. Determine the increase in weight of the tube to within 0.2 mg.

6.15 Calculate the percentage (by weight) of the insolubles in the oil (see 7.1).

6.16 Remove the copper strip from the fluid container, and process it as follows:

a. Wash the strip with petroleum ether, warm n-hexane, and warm VM&P naphtha and allow it to dry in air. See note 1 (see 6.6) and note 2.
b. Determine the weight change of the strip to within 0.2 mg. Calculate the change in weight per square centimeter of total surface.
c. Examine the strip under 20X magnification, and record any visible signs of corrosion.
d. Again wash the strip with petroleum ether, warm *n*-hexane, and warm VM&P naphtha. While washing, scrub the strip with the short-bristled brush. Allow the strip to air dry, and reweigh to within 0.2 mg.
e. Repeat this washing until the strip reaches a constant weight.
f. Determine the weight loss of the strip (after brushing) from its originally polished condition.
g. Calculate the loss in weight per square centimeter of total surface.

7. CALCULATIONS

7.1 Calculate the percentage (by weight) of the insolubles in the oil as follows:

$$Percent\ insolubles = \frac{100(A + B)}{W}$$

where:
A = increase in weight of Gooch crucible.
B = increase in weight of centrifuge tube.
W = original weight of oil sample.

8. REPORTING

8.1 Report the following:

a. weight change of the metal strip before brushing.
b. weight change of the metal strip after brushing.
c. change in acid number of the sample.
d. acid number of the water layer after hydrolysis.

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e. change in viscosity of the oil.
f. weight of insolubles formed during hydrolysis.
g. color of the oil after hydrolysis.
h. general appearance of the sample after hydrolysis.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units. Updated ASTM References.	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT
30 Sep 2019	Replaced 1,1,1-Trichloroethane with VM&P Naphtha	Replacement chemicals are less toxic. They are used by AF PTP laboratory.	3	Army – AT

Method Prepared By:
Army - AR - 1986

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Method 3458.2
September 30, 2019

3458.2 – LOW TEMPERATURE STABILITY TEST FOR OIL

1. SCOPE

1.1 This method is used for determining the tendency of components of finished oil blends to be incompatible or to form gels during low-temperature exposure.

2. SUMMARY

2.1 The test consists of refrigerating the sample for 72 hours, or other specified duration, at the temperature required by specification, and examining it visually for evidence of gelling, separation, crystallization, and clouding.

3. SAMPLE SIZE

3.1 Approximately 100 mL of oil to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Sample bottle, clear-glass 120 mL, with stopper.

4.2 Cold box, capable of maintaining the specified temperature within 6°C.

5. MATERIALS

5.1 No other materials required.

6. PROCEDURE

6.1 Pour sample into bottle, and store bottle for 72 hours, or other specified duration, at the temperature required by the specification ± 6°C.

6.2 At the end of 72 hours, or other duration, remove the container from the cold box, and immediately examine the sample visually for evidence of gelling, separation, or crystallization.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

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8.1 Report the oil as unstable at the specified temperature, if there is any evidence of gelling, separation, or crystallization; if there is no such evidence, report the oil as stable. Also report the presence of dense cloud, if any, that does not settle out.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT
30 Sep 2019	Allowed for other test durations	Deletion of FTM 3459	2	Army - AT

Method Prepared By:
Army - ME - 1986

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Method 3463.3
September 30, 2019

3463.3 – STABILITY OF GREASE IN HOT WATER (WATER IMMERSION)

1. SCOPE

1.1 This method is used for determining the stability of grease in the presence of hot water.

2. SUMMARY

2.1 The method consists of immersing the sample in boiling water for 10 minutes and checking the water for cloudiness or other evidence of emulsification of the sample.

3. SAMPLE SIZE

3.1 Approximately 5 grams of grease to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Beaker, 600 mL.

4.2 Stirring rod, glass.

5. MATERIAL

5.1 Distilled water.

6. PROCEDURE

6.1 Place approximately 500 mL of distilled water in the beaker and heat it to gentle but active boiling.

6.2 Apply a single lump or globule of the sample to one end of the stirring rod.

6.3 Immerse the sample in the boiling water for 10 minutes while visually examining the water, and note the presence of cloudiness, or other evidence of emulsification of the sample.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

7.2 Report the presence of cloudiness, or other evidence of emulsification of the sample.

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9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT

Method Prepared By:
Navy - NADC - 1986

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Method 3467.2
September 30, 2019

3467.2 – STORAGE STABILITY OF LUBRICATING GREASE

1. SCOPE

1.1 This method covers a technique used for measuring the stability of a lubricating grease after a definite storage period at an elevated temperature.

NOTE: The storage period and temperature at which the grease is to be stored must be given in the specification referencing this test method.

2. SUMMARY

2.1 A determination for consistency (unworked and work penetration) is made after the storage interval and results compared with data established prior to the storage interval.

3. SAMPLE SIZE

3.1 Approximately 1 kg of grease, sufficient to fill two standard grease cups (ASTM D217) with test sample.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Two standard grease worker cups, grease worker, penetrometer, etc. (ASTM D217).

4.2 Oven, gravity - convection.

5. MATERIALS

5.1 The grease to be tested.

6. PROCEDURE

6.1 Fill one standard grease worker cup with the test sample and remove excess grease to ensure that the surface of grease is flush with the top of the cup. (Sample for unworked penetration, see 6.6.)

6.2 Fill another cup with sufficient quantity of grease to ensure over filling of the cup. (Sample for worked penetration, see 6.6.)

6.3 Cover the samples with aluminum foil.

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6.4 Place the grease filled cups in an oven at the designated test temperature for the required storage interval.

6.5 After completion of storage interval, remove the cups from the oven and allow to cool to 25°C (77°F).

6.6 Conduct unworked and worked penetration measurements on the grease samples in accordance with ASTM D217.

7. CALCULATIONS

Calculate the change in worked penetration after storage as follows:

$$C = A - B$$

where:
C = change in worked penetration
A = penetration before storage
B = penetration after storage

8. REPORTING

8.1 Report the unworked and worked penetration after storage. Report the difference in worked penetration before and after storage as the change in worked penetration.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT

Method Prepared By:
Navy - NADC - 1986

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Method 3480.3
September 30, 2019

3480.3 – VOLATILITY OF BLENDED OILS

1. SCOPE

1.1 This method is used for determining the volatility characteristics of a blended oil.

2. SUMMARY

2.1 The test consists of heating the sample for 24 hours at 105°C (221°F) and determining its percentage of weight lost as result of evaporation.

3. SAMPLE SIZE

3.1 Approximately 10 grams of test oil.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Oven, gravity-convection, 105°C ± 0.5°C (221°F ± 1°F).

4.2 Evaporating disk, aluminum foil, 5.7 cm (2-1/4 inch) I.D., 1.6 cm (5/8 inch) deep.

4.3 Balance, sensitivity ± 0.1 mg.

5. MATERIALS

5.1 Test oil.

6. PROCEDURE

6.1 Heat oven 105°C ± 0.5°C (221°F ± 1°F).

6.2 Weigh evaporating dish to within 0.2 mg.

6.3 Pour sample into evaporating dish, and weigh dish and contents to within 0.2 mg. Then place dish into heated oven for 24 hours.

6.4 At the end of 24 hours, remove dish, and allow sample to cool to room temperature.

6.5 Determine the evaporation loss of sample within 0.2 mg (see 7.1).

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7. CALCULATIONS

7.1 Calculate the evaporation loss of the sample as follows:

$$Evaporation\ loss, percent\ by\ weight = \frac{S - W}{S} * 100$$

where:
S = initial weight of sample, in grams.
W = weight of sample, in grams after the test.

8. REPORTING

8.1 Report the evaporation loss to the nearest 0.2 mg.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT

Method Prepared By:
Air Force - AFRL/RQTM - 2019

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Method 3500.2
September 30, 2019

3500.2 – MONOBASIC ACID COMPONANTS OF SYNTHETIC ESTER LUBRICANTS BY GAS CHROMATOGRAPHY

1. SCOPE

1.1 This method covers the determination of monobasic acid components of ester lubricants by gas chromatography. The method is intended for use on any neopentyle polyol ester lubricants where the acid components contain from 4 to 10 carbon atoms.

2. SUMMARY OF METHOD

2.1 The lubricant to be analyzed is hydrolyzed with potassium hydroxide, followed by hydrochloric acid treatment to evolve the acids. After ether extraction of the acids, the acids are physically separated by gas chromatography and are identified by comparing them to corresponding acids separated under identical operating conditions from a reference standard acid mixture of known composition. The acids are determined quantitatively by also using the chromatogram of the sample which is interpreted by comparing peak heights with those obtained on the reference standard mixture of pure acids of known concentration.

3. SAMPLE SIZE

3.1 Approximately 20 grams of sample are required for duplicate determination.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Chromatograph and Recorder- A chromatograph with a thermal conductivity detector and a strip –type recorder may be used provided that the butyric and isovaleric acid components in the standard acid mixture (see 5.21 can be satisfactorily resolved when the column of 6.1.1 is used and the conditions of 6.1.2 are met. In addition, the equipment shall have sufficient sensitivity to meet the precision requirements of section 9.

4.2 Two columns, stainless steel. 183 cm (6ft) long, 0.635 cm (0.25in) O.D packed with the mixture and in the manner described in section 6.11.

4.3 Erlenmeyer flask, wide mouth 3.7 cm (1.46 inch) I.D, 250 mL

4.4 Reflux condenser, spiral, block tin

4.5 Electric hot plate

4.6 Steam bath or flash evaporator

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4.7 Crystallizing dish, 9 cm diameter H, 5.0 cm deep

4.8 Graduated cylinders, 50 and 100 mL

4.9 Separatory funnel, pear shaped, 250 mL

4.10 Erlenmeyer flask, narrow mouth, glass stoppered, 300mL

4.11 Erlenmeyer flask narrow mouth, standard taper joint, 300mL

4.12 Vigreux fractionating column, standard taper joint at lower end complementary to joint of flask described in 4.11

4.13 Syringe, 10 µL capacity, graduated in 0.2 µL.

4.14 Analytical balance, 200 grams capacity

4.15 Vacuum oven

4.16 Vacuum pump

4.17 Electric stirrer

4.18 Electric vibrator

4.19 Filter, size 4.7 cm diameter, 10 µL.

4.20 Stainless filter holder and funnel

5. MATERIALS

5.1 Potassium hydroxide, reagent grade [3, 0, 1].

5.2 Ethyl alcohol, 95% [0, 3, 0].

5.3 Distilled water

5.4 Hydrochloric acid, dilute, made by diluting one volume of reagent grade acid, sp.gr. 1.18 with an equal volume of water [3, 0, 0].

5.5 Ethyl ether, reagent grade [2, 4.1]

5.6 Sodium sulfate, anhydrous, reagent grade [u, u, u]

5.7 Diethylene glycol adipate polyester [u, u, u]

5.8 Gas Chrom P, 60 to 80 mesh, acid-based washed, or equivalent

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5.9 Chloroform, reagent grade [2, 2, 0].

5.10 Phosphoric acid, reagent grade, 85% [2, 0, 0].

5.11 Helium, oil free.

5.12 Butyric acid, reagent grade, [2, 2, 0].

5.13 Isovaleric acid, reagent grade [u, u, u]

5.14 Valeric acid, reagent grade [u, u, u]

5.15 Caproic acid, reagent grade [2, 1, 0].

5.16 Heptylic acid reagent grade [u, u, u]

5.17 Caprylic acid, reagent grade [u, u, u]

5.18 Pelargonic acid, reagent grade [u, u, u]

5.19 Capric acid, reagent grade [u, u, u]

5.20 Standard acid mixture prepared as follows:

a. By means of an analytical balance successively weigh into a 125-mL glass-stoppered bottled all the acids listed in 5.13 through 5.20 in the amounts indicated in TABLE I. These listed weights give concentrations of the mixed acids in terms of the indicated individual mole percent.

TABLE I. Weights required to prepare standard acid mixtures.

Acid	Number of carbon [1] Atom in acid	Weight in grams [2]	Mole percent
Butyric	4	4.41	10
Isovaleric	5	5.11	10
Valeric	5	5.11	10
Caproic	6	5.81	10
Heptylic	7	6.21	10
Caprylic	8	10.82	15
Pelargonic	9	11.87	15
Capric	10	17.23	20

NOTES:
[1] For convenience, these acids hereafter in this method will be referred to by the number of their carbon atoms; e.g., butyric as C4, isovaleric as is – C5.
[2] These weights, or some multiple of them are recommended as giving a satisfactory chromatogram of the acids relative to peak heights.

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6. PROCEDURE

6.1 Preparation of apparatus

6.2 Column preparation.

6.3 Packing for chromatographic column prepared as follows.

6.4 Dissolve 6.0 grams of diethylene glycol adipate polyester in 150 mL of chloroform with rapid stirring.

6.5 Add 1.06 grams of phosphoric acid to the mix and continue stirring until the acid is completely dispersed.

6.6 Add 23.1 grams of Gas Chrom P to the mix and continue stirring for 10 minutes.

6.7 Remove bulk of chloroform from mix by flash evaporator or on steam bath.

6.8 Remove remaining chloroform by heating in a vacuum oven for 2 h at 200ºC and 0.1 cm pressure.

6.9 Cool and load into chromatographic column using vibrator on column to ensure uniform and complete packing.

6.10 Chromatograph- Mount the column in the chromatograph and use the programmed temperature operation. For this operation, set the controls of the chromatograph for an initial column temperature of 140ºC (284ºF), a final column temperature of 210ºC (410º F), a temperature rate of 8ºC (46º F)/minutes., a hold time of 20 minutes at the final temperature, an injection port temperature of 220ºC (428ºF), a detector temperature of 220ºC (428ºF), abridge current of 150mA, and the helium flow at 30.5 mL/minutes. The temperature programming must start immediately after injection.

6.11 Test procedure.

6.12 Hydrolysis of lubricant to be analyzed-Weigh 10 grams of the lubricant into a 250 mL wide-mouth Erlenmeyer flask, add a solution consisting of 8 grams of potassium hydroxide and 15 mL water, followed by 60 mL ethyl alcohol. Swirl to mix contents. Insert spiral reflux condenser in mouth of flask, connect condenser to cold water line, place flask on hot plate, adjust heat so that contents boil gently and continue refluxing for 16 hours.

6.13 Recovery of acid. Transfer contents of flask to crystallizing dish, rinse flask 3 or 4 times with 15 mL water and add washing to dish. Take to complete dryness on steam bath[1], add 15 mL water to dish and warm to dissolve salts. Transfer solution to separatory funnel, rinse dish 3 or 4 times with 15 mL water and add washings to funnel. Place funnel in ice bath, add 50 mL of cold dilute hydrochloric acid (1:1), cool for 10 minutes and swirl to complete mixing of contents. Add 50 mL of ethyl ether, shake vigorously for 2 minutes, venting frequently to relieve pressure. Let

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phase separate completely and run the lower (water) phase into a second separatory funnel. Run the upper (ether) layer into a 300 mL glass stoppered Erlenmeyer flask. Add 50 mL of ethyl ether to the second funnel and re-extract the water layer as above, adding the ether layer to the 300 mL flask. Repeat the ether extraction of the water layer twice more in individual funnels, collecting the ether layers in the 300 mL flask. Discard the water layer. Add 35-40 grams of sodium sulfate to the ether extract, shake and let stand a few hours or probably overnight. Filter the ether solution through 10 µm filter membrane (using stainless filter and funnel) into a 300 mL Erlenmeyer flask with tapered joint. Rinse filter and funnel a few times with ethyl ether to ensure complete removal of acids from sodium sulfate. Add 6-10 granules of silicon carbide into the flask to avoid superheating, connect to Vigreux column and distill off the ether in a hot water bath until the remaining solution is 20-25 mL stopper flask and retain for essay. Conduct recovery of acids under hood.

[1]*A flash evaporator may be used as an alternate for removal of the water and alcohol.*
In a similar manner inject 1 µL of the recovered acid, obtain chromatogram and proceed as above until the peak height of the scan is satisfactory.

6.14 Preparation of chromatograms. Using standard acid mixture and the recovered acids, prepare chromatograms in the following manner: Inject 1 µL of the standard acid mixture into the chromatograph, mark for reference on recorder chart paper where recorder pen was located at the time of injection, and complete chromatogram. Examine chromatogram for peak heights of the acids. If the greatest peak height is approximately three quarters of the chart width, the dosage and attenuator settings are satisfactory. If not, adjust dosage and attenuator until the desired peak height is attained. Identification and quantitative determination of recovered acids.

6.15 Identification of the acids recovered from the sample. On the chromatogram of the standard acid mixture, assign C4 acid to first peak from reference mark on chart, designate next peak as is -C5 acid, then the successive peaks to –C5 through C10 acids in numerical order. Measure in mm the distances from reference mark to each peak and record each distance against the corresponding acid. These values are designed as the retention times of the acids. On the chromatogram of the recovered acids, measure distance from reference mark to each peak. Compare with those obtained from the standard acid scan and identify each recovered acid accordingly.

6.16 Quantitative determination of the recovered acids – On the chromatogram of the standard acid mixture, draw a base line from the reference mark to that point on the scan which indicates the C10 acid has been eluted completely. From the base line measure each peak height in mm and sum all peak heights. Normalize each peak height by dividing each by the sum of the heights. Establish a correction factor for each acid by dividing the normalized peak height per mole of C5 acid by the normalized peak height per mole of each component acid. On the chromatogram of the recovered acids, proceed as above up to and including the normalization of each peak height. Multiply the normalized peak height of each acid component by its appropriate correction factor to obtain a “corrected” height for each. To obtain the mole-percent of each acid, divide the “corrected” peak height of each by the sum of all the “corrected” heights times 100. Record each component to the nearest integral value. Examples of the illustrated in TABLE II and TABLE III.

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TABLE II. Standard acid mixture.

Chromatographic retention times and correction factors				
Acid	Retention time (see 6.3.1) In mm (measured)	Peak height In mm (measured)	Normalized peak area Peak height of one component Sum of peak heights Of all components	Correction factor Normalized peak Height of C5 acid per mole normalized peak height of component acid per mole
C4	70	168	11.6	1.14
Iso-C5	74	187	12.9	1.02
C6	84	192	13.2	1.00
C7	98	182	12.6	1.05
C8	115	155	10.7	1.23
C9	136	212	14.6	1.36
C10	162	169	11.7	1.69
	196	184	12.7	2.08

TABLE III. Unknown acid mixture.

Chromatographic Identification and mole percent determination						
Identified acid	Retention time, mm	Peak height Mm (measured)	Normalized peak height	Correction factor (see TABLE II)	A corrected height	Mole percent (rounded)
	See 6.3.1		[1]		[2]	[3]
C4	71	5	1.2	1.14	1.4	1.2=1
iso C5	0	0	0	0	0	0=0
C5	84	165	40	1	40	34.4=34
C6	98	84	20.4	1.05	21.4	18.4=18
C7	115	87	21.1	1.23	26	22.4=22
C8	136	21	5.1	1.36	36.9	5.9=6
C9	162	50	12.1	1.69	20.4	17.6=18

1 $\frac{Peak\ height\ of\ one\ component}{sum\ of\ peak\ heights\ of\ all\ components}$

3 $\frac{100\ times\ "corrected"\ height\ of\ component}{sum\ of\ all\ \text{corrected}\ heights}$

2 $Peak\ height\ times\ correction\ factor$

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7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Report each component to the nearest integral value.

9. PRECISION

9.1 Repeatability- Replicate determinations shall not differ from one another by more than ± mole-percent.

9.2 Reproducibility has not been established.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT

Method Prepared By:
Navy- NAWCAD-1986

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Method 3603.6
September 30, 2019

3603.6 – SWELLING OF SYNTHETIC RUBBERS

1. SCOPE

1.1 This method is used for determining the effect of petroleum products on synthetic rubber.

2. SUMMARY

2.1 The test consists of determining the volume of three standardized test rubber sheets by water displacement, storing them immersed in the sample for 168 hours at 70°C, then determining the average change in volume of the sheets.

3. SAMPLE SIZE

3.1 For liquid products: Approximately 300 mL of the liquid to be tested.

3.2 For greases: Approximately 400 grams of grease to be tested.

4. REFERENCE, STANDARDS, AND APPARATUS

4.1 Container, heat-resistant, consisting of:

a. For liquid sample: Test tube, 5 cm O.D. by 40 cm long. A graduated cylinder with these dimensions may be used provided it doesn't have a pouring spout.
b. For grease sample: Petri dish, 10 cm O.D. by 5 cm deep with cover (watch glass).

4.2 Balance, sensitivity 1 mg, suitable for weighing in both air and water.

4.3 Oven, gravity-convention, 70°C ± 1°C (158°F ± 2°F).

4.4 Spatula (for grease sample).

5. MATERIALS

5.1 Test sheets (3), rubber standard, type as required by the specification, approximately 2.5 cm by 5.1 cm by 0.19 cm. Samples of synthetic rubber stock should not be older than 6 months.

5.2 Alcohol, ethyl or methyl, anhydrous [1, 3, 0].

5.3 Filter paper.

5.4 Distilled water.

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6. PROCEDURE

6.1 Determine the water displacement of each test sheet to the nearest 1 mg by weighing each sheet in air, then in distilled water at 24°C ± 3°C, recording the weights, and computing the difference between the weight in air and the weight in water for each sheet (see 7.1).

6.2 Fill container with sample, using the following:

a. For liquid sample: Approximately 20 mL of sample for each gram of total weight of test sheets.
b. For grease sample: Sufficient sample to fill container completely.

6.3 Blot dry each test sheet with filter paper, then immerse sheets completely in sample in container. For grease samples, immerse test sheets vertically, with 5 cm edge uppermost, and cover the container.

6.4 Store container in oven at 70°C ± 1°C (158°F ± 2°F) for 168 h ± 0.5 h (approximately one week). For liquid samples, maintain a ratio of 20 mL of oil to each gram of test sheet. Make up oil must be the same temperature as the test fluid when added. For grease samples, check daily (except weekends), that the samples are in complete contact with test sheets and close any voids by using a spatula.

6.5 At the end of the storage period, cool the sample to 24°C ± 3°C. Then clean test sheets as follows:

a. Remove test sheets from sample and wipe them to remove most of the adhering sample.
b. Dip sheets in alcohol and wipe or blot them dry, using filter paper (or soft cloths). NOTE THE TIME.

6.6 Within 5 minutes of drying test sheets, re-determine the water displacement of each sheet as specified in 6.1.

7. CALCULATIONS

Compute the percentage change in volume of each rubber sheet as follows:

$$V = \frac{(A - B)}{B} * 100$$

where:
V = change in volume, percent.
A = displacement weight at conclusion of test, mg.
B = displacement weight before immersion in sample, mg.

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8. REPORTING

8.1 Report the average percent volume change of the three rubber sheets.

9. PRECISION

9.1 Repeatability. Test results, obtained by one operator at one laboratory, for each test sheet must not vary from the average by more than the following:

a. If average is 0 to 5% units: 0.5% unit.
b. If average is above 5% units: 1% unit.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial changes to fix spacing and clarify units	Editorial corrections	6	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	6	Army - AT

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Method 3604.3
September 30, 2019

3604.3 – SWELLING OF SYNTHETIC RUBBER BY AIRCRAFT TURBINE LUBRICANTS

1. SCOPE

1.1 This method is used for determining the swelling effect of aircraft turbine lubricants on synthetic rubber. Synthetic lubricating fluids for aircraft turbine engines must be compatible with materials-of-construction, such as elastomers used in the engines. Elastomers are used in seals, O-rings, and related areas. This method is designed to reveal lubricating fluid-elastomer compatibility problems relative to volume swell.

2. SUMMARY

2.1 The volume of three standard test rubber samples is determined by water displacement. The rubber samples are immersed in the test lubricant for a specified time (such as 72, 96, or 168 h) at a given temperature. After storage, the average change in volume is determined.

3. SAMPLE

3.1 Approximately 350 mL of the lubricant to be tested.

4. REFERENCES, STANDARDS AND APPARATUS

4.1 Oven, gravity-convection, 70°C ± 1°C (158°F ± 2°F) or 204°C ± 2°C (399°F ± 3°F).

4.2 Beakers (2), borosilicate, 400 mL.

4.3 Suspension device, glass or stainless-steel (for holding test rubber-sheets completely immersed in specimen in 400 mL beaker so that they do not touch each other or the beaker).

4.4 Hooks, stainless-steel wire (for suspending test rubber sheets from suspension device and from balance).

4.5 Balance, sensitivity, ± 0.1 mg.

5. MATERIALS

5.1 Standard test rubber sheets, type H or F-A, fabricated less than one year prior to use. Test sheets should conform to the properties listed in SAE AMS 3217. If stored for more than one year, the elastomer should be retested before use. Storage should be in a dark place at a temperature of less than 25°C (77°F) away from electrical apparatus.

5.1.1 Standard test rubber sheets, type H, conforming to SAE AMS 3217/1

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5.1.2 Standard test rubber sheets, type F-A, conforming to SAE AMS 3217/4

5.2 Cleaning solution, glass (concentrated sulfuric acid saturated with potassium or sodium dichromate, technical grade).

5.3 Toluene, technical grade.

5.4 Alcohol, ethyl

5.5 Emery cloth No. 60.

5.6 Naphtha, technical grade

6. PROCEDURE

6.1 Clean all glassware as follows:

a. Rinse in toluene, followed by naphtha, and allow to air dry. An alternative procedure is to wash in a commercial type dish washer.
b. Soak in glass cleaning solution for 2 h (minimum).
c. Rinse thoroughly with tap water, and then with distilled water.
d. Dry in an oven at 100°C (212°F), and store in dust-free location.

6.2 Clean suspension device and hooks thoroughly with toluene or equivalent solvent, air dry, and store in a dust-free location.

6.3 Preheat and adjust the oven to maintain the required temperature.

6.4 Prepare standard test rubber sheets as follows:

a. Using No. 60 emery cloth, buff both sides of the rubber sheet, using strokes first in one direction and then at 90° to that direction. Buff just enough so that no gloss can be seen on the sheet when viewed by reflected light.
b. Cut three pieces (3.5 x 5 cm) from the sheet.
c. Punch a 0.3 cm (1/8 inch) (approximate) hole at the center of the 5 cm side, about 0.6 cm (1/4 inch) from the edge.
d. Mark each piece for future identification by cutting small pieces from one or more corners.
e. Attach wire hooks to each piece.

6.5 Weigh each rubber test sheet (and its hook) in air to the nearest 0.1 mg by suspending it from the balance by its hook. Record the weights.

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6.6 Reweigh (to the nearest 0.1 mg) each sheet while suspended in distilled water at room temperature, being careful to remove all air bubbles from the sheet and to prevent the sheet from touching the water container. Record the weights.

6.7 Determine (and record) the displacement weight of each piece by subtracting the weight in water from the weight in air.

6.8 Fill a 400 mL beaker with 350 mL of the sample.

6.9 Attach the rubber sheets to the suspension device, and submerge the sheets in the sample so that the top edges are no less than 1.3 cm (1/2 inch) below the surface. Make certain that the sheets do not touch each other or contact the sides or bottom of the beaker.

6.10 Place the beaker in oven at the required temperature for the specified time. The usual temperature is 70°C for the type H rubber and 204°C for the type F-A rubber. The typical test duration is 72 hours.

6.11 At the end of 72 hours, remove the beaker from the oven and allow it to cool at room temperature for 30 minutes ± 5 minutes.

6.12 Remove the rubber test sheets from the sample and wash them in ethyl alcohol, rubbing them gently with the fingers to remove all oil from the surfaces. Caution - wear protective gloves.

6.13 Wipe the rubber pieces dry with soft lint-free cloth, and allow them to air dry for several minutes.

6.14 Repeat 6.5 to 6.7 to determine the displacement weight of the rubber sheets after immersion in oil.

7. CALCULATION

7.1 Displacement weight

$$Displacement\ Weight = Weight\ of\ test\ sheet\ in\ air - Weight\ of\ test\ sheet\ in\ water$$

7.2 Calculate the percentage change in volume of each rubber sheet as follows:

$$V = \frac{A - B}{B} * 100$$

where:
V = change in volume, percent.
A = displacement weight at conclusion of test, mg.
B = displacement weight before immersion in sample, mg.

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8. REPORTING

8.1 Report the average percent volume change of the three rubber sheets.

9. PRECISION

9.1 Repeatability. Results by one operator using one apparatus must not differ by more than 1%.

9.2 Reproducibility. Results from different operators or different apparatus must not differ by more than 2%.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Added in information for SAE AMS 3217	The U.S. Air Force Memorandum has been cancelled and superseded by AMS 3217	3	NAWCAD
	Naphtha			
30 Sep 2019	Added in a test temperature of 204 C	Reflects correct temperatures for the various polymers	3	NAWCAD
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT
20 May 2021	Removed odd markings after solvents	Does not add additional infroamtion.	3	NAWCAD
20 May 2021	Updated formula for displacement weight	Display correct equation used in calculations	3	NAWCAD

Method prepared by:
Navy – NAWCAD-2019

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Method 3720.3
September 30, 2019

3720.3 – MOLYBDENUM DISULFIDE CONTENT OF LUBRICATING GREASE

1. SCOPE

1.1 This method is used to determine the molybdenum disulfide content of lubricating greases. This method does not apply to lubricating greases that contain non-soap thickeners.

2. SUMMARY

2.1 This method consists of dissolving the base oil and soap thickener in a mixture of oleic acid and *n*-hexane. The MoS_2 is removed by filtration and MoS_2 content determined by weighing the MoS_2.

3. SAMPLE SIZE

3.1 Five (5) grams of lubricating grease to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Desiccator.

4.2 Erlenmeyer flask, 125 mL.

4.3 Filter pump (aspirator).

4.4 Flask, filtering heavy wall.

4.5 Gooch filter crucible and suitable holder.

4.6 Oven capable of 100°C.

4.7 Steam bath.

5. MATERIALS

5.1 *n*-Hexane, ACS grade [1, 3, 0].

5.3 Oleic acid, USP grade [0, 1, 0].

5.4 Petroleum ether (O-E-751) [1, 4, 0].

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6. PROCEDURE

6.1 Perform the test as follows:

a. Weigh 3 to 5 grams of the grease to be tested to the nearest mg and distribute as a thin layer in a 125 mL Erlenmeyer flask prior to adding 90 mL of a 50-50 mixture by volume of *n*-hexane and oleic acid. (See Note 1).
b. Warm the flask for approximately 1/2 h on a steam bath with occasional agitation until all lumps have disappeared and the molybdenum disulfide is dispersed.
c. Filter the contents of the flask through a previously prepared and weighed (to the nearest mg) Gooch filter crucible. Wash the flask and crucible with approximately 10 mL of a 50-50 mixture by volume of oleic acid and *n*-hexane, followed by three washings with petroleum ether. (See Note 1.)
d. Dry the crucible and contents at 100°C for 1/2 h, cool in a desiccator to room temperature and weigh. (See Note 2).

NOTE 1: CAUTION. *n*-Hexane and petroleum ether are both toxic and flammable. Do not breathe their fumes nor allow them to come in contact with the skin. Keep flames away from the *n*-hexane and petroleum ether.

NOTE 2: The purity of MoS_2 powder may be determined by means of spectrochemical analysis.

7. CALCULATIONS

7.1 Calculate the percentage of molybdenum disulfide as follows:

$$Percent\ molybdenum\ disulfide\ (as\ insoluble\ matter) = \frac{Increase\ in\ weight\ of\ crucible}{weight\ of\ sample} * 100$$

8. REPORTING

8.1 Report the percent molybdenum disulfide as insoluble matter.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial changes to fix spacing and other minor corrections	Editorial corrections	3	Army – AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army – AT

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Method 3722.3
September 30, 2019

3722.3 – MOLYBDENUM DISULFIDE CONTENT OF NON-SOAP THICKENED LUBRICATING GREASES

1. SCOPE

1.1 This method is used to determine the MoS_2 content of non-soap thickened greases.

2. SUMMARY

2.1 The test consists essentially of extracting the base oil from the grease with n-hexane and determining the molybdenum disulfide content of the remaining solids by precipitation to sulfate with barium chloride.

3. SAMPLE SIZE

3.1 Approximately 5 grams of grease to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Desiccator.

4.2 Oven capable of 100°C.

4.3 Muffle furnace capable of 760°C.

4.4 Hot plate.

4.5 Steam bath.

4.6 Soxhlet extraction apparatus with 250-mL flask.

4.7 Beakers (2), 600 mL.

4.8 Crucibles (2), 50 mL.

4.9 Funnels (2).

4.10 Burner, Meker (or equivalent).

4.11 Policeman, rubber.

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5. MATERIALS

5.1 Hexane, normal, ACS [1, 3, 0].

5.2 Extraction thimble (double thickness 3.3 by 8.0 cm).

5.3 Filter paper (Whatman No. l or equivalent).

5.4 Filter paper (Whatman No. 40 or equivalent).

5.5 Eschka's mixture - 2 parts of calcined ("light") magnesia thoroughly mixed with one part of sodium carbonate

5.6 Bromine water, saturated [4, 0, 0].

5.7 Barium chloride, 10% [0, 0, 2].

5.8 Sodium carbonate, CP [u, u, u].

5.9 Potassium carbonate, CP [u, u, u].

5.10 Hydrochloric acid

6. PROCEDURE

6.1 Determine the solids content of the grease as follows:

a. Dry an extraction thimble (3.3 8.0 cm) at 100°C, cool in a desiccator and weigh to the 100°C nearest tenth of a mg. Dry thimble to constant weight. Weigh approximately 5 grams of the grease to be tested into the extraction thimble.
b. Place 200 mL of *n*-hexane in the flask, place the thimble containing the grease sample in the Soxhlet extraction tube, and assemble the apparatus.
c. By means of a hotplate bring the hexane to boiling and reflux until the base oil has been completely extracted from the grease (see Note 1). Complete removal of the oil may be determined by sampling small (2 mL) quantities of hexane from the extraction tube and evaporating on a clean evaporating dish. An oily film will remain if oil is contained in the sample.
d. Remove the thimble from the extraction tube and dry in an oven at 100°C. Allow to cool in a desiccator. After cooling, remove and weigh to constant weight. Subtract the weight of the thimble from the latter weight to determine the weight of the solids in the grease. Calculate the percent solids (see 7.1).

NOTE 1. CAUTION. *n*-Hexane is both toxic and extremely flammable. Use only in a well ventilated area. Do not breathe its fumes nor allow it to come in contact with the skin. Keep all flames away from the Hexane.

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6.2 Determine percent molybdenum disulfide in the grease as follows:

a. Ignite a clean crucible (50 mL) in the blue oxidizing flame of a Meker burner. Cool in a desiccator. Remove and add approximately 2 grams of Eschka's mixture and weigh to the nearest 0.1 mg.
b. Add approximately 0.1 grams of the grease solids (6.1 d) and reweigh to the nearest 0.1 mg (thoroughly mix by stirring sample using a wire rod).
c. Place a layer 0.6 cm (0.25 inch) of Eschka's mixture on the sample from 6.2 b. Do not mix.
d. A blank determination using 4 grams of Eschka's mixture is conducted in a separate crucible.
e. Place both crucibles in a cold muffle furnace and slowly raise the temperature to 760°C. Maintain a temperature of 760°C for 2 hours. Remove the crucibles from the muffle furnace and allow to cool.
f. Remove the contents of each crucible into separate 600-mL beakers and place the crucibles in a horizontal position in the beakers. Cover the crucibles with distilled water.
g. Place on heated steam bath for 30 minutes. Remove and rinse each crucible so that the rinse water is deposited in the beakers.
h. Pass the contents of beaker through filter paper (Whatman No. l or equal) and rinse the residue 2 or 3 times with distilled water. Discard residue and paper.
i. Add 10 mL of saturated bromine water to the filtrates. Heat gradually to boiling. Continue to boil until the volume is reduced to 200 mL.
j. Cool the solution and add 5 mL of concentrated hydrochloric acid to each beaker and boil until colorless.
k. While boiling, add at a very slow rate 10 mL of 10% solution of barium chloride. Heat on a steam bath for at least 3 h, and allow to stand for 24 hours.
l. Pass the solutions through filter paper (Whatman No. 40 or equal). Remove all barium sulfate adhering to the surfaces of the beakers by means of a rubber policeman. Wash the precipitates and discard the filtrates.
m. Clean the crucibles by fusing 50-50 mixture of sodium and potassium carbonates. Do not use potassium bisulfate. Dry and cool the crucibles and weigh to the nearest 0.1 mg.
n. Fold the filter paper and precipitates, and place in the clean weighed crucibles.
o. Place the crucibles in a cold muffle furnace and heat gradually to a temperature of 400°C. Hold at 400°C until the filter papers are thoroughly charred. Slowly raise the temperature to 760°C and maintain at this temperature until there is no longer any carbon present. This will require approximately l h at 760°C. The barium sulfate should be pure white.
p. Allow the crucibles to cool and determine the weight of barium sulfate in each one. Subtract the weight of barium sulfate in the blank from that of the sample and record the difference as the weight of barium sulfate. (See note 2).
q. Calculate the percent MoS_2 in the grease solids (see 7.2).
r. Calculate the percent MoS_2 in the grease (see 7.3).

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NOTE 2: A spectrochemical analysis may be conducted to determine purity of MoS_2.

7. CALCULATIONS

7.1 Calculate the percent solids as follows:

$$Percent\ solids = \frac{w_2}{w_1} * 100$$

where:
w_1 = weight of solids in sample
w_2 = weight of sample

7.2 Calculate the percent MoS_2 in the grease solids as follows:

$$Percent\ MoS_2\ in\ grease\ solids\ = \frac{0.343 * M_2}{M} * 100$$

where:
M = weight of solids from grease sample
M_2 = weight of barium sulfate

7.3 Calculate the percent MoS_2 in the grease as follows:

$$Percent\ MoS\ 2\ =\ percent\ solid\ (see\ 7.1) *\ percent\ MoS_2\ in\ grease\ solids\ (see\ 7.2)$$

8. REPORTING

8.1 Report the percent solids, percent MoS_2 in grease solids, and percent MoS_2.

9. PRECISION

9.1 Precision data have not been developed for this method.

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30 Sep 2019	Editorial changes to fix spacing and other minor corrections	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT

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Method 5003.3
September 30, 2019

5003.3 – DEPOSIT-FORMING TENDENCIES OF AIRCRAFT TURBINE LUBRICANTS

1. SCOPE

1.1 This test method describes a procedure for determining the deposit and sludge forming tendencies of aircraft gas turbine lubricants.

2. SUMMARY

2.1 A sample of the lubricant is circulated under controlled conditions for a prescribed period of time through an aerated test chamber containing an aluminum tube held at constant temperature. From the chamber the oil passes through a cooler, a line filter, a circulating pump and back into the chamber. The weight of solid decomposition products on the heated tube and on the line filter are determined. Changes in the viscosity and neutralization number of the oil can be determined if desired.

3. SAMPLE SIZE

3.1 As needed.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Deposition tester, Original device models were from Wright Air Development Center (ERDCO® Engineering Corp., Adison, IL) consisting of an oil circulation pump, coking head with tube, oil feeder, oil cooler, line filter, and the necessary instrumentation and supporting structure (see FIGURE 1 through FIGURE 5). Currently, the only known unit is maintained at Southern Petroleum Laboratories (SPL), Inc.

NOTE 1: Model 1 is equipped with a pyrometer, Model 2 with a temperature recorder, and Model 3 with a pyrometer, a selector switch, and a differential pressure recorder. A "W" following the model number indicates the inclusion of a watt-hour meter. The SPL, Inc unit has an upgraded instrument section.

a. Cabinet. The test cabinet consists of a test section and an instrument section. Ventilation and heat are provided to maintain the test section at 48.9° ± 3°C (120° ± 5°F) and the instrument section at not more than 6°C (10 °F) above room temperature. The sections are separated by a partition.
b. Instrument section. The instruments and their locations are shown in FIGURE 1. Optional items (differential-pressure recorder, watt-hour meter, pyrometer, and selector switch) are included in FIGURE 1. Other configurations are acceptable if they maintain the required testing conditions.

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c. Test section. The test section components and their approximate location are shown in FIGURE 2. A schematic drawing of the entire rig is shown in FIGURE 1. The test section components are mounted on a rack that can be removed from the cabinet to permit easy disassembly for cleaning. A spare test section can be advantageously utilized to keep down-time on the rig to a minimum.

d. Coking tube. The coking tube consists of an aluminum tube with a shrunk-in 1500-watt cartridge heater. Two thermocouple holes are provided. The configuration and dimensions of the coking tube are shown in FIGURE 4.

e. Line filter. The filter consists of a two-piece body, a 100-mesh stainless steel screen, and a screen support. These components are shown assembled in FIGURE 5.

f. Oil cooler. The oil cooler is designed for oil down flow through annular space and water up flow through the inner tube. A continuous water bleed to waste is taken through an orifice just ahead of the inlet to the cooler (see FIGURE 1, item 32). This acts as a drain for the cooler when the water flow is shut off by the control, air being bled back into the cooler through a check valve (see FIGURE 2, item 30).

4.2 Balance, capacity 500 g, sensitivity ±10 mg.

4.3 Stop watch, accuracy ± 0.1 s.

4.4 Brushes, metal; such as 0.195 in diameter, 1/2 in B.P., 10 in; P.A. No. 22 single stem 0.0025 fill (Mill-Rose Co., 1985 East 59th Street, Cleveland, OH).

4.5 Oven, 93.3° to 104.4°C (200° to 220°F).

4.6 Compressed-air source.

4.7 Gooch filter crucible.

5. MATERIALS

5.1 Silicon carbide paper, No. 400.

5.2 Solvent, dry-cleaning (such as Stoddard Solvent or similar).

5.3 Cleaning solution, glass, any suitable laboratory glass cleaning fluid may be used providing it removes all evidence of staining and deposits from previous tests.

5.4 *n*-Hexane, thiophene-free (ACS) [1, 3, 0].

5.5 Isopropyl alcohol (ACS), [1, 3, 0].

5.6 Petroleum ether (ACS) [1, 4, 0].

5.7 Distilled water.

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5.8 Scouring powder, commercial grade.

5.9 Steel wool.

5.10 Pipe cleaners.

5.11 Cleaning compound, Any suitable water based non corrosive cleaner: for cleaning of all reusable components of the test oil system.

5.12 New parts to be installed before each test (available from SPL, Inc.):

a. Pump shaft seal.
b. O-rings for sealing the filter, cooler, and oil-bypass relief valve.
c. Air-diffuser stone for coking head. (The stone in the humidifier is not replaced except as needed).

6. PROCEDURE

6.1 Ascertain periodically that all thermocouples are operating properly.

6.2 Clean thermocouples by swabbing and rinsing in isopropyl alcohol and then air dry.

6.3 Prepare the oil system of the tester as follows:

a. Disassemble all components of the test oil system completely (except for pipe-threaded aluminum joints which cannot be removed without damage).
b. Disassemble pump, relief valve, and all valves completely. Disconnect all lines including those from the pressure gage and from the humidifier to the coking tube.
c. Discard used O-rings, pump shaft seal, and air dispersion stone in the coking head.
d. Soak all reusable components of the test oil system, except the coking tube, in dry-cleaning solvent for 20 minutes.
e. Drain excess solvent, and soak for 4 h in cleaning compound. Clean at 43°C to 79°C (110°F to 120°F).
f. Remove from cleaner, and rinse thoroughly with running water to remove all cleaner, brushing all parts thoroughly to remove all deposits.
g. Rinse all components thoroughly with clean isopropyl alcohol, then with clean petroleum ether, and blow dry with clean, dry air.
h. Inspect the parts for deposits, and repeat the cleaning if necessary (starting with the soak in special cleaning compound), until they all are completely clean.
i. Clean glass coker-assembly body and glass oil-filler tube with scouring powder, rinsing afterwards with water then with isopropyl alcohol. Air dry and immerse both in glass-cleaning solution for at least 1 h. Rinse thoroughly using distilled water for the final rinse, then oven dry at approximately 100°C (212°F).
j. Clean glass feeder flask and its connection by rinsing thoroughly with isopropyl alcohol and air dry.
k. Store all parts in a clean, dust-free location until ready for reassembly.

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NOTE 2: It is advisable to cut open lines and critical fittings periodically to check on the efficiency of cleaning. The most critical section is between the coking head and the cooler.

6.5 Prepare the coking tube as follows:

a. Clean the coking tube by hand abrasion with No. 400 silicon carbide paper holding the tube by the top end in a jig or fixture designed to avoid marring or distorting the tube. (To avoid getting dust in the thermocouple wells it is best to plug them with paper during this operation.)
b. Inspect the surface of the tube for pits. If the tube is pitted, turn it down in a lathe. (However, when the diameter has been reduced from its original 2.984 to 2.883 cm [1.175 to 1.135 in] or less, replace the tube.) Then repolish it by hand.
c. After the final polishing with silicon carbide paper, wipe free of dust with clean rags. Then rinse thoroughly with *n*-hexane (including thermocouple wells). Blow dry with clean, dry air. (Thermocouple wells should also be swabbed out with clean pipe cleaners.)

CAUTION: After final cleaning, the coking tube must be handled only by the top end to avoid contamination of the coking surface.

d. Inspect the thermocouple wells for residual oil. If there is any evidence of oil, repeat the *n*-hexane rinse and swab with pipe cleaners until all oil is removed.
e. Place the coking tube in an oven at approximately 100°C (212°F) for 1 hour.
f. Allow the tube to cool, and store it in a desiccator.

6.6 Weigh a clean coking tube (within 10 mg), and record the weight. Handle by the top to avoid contaminating coking surface.

6.7 Weigh a clean filter screen to the nearest 10 mg, and record the weight.

6.8 Clean a new coking-head diffuser stone, a new pump-shaft seal, and a complete set of new o-rings by rinsing in isopropyl alcohol, and then in petroleum ether, and allow them to air-dry.

6.9 Assemble clean components of the test section, installing the previously cleaned new parts (see 6.8)

6.10 After assembly, check to see that all fittings are tight, all valves are in proper position, bayonet thermocouples are "bottomed" in the coking tube, all thermocouples are registering, and the temperature indicators on all controllers and recorders read correct ambient temperature.

6.11 Level coking head by means of adjusting screws.

6.12 Ascertain periodically that the air flowmeter is calibrated to deliver 300 mL/min of saturated air at atmospheric pressure and 25°C (77°F).

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6.13 Flush the oil system as follows:

a. Charge the circulating system with unused isopropyl alcohol, and circulate for 5 minutes, alternating the flow several times between the filter (see FIGURE 2 and FIGURE 3, item 34), and the pressure-relief bypass valve (see FIGURE 2, item 34). This is accomplished by use of valves (see FIGURE 2, item 35) and (see FIGURE 2, item 37). Drain thoroughly opening valve (see FIGURE 2, item 37), and tubing connections as necessary for complete venting and draining. Discard the used isopropyl alcohol.
b. Repeat step (a) using unused petroleum ether. Drain thoroughly, and discard used petroleum ether.
c. If the first flush of petroleum ether comes out dirty or discolored, repeat until petroleum ether remains colorless. Drain thoroughly, and discard all used solvents.

6.14 Check the adjustment of the system as follows:

a. Open valve (see FIGURE 2, item 29), and charge the system with test oil through oil filler system.
b. Turn on pump, and continue to add oil until level in coking head glass is at or slightly above the mark 3.8 cm ([1-1/2 in] below top of glass).
c. Fill humidifier half full of distilled water, and adjust air flow control valve to give an air flow of approximately 150 mL/minutes.
d. Turn on the coking tube heater, and set controller for 274°C (525°F). (It is imperative that this temperature is not exceeded during flow check since higher temperatures would cause coke formation). As the test oil heats and expands, drain as necessary to maintain a visible level in the coking head. It is desirable to have the oil level close to the top of the coking head glass. Valve (see FIGURE 2, item 29) must be kept open.
e. Turn on the cooling water, and check the operation of the water solenoid valve.
f. Set "oil in" temperature control to 149°C (300°F).
g. As soon as "oil in" temperature is above 121°C (250°F), check the setting of bypass relief valve (see FIGURE 2, item 34) by cutting off filter flow at valve (see FIGURE 2, item 35). Adjust the relief valve to give a pressure of 34.5 kPa gage (5 psig) higher than obtained when flowing through the filter.
h. Open valve (see FIGURE 2, item 35) to direct flow through the filter.
i. Shut off coking-tube heater as soon as the "oil out" temperature reached 138°C (280°F). Check hot flow rate by measuring the time required for a discharge of 100 mL of oil through three-way valve (see FIGURE 2, item 28). Pour the 100 mL of oil back into the filler tube immediately, keeping valve (see FIGURE 2, item 29) open. Calculate the flow rate in mL/minutes.
j. Turn on the coking tube heater.
k. If first hot flow check shows a rate of 300 ± 5 mL/minutes, record the actual flow rate and proceed to step (1). If the flow rate is outside this limit, readjust the variable-speed-transmission speed control (see FIGURE 1, item 14) to correct for the difference, and repeat steps (i) and (j) until a flow rate of 300 ± 5 mL/min is obtained.

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l. Direct oil flow through the bypass valve (see FIGURE 2, item 34) by blocking the flow through the filter, and again check the hot oil flow as in step (i). If the flow rate is above 255 mL/minutes, record this value and proceed to 6.15. If the flow rate is below 225 mL/minutes, the pump is worn or defective and must be replaced or rebuilt before proceeding further. In this event, after replacing or rebuilding the pump, repeat 6.13 and 6.14.

NOTE 3: The entire sequence of flow check procedure (steps j though l) must be completed within 30 minutes to avoid excessive oil deterioration.

TABLE I. Standard operating conditions.

Condition	Value	
Coking-tube temperature, °C (°F)	310° ± 2.5°	(590° ± 5°)
Oil-in temperature, °C (°F)	149° ± 2.5°	(300° ± 5°)
Air temperature (test section), °C (°F)	48.9° ± 2.5°	(120° ± 5°)
Oil flow, mL/min		300 ± 5
Air flow, mL/min		300 ± 5
Oil level (aerated), below top of coking-head glass, cm (in)	3.81 ± 0.32	(1-1/2 ± 1/8)

6.15 Using the standard operating condition listed in TABLE I, conduct a 12 hours ± 10 min run on the test oil used in adjusting the system (see 6.14). Perform the test as follows:

a. While the system is hot, adjust the oil level in the oil filter tube to about 3.8 cm (1-1/2 inch) below the oil level mark on the tube (to allow for an increase in volume as a result of aeration).
b. Start the oil flow at 300 mL/minutes.
c. Adjust the air flow to 300 mL/minutes.
d. Fill the oil feeder flask completely with test oil, stopper tightly, and install in oil filler tube.
e. Set coking tube temperature control to 310°C (590°F).
f. Check setting of "oil in" temperature control for 149°C (300°F).
g. As soon as "oil in" temperature reaches 138°C (280°F) start timing the test period by setting the timer for 12 h. If apparatus is so equipped, set watt-hour meter (see FIGURE 1, item 5) to zero. Read and record all temperatures and pressures.
h. Adjust height of oil-filler assembly to maintain the aerated oil level in the coking head at the mark on the coking-head glass to within ± 0.32 cm (±1/8 inch). Maintain this level for duration of test. Keep valve (see FIGURE 2, item 29) open for duration of test.
i. Keep cabinet door closed throughout the test, opening only when necessary and for as short periods as possible. Check and adjust operation of ventilating and heating equipment to hold test section air temperature at 48.9 °C ± 2.8 °C (120 ° ± 5 °F).

(1) Record the following data every 30 min for the duration of test:

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Oil-in temperature, °C (°F)
Oil-out temperature, °C (°F)
Coking-tube temperature, °C (°F)
Cabinet-air temperature, °C (°F)
Watt-hour meter (total), if equipped
Air flow (rotameter setting and pressure), mL/min
Aerated-oil level (above or below mark), cm
Pump pressure, kPa (psig)
Feeder flask volume, mL

NOTE 4: The feeder flask must be full at the start of the test run (to minimize oil ejection due to air expansion). It is helpful to heat flask and oil to 54.4°C (130°F) prior to its installation. The stopper on the flask must be tight to avoid air leakage after installation.

6.16 At the end of the 12 hour test run, shut off the coking-tube heater, and immediately shut filler valve (see FIGURE 2, item 29). Remove feeder flask, and empty all fresh oil from the filler tube.

6.17 Shut off air flow.

6.18 With pump running, drain the used oil into a clean, dry container through drain valve (see FIGURE 2, item 37). Save used oil if optional analysis is desired.

6.19 Turn off all switches on control panel.

6.20 While the test rig is still hot and as soon after the completion of the test run, check for sludge deposits in the filter as follows:

a. Remove the filter (see FIGURE 5) as a unit, and allow it to drain for 5 minutes with the inlet at the top and the filter housing tilted approximately 30° from the vertical.
b. Being careful not to dislodge sludge particles, disassemble the filter housing, and place the screen (deposit side up) on a clean, absorbent paper towel or on a piece of qualitative-grade filter paper.
c. After 10 minutes replace paper with a new one and allow screen to drain an additional 50 minutes.
d. Weigh the screen and deposits to the nearest 10 mg, and record.
e. Oils with very high sludge values may leave deposits on the upstream half of the filter housing. These deposits should be removed with petroleum ether and a policeman, and filtered through a tared Gooch crucible. The crucible and contents should be washed thoroughly with petroleum ether until the washings are completely free from discoloration. The crucible and contents are than dried in an oven at 96°C (205°F) for 1 hour, cooled at least 1 hour in a desiccator, and reweighed. The resulting deposit weight should be added to that obtained from the filter screen, and the total obtained reported as sludge.

6.21 Check the coking tendency of the oil as follows:

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a. Being extremely careful not to dislodge any deposits, remove the coking tube from the coking head.
b. Allow the tube to cool for at least 1 hour.
c. Immerse the tube up to its flange in *n*-hexane, and allow it to soak for 10 minutes.

CAUTION: Handle carefully to avoid dislodging deposit.

d. Repeat the soaking for 10 minutes in clean *n*-hexane.
f. Place coking tube into an oven at approximately 96°C for 1 hour, then place in desiccator until cool.
g. Remove the thermocouple, allow the tube to cool in air at room temperature for at least 1 hour, weigh to nearest 10 mg, and record.

6.22 Disassemble and clean the deposition test rig. (Refer to 6.3, 6.4, and 6.5.)

7. CALCULATIONS

7.1 Calculate the following:

a. Sludge is expressed as the difference between the final and initial weights of the filter screen.
b. Coke is expressed as the difference between the final and initial weights of the coking tube.
c. Deposit rating is calculated as follows:

$$Deposit\ rating\ =\ Sludge\ +\ 10 * coke$$

8. REPORTING

8.1 Report the weight, in g, of the sludge and the coke separately, and the deposit rating of the oil.

8.2 Report the following operating data:

a. Elapsed time (to within 1/2 hours) to pump-pressure increase of 34.5 gage kPa (5 psig).
b. Oil consumption (difference between initial and final oil volumes in feeder flask).
c. Unusual operational condition (such as foaming of oil in coking head).
d. Elapsed time (and reasons for termination) if test run must be stopped before 12 hours.

9. PRECISION

9.1 Repeatability. Two or more runs on the same oil in the same apparatus and by the same operator should differ from their mean by no more than 0.75 deposit rating.

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9.2 Reproducibility. The average values for the same oil obtained on different apparatus or by different operators (or both) should not differ by more than 0.75 deposit rating.

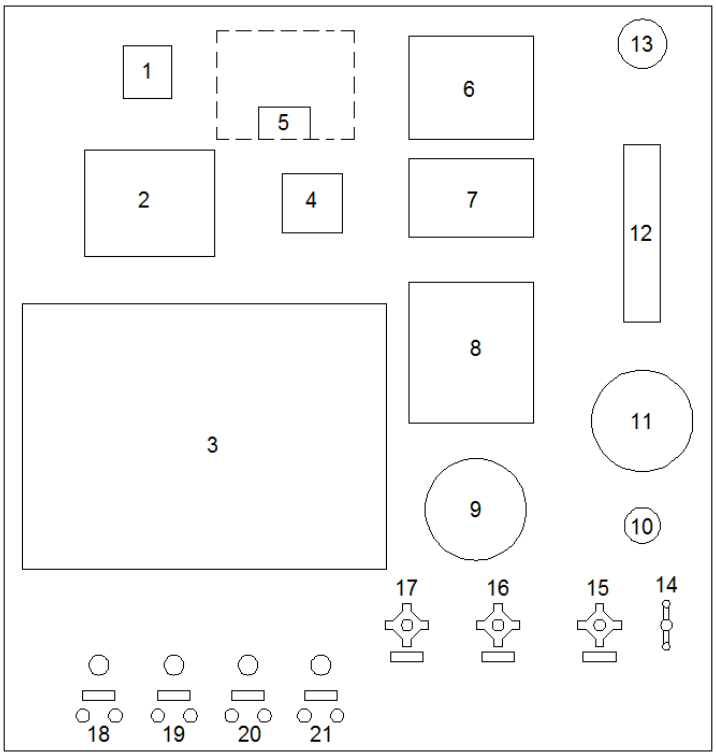


FIGURE 1. Example of deposition tester instrument and control section.

1. Voltmeter, 0 to 300 volts
2. Controller (stepless), coking tube temperature, -18° to 430° C (0° to 800° F)
*3. Strip chart recorder (potentiometer), 6-channel; or differential pressure recorder, 0 to 25 cm (0 to 10) inches of mercury
4. Timer, 24-hour
*5. Watt-hour meter
6. "OFF-ON" controller, oil-temperature, -18° to 320° C (0° to 600° F)
*7. Pyrometer
*8. Selector switch
9. Filter pressure gage, 0 to 69 gage kPa (0 to 10 psig)
10. Pressure regulator, air-flow
11. Air-pressure gage, 69 gage kPa (10 psig) (working)
12. Air flowmeter, 0 to 500 mL/min
13. Air throttle valve
14. Speed control, variable-speed pump drive
15. Humidifier-water valve
16. Water trim valve (flow regulation through solenoid valve)
17. Solenoid-bypass water valve
18. Pump switch and indicating light
19. Coking-tube heater switch and indicating light
20. Air solenoid-valve switch and indicating light
21. Ambient-air heater switch and indicating light (off-on by thermoswitch)

* Optional.

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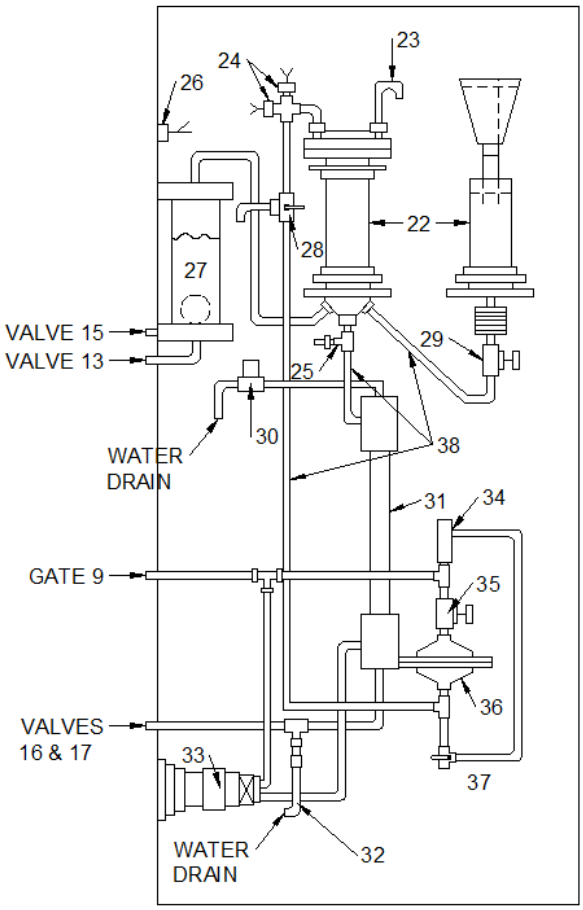


FIGURE 2. Test section components.

22.	Coking head and oil reservoir	31.	Oil cooler
23.	Vent line	32.	Orifice (water drain)
24.	Oil inch thermocouples	33.	Oil pump and variable-speed transmission
25.	Oil out thermocouple	34.	Pressure-relief bypass valve
26.	Test-cabinet ambient temperature thermocouple	35.	Shut-off valve, test filter
27.	Humidifier	36.	Test filter
28.	Bypass valve to coking head	37.	Drain valve, test-oil system
29.	Shut-off valve, oil reservoir	38.	All test-oil tubing shall be 1/4 inch aluminum tubing
30.	Check valve (air bleed)		

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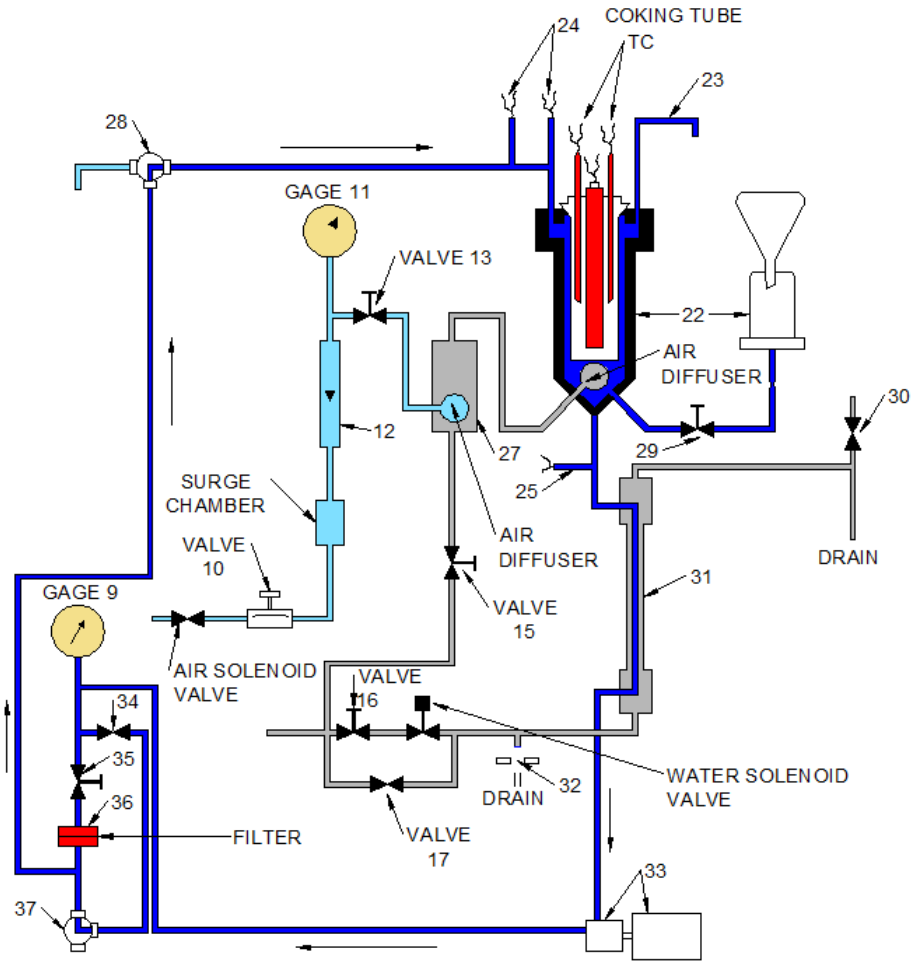
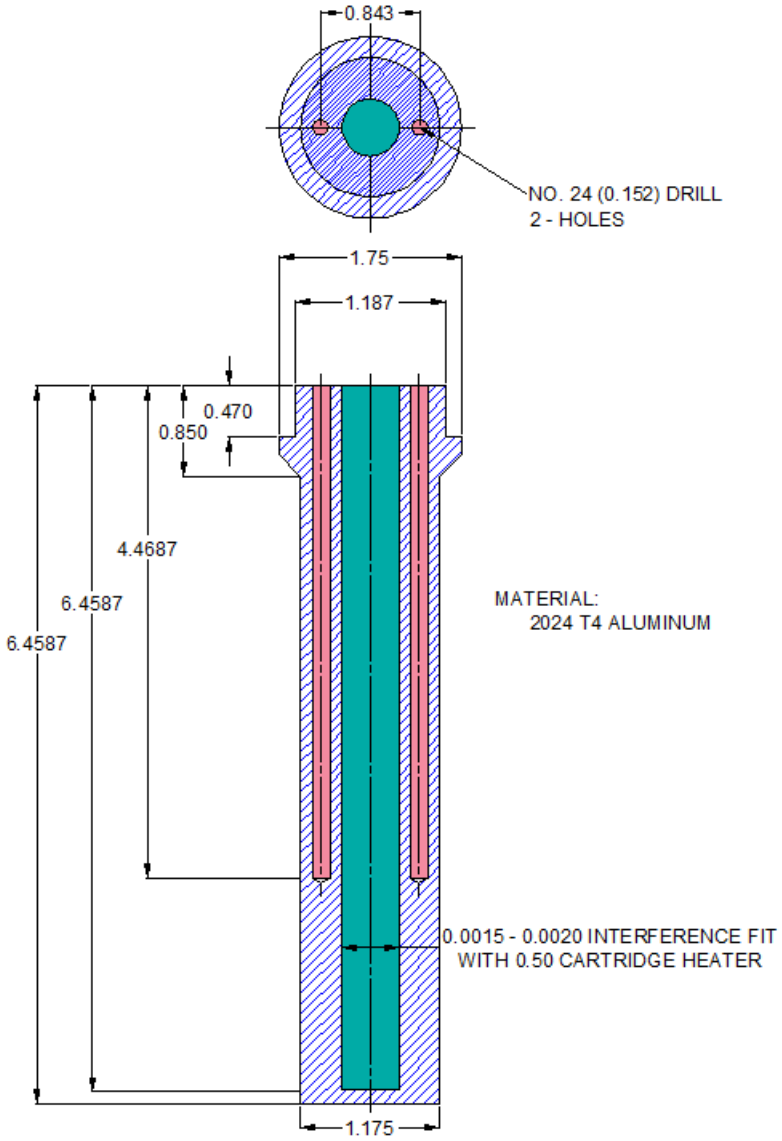


FIGURE 3. WADC deposition tester (schematic).

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NOTE: DIMENSIONS IN INCHES.

FIGURE 4. Coking tube.

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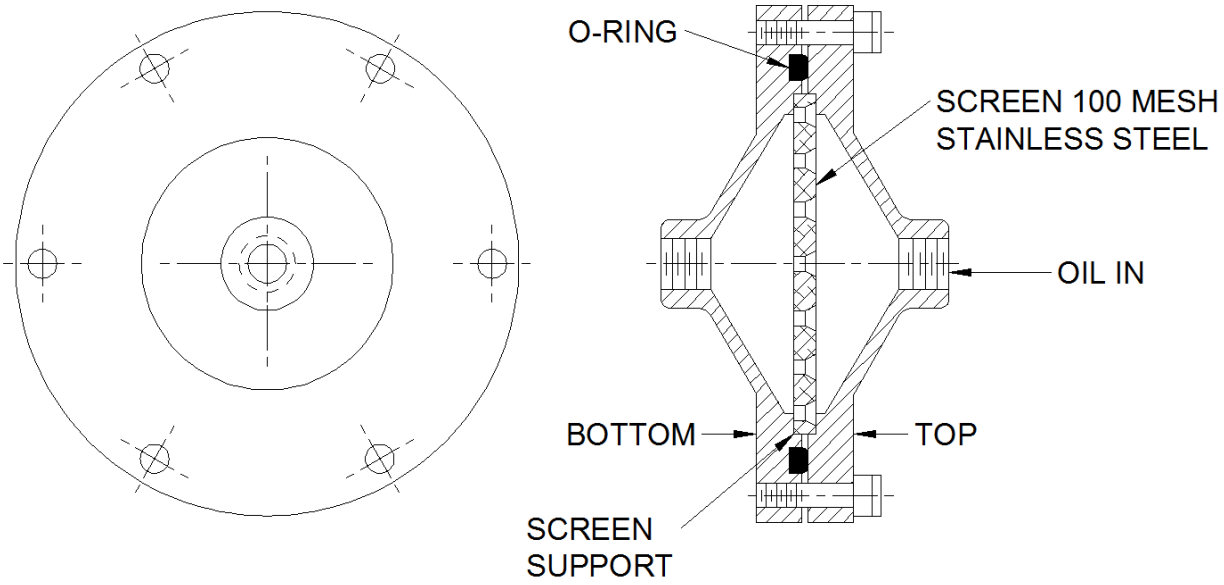


FIGURE 5. Line filter.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Solvent / cleaner substitutions	ODC removal / no longer available	3	Air Force – AFRL/RQTM
30 Sep 2019	Supplier Information	Original Equipment Supplier longer available	3	Air Force – AFRL/RQTM
30 Sep 2019	Oil in / out temperatures incorrect	Oil-Out was noted when Oil-In is the correct temperature	3	Air Force – AFRL/RQTM
30 Sep 2019	Changed drying procedure	Easier logistically to use drying oven	3	Air Force – AFRL/RQTM
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT

Method prepared by:
Air Force-AFRL/RQTM-2019

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Method 5305.2
September 30, 2019

5305.2 – CORROSIVENESS OF LUBRICANTS AT 232°C (450°F)

1. SCOPE

1.1 This method is used for determining the corrosive tendencies of lubricants at high temperatures. It consists of immersing copper and silver strips in the sample at 232°C (450°F) for 50 hours, and then examining them for weight change.

2. SUMMARY

2.1 This method duplicates an extended, high-temperature engine or gear box environment. Copper and silver strips are exposed to the heated lubricant for 50 hours and carefully cleaned and weighed for loss of weight due to corrosive action resulting from corrosive decomposition products being formed in lubricant.

3. SAMPLE SIZE

3.1 Approximately 400 mL of lubricant to be tested.

4. REFERENCES, STANDARDS AND APPARATUS

4.1 Oven, explosion-proof, 232°C ± 3°C (450°F ± 5°F).

4.2 Copper strip (ASTM B152/B152M), 3.8 cm x 3.8 cm x 0.16 cm (1-1/2 x 1-1/2 x 1/16 inch).

4.3 Silver strip, pure electrolytic (MIL-DTL-13282, grade A), 3.8 cm x 3.8 cm x 0.32 cm (1-1/2 x 1-1/2 x 1/8 inch).

4.4 Beakers (2), borosilicate glass, Griffin low-form, 400 mL.

4.5 Supports (2), glass (for holding copper or silver strip vertically in beaker with point contact, so that strip does not touch the sides or bottom of the beaker).

4.6 Balance, sensitivity 0.1 mg.

4.7 Electrolytic bath, carbon-rod anode, 0.5 A (min), containing sulfuric acid solution (5% by weight in distilled water) as the electrolyte.

4.8 Tongs, stainless steel.

4.9 Watch glass (2), 10 cm or larger.

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5. MATERIALS

5.1 Solvent, dry-cleaning (MIL-PRF-680, type I) [0,2,0].

5.3 Acetone (ACS) [1, 3, 0].

5.4 Nitric acid solution (3 parts concentrated nitric acid and 7 parts distilled water) [2, 0, 0].

5.5 Pumice, FFF grade.

5.6 Eraser (Eberhard Faber "Pink Pearl ± No. 100, or equivalent).

5.7 Cleaning solution, glass (concentrated sulfuric acid [3, 0, 2] saturated with ALNOCHROMIX™ [0,1,0]).

NOTE: If the glass-cleaning solution cannot remove all deposits (see 6.1) use a special cleaner, such as Turco Transpo cleaner (Turco Products Co., Chicago, IL).

5.8 Distilled water.

6. PROCEDURE

6.1 Clean two glass supports and two 400-mL beakers as follows:

a. Rinse with dry-cleaning solvent to remove excess oil.
b. Place each support in its beaker, and fill each beaker with glass-cleaning solution. Let them stand for at least 8 hours or until all traces of deposits are removed.
c. Remove from the glass-cleaning solution, and rinse the glassware with running tap water followed by distilled water.

6.2 Handling only with tongs, clean the copper and silver strips in separate bath solutions as follows:

a. Wash with dry cleaning solvent.
b. Immerse in dilute nitric acid solution for 15 s ± 1 s.
c. Rinse the silver strip only in hot 82°C to 100°C (180°F to 212°F) tap water, and then wet polish with pumice. (No polishing of the copper strip is required.)
d. Rinse the strip in hot tap water, and then in hot distilled water.
e. Dip in acetone, and allow to air dry.

6.3 Handling the strips with tongs, weigh each strip to the nearest 0.1 mg.

CAUTION: In succeeding steps the strips must be handled only with tongs until after they have been weighed (see 6.9) at the end of the test.

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6.4 Mount each strip in a glass support then place each support in a separate beaker so that the strip does not touch the sides of the beaker.

6.5 Add 200 mL of the sample to each beaker, and cover with watch glass to avoid evaporation losses.

6.6 Store both beakers in an oven for 50 hours at 232°C ± 3°C (450°F ± 5°F).

6.7 At the end of the heating period, remove both beakers from the oven and allow them to cool to room temperature.

6.8 Remove the copper and silver strips from the supports and clean them as follows:

a. Wash each strip with successive portions of fresh, hot 60°C (approx. 140°F) Solvent, dry-cleaning (MIL-PRF-680, type I), to remove as much discoloration and deposit as practicable.
b. If carbonaceous deposits remain, place the strip (as a cathode) in the electrolytic bath for 10 minutes at 0.5 A.
c. Wash each strip in hot, running water, while rubbing it with an eraser.
d. Rinse each strip with distilled water, then with acetone, and air dry both strips.

6.9 Weigh each strip to the nearest 0.1 mg, and calculate the change in weight of each strip (see 7.1).

6.10 Measure the surface area of each strip (both sides) (see 7.1).

7. CALCULATIONS

7.1 The calculations to determine mg/cm^2 as follows

$$CW = \frac{W_1 - W_2}{2 * W * L}$$

where:
CW = change in mg/cm^2
W_1 = original weight in mg.
W_2 = final weight in mg.
W = width of metal strip in cm.
L = length of metal strip in cm.

8. REPORTING

8.1 Report change in weight in mg/sq cm rounded off to the nearest mg.

9. PRECISION

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9.1 Repeatability. Results of each test obtained by one operator using the same apparatus must not vary by more than that shown in TABLE I.

9.2 Reproducibility. Results of each test obtained by different operators or different apparatus must not vary by more than that shown in TABLE I.

TABLE I. Precision.

Average weight change mg/cm^2	Permissible variation	
	Repeatability	Reproducibility
0 to 0.46	0.046 mg/cm^2	0.09 mg/cm^2
Above 0.46	10%	20%

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT

Method prepared by:
Air Force - AFPET/PTP - 2019

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Method 5309.6
September 30, 2019

5309.6 – CORROSIVENESS OF GREASES (COPPER STRIP, 100°C)

1. SCOPE

1.1 This method is used for determining the corrosive properties of grease at elevated temperatures.

2. SUMMARY

2.1 The test consists of partially immersing a copper strip in the sample at 100°C for 24 hours then visually examining both sample and strip for evidence of corrosion.

3. SAMPLE SIZE

3.1 Approximately 20 mL of grease to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Oven, 100°C.

4.2 Microscope, approximately 60X diameter magnification.

4.3 Test tube, clean, dry, approximately 2.5 cm diameter by 10 cm long.

4.4 Support (beaker) for holding test tube approximately vertical.

4.5 Forceps, stainless steel.

5. MATERIALS

5.1 Copper strip (Copper and Brass Research Association No. 110 Electrolytic Grade), soft tempered, annealed, approximately 1.3 x 7.6 x 0.15 cm or 0.3 cm.

5.2 *n*-Hexane (ACS Grade) non-corrosive (as determined by ASTM D130) [1, 3, 0].

5.3 Acetone (ACS Grade) [1, 3, 0].

5.4 Polishing materials.

5.4.1 Fine silicon carbide or alumina grit paper.

5.4.2 240-grit silicon carbide paper.

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5.4.3 150 mesh silicon carbide grains.

6. PROCEDURE

6.1 Prepare finish of copper strip as follows:

a. Surface preparation - Remove all blemishes from all six sides of the strip with silicon-carbide or alumina grit paper of such degree of fineness as is needed to accomplish the desired results effectively (see Note 1). Finish with 240 grit silicon carbide paper, removing all marks that have been made by other grades of paper used previously. Immerse the strip in acetone (see Note 2) from which it may be with-drawn immediately for final polishing or in which it may be stored for future use.

NOTE 1: As a practical manual procedure for surface preparation moisten it with acetone and rub the strip against the paper with a rotary motion protecting the strip from contact with the finger with an ashless filter paper. Alternatively the strip may be prepared by use of motor driven machines using appropriate grades of dry paper or cloth.

NOTE 2: CAUTION. Acetone is both toxic and flammable. Use only in a well ventilated area. Do not breathe its fumes nor allow it to come in contact with the skin. Keep all flames away from the acetone.

b. Final polishing - Remove a strip from the acetone. Holding it in the fingers protected with the ashless filter paper, polish first the ends and then the sides with 150 mesh silicon carbide grains picked up from a clean glass plate with a pad of absorbent cotton moistened with a drop of acetone. Wipe vigorously with fresh pads of absorbent cotton and subsequently handle only with stainless steel forceps; do not touch with fingers. Clamp in a vise and polish the main surfaces with silicon carbide grains on absorbent cotton. Rub in the direction of the long axis of the strip, carrying the stroke beyond the end of the strip before reversing the direction (see Note 3). Clean all metal dust from the strip by rubbing vigorously with clean pads of absorbent cotton until a fresh pad remains unsoiled. When the strip is clean, immediately immerse it in the prepared sample.

NOTE 3: It is important to polish the whole strip uniformly to obtain a uniformly strained strip. If the ends show wear (surface elliptical) they will likely show more corrosion than the center. The use of a vise will facilitate uniform polishing.

6.2 Fill test tube with sample to depth of 5 cm. Remove air pockets from the grease by tapping the tube on the bench or table as it is being filled.

6.3 Using stainless steel forceps, insert copper strip until it touches the bottom of the tube.

6.4 Place the test tube (supported approximately vertically in 100°C) oven for 24 hours (unless otherwise specified).

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6.5 At the end of the specified time, remove test tube from oven and allow it to cool at room temperature.

6.6 Using forceps, remove copper strip, and wash in non-corrosive ACS grade n-hexane. (See Note 4.)

NOTE 4: CAUTION. *n*-Hexane is both toxic and flammable. Use only in a well ventilated area. Do not breathe its fumes nor allow it to come in contact with the skin. Keep all flames away from the n-hexane.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Report whether the section of the strip immersed in the sample is darker in color than the exposed section.

8.2 Report by describing any corrosion of the strip when examined under a microscope of approximately 60 diameters magnification.

8.3 Report development of any green color in the grease.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	6	Army – AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	6	Army – AT

Method prepared by:
Navy - NADC - 1986

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Method 5321.3
September 30 2019

5321.3 – CORROSION OF LEAD BY LUBRICATING OILS

1. SCOPE

1.1 This method is used for measuring the corrosiveness of lubricating oils on lead in the presence of a copper catalyst.

2. SUMMARY

2.1 It consists of rotating panels of lead and copper in the test lubricant under specific conditions and determining the degree of corrosion from the change in weight of the lead panel.

3. SAMPLE SIZE

3.1 Approximately 500 mL of the oil to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Sample tube, SOD (Standard Oil Development), available from Koehler Instrument Co. Inc., www.koehlerinstrument.com, or may be custom ordered from a laboratory glassware supplier or glassblower. (see FIGURE 1).

4.2 Bath, constant-temperature, 163°C ± 1°C (325°F ± 2°F), minimum depth 31 cm (12 inch).

4.3 Motor, stirring, 600 ± 50 rpm.

4.4 Shaft, stirring, with attaching hardware (see FIGURE 2).

4.5 Air supply, clean, dry, approximately 7 gage kPa (1 psig) to provide flow of 0.940 ± 0.047 L/min (2 ± 0.1 ft^3/h).

4.6 Balance, sensitivity ±1 mg.

4.7 Forceps, stainless steel.

5. MATERIALS

5.1 Test panel, chemical-lead grade (ASTM B29 UNS L51120 or equal) 3.5 cm x 4.13 cm 0.16 cm (1-3/8 in x 1-5/8 in x 1/16 in), conforming to TABLE I.

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TABLE I. Composition of lead test panel.

Element	Content percent	
	Maximum	Minimum
Silver	0.020	0.002
Copper	0.080	0.040
Arsenic, antimony, and tin (total)	0.002	-----
Zinc	0.001	-----
Iron	0.002	-----
Bismuth	0.005	-----
Lead (by difference)	-----	99.9

5.3 Test panel, copper, electrolytic (ASTM B152/B152M), 3.5 cm x 4.13 cm x 0.130 cm (1-3/8 in x 1-5/8 in x 0.051 in [16 gage]).

5.4 Naphtha, petroleum [1,4,0]

5.5 Acetone (ACS) [1,3,0].

5.6 Steel wool, No. 00.

5.7 Pads, absorbent-cotton.

6. PROCEDURE

6.1 Preheat and adjust bath to 163°C ± 1°C (325°F ± 2°F).

6.2 Pour 500 mL of the sample into a clean sample tube, and place tube into the bath so that the bath level is at least 2.5 cm (1 inch) above the sample level.

6.3 Connect air supply line to the aerator intake of the sample tube.

6.4 Allow approximately 30 minutes for the specimen to come to test temperature.

6.5 Drill a 3/16-inch hole in the center of each test panel (see 5.2 and 5.3).

6.6 Polish and clean the test panels (handling only with forceps) as follows:

a. Polish the faces and edges of the panels with steel wool until a uniform bright surface is produced.

CAUTION: When polishing the lead panel, avoid excessive pressure which may cause the steel wool to become embedded in the lead.

b. Remove all loose particles from the panels, using naphtha-soaked cotton pads.

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c. Rinse with fresh naphtha and allow to air dry.

6.7 Using forceps to handle the panel, record the weight of the lead panel to the nearest mg.

6.8 Clean the stirrer shaft and panel attaching hardware by rinsing with naphtha, using naphtha-soaked cotton pads or steel wool as required. Rinse in clean naphtha and allow to air dry.

6.9 Using forceps to handle the panels, mount both panels on the shaft as shown in FIGURE 2.

6.10 Attach stirrer motor to the shaft and lower the shaft into the sample tube so the center of the lead panel is centered in the tube approximately 7.5 cm (3 inch) from the bottom.

6.11 Start the stirrer and immediately adjust the air flow 0.940 L/min ± 0.047 L/min (2 ft^3/h ± 0.1 ft^3/h). Continue the operation for 60 min ± 1 min.

6.12 Remove the stirrer assembly and detach the lead panel from the shaft. Swab the lead panel lightly with naphtha-soaked cotton pads to remove any loose surface deposits. Rinse with fresh naphtha and then with acetone and allow the panel to air dry.

6.13 Reweigh until there is no change in the weight.

7. CALCULATIONS

7.1 Calculate the change in weight in mg/cm^2 (mg/in^2) of the exposed surface area (including the faces and edges, but excluding the areas covered by the washers).

8. REPORTING

8.1 Report the change in weight to the nearest 1.5 mg/cm^2 (mg/in^2).

9. PRECISION

9.1 Repeatability. Results by one operator using the same apparatus must not differ:

a. By more than 0.15 mg/cm^2 (1 mg/in^2) if the change in weight does not exceed 1.5 mg/cm^2 (10 mg/in^2),

or

b. By more than 10% of the average, if the change in weight exceeds 1.5 mg/cm^2 (10 mg/in^2).

9.2 Reproducibility. Results from different operators or different apparatus must not differ:

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a. By more than 0.3 mg/cm^2 (2 mg/in^2) if the change in weight does not exceed 1.5 mg/cm^2 (10 mg/in^2),

or

b. By more than 20% of the average, if the change in weight exceed 1.5 mg/cm^2 (10 mg/in^2).

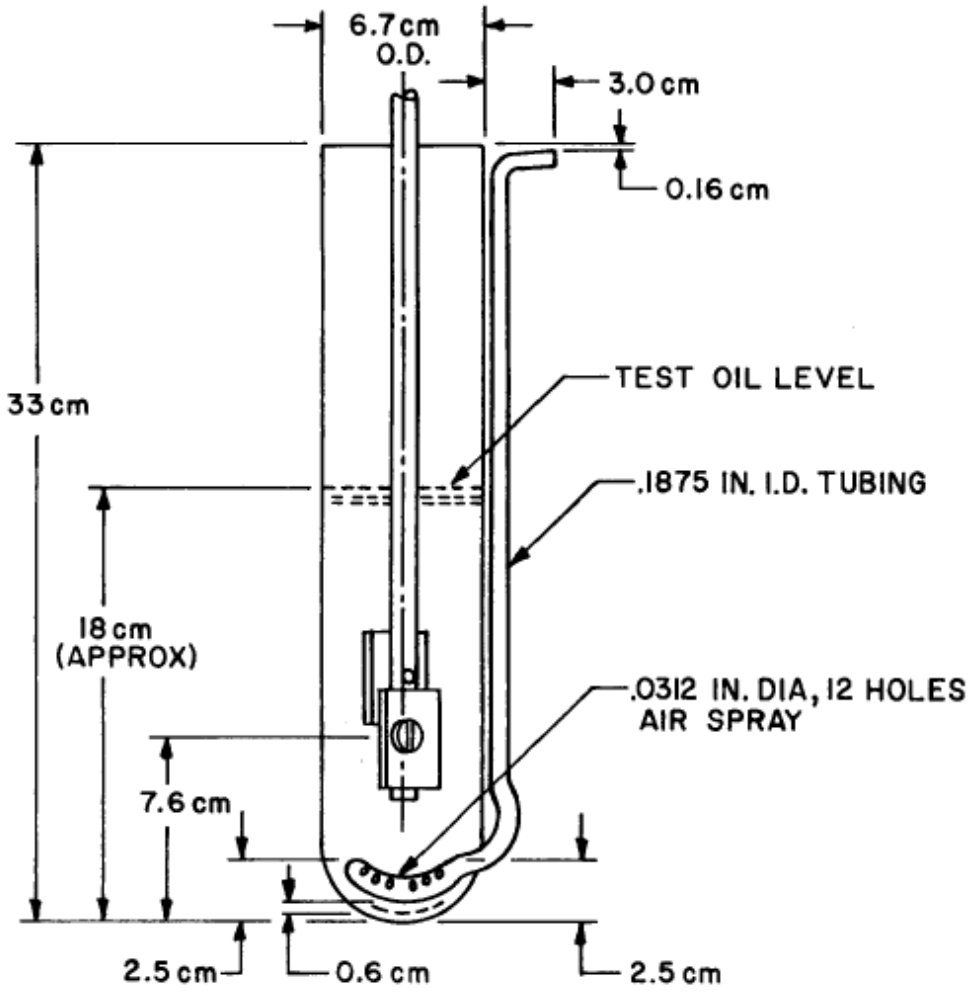


FIGURE 1. SOD sample tube.

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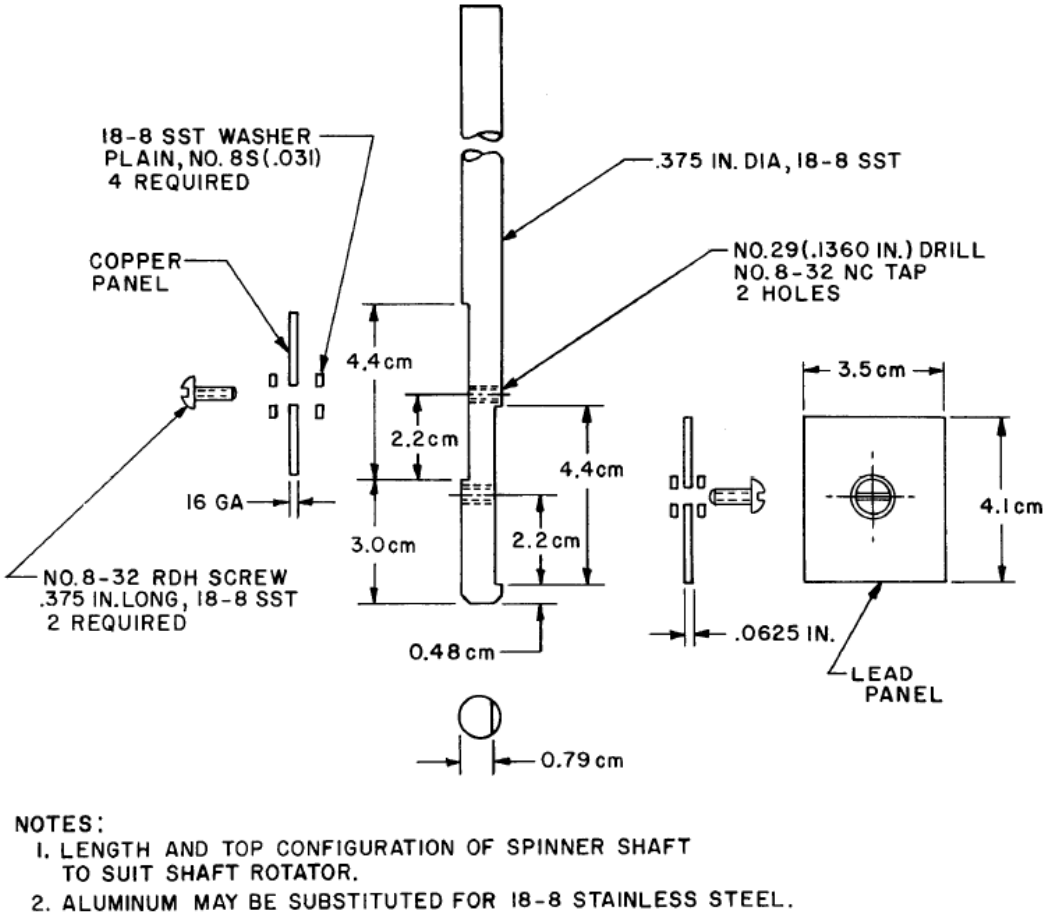


FIGURE 2. Stirring shaft and test panels.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT

Method prepared by:
Air Force – AFPET/PTP – 2019

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Method 5329.3
September 30, 2019

5329.3 – CORROSION-PROTECTION (HUMIDITY CABINET)

1. SCOPE

1.1 This method is used for determining the effectiveness of corrosion inhibiting oils and compounds under conditions of high humidity.

2. SUMMARY

2.1 The test consists of dipping steel test panels in the test sample and then placing them in a humidity cabinet. After the specified exposure time the panels are removed and examined for the presence of rust.

3. SAMPLE SIZE

3.1 Approximately 500 mL of the oil to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Standards.

4.1.1 ASTM D1748 Standard Test Method for Rust Protection By Metal Preservatives In the Humidity Cabinet.

4.2 Humidity cabinet conforming to Appendix I of ASTM D1748 where the following standard conditions must be maintained.

a. Air temperature:
1) Inside the cabinet: 49°C ± 1°C (120°F ± 2°F).
2) Outside the cabinet: 24°C ± 5°C (75°F ± 10°F).
b. Rate of air to the cabinet: 868 ± 28 L (31 ± 1 ft^3) per hour at 25°C (77°F) and 76 cm of mercury.
c. Water in cabinet:
1) Level - 20.3 ± 0.6 cm (8 ± 1/4 inch)
2) pH - 5.5 to 7.5
3) Oil content - clear with no evidence of oil
4) Chlorides - less than 20 ppm
5) Sulfates & sulfites - less than 20 ppm.
d. Speed of rotating stage - 0.33 ± 0.03 rpm.
e. Cover - close fitting.
f. Cloth layers in cover - shall not be torn, contaminated, nor contain droplets of water.
g. Cover opening - to a height of 35.6 cm (14 inch) at the front.

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h. Open the cabinet twice each day, except Saturday and Sunday, once for 15 minutes and again for 5 minutes. (A period of approximately 5 h should elapse between openings.)
i. Check the rate of air to the cabinet, air temperatures, pH, and water level and regulate, if necessary, in the morning and afternoon of each day.
j. Check remaining standard conditions once each week.

4.3 Desiccator, containing an indicating desiccant.

4.4 Panel hook, stainless steel (for handling panels during processing).

4.5 Hooks, Monel or stainless steel (for supporting steel panels).

4.6 Sandblast cabinet (a suitable apparatus for sandblasting test panels).

4.7 Beakers, tall-form, 500 mL.

4.8 Controlled atmosphere (to provide a temperature of 25C ± 3°C [77F ± 5°F] and a maximum relative humidity of 50%).

4.9 Shaded fluorescent light. (A 15-watt balance illuminator type that will permit the panel to be viewed from all angles at a distance of 7.6 cm [3 inches]).

5. MATERIALS

5.1 Naphtha, aliphatic (TT-N-95) [1, 4, 0].

5.2 Methyl alcohol (O-M-232) [1, 3, 0].

5.3 Silica sand, white, dry, sharp. (The size shall be such that it meets the following sieve requirements of ASTM E11.)

a. 100% must pass through a No. 10 sieve.
b. Minimum of 90% must pass through a No. 20 sieve.
c. Maximum of 10% permitted to pass through a No. 50 sieve.

5.4 Test panels, FS 1009 steel, fabricated from open-hearth, "killed ± type low carbon, No. 4 soft-temper, cold-rolled sheet or strip conforming to ASTM A109/A109M. Dimensions of the panels shall conform to Appendix I of ASTM D1748. Cold drawn bar stock is not satisfactory.

5.5 Surgical gauze.

6. PROCEDURE

6.1 Clean the steel panels as follows:

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a. Round the edges and ream out the suspension holes in accordance with Appendix I of ASTM D1748.
b. Wipe the surfaces with solvent soaked rags and scrub with surgical swabs in a beaker of hot naphtha.
c. Rinse in a beaker of hot methyl alcohol.
d. Cool the panels in a desiccator until further processing.
e. Test panels must be handled with a panel hook at all times, avoiding contact with any type of contaminated surface.
f. Maintain the solvents at a temperature high enough to keep the temperature of the panels above the dew point during the cleaning operation.

NOTE 1: CAUTION. Naphtha is flammable. Use only in a well-ventilated area. Keep all flames away from the naphtha.

NOTE 2: CAUTION. Methyl alcohol is both toxic and flammable. Use only in a well-ventilated area. Do not allow it to come in contact with the skin or breathe its fumes. Keep all flames away from methanol.

6.2 Sandblast the panels as follows:

a. Sandblast the edges and lightly sandblast the back of the panels with silica sand.
b. Sandblast the unnumbered side, or test surface, of the panels to a fresh, uniformly abraded surface.
c. Immediately after sandblasting, place the panels in a beaker of anhydrous methanol.
d. Heat the methyl alcohol so that the solvent will evaporate from the panels immediately upon withdrawal from the solvent.
e. Remove remaining residue by holding the panels in a rack at 20° from the vertical and spraying downward with naphtha.
f. Spray the test surface, then the back of the panel and the test surface again.
g. Rinse the panels in hot naphtha and hot methyl alcohol and store in a desiccator until cool.
h. Panels are to be used the same day as prepared.

6.3 Three test panels are processed as follows:

a. Dip the test panels in the test sample at a temperature of 25°C ± 3°C (77°F ± 5°F) and agitate gently for one minute.
b. Suspend the panels by means of Monel or stainless steel hooks and drain for 2 h at 25°C ± 3°C (77°F ± 5°F) and a maximum of 50 percent relative humidity.
c. Place the panels in the humidity cabinet described in Appendix I of ASTM D1748 for the time required by the applicable specification.
d. Dummy panels, fabricated from Monel or stainless steel, are used to fill all spaces not occupied by test panels. They are also placed under the three arm supports of the rotating stage.

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6.4 At the end of the exposure period, remove the panels and rinse in methyl alcohol. Follow with a rinse in naphtha and methyl alcohol and examine the significant area of the test surface as defined in Appendix I of ASTM D1748, using a 15-watt, shaded fluorescent light.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Record the extent of corrosion as follows:

a. The number of rust dots, 0.1 cm in diameter or less.
b. The number of rust spots, 0.1 to 0.2 cm in diameter.
c. The number of rust areas, greater than 0.2 cm in diameter.

9. PRECISION

9.1 Precision data have not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT
30 Apr 2026	Replaced QQ-S-698 with ASTM A109/A109M	QQ-S-698 was cancelled	3	Army - AT

Method Prepared By:
Army - AR - 1986

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Method 5331.2
September 30, 2019

5331.2 – CORROSION PROTECTION OF STEEL AGAINST SULFUROUS ACID- SALT SPRAY BY SOLID FILM LUBRICANTS

1. SCOPE

1.1 This method is used to evaluate the corrosion preventive properties of dry, solid-film lubricant under the combined conditions of salt, sulfurous acid, and humidity environment.

2. SUMMARY

2.1 Four steel disks having solid film lubricant deposited on each surface is subjected to the cycling effect of salt spray and drying. After each cycle of salt spray and drying, the disks are examined for evidence of corrosion. 4 out of 4 or 3 out of 4 disks with no failing criteria depicts a pass. 2 passing and 2 failing requires a re-test.

3. SAMPLE SIZE

3.1 180 mL of solid-film lubricant of a sprayable consistency.

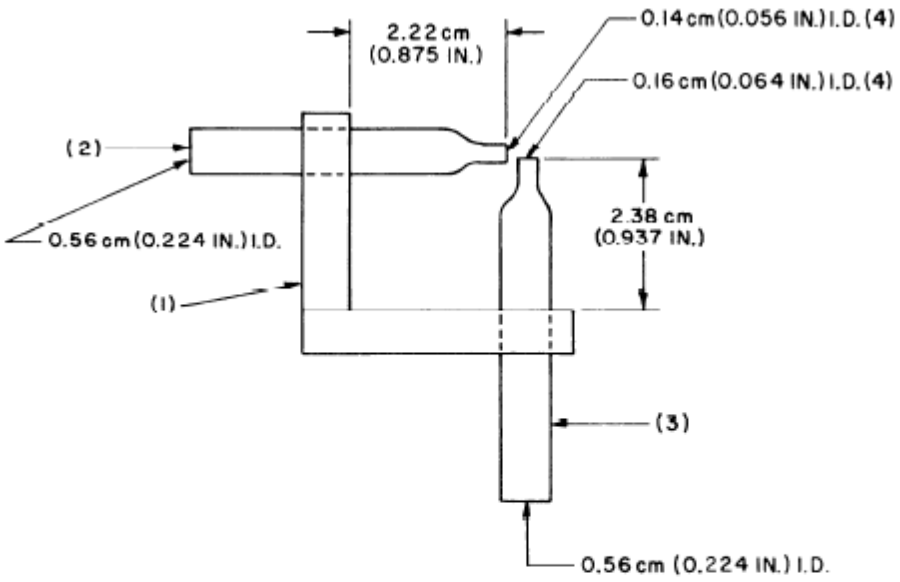
4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Air Supply - filtered, 69 gage kPa to 103 gage kPa (10 to 15 psig).

4.2 Air flow regulator - capable of regulating air flow at 472 mL/s (1 ft^3/min).

4.3 Spray Nozzle (see FIGURE 1) - corrosion resistant with a 13 cm (5 inch) diameter acrylic baffle which prevents direct impingement of the spray but allows a fine mist full contact with the disks.

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NOTES:
1. 0.637 cm (0.25 IN.) A CRYLIC MATERIAL.
2. FILTERED AIR SOURCE.
3. LIQUID PICK-UP TUBE.
4. ORIFICE.

FIGURE 1. Spray nozzle.

4.4 Turntable Assembly which consists of:

a. Turntable - corrosion resistant, 26.7 cm diameter with holes for mounting test holders and specimens.
b. Driving Mechanism - capable of mechanically rotating turntable at 1/3 rpm by an electric motor (slow speed), encased within a protective acrylic box and mounted above jar on an arrangement of flexaframe support rods.

4.5 Specimen Holders - holder cap (see FIGURE 2) and ice jacket (aluminum) that is so placed as to be in direct contact with back of test specimen.

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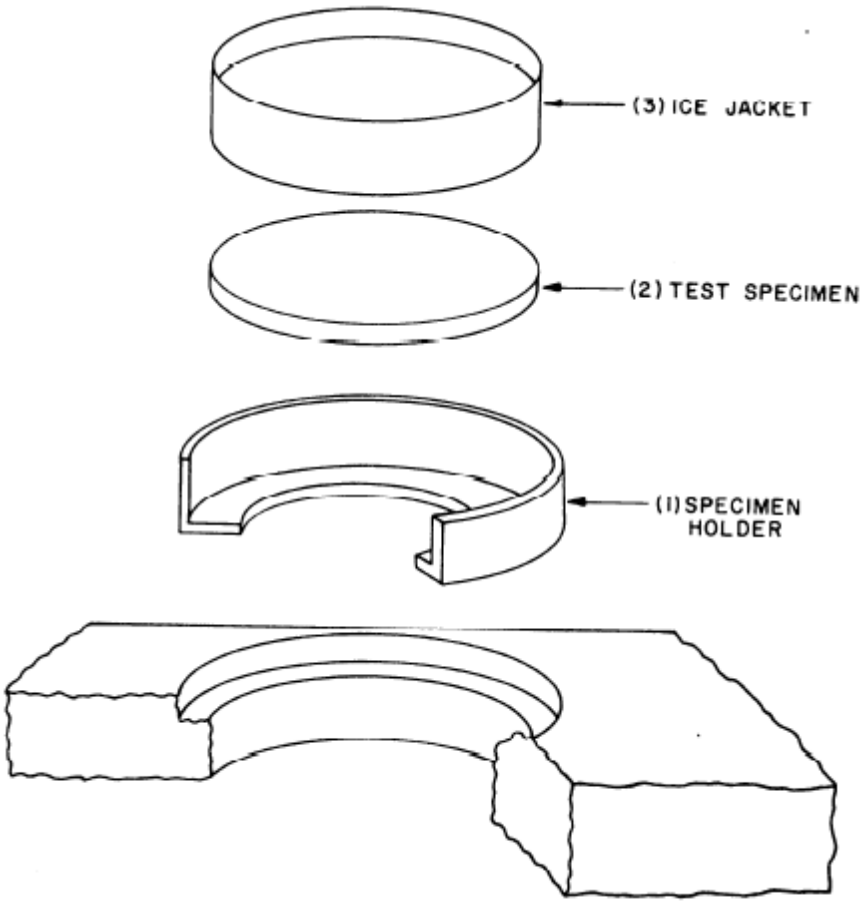


FIGURE 2. Test specimen holder assembly.

4.6 Jar – recommended dimensions of 30.5 cm I.D. and 30.5 cm high.

4.7 Support Rods - flexaframe or equivalent.

4.8 Flexible Tygon Tubing – recommended dimensions of 0.96 cm I.D.

5. MATERIALS

5.1 Steel Specimen - disk with 5.4 cm diameter and thickness of 0.16 cm, made of carbon steel conforming to FS 1010 with a 0.08 cm hole for handling disk with wire.

5.2 Synthetic Sea Water-Sulfurous Acid Test Solution - synthetic sea water (Formula ASTM D1141 - 41.95 g/L, 900 mL with 2 mL sulfurous acid [not less than 6.0% assay as $S0_2$]).

5.3 Aluminum Oxide Cloth - 240 grit.

5.4 VM&P Naphtha.

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5.5 Distilled Water (ASTM D1193 - Type III).

5.6 Sulfurous acid, ACS [3, 0, 1].

6. PROCEDURES

6.1 Prepare test specimens as follows:

a. Degrease test specimens in VM&P Naphtha.
b. Hand polish surface to be tested with 240 grit aluminum oxide cloth to a surface finish of (0.25 to 0.50 µm).
c. Repeat step (a).
d. Allow specimens to dry and store in a dust-free container.
e. Apply solid-film lubricant to surface of steel specimen by spraying to produce a film thickness between 0.0005 cm - 0.0013 cm (0.0002 inch - 0.0005 inch) and cure film in accordance with specification requirements.
f. Handle the specimen at all times with a hook.

6.2 Prepare the apparatus (see FIGURE 3) as follows:

a. Fill bottom 5 cm of jar with the synthetic sea water-sulfurous acid solution and then lower the turntable assembly into jar.
b. Place disk in specimen holder (see FIGURE 2) with ice jacket on top.
c. Mount the complete specimen assembly in one of the holes of the turntable and start the turntable.
d. Turn on the air and regulate rate of flow until a uniform fog is produced.
e. Two minutes after the spray has started, place ice in ice jackets in order to cause a fine condensate to form on the surfaces of disks.

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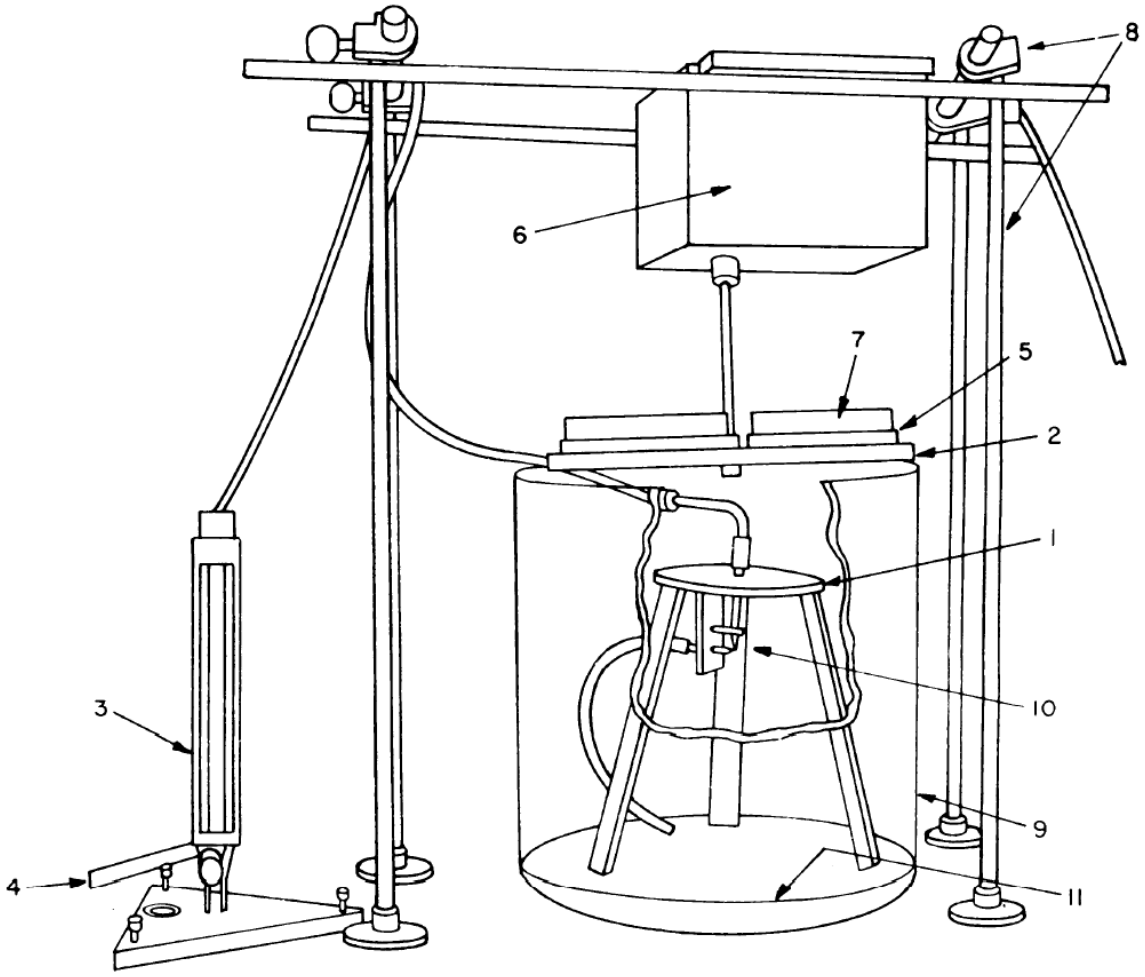


FIGURE 3. Synthetic sea water spray test apparatus.

6.3 Perform test as follows:

a. Continue spraying for 2 h with a rate of flow of approximately 0.4 mL/s or whatever flow maintains the uniform fog mentioned in 6.2 (d).
b. Discontinue the spray at the end of two hours; remove the ice jackets; and raise the turntable above the jar to allow specimens to dry for two hours at room temperature. The two-hour spray and two-hour drying period shall constitute one cycle.

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c. Rinse the test specimens after each full cycle, while still retained in the holder, with distilled water (ASTM D1193 - Type III) and dry with a stream of air.
d. After each cycle examine test specimen for evidence of visible rust colored staining, pitting, or corrosion.
e. Repeat the test cycle until failure occurs. Failure is defined as the first visible sign of corrosion, pitting or staining.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Report passing or railing results according to Paragraph 2.1, or according to the material specification, for the number of cycles required. Report the amount and type of corrosion present.

9. PRECISION

9.1 Precision data have not been developed for this method.

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30 Sep 2019	Editorial changes to fix spacing and other minor corrections	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT
30 Sep 2019	Reworded sections 2.1, 8.1	More user friendly	2	Army - AT
30 Sep 2019	Changed sections 4.6 & 4.8 to read “recommended dimensions”	The dimensions of the materials does not have to be exact for the test to run accurately	2	Army - AT
30 Sep 2019	Changed 1-1-1 trichloroethane to (VM&P Naphtha)	Safer degreaser with no effect on results	2	Army - AT
14 Jun 21	Removed MIL-T-83153	Specification was cancelled without replacement	2	Army - AT

Method Prepared By:
Navy - NADC - 1986

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Method 5414.5
September 30, 2019

5414.5 – RESISTANCE OF GREASE TO FUEL

1. SCOPE

1.1 This method is used for determining the resistance of grease to the solvent action of fuel.

2. SUMMARY

2.1 The test consists of determining the solubility of the grease in a standard test fluid, and observing for physical changes caused by an 8 hour immersion in the test fluid.

3. SAMPLE SIZE

3.1 Approximately 4 grams of grease to be tested (duplicate 2 grams samples).

4. REFERENCE, STANDARDS, AND APPARATUS

4.1 Centrifuge, approximately 650 rpm, swing diameter (tip to tip of whirling tubes) approximately 40 cm, or equivalent.

NOTE: An equivalent centrifuge must be capable of having its speed adjusted according to the following formula:

$$rpm = \sqrt{\frac{40}{d}} * 650$$

where:
d = diameter (tip to tip) of whirling tubes centimeters

4.2 Shaker, mechanical, stroke approximately 3.8 cm (1-1/2 inch), speed approximately 1350 rpm.

4.3 Centrifuge tube, 200 mL, with stopper.

4.4 Evaporation dish, 100 mL.

4.5 Pipet, 50 mL.

4.6 Test tube, approximately 1.5 cm (1/2 inch) O.D. by 7.5 cm (3 inch) long, with stopper.

4.7 Steam bath (to accommodate evaporating dish).

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4.8 Oven, approximately 100°C (212°F).

4.9 Balance, sensitivity 0.2 mg.

4.10 Spoon type spatula.

5. MATERIALS

5.1 Standard Test fluid (ASTM D471, Type III).

5.2 Aluminum strip (ASTM B209/B209M, tempers T-3 or T04)

6. PROCEDURE

6.1 Determine the solubility of the sample in the standard test fluid as follows:

a. Clean, dry and weigh a centrifuge tube to within ± 0.2 mg accuracy. With a spoon spatula spread approximately two grams of grease in a thin film inside the tube and weigh to within ± 0.2 mg; add 100 mL of the test fluid and stopper the tube.
b. Using the mechanical shaker, shake tube until sample appears to be thoroughly dispersed (approximately 1/2 h).
c. Centrifuge the tube until fluid and sample are completely separated, and the fluid portion is clear and transparent.
d. Weigh evaporating dish to within 0.2 mg.
e. Pipet 50 mL of the clear fluid from the centrifuge tube into evaporating dish. Place dish on steam bath until liquid is evaporated. Then dry dish in oven for 1/2 h at approximately 100°C.
f. After drying, remove dish from oven and reweigh to within 0.2 mg. Determine the weight of residue, and calculate the percent of sample soluble in test fluid (see 7.1).

6.2 Determine the deterioration of the sample due to the test fluid as follows:

a. Fill test tube to a depth of approximately 2.5 cm with test fluid. Spread the grease evenly on both sides of the aluminum strip, and place the strip in the test tube so that it is half immersed in the test fluid. Stopper test tube, and allow it to stand for 8 h at room temperature (approximately 25°C).
b. After 8 h, remove strip from test tube (being careful not to disturb sample), and dip once into fresh test fluid. Examine sample while wet for swelling, loss of adhesion, blistering, cracking, or deterioration, and report observations.
c. Allow strip to dry for 24 hours at room temperature (approximately 25°C), then reexamine the sample for the signs of deterioration, and report observations.

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7. CALCULATIONS

7.1 Calculate the percent of sample soluble in test fluid as follows:

$$Percent\ solubility\ = \frac{200\ A}{B}$$

where:
A = weight of sample, mg (see 6.1 f).
B = weight of original sample, mg.

8. REPORTING

8.1 Report percent solubility as determined under 6.1 f.

8.2 Report condition of grease on strip as determined under 6.2. b and c.

9 9. PRECISION

9.1 Precision data have not be developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	5	Army - AT
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Method prepared by:
Navy - NADC - 1986

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Method 5415.2
September 30, 2019

5415.2 – RESISTANCE OF GREASE TO WATER AND A 1:1 WATER-ETHANOL SOLUTION

1. SCOPE

1.1 This method is used for determining the resistance of a grease to the solvent action of water.

2. SUMMARY

2.1 The test consists of storing two portions of grease, one in distilled water and the other in an ethyl alcohol-water solution, for one week, and examining the grease for disintegration.

3. SAMPLE SIZE

3.1 Approximately 4 grams of grease to be tested.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Containers (2), glass, 250 mL, with stoppers.

5. MATERIALS

5.1 Alcohol, Ethyl, Denatured.

5.2 Distilled water.

6. PROCEDURE

6.1 Place 200 mL of distilled water into one container, and 200 mL of an alcohol-distilled water solution (1:1 by volume) into the other.

6.2 Place a single globule of grease (approx. 2g) into each of the two containers, stopper the containers tightly, and allow both to stand for one week.

6.3 At the end of one week, shake each container once or twice, and then visually examine each portion of grease for disintegration.

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

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8.1 Report the degree of disintegration of the grease in each container.

9. PRECISION

9.1 Precision data have not been developed for this method.

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30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	2	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	2	Army - AT

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Method 6052.2
September 30, 2019

6052.2 – HIGH-TEMPERATURE--HIGH-PRESSURE SPRAY IGNITION

1. SCOPE

1.1 This method is used for determining the relative flammability of liquids.

2. SUMMARY

2.1 The method consists of forcing the test fluid sample through a 0.0368 cm (0.0145 inch) orifice at 6.98 MPa gage (1000 psig), attempting to ignite the spray with a torch and noting the characteristics of the resulting flame.

3. SAMPLE SIZE

3.1 Sufficient liquid to be tested to fill cylinder of spray test setup.

4. REFERENCES, STANDARDS AND APPARATUS

4.1 Apparatus.

4.1.1 Spray test setup. A setup similar to FIGURE 1 shall be assembled so as to supply a constant pressure of 6984 ± 69 kPa gage (1000 ± 10 psig) to the sample.

4.1.2 Torch. An oxyacetylene torch equipped with a No. 3 tip or equal.

4.1.3 Nozzle. A nozzle 0.163 cm (0.064 inch) thick by 0.965 cm (0.38 inch) minimum diameter with a centered orifice approximately 0.0368 cm (0.0145 inch) diameter having sharp, square edges (see FIGURE 1).

5. MATERIALS

5.1 Materials. CAUTION: Some materials are toxic and hazardous. The chemical materials listed in this section must be handled carefully.

5.2 Nitrogen. A cylinder of nitrogen fitted with a regulator capable of supplying 6890 kPa gage (1000 ± 10 psig) of pressure to the system.

6. PROCEDURE

6.1 Fill the fluid cylinder of the spray test setup with test fluid at 15.6°C to 37.8°C (60°F to 100°F) and adjust nitrogen pressure to produce a liquid pressure of 6894 ± kPa gage (1000 ± 10 psig).

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6.2 Adjust the torch to deliver a neutral flame.

6.3 Open the quick-opening valve at the orifice and attempt to ignite the spray with the torch at the nozzle.

6.4 If the spray does not ignite gradually move the torch away from the nozzle until ignition takes place (45.7 cm max.)

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

8. REPORTING

8.1 Report whether the spray ignites. If the spray does ignite, also report:

a. The distance from torch to nozzle at ignition.
b. Whether the spray flashes readily or with difficulty.
c. Whether the flame produced is self-extinguishing or sustained.

9. PRECISION

9.1 Precision data have not been developed for this method.

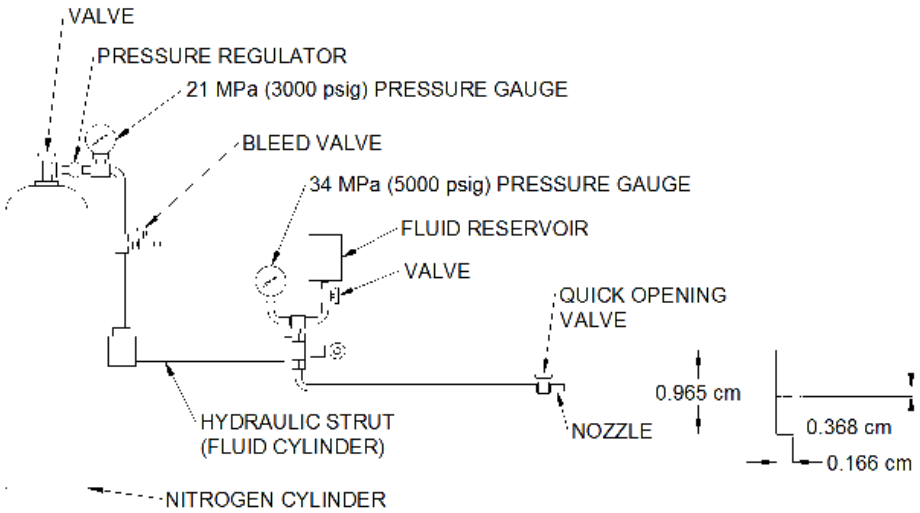


FIGURE 1. Spray test setup.

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30 Sep 2019	Editorial changes to fix spacing and other minor corrections	Editorial corrections	3	Army – AT

Method Prepared By:
Army - AT - 1986

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Method 6053.2
September 30, 2019

6053.2 – MANIFOLD IGNITION TEST

1. SCOPE

1.1 This method is used for determining the relative flammability of a liquid in contact with a hot surface.

2. SUMMARY

2.1 The test method consists of dropping the flammable fluid on a hot surface and observing the presence of a flame.

3. SAMPLE SIZE

3.1 15 mL per test run.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Simulated manifold test set up (see FIGURE 1), fabricated from 7.6 cm (3 inch) O.D. corrosion-resistant steel tubing of 0.114 (0.045 inch) wall thickness. The exterior of the tube shall be sandblasted. A 1/8 inch diameter 18-8 corrosion-resistant rod 25 cm (10 inch) long shall be tacked to the stack on the exterior surface. Opposite the steel rod, tack, weld or suitably attach a thermocouple, and insulate the leads. Care should be taken in this attachment to provide a minimum additional radiating surface.

4.2 Heating element, electrical, “globar” type or equivalent, (Carborundum Company, AT, 31 x 12 x 1, 0.633 Σ unit, or equal), with suitable means for controlling the temperature of the tube to 704°C (1300°F).

5. MATERIALS

5.1 Steel wool.

6. PROCEDURE

6.1 Clean the outside surface of the tube of the test setup (see FIGURE 1), using steel wool.

6.2 Mount the heating element centrally in the tube, and adjust the voltage so that the temperature of the tube is maintained at 704°C (1300°F).

6.3 When the tube has reached the correct temperature, drop 10 mL portions of the test liquid at a rate of 10 mL in 40 to 60 s from various heights onto various points on the tube, and observe

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the ignition characteristics of the liquid. (Clean the tube with steel wool before each 10 mL application of the liquid.)

7. CALCULATIONS

7.1 This section is not applicable to the test procedure.

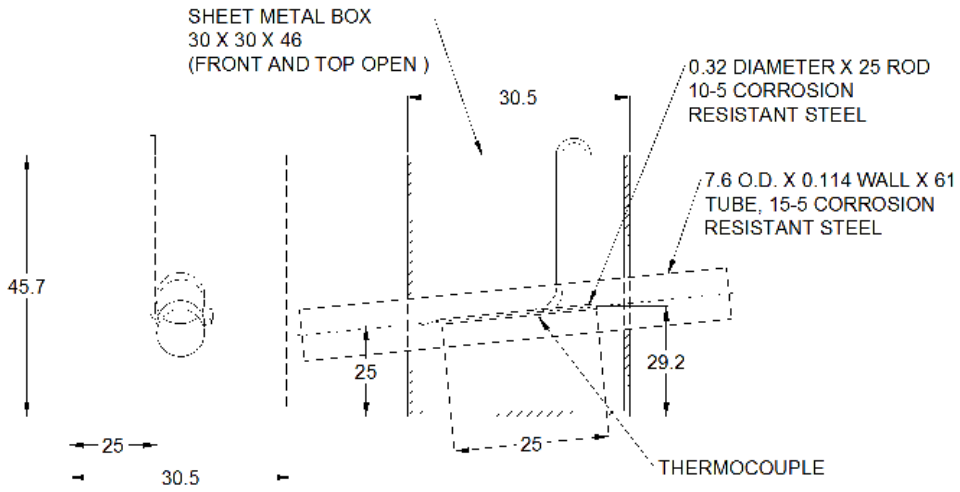
8. REPORTING

8.1 Report results as follows:

a. Flashes or burns on the tube, but does not after dripping from the tube.
b. Does not flash or burn on tube, but does after dripping from the tube.
c. Does not flash or burn on the tube or after dripping from the tube.

9. PRECISION

9.1 Precision data have not been developed for this method.



NOTE: Dimensions in centimeters.

FIGURE 1. Simulated manifold test setup.

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30 Sep 2019	Editorial changes to fix spacing and other minor corrections	Editorial corrections	2	Army - AT
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Method Prepared By:
Army - ME - 1986

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Method 6508.3
September 30, 2019

6508.3 – LOAD CARRYING CAPACITY OF LUBRICATING OILS (U.S. NAVY RYDER GEAR MACHINE)

1. SCOPE

1.1 This method describes a procedure for determining the load carrying ability of lubricating oils with reference to gears.

2. SUMMARY

2.1 The test consists of incremental increasing the pressure between gears to measure the resultant scuffing. The ability of the test oil to impair scuffing is measured by the maximum pressure when operating at 10,000 rpm.

3. SAMPLE SIZE

3.1 Approximately 2 L of test oil.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Test unit. The apparatus used in this method is the WADD gear machine, or Ryder gear-ERDCO® universal tester (see FIGURE 1), and test gears. This apparatus consists of a WADD gear machine adapted to a modified ERDCO® universal tester drive stand, a drive system, a support and load oil system, a test oil system, and the necessary instruments and controls.

4.1.1 U.S. Navy Ryder gear machine. The U.S. Navy Ryder gear machine operates on the so called “four square” principle. As shown in FIGURE 1, two parallel shafts are connected by two slave gears and two test gears to form a "square", so that the power required to operate the machine is only that required to overcome the friction losses in the gear and bearings. The slave gears are helical gears made integral to the shafts. The replaceable test gears are spur gears, shown installed on the shaft in FIGURE 4. Load on the test gears is obtained through the application of a controlled oil pressure in the load chamber which causes an axial movement of one shaft relative to the other. Because of the integral helical slave gears, a torsional load is thus applied on the shafts thereby loading the test gears. The relation between the tooth load on the test gears and the load oil pressure is obtained by arithmetic calculation, calibration, or both. Arithmetic calculation is used herein.

4.1.2 Test gears. The test gears are special spur gears, made of AMS 6265 steel, case hardened and ground, having 28 teeth, 8.88 cm (3.5 inch) pitch diameter, 8 diametral pitch, 22.5° pressure angle, and zero tip relief. Each set of test gears comprises gears of two different widths. The narrow test gear is used for rating purposes. The mating gear is wider in order to maintain a constant contact width between the two gears at different tooth loads. Both sides of the test gear

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are used for testing. The “A” side of the gear refers to the side obtained by installing the tests gears with the serial numbers facing the end cover. The “B” side is obtained by reversing the gear (serial numbers facing away from the end cover). Test gears can be used once only on each side and then discarded.

4.1.3 Support and load-oil system. The support and load-oil system serves two purposes: to provide lubrication to all parts except the test gears and to apply load to the test gears. The system is charged with approximately 30 L (8 gal) of lubricating oil, MIL-PRF-85734. It is recommended that the oil be changed every 200 hours of running time under normal circumstances. A support oil pressure pump and a relief valve furnish oil to the distributing manifold at a specified support oil pressure of 758 ± 69 gage kPa (110 ± 10 psig). The support oil temperature, also measured at the distributing manifold, is maintained at a specified value of 73.9 ± 2.5°C (165 ± 5°F) by electric heaters located in the sump and by a water cooler located after the pump, through the action of thermoswitches. Pressure lines from the distributing manifold lubricate the bearings and helical gears of the machine as well as the adapter housing. The pressure to the latter is reduced to 172 to 207 kPa gage (25 to 30 psig) by a throttle valve. Oil from the distributing manifold enters the Ryder gear machine, and after lubricating the bearings and slave gears, the oil enters the load chambers. For a given pressure in the distributing manifold, the pressure in the load chambers is dependent on the flow restriction offered by the circuit downstream of the load chambers. Upon leaving the load chambers the oil must pass through a control in the load oil circuit before returning to the support oil sump. Two pressure gages (a low range and a high range) located near the outlet points read the support oil outlet pressures. The load oil circuit contains three valves. Of these the load bypass valve is an open or shut valve which is closed during the test run. The load oil pressure is controlled by regulating the load oil adjusting valve. The bleeder valve is normally slightly open, its function being to decrease the sensitivity of the load adjusting valve, thus facilitating the control of load oil pressure. The load bypass valve is opened when it is desired to quickly release the load on pressure, such as at the end of a test run.

4.1.4 Test oil system. The capacity is 1000 mL. Test oil is supplied to the test gears by a pressure pump through a jet located on the unmeshing side of the gears. Oil temperature to test gears is maintained at 74 ± 2°C (165 ± 5°F) with a flow rate of 270 ± 4 ml/min by a line heater actuated by a thermo switch and controlled by a thermocouple close to the oil jet. The oil is returned through a scavenge line to the sump.

4.1.5 Drive system. The U.S. Navy Ryder Gear machine is driven by a 30 hp induction motor through a jackshaft coupling. The test gear speed is controlled by adjusting the variable frequency drive controller. By this means, accurate speed control from standstill to well over the specified test speed of l0,000 ± 100 rpm can be attained.

4.2 Instrumentation and controls. All of the instruments and controls that require constant attention during test are located on the control console. Items that do not require frequent attention are located on the test stand. The functions of most of the instruments and controls are apparent. Therefore, only selected items will be explained further.

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4.2.1 Speed measurement and control. An electronic counter, located on the control panel, measures the speed. This instrument is actuated by pulses supplied by a variable reluctance pickup (the variable reluctance being provided by a 60- tooth gear operated from the driven shaft). The rpm adjustment control, located on the console, regulates the field excitation of the dynamic coupling and thus the gear speed.

4.2.2 Support-oil controls. The support-oil pump buttons on the console are connected to the support-oil pressure pump as well as the scavenge pump. Pressing the start or stop button controls all two pumps at the same time. The support-oil heater buttons control both the electric heaters and the water cooler through the action of thermoswitches.

4.2.3 Control of load oil pressure. The control of load oil pressure has been explained in 4.1.3. Of the three valves in the load oil circuit only the bleeder valve is not remote controlled. However, this valve need only be set to give the desired sensitivity for load control during installation of the tester. As a rule, adjustments are seldom necessary thereafter. The load adjusting valve is air actuated and is controlled by the load adjust control located on the console. The load bypass valve is solenoid actuated and is controlled by the load bypass switch on the console. During the test run the load bypass switch is in the “close” position and the load bypass valve is closed. At the end of a run, the solenoid circuit will be opened. This opens the load-bypass valve which relieves the load from the test gears.

4.2.4 Emergency stop. The emergency stop button on the console stops the drive motor and turns off all the controls except those for instruments.

4.3 Scuff rating equipment. A CCD camera and a light source for visual inspection of the narrow test gear are used to inspect the amount of scuffing present on the narrow gear. The camera is streamed to a computer in the control room where a monitor is used to rate the gears. A 10x10 grid that conforms to the active area of the test gear is used to inspect the scuffing damage to the nearest 1% for each gear tooth. To inspect the narrow test gear the machine is stopped, but the test oil is kept circulating. The inspection can then be accomplished through special inspection holes provided in the gear case without taking off the end cover. These holes are closed with corks during the test run.

4.6 Gear tooth indexing. The electric VFD is used to rotate the test gears so that the teeth of the narrow gear can be brought into position one by one for inspection or photography.

4.7 Additional instruments.

a. Stop watch for measurement of test oil flow rate.
b. Micrometer 0.0 cm to 2.5 cm (0 to 1 inch) for measurement of the gross tooth width of the narrow test gear.
c. Set of conventional mechanical tools as necessary for assembly, disassembly, maintenance and repair.

5. MATERIALS

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5.1 Lubricating oil, DOD-PRF-85734.

5.2 Flushing solvents.

5.2.1 Dry cleaning (A-A-59601D).

5.2.2 Trimethylpentane (540-84-1)

5.2.3 Oakite LRS (REL-4576)

5.3 Test gear retaining nut lock washer, Fafnir WO-6 (ERDCO® Engineering Corp- oration, Addison, ILL, or any commercial supplier of Fafnir parts).

6. PROCEDURE

6.1 Operating condition.

6.1.1 Throughout the entire test, the variables listed in TABLE I should be controlled within the limits specified. With the tester operating within the specified limits, the test gears are loaded first to 34 kPa gage (10 psig) load oil pressure, and then at successive increments of 34 kPa (5 psi). The duration of each loading period is 10 minutes ± 5 seconds. At the end of each loading period, the tester is stopped and each tooth of the narrow test gear is examined to determine the percent of tooth area scuffed.

TABLE I. Operating conditions.

Condition	Value	
Test gear speed, rpm	10,000 ± 100	
Test oil inlet temperature	74 ± 2°C	165 ± 5°F
Test oil flow rate	270 ± 4 mL/min	
Support oil inlet temperature	73.9 ± 2.5°C	165 ± 5°F
Support oil pressure	758 ± 69 kPa gage	110 ± 10 psig

6.2 Preparation for test.

a. Record pertinent information such as test number, test oil code, operator's name, etc., on suitable data sheets (see FIGURE 5).
b. Turn on main electric switch to test cell.
c. Turn on main water valve to test cell.
d. Push instruments button "ON".
e. Push support oil heaters button "ON".
f. Make sure that the test oil system has been thoroughly flushed with Stoddard solvent and iso-octane after the last test. See 6.3.2 steps (17) through (25) for the flushing procedure. Use the flushing solvent once and discard.
g. Remove gear case end cover.

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h. Clean a set of new test gears with Oakite LRS and discard solvent. Measure gross tooth width of the narrow test gear, and record the measurement.
i. Install test gears with the serial number facing outward (if side A), matching timing marks on shafts and gears. (see FIGURE 3). Check for proper seating of the retaining nut, and lock the lock washer.
j. Reinstall gear case end cover.
k. Start test oil pressure pump.
l. Flush test oil system with test oil until the test oil temperature is reached at the supply jet.

NOTE 1: The test oil for flushing is to be discarded after use. If the supply test oil is limited, a reduced quantity may be used for flushing.

m. Turn valves L, M, and N to running position.
n. Fill test oil sump with a new charge of test oil (946 mL).
o. Push test oil heaters button "ON".
p. Close valve to support oil chamber vacuum manometer.
q. Allow at least 20 minutes to elapse since the support oil heaters were turned on (step 5). Then start support oil pumps.
r. Turn timer switch to “STOP” position.
s. Set electric timer to 10 minutes and 0 s.
t. Turn rpm adjust control to “ZERO” position.
u. Turn load bypass control to "OPEN ± position.
v. When the test oil and support oil temperature are up to the specified value of 73.9C ± 2.5°C (165°F ± 5°F), check support oil check the test oil flow rate.
w. Check test oil flow rate as follows:

1) Turn 3-way valve L at the discharge valve flow from the Ryder gear machine. Use a graduated cylinder to measure the amount of oil that flow out of the valve in 1 minute. The flow rate should be 270 ml/min.
2) Adjust the variable speed transmission to vary the speed of the test oil pressure pump, until the specified flow rate at 270 ± 4 mL/min is obtained.

6.3 Test procedure.

6.3.1 Test on “A” side of test gears.

NOTE 2: The “A” side of the gear refers to the side obtained by installing the test gears with the serial numbers facing the end cover. The “B” side is obtained by reversing the gear (serial numbers facing away from the end cover).

a. Check all temperatures and pressures for the specified limits. (See paragraph 3). Recheck test oil flow rate if necessary.
b. Make sure that the load adjust control is wide open (counterclockwise).
c. Start drive motor.

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d. Increase gear speed by turning the rpm adjust control clockwise until the specified 10,000 ± 100 rpm is obtained. Measure speed with the electronic rpm counter.
e. Apply load by using the PLC system.

NOTE 3: In adjusting load oil pressure, take care not to overshoot the desired pressure.

f. Check gear speed on control console.
g. Start the 10-minute timer when the full load is reached.
h. Record all pertinent data about midway into the 10-minute test run. At the end of the 10-minute run, use the PLC to unload the gears. Set the speed of the electric drive motor to 0 RPM.

i. Make visual scuff inspection of the narrow test gear, as follows:

1) Remove corks from inspection holes on the gear case.
2) Blow off excess oil from the test gear with compressed air through the inspection hole.
3) Install microscope and inspection light.
4) Set indexing to “No. 1” position.
5) Estimate the percent scuff on No. l tooth to the nearest 1%, and record the value (see FIGURE 6).
6) Operate VFD to index test gear to No. 2 tooth, then No. 3 tooth, etc., repeating step (5) until all 28 teeth have been examined
7) Remove microscope and light. Replace corks in the holes.

j. Repeat steps (3) through (11) at the next test load of l70 kPa (10 psig) load oil pressure.
k. Continue the test at 34 kPa (5 psi) load oil pressure increments, until the average scuffed area for all 28 teeth is closest to 22.5%. At this juncture, visually estimate the percent effective tooth width to the nearest 1 percent and record the value.
l. Push test oil heaters button "OFF".
m. Close all drains.
n. Stop test oil pressure pump.
o. Push support oil heaters button "OFF".
p. Stop support oil pumps.

NOTE 4: Leave other controls on if test on “B” side will proceed after a short midday break. Otherwise, push instruments button "OFF", and turn off main electric switch and main water valve to test cell.

6.3.3 Test on “B” side of test gears.

NOTE 5: The “B” side refers to the side obtained by reversing the gear (serial numbers away from end cover). The same oil should be tested on both side A and side B of the test gear set.

a. For side B, repeat steps for preparation (6.2), and continue through the complete test sequence (6.3)).

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6.3.4 Emergency stop.

a. Push emergency stop button.
b. Push instruments button "OFF".
c. Turn off main electric switch to test cell.
d. Turn off main water valve to test cell.
e. Investigate and take appropriate action.

6.4 Inspection and rating.

6.4.1 Scuffed area. The scuffed area of a gear tooth is that area from which the axial grinding marks have been removed by scratching, scoring, abrasion, and wear. As shown in FIGURE 4, areas A and B are scuffed; area C is not scuffed.

NOTE 6: Only the actual working portion (areas A plus B plus C) of the tooth surface is considered in the definition of areas.

6.4.2 Percent of tooth area scuffed. The percent of tooth area scuffed is that portion of the working surface of a gear tooth that has been scuffed. Referring to FIGURE 4, the percent of tooth area scuffed is estimated visually as accurately as possible, but at least to the nearest 1%, for each individual tooth.

$$Percent\ tooth\ area\ scuffed\ = \frac{scuffed\ area * 100}{working\ area} = \frac{A + B}{A + B + C} * 100$$

6.4.3 Average percent of tooth area scuffed. The average percent of tooth area scuffed is the algebraic average of the percent of scuffed area of all 28 teeth of the narrow test gear. Calculate this value to the nearest 1%.

6.4.4 Effective tooth width. The effective tooth width is the actual width of the narrow gear tooth that is in contact with the tooth of the wide gear. To determine the effective tooth width, measure the gross tooth width by means of a micrometer, and estimate the ratio of effective tooth width to gross tooth width by visual examination or with the aid of a suitable scale, referring to FIGURE 4, and 7.1.

6.4.5 Load carrying ability. The load-carrying ability, or the scuff limited load, of a lubricant is defined as the gear tooth load at which the average percent of tooth area scuffed is 22.5% for the narrow test gear as determined by the method outlined herein.

6.4.6 Data sheets. The suggested data sheets are shown in FIGURE 5 and FIGURE 6. The gear teeth inspection data sheet (see FIGURE 6) has provision for computing the average percent of tooth area scuffed.

6.4.7 Determination of load carrying ability.

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a. Plot the average percent of tooth area scuffed for the narrow test gear versus the load oil pressure. The use of a semi logarithmic paper is recommended for this purpose. FIGURE 7 is a suggested plot and calculations sheet, showing typical results from tests on both sides of a test gear.
b. Determine from the plotted curve the load oil pressure at which the average percent of tooth area scuffed is 22.5%. In FIGURE 7, this condition is represented by point X for "A" side and point Y for "B" side of the test gear.
c. Calculate load carrying ability (see 7.2).

6.4.8 Relative rating procedure. This procedure consists of reporting the load carrying ability of an oil under test as a percentage of the load carrying ability of a standard reference oil, both ratings being determined by the procedure outlined in 6.4.7. Details of this comparative procedure are given in the following paragraphs.

NOTE 7: The standard reference oil is defined in the latest version of MIL-PRF-23699.

6.4.9 Determination of the average rating on standard Ryder gear test reference oil.

a. For a new rig, at least six determinations (three gear sets) shall be made, according to the foregoing procedure, on the standard reference oil. If the high and low values obtained do not differ by more than 1400 N/cm (800 lb/in) the average is taken as the reference oil rating. If the high and low values differ by more than 1400 N/cm (800 lb/in) four additional determinations (two gear sets) shall be made and the average of the 10 determinations taken as the reference oil rating.
b. After a major overhaul to a rig such as a bearing or shaft replacement, four determinations (two gear sets) shall be made on the standard reference oil. If the average of these ratings does not differ from the average obtained prior to the overhaul by more than 525 N/cm (300 lb/in) and the high and low values obtained do not differ by more than 1400 N/cm (800 lb/in) this average will be taken as the reference oil rating. If the ratings obtained do not meet the above criteria, four additional determinations (two gear sets) shall be made and the average of all eight determinations taken as the reference oil rating.
c. At least two determinations (one gear set) shall be made on the standard reference oil for each 20 determinations made on oils under test. The reference oil rating is a progressive rating in that it will consist of the average value obtained on the last eight determinations, i.e., when two new determinations are made, the two oldest determinations of the eight determinations used in determining the previous reference oil rating are dropped, the two new ratings replacing them and a new reference oil average rating obtained. The sole exception to the above is where it is necessary to run 10 determinations to establish an average on a new rig. (See item l, 6.9).

NOTE 8: Maximum and minimum acceptable average ratings for the reference oil are 4550N/cm (2600 lb/in) and 3680 N/cm (2100 lb/in), respectively. Rigs which do not give an average reference oil rating within these limits shall not be considered as being satisfactory. Faulty maintenance practices, improper test procedures, and/or the need for major overhaul can be

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indicated if the average on the reference oil does not fall within these limits.

d. Determinations, as necessary, shall be made on the oil under test and an average obtained. Should the high and low values of these determinations differ by more than 1400 N/cm (800 lb/in) additional determinations should be made to establish an acceptable average.

7. CALCULATIONS

7.1 Calculate the effective tooth width (cm, inch)

$$W_2 = W_1 * R$$

where:
W_2 = effective tooth width in cm or in.
W_1 = measured gross tooth width in cm or in. R = W_2/W_1 = estimated ratio.

7.2 Calculate load carrying ability by the following formula:

$$P = \frac{KL}{W_2}$$

where:
P = load carrying ability (or scuff limited load) of the lubricant, N/cm (lb/in).
K = Ryder gear machine constant (4.1.1).
L = Load oil pressure, gage Pa (psig).
W_2 = effective tooth width, cm (in).

7.3 Calculate the percent relative rating using the following formula:

$$Relative\ rating, percent\ = \frac{test\ oil\ average\ rating}{reference\ oil\ average\ rating} * 100$$

where:
P = load carrying ability (or scuff limited load) of the lubricant, N/cm (lb/in).
K = WADD gear machine constant (11.5 by definition).
L = Load oil pressure, gage Pa (psig).
W_2 = effective tooth width, cm (in).

8. REPORTING

8.1 Report the percent relative rating (see 7.3).

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9. PRECISION

9.1 Repeatability. Results from one apparatus and operator should be considered suspect if the relative ratings differ from their mean by more than 5%.

9.2 Reproducibility. Results from different apparatus and/or operators should be considered suspect if the relative ratings from each apparatus and/or operator differ from their mean by more than 5%.

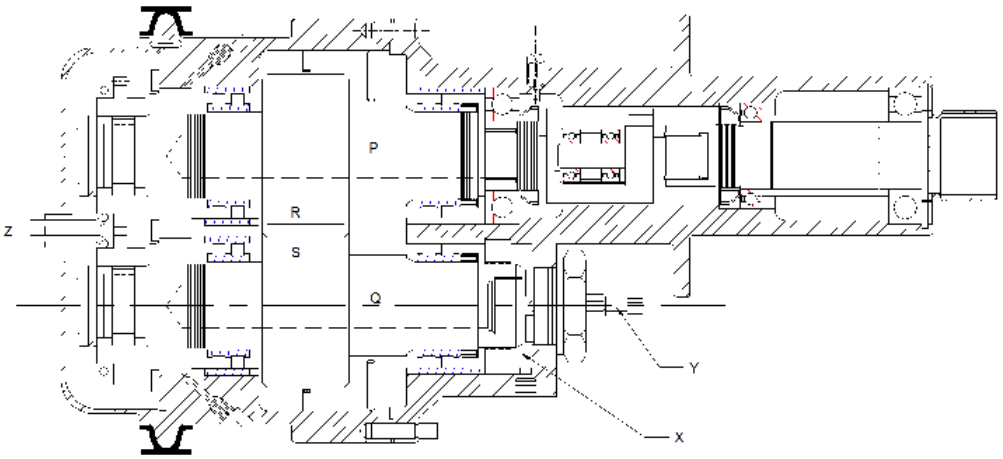


FIGURE 1. U.S. Navy Ryder gear test machine (cross section).

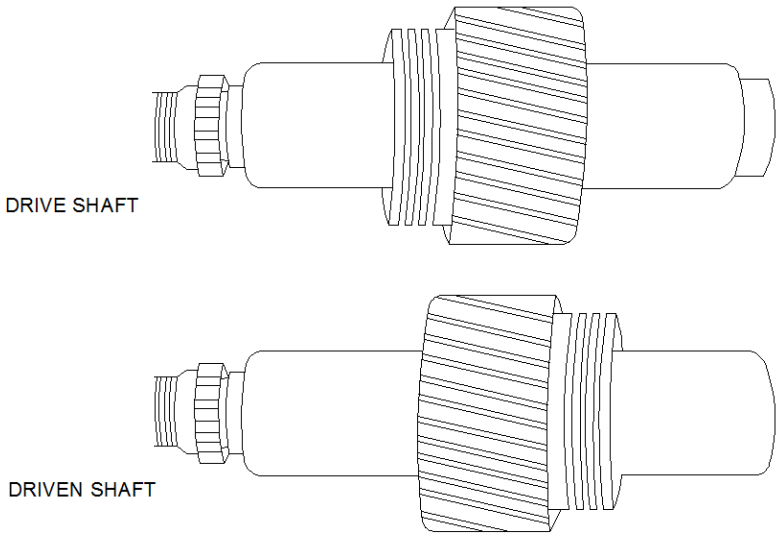


FIGURE 2. Slave gears.

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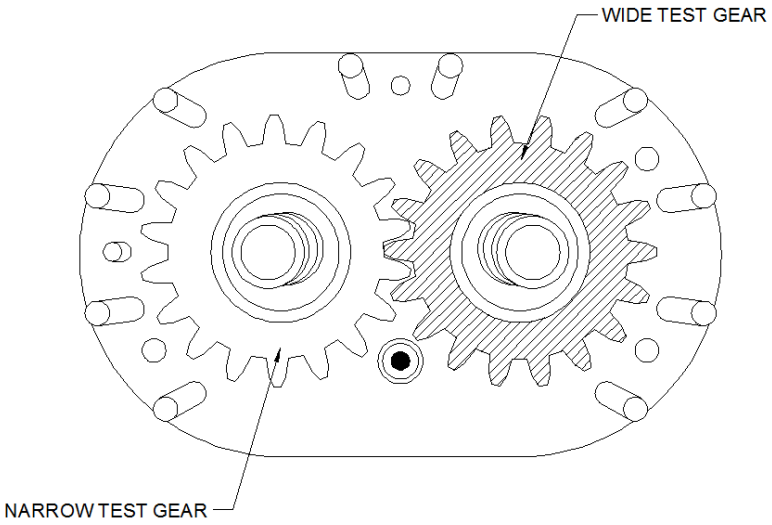


FIGURE 3. Test gears installed.

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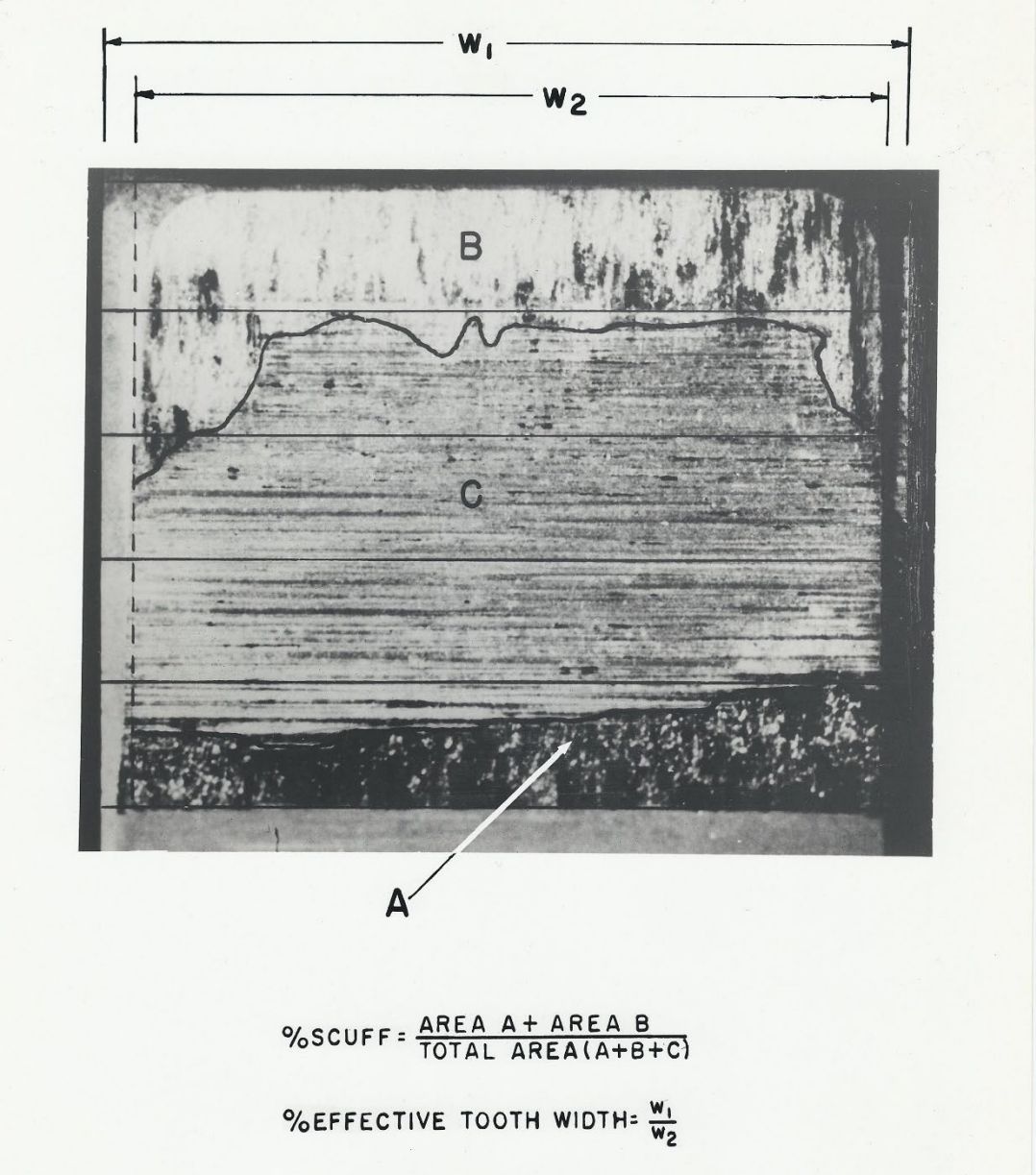


FIGURE 4. Scuffed tooth (enlarged).

Area "A". . . . . Scuffed
Area "B". . . . . Scuffed
Area "C". . . . . Not Scuffed

W_1 = Gross tooth width
W2 = Effective tooth width

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	Test No. _____ Sheet No. _____

LOAD-CARRYING ABILITY TEST
OPERATING CONDITIONS DATA SHEET

OIL CODE _____	TEST SPECIFICATION _____
DATE _____	OPERATOR _____
TESTER NO. _____	MACHINE NO. _____
TEST GEARS: NARROW GEAR T-116286 _____	SERIAL NO. _____
WIDE GEAR T-116287 _____	SERIAL NO. _____ GEAR SIDE: _____ "A" SIDE _____ "B" SIDE

Record the following after 5 minutes operation at each load

Load Oil Pressure Gage kPa (psig)	34 (5)	69 (10)	103 (15)	138 (20)	172 (25)	207 (30)	241 (35)	275 (40)	310 (45)	345 (50)	379 (55)	414 (60)	448 (65)	483 (70)	517 (75)
Time															
R.P.M.															
Support Oil Press. Gage kPa (psig)															
Test Oil Pressure Gage kPa (psig)															
Test Oil "In" Temp. °C (°F)															
Test Oil "Out" Temp. °C (°F)															
Support Oil "In" Temp. °C (°F)															
Support Oil "Out" Temp. °C (°F)															

TEST OIL FLOW TIME FOR 270mL _____ MINUTES _____ SECONDS_____

TEST OIL SUMP LEVEL, mL: START OF TEST _____
END OF TEST _____
CONSUMPTION _____

CHAMBER VACUUM, cm: (inches) OF WATER: START OF TEST _____
END OF TEST _____

REMARKS: _____

_____

FIGURE 5. Operating conditions data sheet.

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LOAD-CARRYING ABILITY TEST
GEAR TEETH INSPECTION DATA SHEET

Test No. _____
Sheet No. _____

OIL CODE _____ TEST SPECIFICATION _____

DATE _____ OPERATOR _____

GROSS TOOTH WIDTH _____ cm (INCHES) GEAR SIDE ____ "A" SIDE

PERCENT EFFECTIVE TOOTH WIDTH _____ GEAR SIDE _____ "B" SIDE

Tooth No.	Percent of Tooth Area Scuffed at Load Oil Pressure Gage kPa (psig) of															Photo-graphs
	34 (5)	69 (10)	103 (15)	138 (20)	172 (25)	207 (30)	241 (35)	276 (40)	310 (45)	345 (50)	379 (55)	414 (60)	448 (65)	483 (70)	517 (75)	
1																
2																
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25																
26																
27																
28																
Sum of Percent																
Average Percent																

FIGURE 6. Gear teeth inspection data sheet.

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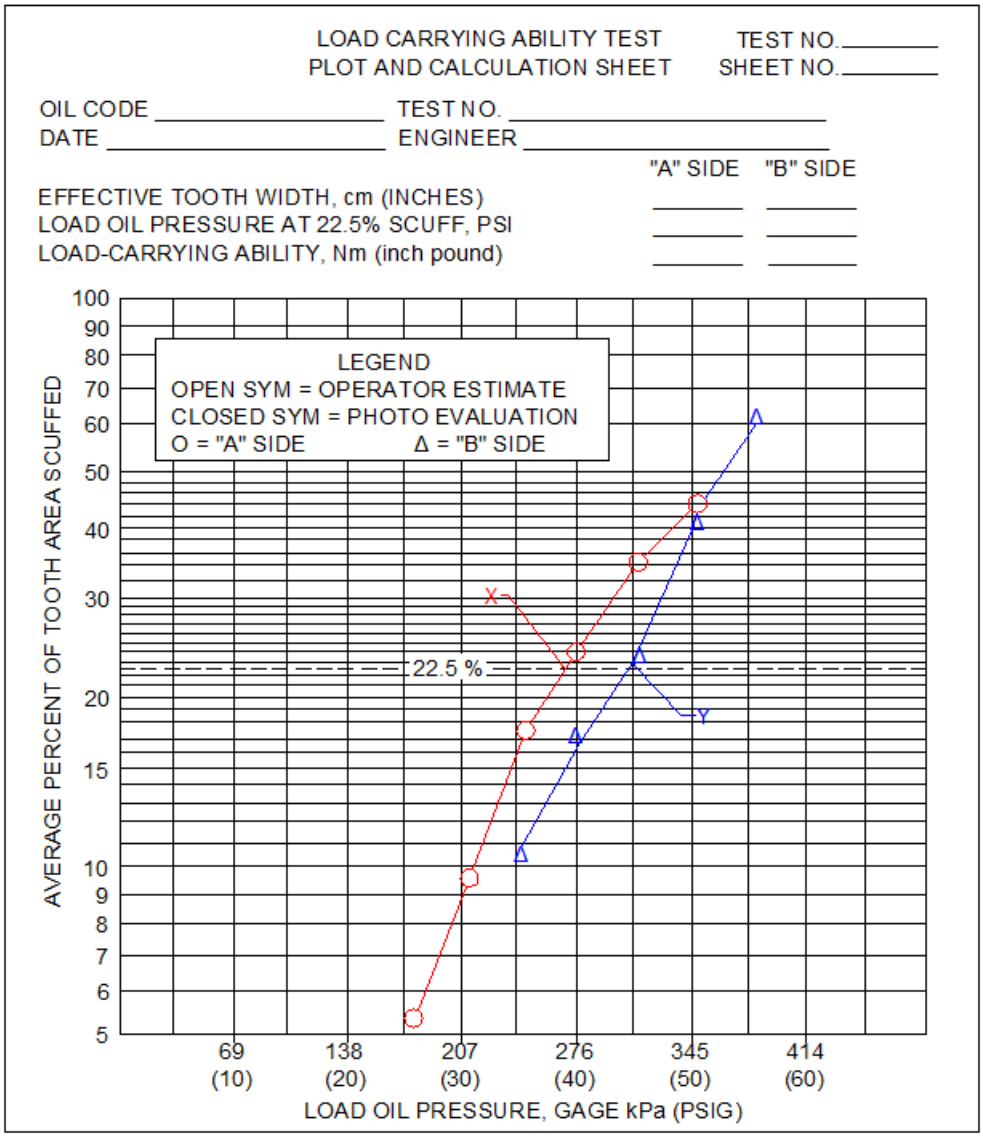


FIGURE 7. Plot and calculation sheet.

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DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Updated the name in parenthesis of the document from “Ryder Gear Machine” to “U.S. Navy Ryder Gear Machine”	Other organizations used to operate the test and now only the Navy operates the test	3	NAWCAD
30 Sep 2019	Updated affiliation from NAPC to NAWCAD	The organization responsible for the test methods has moved since the last revision and is now NAWCAD	3	NAWCAD
30 Sep 2019	Updated All Figures	Original method was written around a test head that is no longer used for the test	3	NAWCAD
30 Sep 2019	Updated Solvents Used	Current solvents are different from original method	3	NAWCAD
30 Sep 2019	Updated test methods section with various items	The test rig has changed over time	3	NAWCAD
30 Sep 2019	Editorial Changes to fix spacing and clarify units	Editorial corrections	3	Army - AT
30 Sep 2019	Removed the Caution Statement referencing FTM 1000 from the Materials section	FTM 1000 was removed from this standard	3	Army - AT
17 May 2021	Removed a portion of section 4.2.2	No longer accurate	3	NAWCAD
17 May 2021	Updated Figure 29	Make the figure legible	3	NAWCAD
17 May 2021	Updated Note 8 for the reference oil.	Reference the source for the proper oil.	3	NAWCAD
17 May 2021	Updated notes in section 6 to be correct values	Values were not in sequential order	3	NAWCAD

Method Prepared By:
Navy - NAWCAD - 2019

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Method 7501.1
September 30, 2019

7501.1 – LOW TEMPERATURE TORQUE TEST METHOD FOR LUBRICATING GREASES

1. SCOPE

1.1 This test method determines the torque values of lubricating greases using tapered roller bearings when subjected to low temperatures. The method was developed using greases giving torques of less than 80 Nm at -54°C. This method was developed to assess the low temperature performance of greases at -54°C and clearly distinguish between differing grease formulations. The torque data obtained from this test method agree with the low temperature performance of military greases in the field. A correlation was found between the breakaway torque and the apparent viscosity range of 5,000 - 30,000 poises at a shear rate of 25 reciprocal seconds.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

MILITARY SPECIFICATIONS

MIL-PRF-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2. SUMMARY

2.1 A worked sample of the test grease is packed into four test specimen assemblies which simulate the automotive front wheel bearing system. These assemblies are mildly heated and then rotated one revolution. The test assemblies are soaked for 16 hours at -54°C. At the end of this time, the two test assemblies are rotated consecutively at 1 rpm and both breakaway (starting torque) and torque after 5 minutes of rotation (running torque) are then determined. The two remaining test assemblies are tested after the additional four hours cooling to restore the test temperature.

4. DEFINITIONS

4.1 Breakaway torque. The maximum torque measured at start of rotation.

4.2 Running torque. The torque value after rotation for a specified period of time (5 minutes).

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5. APPARATUS

5.1 Low temperature torque test unit. This test unit consists of an ASTM D4693 test apparatus which is designed with a cold chamber and two spindle drive systems, torque measuring system, test specimen assembly, and data acquisition and control system.

5.1.1 Low temperature test chamber. The low temperature test chamber consists of a cascade mechanical refrigeration system, a blower, and a solid-state electronic temperature controller. This chamber is designed to maintain the test temperature down to -58°C. The chamber temperature is monitored with a T type thermocouple.

5.1.2 Spindle drive. The spindle drive system consists of a 1/3 hp electric motor, a timing belt and pulley, a gear reducer and a solid stainless steel extension shaft (diameter: 2.2 cm, length: 25.6 cm). The gear reducer provides 84 Nm of maximum output torque and approximately 1 rpm of fixed speed. Two spindle drive systems are mounted externally and connected to the inside of the cold chamber using a solid coupling and extension shaft, respectively. These shafts are supported by two sealed roller bearings which are lubricated with MIL-PRF-23827 grease. Only the outside supporting bearings are rotated with the extension drive shafts during the test time.

5.1.3 Torque measuring system. The torque measuring system consists of a strain-gage load cell (capacity: 136 kg), a cylindrical rod (diameter: 1.9 cm length: 22.5 cm), a load button and a DC power supply (capacity 20 V). Two torque measuring systems are mounted on a plate which is situated below and external to the cold chamber. To measure the grease torque, rods are connected to the inside of the cold chamber with the load cell. A spherical load button is attached to the top of the rod to allow for the adjustment of the level of the torque arm. A DC power supply is connected to both load cells and applies 10 volts for the input bridge excitation.

5.1.4 Test specimen assembly. The test specimen assembly is as specified in the ASTM D4693 test method.

5.1.5 Data acquisition and control system. The data acquisition and control system consist of a HP data acquisition and control unit, a HP computer as system controller and the user operation program (LOWTEMP). The HP data acquisition and control unit also contains three plug inch assemblies: relay multiplexer assembly for measuring the torque value, relay multiplexer assembly with thermocouple compensation, and high voltage actuator to control the drive system.

5.2 Laboratory oven. Laboratory oven, gravity convection, 70°C ± 3°C.

5.3 Grease packer. Grease packer, as described in the ASTM D3527 test method.

5.4 Spindle holder. Spindle holder or test preparation stand.

5.5 Ultrasonic cleaner. Ultrasonic cleaner.

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6. CALIBRATION

6.1 Torque calibration. Attach a support disk (diameter: 4 cm), for holding the dead weight, on the top of the rod. Measure output (millivolt) of load cell as a known force (dead weight: 0.5 kg, 1-10 kg) is applied to load cell at the room temperature. Then, the calibration equation shall be determined using the statistic method. The Method of Regression analysis is very suitable to develop the torque calibration equation. This calibration needs be done only at the time of initial setup and when occasional checks indicate that it is required.

6.2 Temperature calibration. Temperature calibration, as described in ASTM D4693, or equivalent.

6.3 Spring calibration. Spring calibration, as described in ASTM D4693, or equivalent.

7. TEST BEARINGS.

7.1 Test bearings. Use LM67010-LM67048 and LM11910-LM11949 inboard and outboard bearings, respectively. Timken bearings are suitable. Prior to use in this test, new bearings shall be conditioned by installing any suitable wheel bearing grease in the bearings and running them at room temperature for 48 hours at 1,000 rpm under a normal thrust load of 111 N. ASTM D3527 test apparatus is suitable for conditioning the bearings.

Note 1. The bearings must be keyed to prevent race rotation on spindle.

8. PROCEDURE

8.1 The following procedures are used to clean test bearing:

1. Clean the test bearings with heptane or equivalent reagent using an ultrasonic cleaner and dry with compressed air.
2. Work a grease sample for 60 double strokes using the full scale grease or one-half scale grease worker at room temperature.
3. Weigh an inboard and outboard bearing cone to the nearest 0.1g.
4. Pack a worked grease sample into the test bearings (inboard: 3.0 g ± 0.1 g. outboard: 2.0 g ± 0.1 g) using the following bearing packing procedure: Fill the bearings with the worked test grease using a grease packer utilized in ASTM D3527 test method. Use care to prevent relative rotation of rollers and bearing components while removing the cones from cups and in all subsequent wiping and handling steps. Strike off excess grease flush using a small spatula. Wipe all grease from cone bore, cone back face, exterior cage surfaces and exposed roller surfaces with a clean lint free cloth or towel and reweigh. Adjust the grease weight in the inboard and outboard cones to 3.0 g and 2.0 g ± 0.1g, by wiping or adding grease to the groove between the cage and cone back face.
5. Clean the bearing cups installed in the hub using heptane and apply a thin film of the test grease to the races.

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6. Set-up a test specimen assembly according to the following procedure: Without any rotation of the test bearing, the inboard bearing is installed on the spindle. Then, this spindle is mounted on the test preparation stand which was designed for holding it in a vertical position. The hub is gently set on the inboard bearing with the torque arm aligned with the spindle set screw (i.e., pointing in the same direction). Then, the outboard bearing is placed on the hub. The thrust force (400 N) is applied to the test bearings with a calibrated spring-loading device.
7. Prepare four test specimen assemblies using the above test assembly preparation procedure.
8. Without disturbing the test assemblies, place them in an oven preheated to 70°C ± 3°C for 1 hour.
9. Rotate the test specimen assemblies for 1 revolution using the test apparatus at room temperature.
10. Install two test specimen assemblies on each drive shaft for the first two tests. Without any handling of the test assemblies, the torque arms should be placed vertically with the drive motors.
11. Place the other two test assemblies inside the cold chamber in preparation (cold soak) for the next tests.
12. Insert two thermocouples (T type) in the test spindles to monitor the test temperature.
13. Start 16-hour cold soak at -54°C ± 0.5°C.
14. Following a 16-hour cold soak, the two torque tests shall be consecutively performed using the LOWTEMP computer application program. The following additional test conditions are applied to the torque test runs:
    a. Test temperature: -54°C ± 0.5°C Test time: 5 minutes
    b. Spindle speed: 1 rpm
    c. Data scanning interval: 20 centiseconds
    d. The torque reading starts when torque value is >0.1 Nm
15. Store the test results on a floppy disk and plot them on the formatted papers.
16. Cut off the cooling system and remove the tested specimen assemblies from the drive shafts and place them in the preheated oven until completely dried in order to preclude rust formation.
17. For the second series of tests, the two remaining test specimen assemblies shall be set up on the drive shafts identically to the previous tests.
18. Resume the cooling for the next four hours to restore the test temperature.
19. After four hours, two additional torque data shall be generated using the above torque measurement procedure.

9. REPORT

9.1 Breakaway torque (Nm).

9.2 Running torque (N·m) at 5 minutes.

Note 2. Both determinations shall be calculated by the average of two test runs.

10. PRECISION

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10.1 Repeatability Breakaway torque: 10%

10.2 Running torque at 5 minutes: 7%

Note 3. Duplicate results by the same operator should be considered suspect if they differ from the mean by more than the above amounts at the 95 percent confidence level. Reproducibility has not been established.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial changes to fix spacing and other minor corrections	Editorial corrections	1	Army - AT

Method prepared by
Army – AT - 2006

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Method 7502.1
September 30, 2019

7502.1 – SOLVENT CLEANING POWER BY A SOIL TEST METHOD

1. SCOPE

1.1 This test method is used for determining the cleaning power of solvents that have been subjected to soiled mechanical parts. This test method is a mandatory part of the specification. This test method measures the solvency of all types of cleaners and differentiates their cleaning powers.

2. REFERENCED DOCUMENTS

MILITARY STANDARDS

MIL-PRF-10924	-	Grease, Automotive, Artillery (GAA)

COMMERCIAL STANDARDS

ASTM

ASTM D4636	-	Standard Test Method for Corrosiveness and Oxidation Stability of Hydraulic Oils, Aircraft Turbine Engine Lubricants, and Other Highly Refined Oils

3. SUMMARY OF METHOD

A small amount of grease applied to a steel coupon is immersed in the test solvent and then placed in an ultrasonic cleaner maintained at up to 50˚C. The grease is dissolved by the solvent and removed from the steel coupon by the propagation of sonic sound. The solvent power is determined based on the time to completely remove grease from the coupon.

4. APPARATUS

4.1 The following apparatus are to be used:

a. Ultrasonic Cleaner
b. Beaker, 200ml
c. Test specimen assembly
d. Steel metal square 0.081 cm thick by 2.5 cm square with holding wire (see FIGURE 1). This specimen has been used in ASTM D4636.

(Note: Paper clips have been successfully used for holding metal square)

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Check the source to verify that this is the current version before use.

e. Timer
f. Thermometer, having a range of 50°C.

5. MATERIALS

5.1 Toluene, CP, at least 99% (Caution- Flammable, vapor harmful)

6. PROCEDURE

6.1 The following procedures are to be used:

1. Wash the test assembly with toluene until free from any soil or oil film. Dry the test assembly using laboratory air system. Weight the test assembly to the nearest 0.1 g. Approximately, 0.4 grams of MIL-PRF-10924 grease is uniformly applied to both sides of the metal coupon. The test assembly is placed in a beaker using a holder. Add the test solvent into the beaker until the test assembly is completely immersed.
2. Place the beaker in the ultrasonic cleaner that is maintained water temperatures of 50 °C. Start the test using a timer. Observe the test assembly until all grease is removed from metal coupon. Record the time in minutes. If the portion of grease still remains on the metal coupon after 100 minutes, the test is immediately terminated with the testing time being recorded as 100 minutes.
3. Repeat two more tests in accordance with 7.1 and 7.2.

7. CALCULATION

7.1 Calculate the cleaning power as follows:

$$Solvent\ cleaning\ power, percent = \frac{100 - A}{100} * 100$$

where:
A - is average time of three runs obtained from the tests

(Note 2. The cleaning power of water is found as zero % while toluene is 92- 96%)

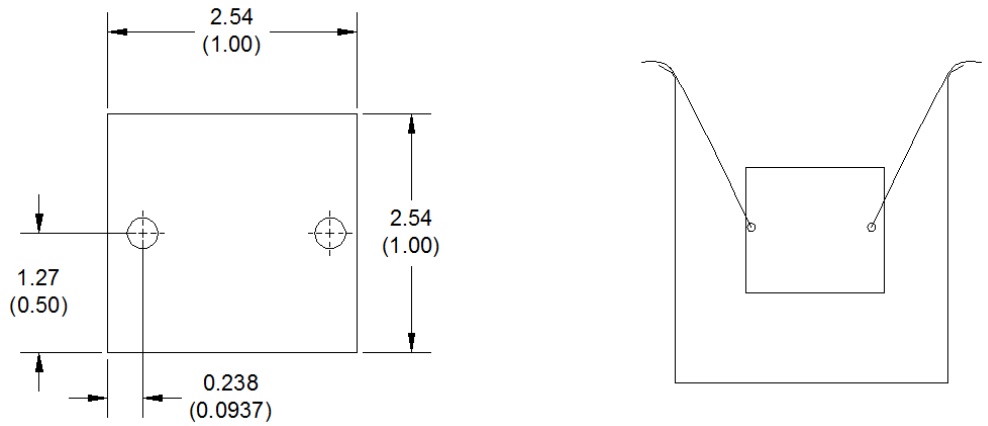
8. REPORT

8.1 Report the calculated solvent cleaning power in percentages to the nearest 0.1%.

9. PRECISION

9.1 Precision data have not yet been developed for the method.

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(a) Metal square

(b) Test assembly

DRILL 2 HOLES
0.159 cm (0.0625 INCH)
DIAMETER

FIGURE 1. Metal square dimensions and test assembly.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
30 Sep 2019	Editorial changes to fix spacing and other minor corrections	Editorial corrections	1	Army - AT

Method prepared by:
Army – AT – 2006

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Method 7503.0
September 30, 2019

7503.0 – HIGH-TEMPERATURE ROLLING CONTACT FATIGUE LIFE TESTING (RCF) OF LUBRICATING OILS AND BEARING MATERIALS (BALL-ON-ROD RCF TESTER)

1. SCOPE

1.1 This method describes a procedure for determining the rolling contact fatigue (RCF) life of lubricating oils and bearing materials at test temperatures of 22°C (72°F) through 371°C (700°F) using a ball-on-rod tester. The effect of either bearing material or lubricating oil on RCF life can be examined with this method.

2. SUMMARY

2.1 The method consists of conducting RCF tests on the lubricant/ bearing material at a high temperature under simulated bearing geometry using a ball-on-rod type tester. Each test is conducted until a fatigue spall occurs on the rod specimen or failure of ball/ race occurs detected by increased vibration. The rolling contact fatigue life is determined using Weibull analysis.

3. SAMPLE SIZE

3.1 Sufficient lubricant amount to conduct 20 RCF tests. Using baseline materials (4.1.2 and 5.1) and TABLE I conditions, approximately 6 L of test oil.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1 Test unit. The apparatus used in this method is a high temperature ball-on-rod RCF tester (see FIGURE 1). This tester comprises of a test chamber, a heating system, a drive system, a test oil system, and the necessary instrumentation and controls. FIGURE 2 illustrates the control console.

4.1.1 Ball-on-Rod RCF Tester: The basic principle of the ball-on-rod RCF tester is shown in FIGURE 1. The ball-on-rod tester consists of a rotating cylindrical test specimen (9.525 mm in diameter x 127 mm long rod) stressed by rolling contact with three radially loaded balls (12.7 mm in diameter) held apart in a retainer (cage). The balls are loaded against the rod by two stationary tapered races at a predetermined calibrated thrust load applied by three compression springs. These test components simulate the components of a rolling element bearing. The test rod specimen simulates the inner race; the balls, the rolling elements; the retainer, the cage; and the tapered races, the outer race. A set of test specimens are shown in FIGURE 2. A cross section schematic of the tester is shown in FIGURE 3. A photograph of a multi head high temperature RCF tester is shown in FIGURE 4.

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NOTE 1: The procedure for the cycle calculation for the tester is given in detail in Reference: Glover, D., “A Ball on Rod Rolling Contact Fatigue Tester”, ASTM STP771, 1982, p. 107-124.

4.1.2 Test Specimens: The standard test specimens (rod, races and balls) for baseline RCF tests are made of AMS 6491 Rev E steel (VIM-VAR M50) and heat treated in accordance with AMS 6491 Rev E sec. 3.4.3. The retainer (cage) is made from AMS6415 (4340) steel. Other materials may be used to make test specimens to compare against the baseline materials.

4.1.3 Test oil system. The test oil lubrication system (model B1589-6 501 x00) is supplied by Oil-Rite Corporation, Manitowoc, MN. The capacity is 4000 mL. The test specimens are lubricated by a gravity fed test lubricant at a flow rate of 22 ml/hr. The oil is not recirculated through the system once it is fed through the test head. For standard test protocol, the test lubricant is at room temperature prior to flowing into the test section. The oil tank is fitted with a cartridge heater to preheat the oil up to 232°C (450°F) if needed to test high viscosity lubricants. The post-test oil is drained to collection container below the drive motor.

4.1.4 Drive system. The RCF tester is driven by a direct drive 3/4 hp motor mounted in line with the rod specimen. The rod specimen is rotated at a constant speed of 3600 rpm resulting in surface speed 1.8 m/s. The test chamber and drive motor forms a compact unit which is suspended through two sets of shock mounts on the table.

4.1.5 Test Heating system. The test housing can be heated up to 400ºC using band heaters. The temperature controlling thermocouple is mounted on the upper tapered race.

4.2 Instrumentation and controls. All of the instruments and controls for this tester are automated and do not require constant attention during a test. The instrument panel for the rig is shown in FIGURE 5.

4.2.1 Temperature measurement and control. The temperature controller for the test head is mounted on front panel as shown in FIGURE 5. It displays the test temperature and controls the band heaters to maintain test head temperature. The controller is also connected to the drive motor and shuts down when occurs when vibration exceeds set level (see 4.2.3) signifying test failure/end of test.

4.2.2 Control of test run duration. An electric timer (see FIGURE 5) located on the panel, is used to record the duration of each test in hours. When vibration limit is reached, the timer turns off automatically and the drive motor shuts down.

4.2.3 Automated Detection of Test Vibration. An accelerometer is mounted on the top of the test housing. It monitors and displays vibration on the control panel. It is tied into the control system of the test rig. When vibration exceeds baseline level by 1G, indicating a test specimen surface crack or fatigue spall, the drive motor is automatically stopped along with the heating system and oil supply.

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4.6 Additional instruments.

Set of conventional mechanical tools as necessary for assembly, disassembly, maintenance and repair.

5. MATERIALS

5.1 Lubricating oil: A MIL-PRF-23699 HTS lubricant is used for baseline testing or when comparing different bearing materials. Other lubricants may be used when it is the lubricant performance that is of interest.

5.2 Flushing solvents.

5.2.1 Isopropyl Alcohol: for cleaning test specimens and test rig

Note: Some high temperature lubricants may require a manufacturer recommended solvent for cleaning. For these lubricants the manufactured recommend solvent should be used and all appropriate safety precautions and disposal methods followed.

5.2.2 Any suitable water based non-corrosive cleaner: for ultrasonic cleaning of test head.

6. PROCEDURE

6.1 Operating condition.

6.1.1 Throughout the entire test, the variables listed in TABLE I should be controlled within the limits specified.

TABLE I. Standard operating conditions.

Condition	Value	
Test speed, rpm	3600 ± 10	
Test oil inlet temperature	23 ± 2 °C	74 ± 4°F
Test oil flow rate	22 ± 1 mL/hr (10 drops/min)	
Test temperature	218 ± 3 °C	425 ± 5°F
Maximum Hertzian Pressure (between ball & rod)	4.826 ± 0.15 Giga Pascal	700 ± 20 ksi

6.2 Test Procedure.

a. Record pertinent information such as test number, test material (rod, raceways, rolling element, and cage), test oil, test oil code, operator's name, date, etc., on suitable data sheets such as in FIGURE 6 and FIGURE 7. Record current hour meter reading.
b. Clean test specimens (rod, races, balls and cage) with isopropyl alcohol by rinsing and wiping.

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c. Install race in each half of the test head.
d. Assemble the rod using collet and nut on the test motor.
e. Place bottom half of the test head and assemble cage and ball around the rod.
f. Install top half of the test head and assemble top and bottom using three bolts. Alternately tighten the bolts until they are snug.
g. Connect thermocouple and plug in heaters.
h. Place accelerometer on the top of the test assembly (see FIGURE 4).
i. Install oil drip tube.
j. Turn on the control unit and start oil drip feed. Adjust the drip rate of 10 drops/minute.
k. Turn motor power on and set the vibration to 1 G setting to begin the test.
l. Check function of test rig each working day until vibration limit is reached. Monitor oil level in tank, test temperature and vibration.
m. Test will run until vibration triggered motor shutdown occurs indicating a test component surface crack or fatigue spall (end of test).
n. To confirm end of test, restart test by pressing reset button on vibration unit (see FIGURE 4). The test should shut down due to vibration limits in less than a minute. If test does not shut down and continues to run at low vibration, the test shutdown may have been a false alarm. Allow test to continue until automatic shutdown.

6.3 Post Test Procedure.

a. Record hour meter reading on the data sheet.
b. Turn power switch unit to the “OFF” position (see FIGURE 4).
c. Disconnect band heaters.
d. Disconnect thermocouple
e. Remove accelerometer.
f. Remove oil drip tube.
g. Loosen bolts between middle and top plate.
h. Remove top assembly of the rod.
i. Remove cage & balls.
j. Loosen the nut.
k. Remove test head bottom assembly.
l. Measure rod height from collet using depth gauge.
m. Remove rod from collect.
o. Examine rod for spall & note a rod spall on the data sheet.
p. If no spall is observed on the rod, examine balls and raceways for possible failure modes and record ball or raceway failure on data sheet.
q. If there is a spall on one of the raceways, replace it

6.4 Test Procedure to Run a Set of RCF Tests

a. Replace balls after each test.
b. Clean test specimens (rod, balls, races & cage) using isopropyl alcohol.
c. Move rod up by 3.12 mm (1/8”) using height measured using height gauge (Sec. 6.3.l)

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d. Repeat test using procedure described in 6.2 & 6.3.
e. Run 10 tests on one side. After ten tests flip the rod and run an additional ten tests.
f. Replace raceways: after twenty tests, testing involving a different set of material, tests involving a different lubricant, or if failure is detected (see 6.3.q).

NOTE 2: A set of twenty (20) tests can be conducted on each rod specimen (ten on each side of the rod).

NOTE 3: Use manufacturer recommended cleaning solvent to remove lubricant residue if alcohol is not effective.

6.5 Rig Cleaning Procedure for Testing Different Lubricant

a. Clean top and bottom of test head, collect, nut, drip tube, bottom oil collection dish using alcohol
b. Ultrasonically clean all the component listed above for 240 minutes at 40ºC using non-corrosive water based cleaner.
c. Drain all the oil from previous test and remove glass cylinder. Clean the reservoir dome with alcohol. Run alcohol through solenoid valve.
d. Clean oil drain using isopropyl alcohol.
e. Assemble test rig without test specimens and heat it at 316ºC for 24 hours.
f. Replace tested oil collection container at bottom of the test rig.

NOTE 4: Use manufacturer recommended cleaning solvent to remove lubricant residue if alcohol is not effective.

6.6 Rig Calibration Procedure

a. Assemble the test rig with test specimens.
b. Bolt the top and bottom heads firmly together.
c. Accurately measure the gap between middle and bottom plates using feeler gauges (three sets of gauges are recommended).
d. Disassemble the test head assembly.
e. Weigh upper test head, lower head without bottom plate, spacers, bolts and springs.
f. Set bottom head assembly (with middle and lower plate with compression springs) on a solid surface with center hole exposed for hanging weights.
g. Place a tapered cone and roller assembly in the lower race to hang the weight.
h. Hang weight (calculated load for desired stress). The springs will carry this load.
i. Tighten the bolts (through springs) to obtain the proper gap between the middle and lower plate. Make sure all the gap is identical for all three springs for even load distribution.

7. Determination of RCF Life

a. Calculate the RCF life using 2 parameter Weibull Analysis at 90 percent confidence level.

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b. Ball and raceway spalls are treated as suspensions for the Weibull analysis. Only rod spalls are used as failure data points for RCF life.
c. Report total number of tests, number failures, number of suspensions, L_{10} and L_{50} life and slope.

7.1 Determination of RCF Life using standard Bearing Material and Lubricant

a. For a new rig at least twenty baseline RCF tests shall be conducted, according to the foregoing procedure, with standard test specimens, baseline lubricant, and using the standard operating conditions. Typical time to a spall for a single baseline test is 6 hours.
b. After a major overhaul to a rig such as a test housing component failure (disassemble of lower housing) or recalibration of test rig for different stress condition or different test temperature, twenty baseline RCF tests shall be conducted using standard test specimens described above.

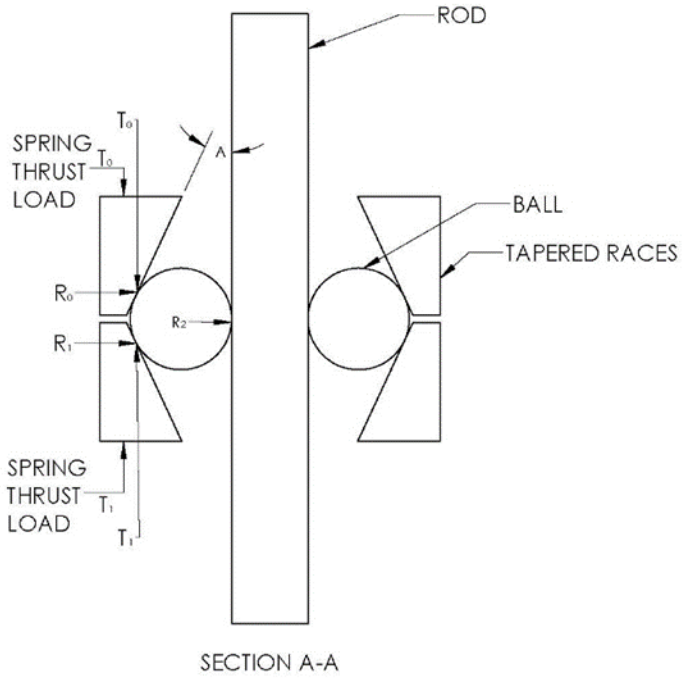


FIGURE 1. Concept of ball-on-rod RCF tester.

Where:

A= Race Angle, deg.
$T_0 = T_1$ = Spring Thrust Load
$R_0 = T_0 / \tan A$
R_1 = radial load between lower race and ball
$R_1 = T_1 / \tan A$
$R_0 = R_1$
$R_2 = R_0 + R_1$

Number of stress cycles/minute = motor speed x 2.389

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FIGURE 2. Ball-on-rod RCF test specimens – rod, races, rolling elements and a cage.

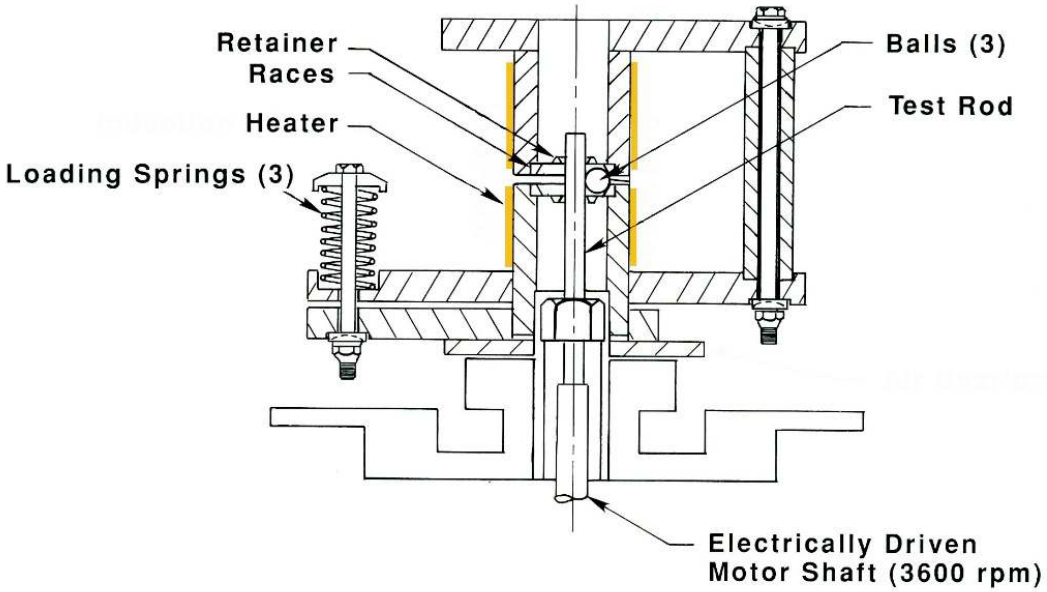


FIGURE 3. Cross sectional schematic of ball-on-rod RCF tester.

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FIGURE 4. Multi head ball-on-rod RCF tester set-up.

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FIGURE 5. Control panel for RCF tester.

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RCF RIG CALIBRATION DATA SHEET

Test Head #: **Date:** _____

1. **Rod Material:** _____

   Modulus of Elasticity (E, KSI): _____

   Poisson's Ratio (ν): _____

2. **Ball Material:** _____

   Modulus of Elasticity (E, KSI): _____

   Poisson's Ratio (ν): _____

3. **Maximum Hertzian Stress (Ksi) :** _____

   Total Load (lb.): _____

   Load at Contact (lb.): _____

4. **Test Head Total Weight (lb.):** _____

   (Without bottom plate)

   Spring Load (lb.): _____

5. **Gap (in.):** _____

Remarks:

_____

_____

_____

FIGURE 6. Sample calibration data sheet.

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RCF DATA SHEET

Date:
Rod Material:
Race Material:
Ball Material:
Cage Material:

Head Number:
RCF Test Number:
Lubricant:
Stress:
Temperature:

Rod Number:

Test No.	*Date*	*Time*		*Life (hours)*	*Life x 0.516 (Cycles x 10^6)*	*Spall*	*Remarks*
		Start	**Stop**				
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							

FIGURE 7. Sample RCF test data sheet.

Method Prepared By:
Air Force—AFRL/RQTM – 2019

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Method 7504.0
September 30, 2019

7504.0 – EFFICIENCY OF AXLE GEAR LUBRICANTS

1. SCOPE

1.1 This method is used for determining the relative mechanical efficiency improvement provided by a candidate axle gear lubricant against a reference lubricant under controlled laboratory conditions.

2. SUMMARY

2.1 A medium tactical vehicle axle (see 4.3.1 for more details) is used to measure the efficiency change between axle gear lubricants under controlled laboratory conditions. At the start of the test segment, a minimum of 5 baseline runs are conducted using the reference fluid to determine baseline axle efficiency. The axle is then double flushed with the candidate axle gear lubricant, and a minimum of 5 candidate runs are conducted following the same operating procedure. At the completion of the candidate testing, the axle is double flushed with the reference lubricant, and a single test run is conducted using the reference oil to determine test validity. A test is considered valid if the post-candidate reference run returns data that is statistically equivalent to the pre-candidate reference runs. If valid, the pre-candidate baseline reference test data is compared to the candidate test data to determine the change in measured efficiency of the axle. Combined statistical analysis is conducted on all of the reference and candidate runs to determine efficiency change and confidence interval for each operating condition.

3. SAMPLE SIZE

3.1 A total of approximately 38 L (10 gal) of candidate lubricant is required to double flush the axle and oil temperature control loop. Approximately twice this amount (i.e., 76 L (20 gal) of reference fluid is required.

4. REFERENCES, STANDARDS, and APPARATUS

4.1 References

4.1.1 SAE Standard J2360, Automotive Gear Lubricants for Commercial and Military Use.

4.2 Test Apparatus

4.2.1 The axle test apparatus is a T-type test stand where the axle is driven by an electric alternating current (AC) motor, and the two outputs of the axle are coupled using speed increasing gear boxes and absorbed by an identically sized AC motor/generator. Speed control is provided by the input motor, while load control is provided by the output absorber (controlling against input torque measurement). Speed is measured through incrementing encoders mounted at the input and

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output motors, and torque is measured using high accuracy digital torque flanges mounted directly at the axle's input and output interfaces to reduce any outside influence from the remainder of the test stand on the resulting efficiency calculation.

4.3 Test Axle Description

4.3.1 The axle used for testing is the rearmost axle from the 5-ton M1083A1 cargo truck of the Family of Medium Tactical Vehicles (FMTV). The axle is produced by Meritor and is identified by the part number: RR15611NFDF32-780.

4.3.2 The axle is a typical beam type axle with an open differential and an overall gear ratio of 7.8:1 (3.9:1 ring and pinion ratio, and 2:1 wheel end reduction). The axle has a common oil sump between the center differential section and wheel end hub reductions.

4.3.3 The factory input pinion yoke is replaced by a custom machined input yoke that allows for direct mounting of the input torque flange.

4.3.4 Special output hubs are machined to bolt to the axle's wheel studs and provide mounting for the output torque flange.

4.4 Test Stand Detailed System Description

4.4.1 Axle input power and output absorption is provided by two 250hp AC electric motor/generators controlled through two variable frequency drives (VFD).

4.4.2 A total of two speed increasing gear boxes are utilized to increase each axle wheel end output speed and reduce torque prior to the absorbing motor. The gearboxes have a mechanical gear ratio of 7.259:1, and contain an integrated lubrication and cooling circuit.

4.4.3 Input torque is measured using a 1 kNm digital torque flange with a minimum 0.05 accuracy class.

4.4.4 Output torque at each wheel end is measured using 3 kNm digital torque flanges with a minimum of 0.05 accuracy class.

4.4.5 A three-way ball valve is installed into the rear axle fill port to allow the test stand operator to select if the differential housing is open to atmosphere for fluid level setting, or in the recirculate position for the heater control loop return flow. This valve must be installed in such a way that when vented for fluid level setting, the resulting fluid level in the axle is level with the lower portion of the original fill port (see FIGURE 1).

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FIGURE 1. Rear axle fill port three-way ball valve.

4.4.6 Differential oil temperature control is provided through the use of an external heater control loop with the ability to heat or cool the axle gear oil during operation depending on loading conditions. The heater control loop has a nominal recirculation flow rate of 5 gpm. Gear oil is removed from the bottom of the differential at the drain port, circulated through heater and trim heat exchanger, and returned back to the axle at the rear fill port during operation.

4.4.7 Gear oil temperature measurement is captured using a closed tip thermocouple entering into the center housing of the axle at the drain port location. The thermocouple should enter into the differential housing at a distance of 1 in ± 0.25 in referenced from the flat external boss of the drain port on the lower differential housing (see FIGURE 2).

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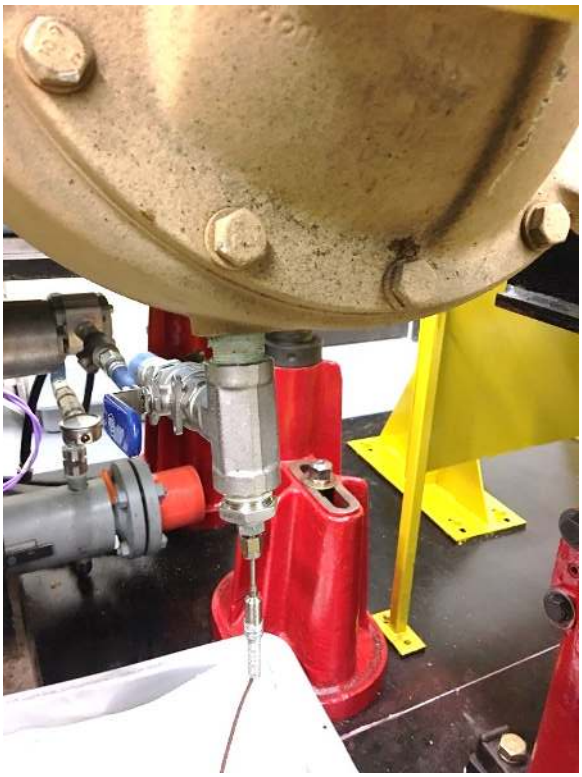


FIGURE 2. Axle drain port location.

4.5 Instrumentation

4.5.1 The following parameters are the minimum required measurements to be recorded during testing:

a. Input speed
b. Output speed
c. Input torque
d. Left output torque
e. Right output torque
f. Differential gear oil temperature
g. Heater control loop return temperature
h. Ambient temperature

4.6 Data Acquisition

4.6.1 A data acquisition and control system must be utilized to simultaneously record all testing parameters and provide speed and torque control for the axle under test. The data acquisition system must be capable of logging data at the specified 2 Hz for the stabilized data recording steps during the efficiency test. It is recommended that the data acquisition system be able to monitor and control the system at an update rate of 100 Hz to ensure precise control of the system.

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4.6.2 The data acquisition and control system must also be capable of controlling limits for over/under oil temperature, and over/under torque and speed.

4.7 Test Stand Diagram

4.7.1 A detailed diagram of the test stand and heater control loop is provided in FIGURE 8.

5. MATERIALS

5.1 A sufficient volume of reference lubricant is required to double flush the axle and heater control loop for the pre-candidate and post-candidate test baseline runs. This is equivalent to approximately 8 gallons, plus two times the capacity of the external heater control loop volume.

6. PROCEDURE

6.1 Fluid Change:

NOTE: If the axle fluid is being changed from one lubricant to another, a double flush procedure should always be used! The procedure below outlines the process to complete a single flush. Complete steps 6.1.1 through 6.1.6 twice to complete a double flush.

6.1.1 If starting with the axle already drained, proceed to step 6.1.3. If starting with the axle full, start the test stand and operate the axle at approximately 1500 rpm and 200 lbf·ft until the differential fluid temperature is ≥ 175°F.

6.1.2 Once the temperature has reached 175°F, bring the test stand to a stop while keeping the fluid temperature elevated (i.e. do not apply cooling). Once stopped, drain the axle and heater control loop and dispose of the drained fluid. Refer to FIGURE 2 for drain port location.

6.1.3 Ensure that the circulation pump is off and position the three-way valve at the rear fill port of the axle to the vent position (open to atmosphere). Refer to FIGURE 1 for three-way valve location. Add fluid to the axle housing through the upper vent port until the fluid level is even with the rear housing fill port (this is noted by a trickle of fluid from the vented three-way valve on the axle).

6.1.4 Move the three-way valve at the rear fill port of the axle to the recirculation position (see FIGURE 1) and turn on the circulation pump and allow the fluid to flow through the heater system for a minimum of 2 minutes to purge air from the heater system.

6.1.5 Turn off the circulation pump and reposition the three-way valve at the rear fill port of the axle to the vent position and add fluid to the axle housing through the upper vent port until the fluid level is topped off to the rear housing fill port level.

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6.1.6 Once complete, position the three-way valve at the rear fill port of the axle back to the recirculation position for testing.

6.2 Efficiency Testing:

6.2.1 Start the test stand and ramp the axle to step one test conditions (see TABLE I) and hold until the axle oil temperature reaches 175°F ± 1°F.

6.2.2 Once the axle oil reaches temperature, progress the axle through the speed, load, and temperature points outlined in TABLE I while logging (at a minimum) input torque, output torque left and right, input speed, output speed, and axle fluid temperature.

6.2.3 Note that the axle fluid temperature set point is reduced from 175°F to 140°F for the low torque operating conditions.

6.2.4 Operation at each step should consist of two sub-steps, first a stabilization sub-step to allow the axle to stabilize at the specified test conditions, then second, a data recording sub-step.

6.2.5 During the stabilization sub-step, the pinion speed should be held at ± 2 rpm of the speed set point, and the pinion load should be held at ± 5 lbf·ft of the load set point.

6.2.6 During the stabilization sub-step, the axle should be operated at the specified test condition in TABLE I for a minimum of 5 minutes, and the moving average of the axle differential fluid temperature should be at the temperature set point ± 0.25°F before continuing to the recording step. The moving average should be calculated over a 60 second interval with a 1 second sample time. Overall data logging rate during the stabilization step should be 0.2 Hz (i.e., every 5 seconds).

6.2.7 During the data recording sub-step, the axle should be operated at the specified test condition for 60 seconds at a logging rate of 2 Hz. Once complete the axle can be ramped to the next step/test condition for stabilization.

TABLE I. Federal test method speed & load points.

Step	Nominal Speed [mph]	Pinion Speed [rpm]	Pinion Load [lbf·ft]	Axle Differential Fluid Temperature [°F]
1	25	1469	450	175
2	35	2100	250	175
3	25	1469	325	175
4	45	2600	175	175
5	15	865	400	175
6	55	3207	104	175
7	35	2100	67	140
8	25	1469	54	140
9	15	865	45	140
10	5	294	80	140

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6.2.8 The axle should be operated through all 10 steps to complete 1 cycle. For each individual test, 10 cycles should be completed for data averaging.

7. CALCULATIONS

7.1 Efficiency of Individual Test Run

7.1.1 All final calculations are completed using the data from the data recording sub-step captured after stabilization at each speed and load. To determine efficiency results for the baseline or candidates tests, input and output power and efficiency for each step of all 10 cycles should be calculated as follows:

$$Power_{In} = \frac{(Speed_{In} * Torque_{In})}{5252}$$

$$Power_{Out} = \frac{Speed_{Out} * (Torque_{Out\,Left} + Torque_{Out\,Right})}{5252}$$

$$Efficiency = \frac{Power_{Out}}{Power_{In}} * 100$$

$$where: Power\ [hp], Speed\ [rpm], Torque\ [lbf \cdot ft]$$

7.1.2 From these calculations, a matrix of resulting efficiency can be tabulated for each baseline or candidate run. An example table is shown below (see FIGURE 3), where "xx" denotes the calculated efficiency for all 10 cycles of the baseline or candidate run. (Note: cycle 1 is not included in the final analysis to reduce any impacts in efficiency measurement from long thermal transients that persist in the tested hardware after the initial warm-up).

	Baseline or Candidate Single Run Results								
	Cycle								
STEP	2	3	4	5	6	7	8	9	10
FTM_25_450r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_35_250r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_25_325r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_45_175r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_15_400r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_55_104r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_35_67r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_25_54r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_15_45r	xx	xx	xx	xx	xx	xx	xx	xx	xx
FTM_5_80r	xx	xx	xx	xx	xx	xx	xx	xx	xx

FIGURE 3. Baseline or candidate single run results.

7.1.3 The composite result for a single baseline or candidate run is then calculated by taking the average efficiency over cycles 2 through 10 for each step of the individual run. An example composite result table for a single run is shown below (see FIGURE 4).

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Baseline or Candidate Run Composite Result	
STEP	Run #
FTM_25_450r	xx
FTM_35_250r	xx
FTM_25_325r	xx
FTM_45_175r	xx
FTM_15_400r	xx
FTM_55_104r	xx
FTM_35_67r	xx
FTM_25_54r	xx
FTM_15_45r	xx
FTM_5_80r	xx

FIGURE 4. Example of baseline or candidate run composite results.

7.1.4 Recall, a minimum of 5 runs must be conducted for each baseline or candidate test. After all 5 runs are completed, a baseline or candidate test results table can be formed. An example is shown below in FIGURE 5.

	Baseline or Candidate Test Result				
STEP	Run 1	Run 2	Run 3	Run 4	Run 5
FTM_25_450r	xx	xx	xx	xx	xx
FTM_35_250r	xx	xx	xx	xx	xx
FTM_25_325r	xx	xx	xx	xx	xx
FTM_45_175r	xx	xx	xx	xx	xx
FTM_15_400r	xx	xx	xx	xx	xx
FTM_55_104r	xx	xx	xx	xx	xx
FTM_35_67r	xx	xx	xx	xx	xx
FTM_25_54r	xx	xx	xx	xx	xx
FTM_15_45r	xx	xx	xx	xx	xx
FTM_5_80r	xx	xx	xx	xx	xx

FIGURE 5. Example of baseline or candidate test results.

7.2 Statistical Analysis – Comparison of Baseline Test to Candidate Test.

7.2.1 Once all baseline and candidate test data is gathered, statistical analysis can be conducted to determine overall efficiency change and confidence interval.

7.2.2 For both the baseline and candidate test results, calculate the mean, standard deviation, and variance from each run at each individual step. Example calculation table provided in FIGURE 6.

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	Baseline				Candidate		
STEP	Mean	Std Dev	Variance		Mean	Std Dev	Variance
FTM_25_450r	xx	xx	xx		xx	xx	xx
FTM_35_250r	xx	xx	xx		xx	xx	xx
FTM_25_325r	xx	xx	xx		xx	xx	xx
FTM_45_175r	xx	xx	xx		xx	xx	xx
FTM_15_400r	xx	xx	xx		xx	xx	xx
FTM_55_104r	xx	xx	xx		xx	xx	xx
FTM_35_67r	xx	xx	xx		xx	xx	xx
FTM_25_54r	xx	xx	xx		xx	xx	xx
FTM_15_45r	xx	xx	xx		xx	xx	xx
FTM_5_80r	xx	xx	xx		xx	xx	xx

FIGURE 6. Example of calculation table for baseline and candidate test results.

7.2.3 For each step, conduct an F-Test to determine if the baseline and candidate tests have equal variances.

7.2.4 Based on the results of 7.2.3, conduct an appropriate T-Test for each step between the baseline and candidate test results to establish the statistical significance (95% confidence interval) of the difference in means between the reference and candidate results. The results should be complied in a single table as shown in FIGURE 7.

	Candidate Improvement		
STEP	% change	Statistically Significant?	Confidence Interval
FTM_25_450r	xx	Y/N	±xx
FTM_35_250r	xx	Y/N	±xx
FTM_25_325r	xx	Y/N	±xx
FTM_45_175r	xx	Y/N	±xx
FTM_15_400r	xx	Y/N	±xx
FTM_55_104r	xx	Y/N	±xx
FTM_35_67r	xx	Y/N	±xx
FTM_25_54r	xx	Y/N	±xx
FTM_15_45r	xx	Y/N	±xx
FTM_5_80r	xx	Y/N	±xx

FIGURE 7. Example of statistical significant showing candidate improvement.

8. REPORTING

8.1 At the completion of testing, report the baseline and candidate test result tables calculated in step 7.1.4, the candidate improvement table calculated in step 7.2.4, and the post-candidate reference run result to document test validity.

9. PRECISION

9.1 Precision data has not been developed for this method. Since candidate testing includes the completion of baseline runs using a reference fluid for data comparison, statistical analysis of the

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reference and candidate results effectively captures and reflects the repeatability of the test stand measurement.

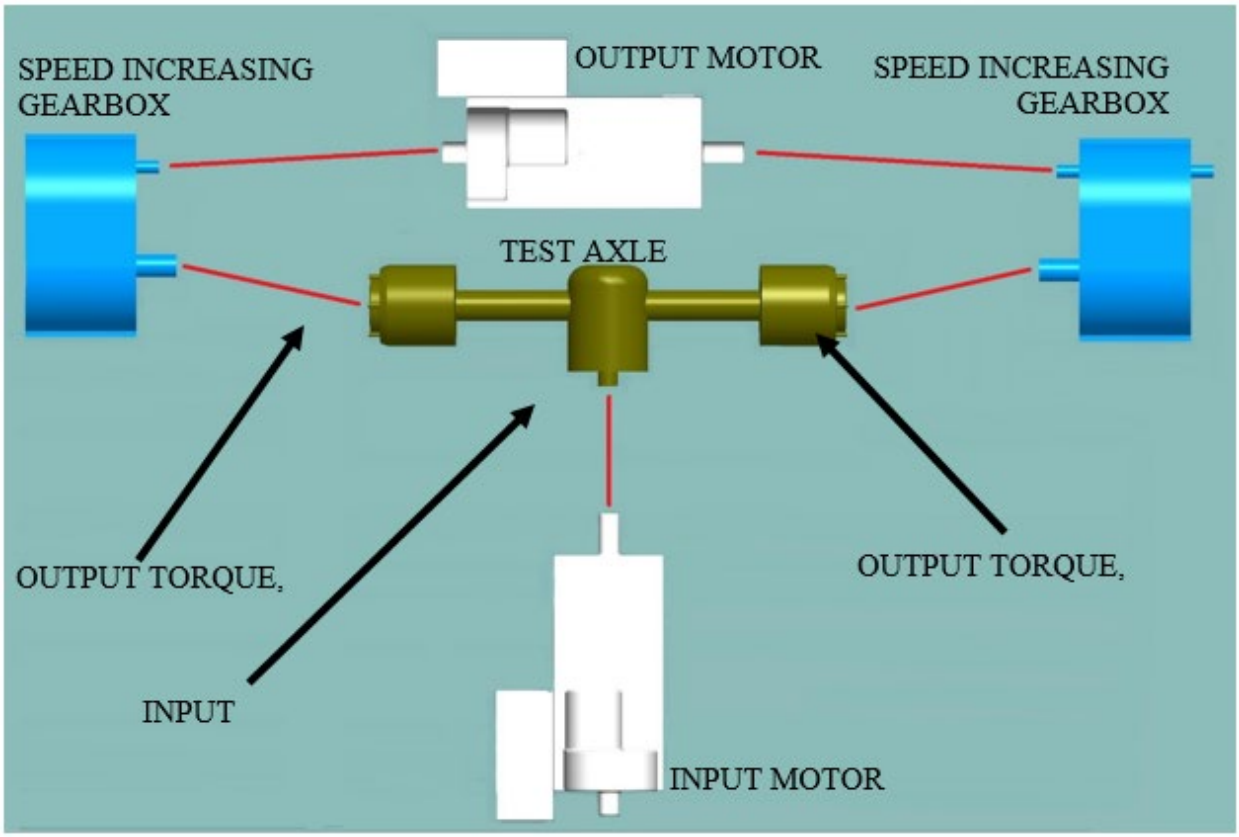


FIGURE 8. Axle test stand block diagram.

Method Prepared By:
Army - AT – 2019

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Method 7505.0
September 30, 2019

7505.0 – LOAD CARRYING CAPACITY OF LUBRICATING OILS USING A MODIFIED FZG GEAR TEST RIG
(FZG-Ryder Rig)

1. SCOPE

1.1. This method describes a procedure for determining the load carrying ability of lubricating oils with reference to gears. The method uses a conventional FZG Gear Test Rig which has been modified to match, as closely as possible, the operating conditions and gear geometry of the Ryder Gear Test Rig (Method 6508.3).

2. SUMMARY

2.1. The test consists of a gear pair in which the teeth are loaded against each other at set load stages. Each stage is run for 10 minutes at 9700 RPM. After each load stage the teeth on the rated gear are inspected and the amount of scuffing measured. The test continues with increasing load stages until the test gear pair reaches the failure criteria. A lubricant's load stage at failure can then be compared (rated against) a reference lubricant.

3. SAMPLE SIZE

3.1. Approximately 7 L of test oil are needed for six determinations (tests) assuming the test oil is able to be reused for each determination. An additional quantity of oil will be required if oil is not reused for each determination.

4. REFERENCES, STANDARDS, AND APPARATUS

4.1. Test unit. The apparatus used for this testing is a modified FZG Gear Test Rig. This modified rig is referred to as an FZG-Ryder Gear Test Rig. The modifications include increasing the speed of the power circuit to 9700 RPM through a set a speed-up gears, adding additional oil heaters for high temperature testing, and a new test gear design. A detailed description of the FZG-Ryder Gear Test Rig can be found in *Development of a High Temperature FZG-Ryder Gear Lubricant Load Capacity Machine* by Michaelis and Winter. As described in this report the FZG-Ryder Rig operates on the "four square" principle. The test gears and slave gears are connected by two split parallel shafts, the load clutch shaft and measuring clutch shaft. The load clutch divides the load clutch shaft into two halves that can be twisted against each other by means of a spindle load device. The resulting torque loads the test and slave gears against each other. The measuring clutch shaft contains a calibrated 18 mm torsional shaft. One end of this shaft is directly connected to the wide test gear and twists as the test gear pair are loaded against each other. This resulting torsion is translated as a semi-linear deflection in the measuring clutch. A dial gauge is placed in the measuring clutch to measure this displacement. Therefore it is

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possible to know the tooth load on the test gear is by reading the displacement of the load clutch. More information about the loading process is given in section 6.3.6. The other end of the two parallel shafts (motor side) are connected inside the slave gear box with two slave gears which completes the mechanical power circuit. The power circuit as a whole is driven by the electric motor and a set of speed up gears. FIGURE 1 shows a labeled picture of the FZG-Ryder Rig.

4.2. Gear Sets.

4.2.1. Test Gear Box Gears. Two gears are inside the test gear box. These two gears are the main test article. These test gears are spur gears made from 9310 VAR steel, case hardened. Each set of test gears is composed of a narrow gear which is called the rated gear and a wider gear referred to as the wide gear. The number of teeth for both gears is 30, pressure angle of 20°, and pitch diameter of 90 mm. The tooth width is 6.35 mm for the test gear and 26 mm for the wide gear. Drawings for the test and wide gears may be obtained by contacting AFRL/RQTM. Both the rated and wide gears are stamped on one side with a serial number. The gears should be mounted in the test gear box so that the serial numbers are facing the same direction. When the serial numbers are facing the end cover of the test gearbox (left side of FIGURE 1) this is referred to a Side A of the gears. When the serial numbers face the motor (right side of FIGURE 1) this is referred to as Side B. These gear sets are only used for two tests, once by running side A and once by running side B. FIGURE 2 shows the test gear pair.

4.2.2. Slave Gear Box Gears. The two gears in the slave gear box are identical to the wide gear from the test gear box. As they do not experience much loading during testing they should not need to be changed as long as proper lubrication is maintained.

4.2.3. Speed-Up Gear Box Gears. The standard FZG motor supplied with a FZG test rig does not have the capability to reach the test speed of the FZG-Ryder Test (9700 RPM). Therefore a set of helical cut speed up gears are placed in a separate gear box between the motor and the support gear box. The gear ratio for this set should be chosen so that the motor speed is increased to the test speed. For example a motor with a speed of 3600 RPM would need a set of speed-up gears with a ratio of about 2.7 to supply the necessary 9700 RMP.

4.3. Oil systems. There are two separate oil systems for the FZG-Ryder Rig. One system supplies oil to the test gear box and the other system supplies oil to both the slave gear box and the speed-up gear box.

4.3.1. Test Oil System. The test oil system supplies oil to the test gear set and test gear box bearings. It consists of an oil pump, oil filter, oil heat exchanger section, and flowmeter. The oil pump is a variable speed gear pump which scavenges oil from the outlet of the test gear box and pumps it through the oil filter, oil heater, flowmeter, and finally through the oil jet in the test gear box onto the gear mesh. Each component in the test oil system should be designed to handle the temperature requirements of the high temperature test protocol (200°C). The variable speed pump must be capable of delivering oil to the gear mesh at 1 L/min at 74°C and 200°C. A drain valve is located downstream of the pump at the lowest point in the system to facilitate easy

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drainage of the test oil. The oil filter is a reusable wire mesh filter with a filtration rating of between 10 and 60 microns. The oil heat exchanger uses electric elements of a low watt density (to avoid localized overheating of the oil) and needs to be capable of heating the oil to and maintaining to the test oil temperature at the test flow rate. The oil used in this system is the lubricant which is being evaluated/rated.

4.3.2. Slave Oil System. The slave oil system supplies oil to both the gear meshes in the slave gear box and speed-up gear box, as well as the bearings in each of these boxes. The system is composed of a 30 L oil tank with an immersion heater, oil pump, oil filter and flowmeter. The oil pump should be capable to providing an oil flowrate of 3 to 4 L/min to both gearboxes at temperatures up to 100°C and the oil heater need to be able to heat the oil to this temperature. The oil filter can be of a similar type to the one used for the test oil system. The oil used in this system can be any low viscosity lubricant but is typically a MIL-PRF-7808 Grade 3 oil. This oil only needs to be changed occasionally. The recommended change interval is at least once every two years.

4.4. Inspection/ Rating Equipment. Each tooth on the rated test gear is inspected for scuffing wear at each load stage during testing. This is accomplished using a microscope with a digital camera, which is mounted to or near the rig in such a way to allow it to be swung over the inspection port in the test gear box. A software package with microscope digital imaging capture and measurement capabilities should be installed on a computer to allow the operator to take images of each test gear tooth and make measurements as to the scuffed area of each tooth.

5. MATERIALS

5.1. Solvents

5.1.1. Dry cleaning solvent (such as Stoddard Solvent or similar)

5.2. Test Gear Pair

5.3. Dial indicator, capable of measuring displacements in 0-6 mm range with a graduation of 0.01 mm

5.4. Spindle loading device for load clutch

5.5. Wrench or socket

5.6. Oil dip stick

5.7. Ruler in mm

5.8. Oils

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5.8.1. Tested/ Rated oil

5.8.2. Slave gear oil such as a MIL-PRF-7808 oil

6. STANDARD PROCEDURE

6.1. Operating conditions

6.1.1. The operating conditions of the standard FZG-Ryder test are listed in TABLE I. These conditions should be maintained within the given limits.

TABLE I. FZG-Ryder standard test conditions.

Condition	Value
Test gear speed, rpm	9700 ± 20
Test oil inlet temperature	74°C ± 3°C
Test oil flow rate	1 ± 0.1 l/min
Slave oil flow rate	3 ± 1 l/min
Slave oil inlet temperature	74°C ± 10°C

6.2. Test Preparation

6.2.1. Note about the test oil. The test requires approximately 3.2 L of the test oil. This oil may be used for up to six tests/determinations before discarding. If the test oil is taken from a larger container, such as a 5 gallon drum, the oil in the larger container should be thoroughly mixed or shaken before measuring out the 3.2 L. To reuse the oil from a previous test, collect the oil in a large glass beaker placed under the drain valve, open the drain valve and run the test oil pump until all the oil is drained. If the oil is not going to be used within one week then it should be discarded and fresh oil used for additional tests. Note that an additional 3.5 to 4 L of oil will be required when cleaning the test oil system in accordance with 6.2.3.

6.2.2. Slave gear oil system. Ensure the slave gear oil system tank has been filled in accordance with 4.3.2.

6.2.3. Test oil system Cleaning Procedure. Whenever switching to a new lubricant the test oil system needs to be thoroughly cleaned with the oil solvent. If the same test oil is going to be used as the previous test this step can be skipped. Drain the previous oil from the test oil system by opening the drain valve and running the test oil pump. Turn the test oil heaters off. Once the oil is drained, close the drain and pump 3 to 4 L of oil solvent through the test oil system. Drain the solvent in the same manner as the oil and discard. Flush the system with oil solvent three more times, draining the solvent after each flushing and using fresh solvent for the next flush.

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Once the test oil system has been flushed with solvent four times, open the drain valve and allow the system to dry for at least one hour then close the drain valve. Clean and mount the test gears in accordance with 6.2.4. Add 3.5 to 4 L of the test oil to the test gearbox and run the test oil pump to circulate the oil throughout the test oil system. Allow to oil to circulate for at least one minute. Open the drain valve and run the pump to drain this oil. Discard this oil as it may contain a small amount of any residual oil solvent. Close the drain valve.

6.2.4. Test gear pair installation. Rinse the gear set in oil solvent to remove any preservative oil from the gears. If it is difficult to remove all traces of preservative oil, the gears may be placed in an ultrasonic cleaner filled with oil solvent. Once cleaned allow the gears to dry completely. This can be accelerated by blowing compressed air or nitrogen gas on the gears. Mount the gears inside the test gear box. The rated gear is mounted on the load clutch shaft and the wide gear is mounted on the measuring clutch shaft. The gears will likely need to be heated first due to the tight fit on the shafts. This can be performed in a laboratory oven at a temperature up to 120°C. If running the “A” side of the gear pair, each side should be mounted so the serial number marking is facing away from the motor. If running the “B” side of the gear pair, each side should be mounted so the serial number marking is towards the motor. FIGURE 3 shows the test gear pair mounted in the test gearbox with the front faceplate removed. The gear on the left is the wide test gear and the gear on the right is the rated gear. Once the gears are mounted and the test gear box closed the test oil can be added to the test gear box.

6.3. Test Procedure.

6.3.1. Add approximately 3.2 L of test oil to test oil gear box

6.3.2. Turn on test gear oil pump and slave gear oil pump

6.3.3. Ensure that temperature set points are correct for both the test oil heaters and slave oil heaters and turn on each.

6.3.4. When each oil system is at the required temperature, verify their flow rates are correct and make any changes as needed.

6.3.5. Check the test oil level with the dipstick and ruler in the test oil gearbox. The oil level should be 3.5 to 4 cm from the bottom of the gearbox. If the oil level is too low add more oil. If the oil level is too high drain some out of the system using the drain valve.

6.3.6. Apply the test load.

a. Insert the locking pin into the load clutch to lock the load clutch in place.
b. Loosen but do not remove all of the load clutch bolts.
c. Attach the load spindle to the load clutch and insert the dial gauge into the measuring clutch. FIGURE 4 shows the spindle and dial gauge placed on the appropriate clutches.
d. Zero out the dial gauge.

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e. Twist the spindle in the appropriate direction to apply the load. The dial gauge reading should start to increase in number. Continue to twist until the dial gauge is reading the appropriate displacement for load stage 1 (see TABLE II).
f. Tighten all the bolts on the load clutch in a crosswise pattern using the wrench or socket.

6.3.7. Remove the spindle, dial indicator, and locking pin.

6.3.8. Ensure all the necessary safety shrouds are in place and make sure the oil cap is placed over the test gear box inspection port.

6.3.9. Double check all oil flow rates and temperatures.

6.3.10. Leave the room and make sure no other personnel are in the room.

6.3.11. Set the automatic revolution shut off so that the motor will automatically shut off after 10 minutes at 9700 RPM.

6.3.12. Start the motor and wait for the 10 minute load stage to complete.

6.3.13. Once the motor has shut off turn off all oil pumps and heaters.

6.3.14. Wait approximately 10 minutes for the rig to cool.

6.3.15. Remove the test load
a. Insert the locking pin into the load clutch.
b. Loosen but do not remove all the bolts on the load clutch.
c. Remove the locking pin.

6.3.16. Photograph each rated gear tooth.
a. Remove the oil cap from the inspection port. Wait for any oil vapors to dissipate.
b. Position the microscope over the inspection port and adjust to focus in on one gear tooth on the rated gear.
c. Capture images of all 30 gear teeth. See FIGURE 5 for a sample image of one gear tooth.

6.3.17. If the failure criterion has not been reached, repeat the steps above starting with 6.3.2. However, when applying the load, load the gears to load stage 2 (see TABLE II for load settings for all load stages). Keeping repeating steps 6.3.2 through 6.3.16.a, each time increasing the load to the next load stage until the failure criterion is met. Once the failure criterion has been met the test is completed and no further load stages are run. The test is suspended if load stage 16 has been run and the failure criterion has not been met or for a catastrophic failure such as a tooth breakage. See section 6.4 for the failure criterion.

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TABLE II. FZG-Ryder load stage settings.

	FZG Ryder			
Load Stage	Dial Gauge Setting (mm)	Torque (Nm)	Tooth Load (N/mm)	Tooth Load (lb/in)
1	0.33	18	66	375
2	0.66	35	131	750
3	0.99	53	197	1125
4	1.32	70	263	1500
5	1.66	88	328	1875
6	1.99	105	394	2250
7	2.32	123	460	2625
8	2.65	140	526	3000
9	2.98	158	591	3375
10	3.31	175	657	3750
11	3.64	193	723	4125
12	3.97	210	788	4500
13	4.31	228	854	4875
14	4.64	245	920	5250
15	4.97	263	985	5625
16	5.30	280	1051	6000

6.4. Failure Criterion. The test gears have “failed” when the average scuffed area for all 30 gear teeth has reached 22.5 percent of the tooth flank area. See FIGURE 6 for an example of scuffing. The operator should make a note of the approximate scuffed area while imaging all the gear teeth in 6.3.16.c. When the operator is confident that the average scuffed area has exceeded 22.5 percent no more load stages should be run. If it is uncertain whether the gear has exceeded 22.5 percent scuffing additional load stages can be run.

6.5. Measurement of the scuffed area. The scuffed area of the gear teeth are measured using computer software which can measure areas on digital image. Each gear tooth image is opened and the tooth flank area of each tooth is measured and recorded. The average tooth flank area is determined by taking the average of all 30 measurements of the tooth flank area (equation 1). The scuffed area for each tooth is then measured and recorded. Scuffing typically occurs at the tip and root of the gear teeth. FIGURE 7 shows a typical scuffed gear tooth with the tooth flank area and scuffed area noted. The percent scuffed area for a tooth is calculated by dividing that tooth’s scuffed area by the average tooth flank area and multiplying by 100 (equation 2). The

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percent scuffed area is determined for all 30 gear teeth. The average of all 30 percent scuffed area measurements yields the average percent scuffed area for that load stage (equation 3). As stated in 6.4, when this number has exceeded 22.5 percent the gear is said to have failed and no more load stages need to be run. Logarithmic interpolation is used to determine the exact point between two load stages where the gear was scuffed at 22.5 percent. Equation 4 is used for this interpolation which is call the load stage at failure, S_S.

EQUATION 1

$$\text{Avg. Tooth Flank Area} = \frac{\text{Flank Area Tooth 1} + \text{Flank Area Tooth 2} + .. \text{Flank Area Tooth 30}}{30}$$

EQUATION 2

$$\text{Percent Scuffed area for a given gear tooth} = \frac{\text{Gear Tooth's scuffed area}}{\text{Avg. Tooth Flank Area}}$$

EQUATION 3

$$\text{Avg. \% Scuffed Area for a given load stage} = \frac{\text{\% Scuffed Area Tooth 1} + \text{\% Scuffed Area Tooth 2} + .. \text{\% Scuffed Area Tooth 30}}{30}$$

EQUATION 4

$$S_s = \frac{\log 22.5 - \log A_1}{\log A_2 - \log A_1} + S_1$$

S_1 – load stage (integer) prior to 22.5% scuffed area
S_2 – load stage (integer) over 22.5% scuffed area
S_s – load stage (decimals) where exactly 22.5% scuffed area occurs (load stage at failure)
A_1 – percent scuffed area in load stage S_1
A_2 – percent scuffed area in load stage S_2

7. ALTERNATIVE PROCEDURE

7.1. High Temperature test. The procedure for the high temperature test is identical to the standard test procedure given in section 6 except for increased oil temperatures. The test oil inlet temperature is raised to 200°C ± 10°C. The slave gear oil inlet temperature is raised to 95°C ± 10°C. The results from the high temperature test may differ from and should not be compared to tests performed at the standard temperature.

8. BEST PRACTICES

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8.1. For a given oil to be rated, typically six tests are conducted by using the A and B sides of three gear sets. This gives six different load stage at failure measurements, each of which is referred to as a determination. These determinations can then be averaged to give an average load stage at failure.

9. PERCENT RELATIVE RATING

9.1. The average load stage at failure can be compared to another oil's average load stage at failure. This percent relative rating of an oil is determined by taking the oil's average load stage at failure and comparing it to a reference oil's load stage at failure. Both the average load stage at failure for the tested oil and for the reference oil should be made from at least six determinations. Typically the reference oil is a non-additized base stock. Equation 5 shows how to calculate an oil's rating. Over time a database of oil ratings can be compiled. The higher an oil's percent relative rating, the higher its load carrying capacity is said to be. Whenever a newly manufactured lot of test gears are used, the reference oil should be re-tested on the new lot and a new average load stage at failure for the reference oil made with at least six determinations. This is because the average load stage at failure often shifts from lot to lot of test gears due to slight manufacturing differences.

EQUATION 5

$$\text{Oil Rating of Tested Oil} = \frac{\text{Avg. Load Stage at Failure tested oil}}{\text{Avg. Load Stage at Failure reference oil}} * 100$$

10. REPORTING

10.1. Report the load stage at failure for each determination of the test lubricant. Also report the reference oil used and the average load stage at failure for each determination of the reference oil. Calculate and report the percent relative rating of the test lubricant in accordance with section 9.1.

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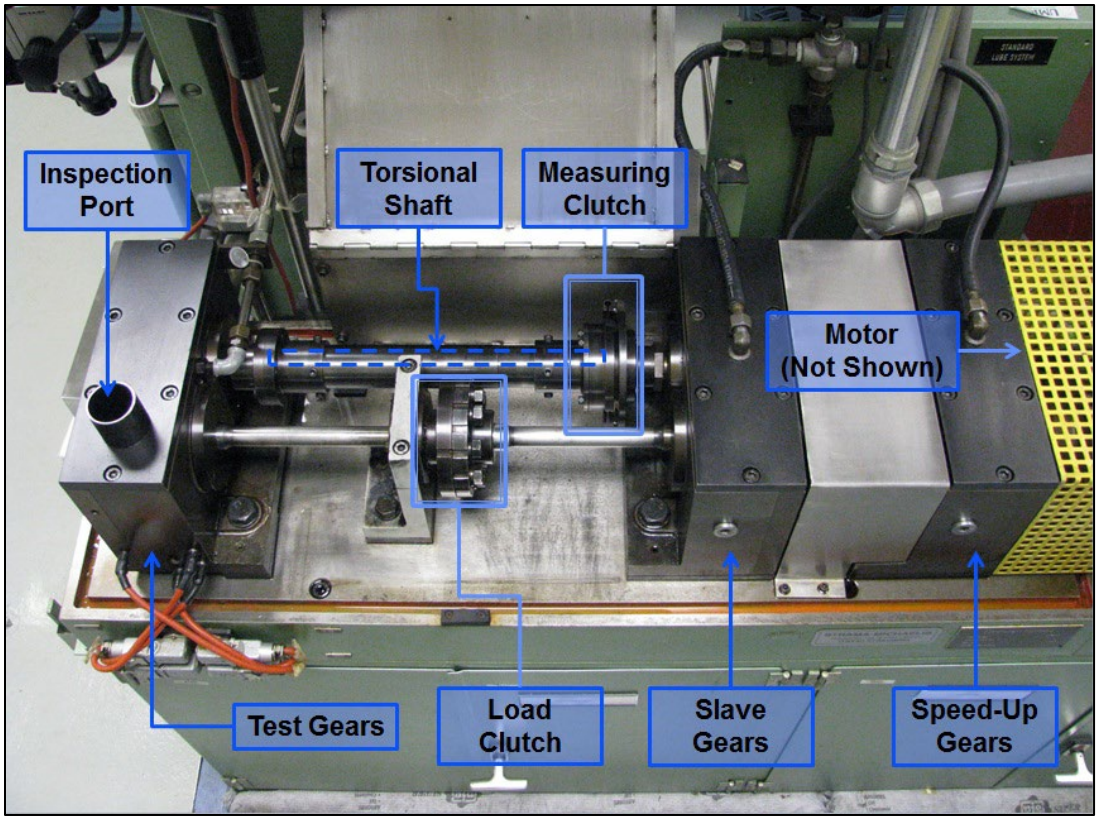


FIGURE 1. FZG-Ryder gear test rig.



FIGURE 2. FZG-Ryder test gear set, wide gear top, rated gear (narrow) bottom.

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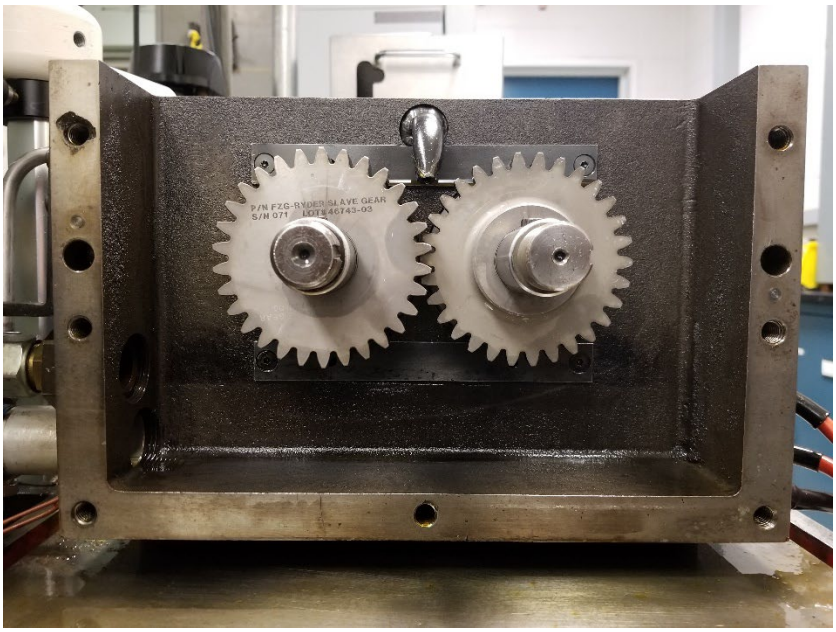


FIGURE 3. Test gear set mounted inside test gearbox.

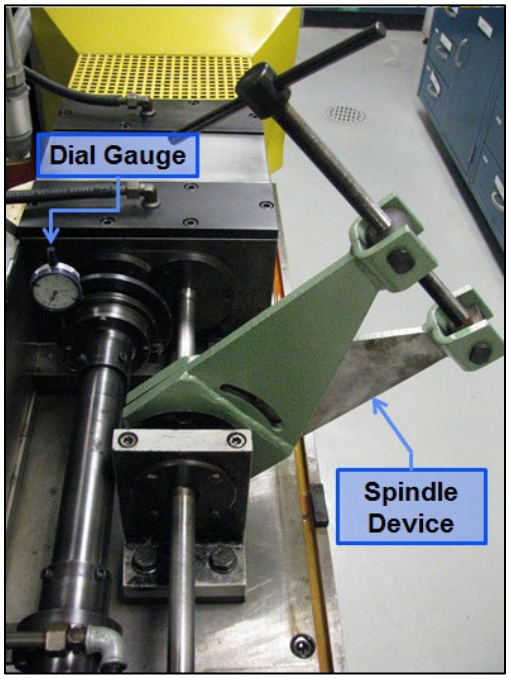


FIGURE 4. Placement of spindle and dial gauge.

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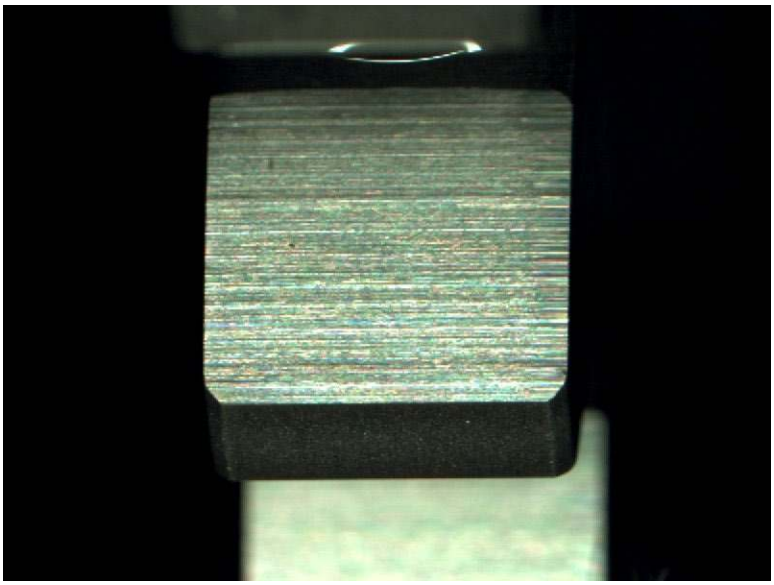


FIGURE 5. Microscope photograph of a single gear tooth.

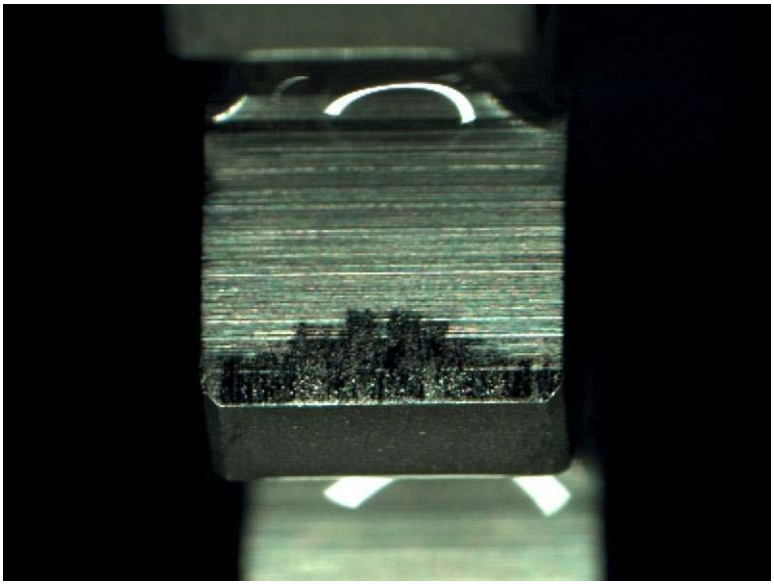


FIGURE 6. Scuffed gear tooth.

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FIGURE 7. Total tooth flank area compared to scuffed area.

Method Prepared By:
Air Force—AFRL/RQTM – 2019

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Method 7506.1
December 18, 2025

7506.1 – GEAR LUBRICANT COMPATIBILITY FOR STRYKER VEHICLE LIMITED SLIP DIFFERENTIAL CLUTCHES

1. SCOPE

1.1 This method is used for determining the compatibility of axle gear lubricants with the wet clutch limited slip differential (LSD) used in the General Dynamics Land Systems Stryker vehicle. Compatibility is assessed through frictional performance and the lubricants ability to mitigate stick-slip or chatter during clutch slip events.

2. SUMMARY

2.1 An SAE No. 2 Friction Test Machine is used to drive a single friction element (1 friction disk against 2 steel reaction plates) at a controlled temperature, speed, and application pressure. The friction element is operated at steady state conditions for 1 hour intervals to age the fluid and friction disk. At the completion of each 1 hour aging segment, the drive motor is stopped and a performance test is completed. The performance test is comprised of an increasing-decreasing speed ramp where the friction response and reaction torque is monitored to assess frictional stability. A cycle consists of one aging and one performance test. Candidate tests consist of 100 cycles, or until failure, whichever occurs first.

3. SAMPLE SIZE

3.1 A total of 1.9 liters (0.5 gallons) of candidate lubricant is required to both pre-soak the friction disk prior to assembly and fill the test head of the SAE No. 2 friction machine.

3.1.1 The SAE No. 2 friction machine requires 1 liter (0.26 gallons) of test lubricant.

4. REFERENCES, STANDARDS, and APPARATUS

4.1 References

4.1.1 SAE Standard J2360, Automotive Gear Lubricants for Commercial and Military Use

4.2 Test Apparatus

4.2.1 The test apparatus consists of an SAE No. 2 Friction Test Machine fitted with specially machined adaptors that enable the installation of a single friction element representative of those used in the GDLS Stryker vehicle. A friction

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element is defined, in this case, as a single friction disk, and two steel reaction plates. The SAE No. 2 Friction Test Machine consists of a drive motor, test head assembly, and the appropriate ancillary equipment to control temperature, speed, and application pressure of the friction element. The friction element is installed into the test head with the inner diameter splines of the friction disk coupled to the input shaft, and powered by a variable speed AC motor. The steel reaction plates are installed into the test head on both sides of the friction disk to create the working friction surface, and the outer lugs of the reaction plates are coupled to the outer diameter of the test head housing. The test head contains a piston that is used to apply controlled application pressure to compress the reaction plates and friction disk together. The test head is mounted to the test stand through bearings which allow the input shaft and test head to rotate relative to one another, while the test head is coupled to the test stand through a reaction arm and load cell to measure the resulting reaction torque from the friction element. This reaction torque and the physical geometry of the friction elements are used to calculate the resulting coefficient of friction, and assess friction performance/stability of the friction elements during slip events.

4.3 Test Stand Detailed System Description

4.3.1 The SAE No. 2 Friction Test Machine used for the LSD compatibility evaluation has a total of three different input motors that can be used to rotate the friction components under test, depending on the input speed and power requirements. For the LSD evaluation, the medium speed motor is used exclusively. It is a 3 hp variable frequency drive controlled AC motor with a maximum speed of 3450 rpm, and is coupled to the input shaft through a 15:1 reduction gear box, giving the friction components an effective speed range of 0 to 230 rpm.

4.3.2 Input speed is measured using a 200 pulse per revolution incrementing encoder (model # H210200104013) coupled to the input shaft of the clutch. The frequency output signal of the encoder is processed through a 2.5 kHz analog to digital converter module (model # SCM5B45-1658).

4.3.3 Reaction torque is measured using a 500 pound rated load cell coupled between the friction machines test head reaction arm and the test stand base.

4.3.4 Temperature control of the test head is provided by a total of 3 kW internally housed fire rod heater (12 individual 250 W heaters), and cooling passages that allow for the circulation of controlled flow chilled process water. A control system built into the data acquisition system regulates the heaters and process water flow to maintain the desired test temperature.

4.3.5 Clutch application pressure is regulated through a pneumatic control system that is maintained at the specified pressure to achieve the desired clutch application pressure based on the surface area of the apply piston. Application and release of the apply piston is controlled through electro-pneumatic solenoids.

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4.4 Instrumentation

4.4.1 The following parameters are the minimum required measurements to be recorded during testing:

- Input speed
- Application Pressure
- Fluid Temperature
- Front & Rear Reaction Plate Temperature
- Reaction Torque

4.5 Data Acquisition

4.5.1 A data acquisition and control system must be utilized to simultaneously record all testing parameters and provide speed, application pressure, and temperature control for the friction components under test. The data acquisition system must be capable of logging data at the specified 1 Hz for the steady state aging segment of the test, and 500 Hz for the performance segments. It is recommended that the data acquisition system be able to monitor and control the system at an update rate of 1000 Hz to ensure precise control of the system.

4.5.2 The data acquisition and control system must also be capable of controlling limits for over/under temperature, and over/under torque and speed.

4.6 Test Stand Setup

4.6.1 A detailed description of the SAE No. 2 friction machine can be found in SAE J286. A picture is included in Figure 1 for reference. The test head is pointed toward the operator.

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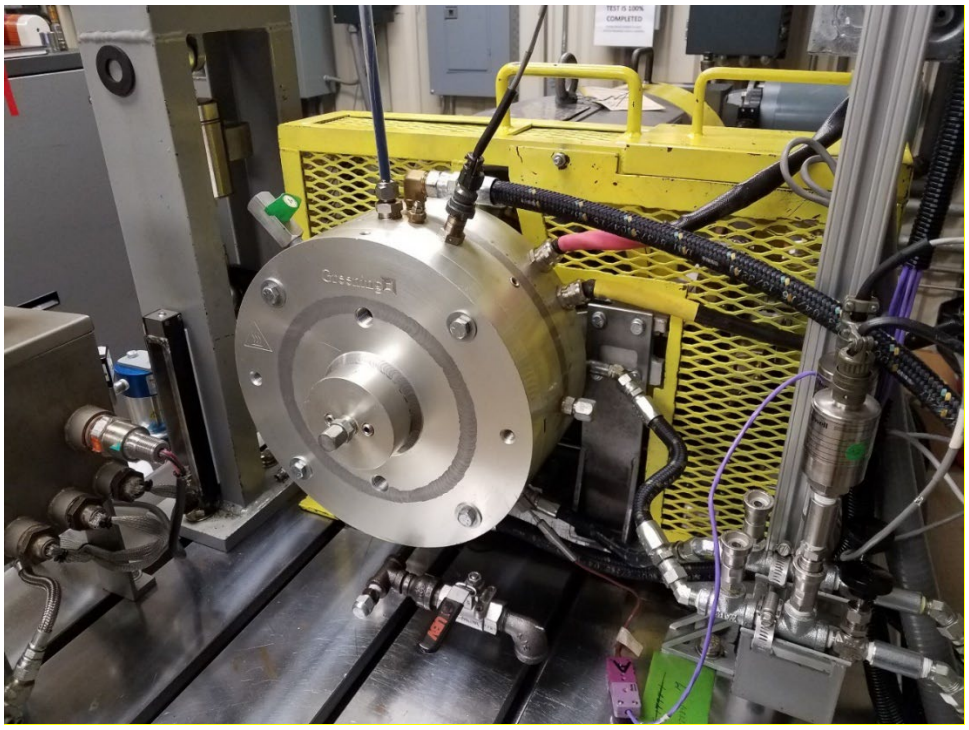


Figure 8: Front view of SAE NO. 2 friction machine, including test head

4.6.2 The test stand utilizes a gravity flow oil circuit setup for this testing. The lubricant reservoir should be located above the test head.

4.6.3 An optional thermocouple may be mounted into a drilled hole on each steel reaction plate using oil resistant epoxy. The wires are routed out of the test head through the lubricant reservoir cap. Figure 2 shows the location of the thermocouple in relation to the reference tang.

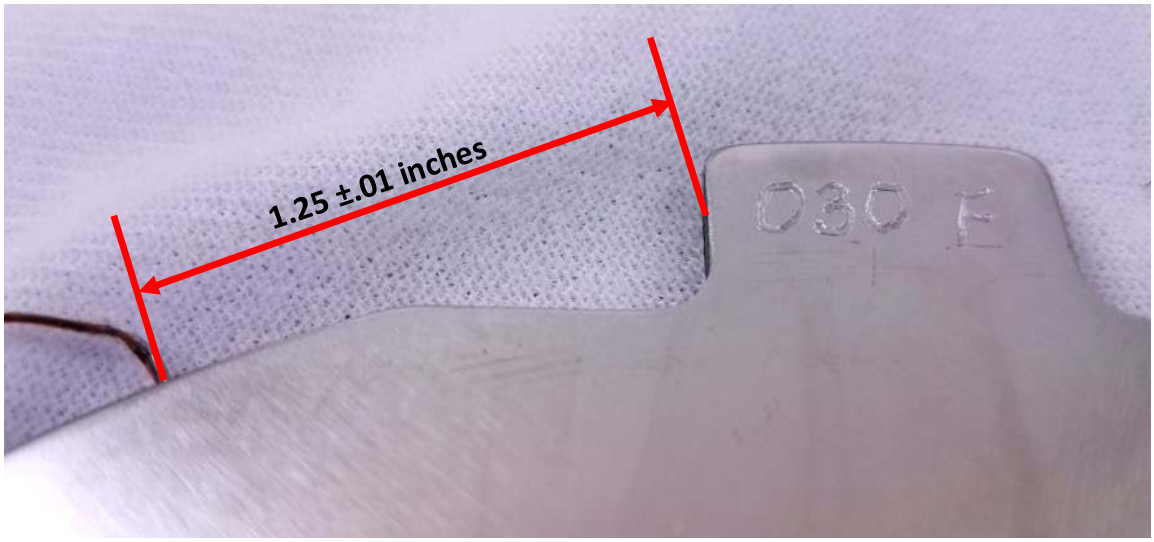


Figure 9: Thermocouple location in reference to labelled tang on Front Plate (Optional)

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4.6.4 The top-most tangs should be marked on each reaction plate. As a result of the stamping process that is used to manufacture the reaction plates, the side to be marked is the side with the rolled edge. Historically, this is the smoother side, but it can have higher waviness. In order to improve test consistency, the flat (fractured) sides are run against the friction disk. These markings and orientations can be seen in Figures 2-5.



Figure 10: Thermocouple location in reference to labelled tang on Rear Plate (Optional)

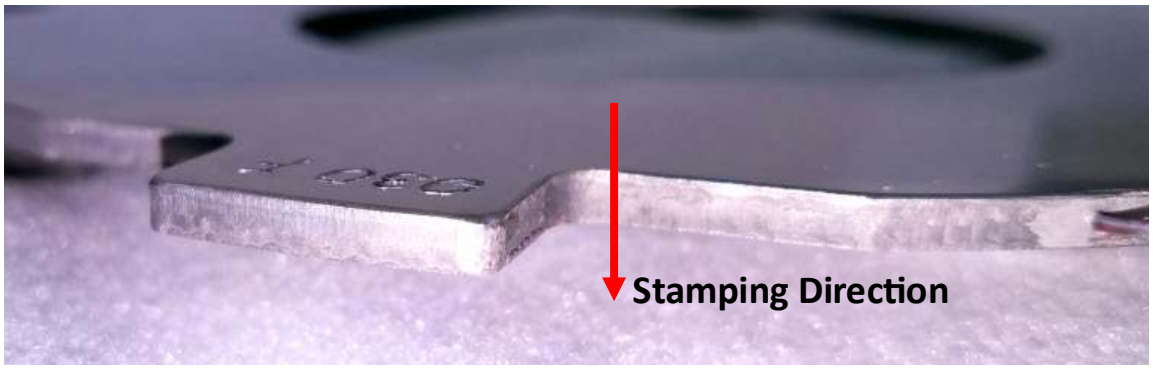


Figure 11: Front Plate with labelled tang and orientation noted

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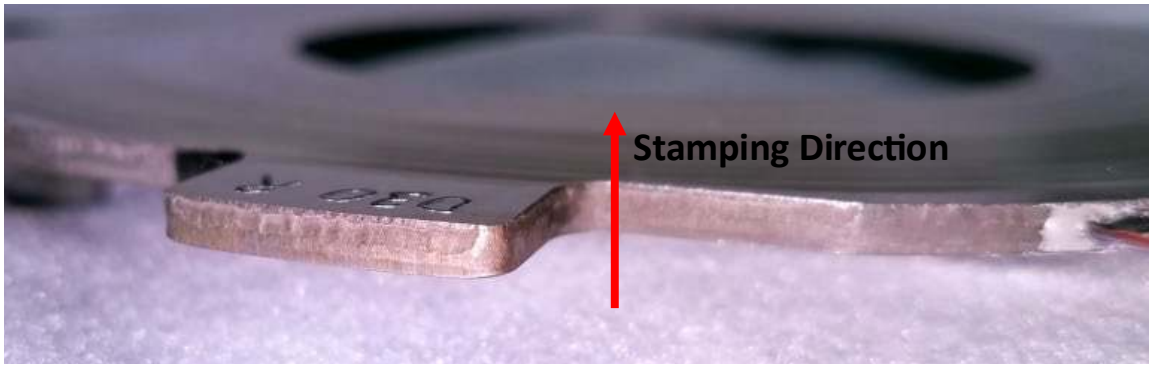


Figure 12: Rear Plate with labelled tang and orientation noted

5. MATERIALS

5.1 Friction Element Description

5.1.1 The friction components used for this evaluation are commercially equivalent components to those used in the Stryker vehicle axle produced by Eaton Corporation. Alternative components were required as the OE Stryker axle components inner diameter is too small to fit over the input shaft of the SAE No. 2 Friction Test Machine.

5.1.2 The friction plate is Eaton PN 91393, with a minor diameter of 75.20 mm. One friction plate is required per test.

5.1.3 The steel reaction plate is Eaton PN 90667, with a major diameter of 157 mm, and an overall outer lug diameter of 173 mm. Two reaction plates are required per test.

5.2 Reference Fluid

5.2.1 Reference fluid is available from the U.S. Army Combat Capabilities Development Command - Ground Vehicle System Center, 6501 E. 11 Mile Road, Warren, MI 48397-5000, ATTN: Fuels and Lubricants, FCDD-GVS-IEE MS110 or by emailing usarmy.detroit.ccdc-gvsc.mbx.pol-help@mail.mil.

6. PROCEDURE

6.1 Start of Test

6.1.1 Install thermocouples into the steel reaction plates with epoxy cement, as shown in Figures 2 and 3. While this step is optional, recording the plate temperatures can provide additional data for understanding oil failures and frictional anomalies.

6.1.2 Measure and document the outer and inner thickness of the friction disk at 3 equidistant locations each (e.g., 0°, 120°, 240°).

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6.1.3 Solvent clean the friction and steel plates to ensure no residue is left behind from manufacturing or handling. It is appropriate to use heptane, hexane, or a mixture of iso-octane and propanol. Stoddard solvent is not acceptable for this step.

6.1.4 Air dry the parts after cleaning.

6.1.5 Soak the friction disk in the test lubricant, at room temperature, for a minimum of 1 hour prior to assembly in the SAE No. 2 machine.

6.1.6 Rinse the adaptors and test head with heptane prior to assembly and allow to air dry.

6.1.7 Measure and document the clutch pack clearance upon installation.

6.1.7.1.1. To ensure proper orientation, the plate marked “xxxR” should be the rear plate, and is the rearmost plate which is installed into the test head first. The numbers engraved on the tang should face outward from the head, and be visible upon installation.

6.1.7.1.2. Similarly, the plate marked “xxxF” should be the front plate, and is the outermost plate which installed into the test head last. The numbers engraved on the tang should face outward from the head, and be visible upon installation.

6.1.8 Fill SAE No. 2 head with 1.0 liter of the test lubricant and mark the fill level on the gravity-fed lubricant system. Document the lubricant charge volume.

6.2 Cycle Testing

6.2.1 Each test runs for 100 cycles, or until failure, whichever occurs first.

6.2.2 Cycle Definition

6.2.2.1 Aging segment

6.2.2.1.1. Set clutch application force to 1000 lbf.

6.2.2.1.2. Run for 60 minutes at the following conditions: 40 rpm, 93.3 °C (200 °F), data acquisition rate 1 Hz.

6.2.2.1.3. Ramp to 0 rpm and hold for 2 minutes.

6.2.2.2 Performance segment

6.2.2.2.1. Maintain clutch application force and temperature (i.e., 1000 lbf and 93.3 °C).

6.2.2.2.2. Ramp from 0 rpm to 80 rpm to 0 rpm at 4 rpm/s. Total step time should be 40 seconds, data acquisition rate 500 Hz.

6.2.2.2.3. Hold for 2 minutes at 0 rpm.

6.2.2.2.4. Maintain clutch application force and temperature (i.e., 1000 lbf and 93.3 °C).

6.2.2.2.5. Motor instant on 0.7 rpm to measure static breakaway

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6.2.2.2.6. Hold for 2 minutes at 0 rpm.

6.3 End of Test

6.3.1 The test shall be stopped if the chatter, as defined in section 7, during the aging segment exceeds a value of 0.83.

6.3.2 At the completion of the test, the stand will stop and release pressure on the clutch. Once stopped, the stand can be locked out safely for post-test activities.

6.3.3 Prior to disassembly, remove an 8 oz. sample of test oil and label it with the appropriate test markings. The remaining oil can be discarded.

6.3.4 Measure and document the post-test clutch pack clearance.

6.3.5 Measure the friction disks and document the post-test thickness. Measure and document the outer and inner thickness of the friction disk at 3 equidistant locations each (e.g., 0°, 120°, 240°).

6.3.6 Remove the post-test steel plates and friction disk, and bag them for retention. Tag the parts with the appropriate date and test information.

6.3.7 Clean the residual oil from the test head with solvent to prepare for the next test.

7. CALCULATIONS

7.1 Chatter

7.1.1 The chatter of the clutch pack is a function of the static and dynamic coefficient of friction values, and is indicated by variability in the torque signal at a given set of conditions. The chatter value during the aging segment for this test should be calculated as follows:

$$Chatter = \ Moving\ Average\ (Chatter\ Indicator), period\ is\ 60\ seconds, rate\ is\ 1\ Hz$$

$$Chatter\ Indicator = 1\ if\ Moving\ Standard\ Deviation\ (Torque)\ is > 3$$
$$Chatter\ Indicator = 0\ if\ Moving\ Standard\ Deviation\ (Torque)\ is\ \leq 3$$

$$Moving\ Standard\ Deviation\ (Torque), period\ is\ 10\ seconds, rate\ is\ 1\ Hz$$

$$Torque\ has\ units\ of\ N \cdot m$$

7.2 Coefficient of Friction

$$COF = \frac{Torque}{Apply\ Pressure * .0151 * \frac{4}{3} * \frac{(\frac{Plate_{OD}}{2})^3 - (\frac{Plate_{ID}}{2})^3}{(\frac{Plate_{OD}}{2})^2 - (\frac{Plate_{ID}}{2})^2}}$$

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$Apply\ Pressure\ has\ units\ of\ kPa$

$Plate_{xx}\ have\ units\ of\ mm, and\ are\ in\ reference\ to\ the\ running\ surface\ of\ the\ clutch$

7.3 Other calculations that may be performed for further analysis are:

7.3.1 Moving Standard Deviation of Torque for the speed sweep

$Moving\ Standard\ Deviation\ (Torque), period\ is\ 1\ second, data\ acquisition\ rate\ is\ 500\ Hz$

8. REPORTING

8.1 Report the cycle number at which the chatter indicator first exceeds 0.83. If the chatter indicator does not exceed 0.83 during any cycle, report "No Chatter".

8.2 Provide a graph of Coefficient of Friction (COF) vs speed during the last half of the speed sweep (i.e., ramp down) for all cycles.

8.2.1 From the tabulated data for the graph, report the static and dynamic COF values. Report the average dynamic COF for the speed range of 40 to 80 rpm.

8.3 Provide a graph of torque and speed vs time during the speed sweep for all cycles.

8.4 Provide a graph of Moving Standard Deviation of Torque vs time may for all cycles.

9. PRECISION

9.1 Precision data has not been developed for this method.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
18 Dec 2025	Revised section 4.6.2 to make the installation of thermocouples in the steel reaction plates optional.	Thermocouples are not essential requirement for conducting this test. They can be installed as an option.	1	Army - AT
18 Dec 2025	Revised figure 2 and figure 3 label to include "(Optional)".	To be consistent with the changes to section 4.6.2 and to clarify that the installation of the thermocouples is optional.	1	Army - AT
18 Dec 2025	Revised sections 6.3. Removed section 6.4 and added these items to revised section 6.3.	Clarify end of test requirements.	1	Army - AT

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18 Dec 2025	Revised section 7 Calculations, changing COF calculation to section 7.2, making it a mandatory reporting item.	Clarify calculation requirements. Improved alignment with requirements of SAE J2360.	1	Army - AT
18 Dec 2025	Revised section 8 Reporting. Made the graph of COF vs speed and the average of the dynamic COF for the speed range of 40 to 80 rpm mandatory reporting items.	Clarify reporting requirements. Improved alignment with requirements of SAE J2360.	1	Army - AT

Method Prepared By:
Army - AT – 2025

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Method 7507.0
April 1, 2015

7507.0 – ISENTROPIC BULK MODULUS OF AVIATION TURBINE FUELS AND DIESEL FUELS

1. SCOPE

1.1 This method describes the measurement of isentropic bulk modulus via speed-of-sound and density at temperatures ranging from 30°C - 80°C and applied pressures of 1000 - 18,000 psi. This method has been applied successfully to aviation turbine fuels and diesel fuels composed of petroleum, synthetic, and alternative feedstocks.

2. SUMMARY

2.1 This method utilizes a custom built apparatus that simultaneously measures speed-of-sound via a pulse/echo technique and density via a high temperature/pressure process densitometer. The test fuel is pumped into the system, purged of air, and sealed in a closed loop. An oven is used to equilibrate the fuel to the desired test temperature and then a pressure curve is generated by slowing reducing the volume of the system using a pressure generator. An oscilloscope is used to measure the time-of-flight of the acoustic signals generated by an acoustic transducer. Density is simultaneously measured and the bulk modulus is computed.

3. SAMPLE SIZE

3.1 The minimum recommended sample size is 500 mL.

4. STANDARDS AND APPARATUS

4.1 CAUTION – New operators should review the operating manual thoroughly before use of this equipment. This system operates at high pressures and temperatures on combustible samples.

4.2 Isentropic Bulk Modulus Apparatus consisting of the following components.

4.2.1 Oven – range of 30°C - 100°C, stable within 1°C.

4.2.2 Data Acquisition System (DAQ) – for recording pressure, volume, temperature, and density data.

4.2.3 Bulk Modulus Excel Toolkit.

4.2.4 Digital Oscilloscope – for measuring pulse echoes.

4.2.5 High Pressure Generator (rated for 60K psi).

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4.2.6 Flow loop and valving (rated for 60K psi).

4.2.7 High Pressure Cell (rated for 30K psi max).

4.2.8 Pressure Transducer (50K psi max).

4.2.9 Pulser/Receiver Module – to drive the acoustic transducer.

4.2.10 Acoustic transducer.

4.2.11 Process Densitometer (rated for 20K psi at 100°C max).

4.2.12 Densitometer Control/Display.

4.2.13 Resistive Temperature Device (RTD).

4.2.14 Two (2) 500 mL beakers or sample bottles for sampling and waste.

4.2.15 Secondary containment for beakers/sample bottles.

4.2.16 Peristaltic pump and fuel compatible hose – for pumping fluid into the system.

4.2.17 Sonicator – optional, if de-gassing sample prior to testing is preferred.

5. MATERIALS

5.1 Heptanes, technical grade – for flushing the system.

5.2 n-Pentane, HPLC grade – for calibration/validation of density and speed-of-sound.

5.3 Isopropyl Alcohol, Certified ACS Plus - for flushing the system after the use of water.

5.4 Water, deionized, ultra-filtered (DIUF) – for calibration of the densitometer.

5.5 Nitrogen, dry – for leak checking or purging and drying the system (typically following an isopropyl alcohol rinse).

6. PROCEDURE

6.1 Calibration

6.1.1 System Calibration/Validation – The bulk modulus apparatus is generally calibrated/validated as a complete system. The densitometer is calibrated using n-pentane and water at multiple temperatures and pressures according to the operating manual. The speed-of-sound is calibrated using n-pentane. Overall, the system is validated by measuring the speed-of-sound and density of n-pentane at multiple pressures and a fixed temperature and plotting the

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results against the expected reference values. Most of the components are not user serviceable and must therefore be returned to the manufacturer for major repairs.

6.1.2 Components

6.1.2.1 Oven – Once every six (6) months, compare the oven output to a calibrated secondary reference at several temperatures.

6.1.2.2 Data Acquisition System (DAQ) – adjust engineering units as needed following individual calibration of pressure and temperature components.

6.1.2.3 Digital Oscilloscope – per the manufacturer's recommendations.

6.1.2.4 Pressure Transducer (50K psi minimum) - Once every twelve (12) months, it is recommended that the transducer be removed and verified/calibrated by a qualified calibration lab.

6.1.2.5 Pulser/Receiver Module – per the manufacturer's recommendations.

6.1.2.6 Process Densitometer - per the manufacturer's recommendations.

6.1.2.7 Densitometer Control/Display - per the manufacturer's recommendations.

6.1.2.8 Resistive Temperature Device (RTD) - Once every twelve (12) months, it is recommended that the RTD be removed and verified by a qualified calibration lab.

6.1.3 System Operation

6.1.3.1 Turn on the main instrument cabinet and allow the density display to initialize.

6.1.3.2 Turn on the oscilloscope and allow it to initialize.

6.1.3.3 Start the data acquisition process on the oscilloscope.

6.1.3.4 Load the sample into the instrument.

6.1.3.4.1 Ensure that the system is at atmospheric pressure.

6.1.3.4.2 Open all six valves (see FIGURE 1).

6.1.3.4.3 Collect approximately 250 mL of sample in a beaker or other suitable container.

6.1.3.4.4 With the oven turned off and the door open, connect the tubing from the peristaltic pump to the inlet fitting. The other end of the tubing connected to the peristaltic pump will contain a stainless siphon tube. Place this in the beaker of fuel.

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6.1.3.5 Connect another piece of tubing to the outlet fitting. If purging the system is desired, then place the stainless outlet tube into a waste container; otherwise, place the stainless outlet tube into the same beaker of fuel for recirculation. The following assumes that a purge of the previous sample with the next sample is necessary so the outlet tube should be placed in a waste container.

6.1.3.6 Turn on the peristaltic pump at full speed and ensure that sample is flowing in the proper direction toward the inlet valve (note the pump is reversible).

6.1.3.7 Close Valve C to force sample through the densitometer leg and wait for sample to exit the other end.

6.1.3.8 Once there is sample coming out of this leg, let the pump run for a few seconds to purge out the previous sample.

6.1.3.9 Close Valve A and then immediately open Valve C. This will now purge the other leg of the system.

6.1.3.10 Once there is sample coming out of this leg, let the pump run for a few seconds to purge out the previous sample.

6.1.3.11 Repeat these steps several times, alternating between one leg and the other always making sure to close one valve before opening the other.

6.1.3.12 Open Valve A and Valve C allowing sample to flow through both paths.

6.1.3.13 Cycle the high pressure generator slowly (~1 full turn every 4-5 seconds) until the piston is fully retracted (rising stem extended rearward) and then cycle the piston again to the forward position. This will flush out the piston chamber. Repeat once.

6.1.3.14 The system should now be adequately flushed. Place the stainless outlet tube into the 250 mL beaker containing the sample. The sample will now recirculate. Add additional sample to the beaker as needed if additional flushing is warranted.

6.1.3.15 At this point, additional purging/flushing can be achieved by alternate flushing of the individual legs and/or cycling of the pressure generator as described above. The key is to ensure that no air bubbles can be seen exiting the outlet fitting.

6.1.3.16 When ready to continue, slowly cycle the pressure generator until the piston is approximately 3/4 retracted (rising stem extended).

6.1.3.17 Once all of these steps have been carried out and no air bubbles are seen exiting the system, close the outlet valve, then immediately close the inlet valve, and turn off the peristaltic pump in that order.

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6.1.3.18 To test whether the sample fill and air purge was successful, very slowly cycle the piston forward (clock-wise) while monitoring the pressure on the mPDS-5 display. If the system is sufficiently packed, then an immediate increase in pressure will be observed (having turned the pressure generator wheel only a few degrees). If not, then additional purging is required (re-open the inlet and outlet valves, turn on the peristaltic pump, and continue the purge process in recirculation mode).

6.1.3.19 If satisfied that the system is ready to run, with the inlet and outlet valves closed, disconnect the tubing from the inlet and outlet valves and place a Swagelok cap on each fitting.

6.1.3.20 Close the oven door.

6.1.4 Turn on the oven, set the temperature, and allow the system to equilibrate (~ hours).

6.1.5 Turn on the Pulser/Receiver.

6.1.6 Open a copy of the Bulk Modulus Excel Toolkit and begin acquiring data.

6.1.7 Increase the system pressure to the target value and allow to stabilize (~ minute(s)).

6.1.8 Measure the time-of-flight of the acoustic pulse on the oscilloscope and record the data into the Excel Toolkit.

6.1.9 Repeat steps 8-9 for each target pressure.

6.1.10 When complete, return the system pressure to baseline conditions (~atmospheric).

6.1.11 Turn off the Pulser/Receiver (when not in use).

6.1.12 If desired, change the oven temperature to the next target value, allow to equilibrate, and repeat steps 6-12.

6.1.13 Otherwise, set the oven temperature to ~25 °C and allow the system to cool (~ hour(s)).

6.1.14 Flush the system.

6.1.15 Exit out of all software and power down the oscilloscope.

6.1.16 Turn off the main instrument cabinet.

7. CALCULATIONS

7.1 Isentropic Bulk Modulus is calculated as $\beta=\rho c^2$, where β is the bulk modulus in Pa, ρ is the density in kg/m^3, and c is the speed-of-sound in m/s.

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8. REPORTING

8.1 For a given temperature and pressure, report the following:

8.1.1 Test Temperature in °C to the nearest 0.1°C.

8.1.2 Isentropic Bulk Modulus in psi to the nearest integral value.

8.1.3 Speed-of-Sound in m/s to the nearest 0.1 m/s.

8.1.4 Density in kg/m^3 to the nearest 0.1 kg/m^3.

9. PRECISION

9.1 Precision data has not been developed for this method.

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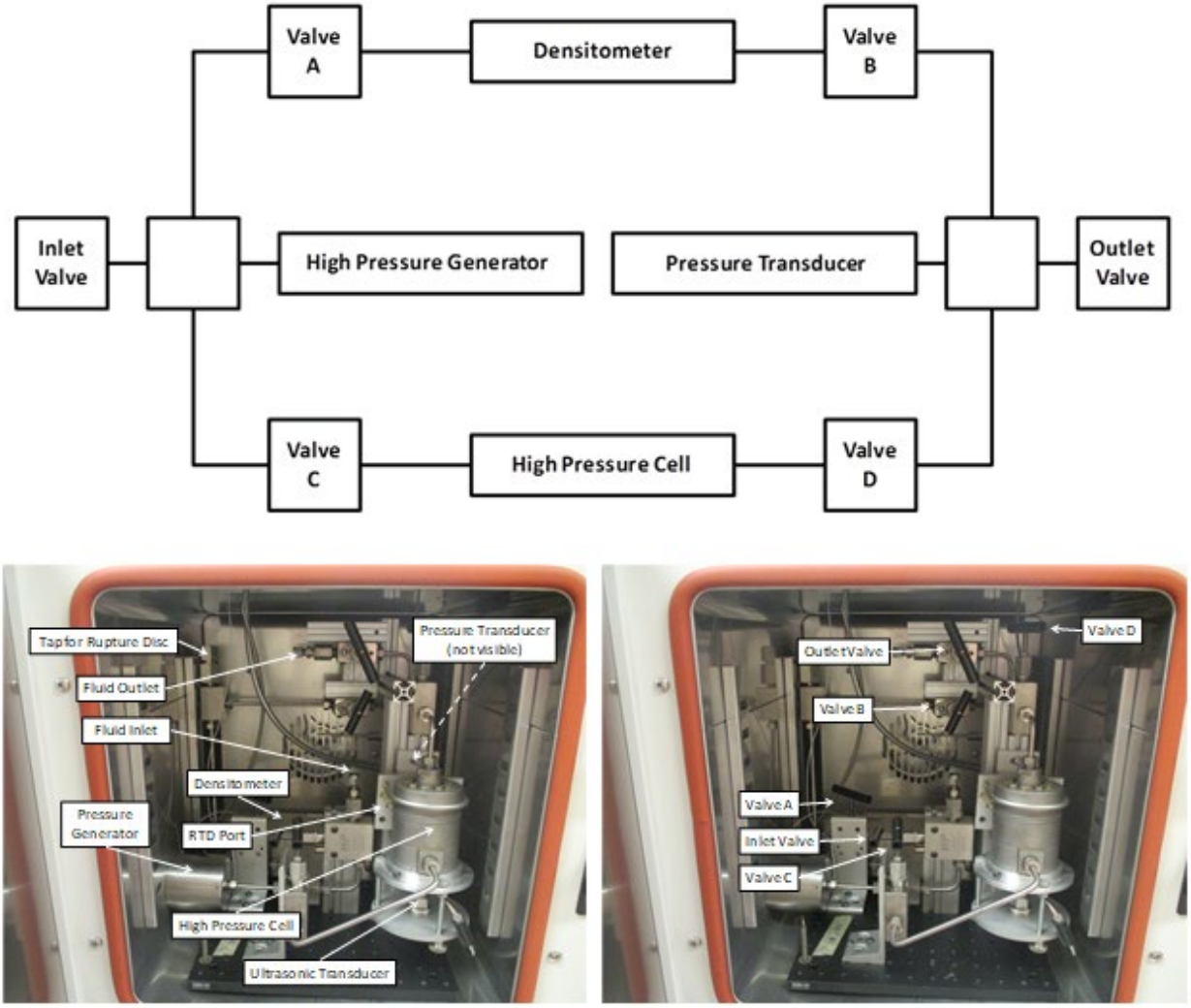


FIGURE 1. Instrument flow loop schematic.

Method Prepared by:
Army – AT- 2019

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Method 7508.1
December 23, 2025

7508.1 – METHOD FOR DETAILED HYDROCARBON ANALYSIS OF MIDDLE DISTILLATE FUELS BY TWO-DIMENSIONAL GAS CHROMATOGRAPHY

1. SCOPE

1.1 This method describes detailed hydrocarbon analysis of finished middle distillate fuels (e.g., aviation turbine fuel, diesel fuel, and marine gas oil) according to an analytical method based on two-dimensional gas chromatography with either flame ionization detection (GC×GC-FID) or simultaneous FID and mass spectrometry (MS) (GC×GC-FID/MS). The hydrocarbon composition is reported for a series of distinct compound classes (Tier 2), which are further characterized into subclasses based on carbon number (Tier 3). The compositional data gathered from this method is reported in relative mass % at Tier 2 and Tier 3 levels. Tier 1, defined as paraffins, cycloparaffins and aromatics, is the broadest level of categorization and aims to characterize by chemical family, thus, is not a reporting requirement for this test method. Table 1 lists the anticipated Tier 2 concentration ranges for typical petroleum derived middle distillate fuels based on previously analyzed fuels across multiple DoD labs over a 12 year period. Synthetic aviation turbine fuels (SATF) can also be analyzed per this method with additional instrumental and analytical considerations (see Appendix A).

Table 3. Typical Concentration Ranges for Tier 2 Compound Classes

Hydrocarbon Compound Class	Range, Mass %
n-Alkanes	13 - 24
Isoparaffins	22 - 34
Monocycloparaffins	20 - 31
Dicycloparaffins	6 - 17
Tricycloparaffins	0 - 4
Alkylbenzenes	8 - 15
Cycloaromatics	1 - 12
Diaromatics	0 - 5
Triaromatics (PAHs)	0 - 2

2. SUMMARY

2.1 A fuel sample is injected into a two-dimensional gas chromatograph (GC×GC), where it is vaporized and separated using two (2) GC capillary columns connected sequentially by a modulation system. The modulation system, either thermal or pneumatic/flow, captures fuel components as they elute from the primary column and reinjects them onto the second column. As fuel components elute from the second column, they are detected by a FID. Orthogonal stationary phases are chosen for the column set to provide appropriate separation of the fuel components within the chromatogram. A

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template is superimposed over the chromatogram to delineate regions corresponding to each reportable subclass of chemical compounds. Chromatographic peaks falling within each template region are integrated and summed to provide the total FID response area (volume) associated with each compound subclass. These subclass areas (volumes) are corrected using a unit conversion factor for differences in the carbon/hydrogen ratio so that the relative mass percent of the compound is reported.

3. SAMPLE SIZE

3.1 The minimum recommended sample size is 1 mL.

4. APPARATUS

4.1 Commercially available gas chromatograph equipped with the following components:

4.1.1 Inlet, standard split/splitless

4.1.2 Oven, –40 to 350˚C at 1.5 to 10˚C min^{-1} programming rate

4.1.3 Liquid autosampler

4.1.4 Pressure control module

4.1.5 GC columns, two capillary columns with orthogonal stationary phases in either normal (nonpolar to polar) or reverse (polar to nonpolar) phase configuration.

4.1.5.1 Nonpolar, a column with 95-100% polydimethylsiloxane (PDMS) stationary phase (such as Agilent DB-5, DB-1, Restek Rtx-1, Supelco SPB-1, Phenomenex ZB-1, etc.) or other similar phase that separates hydrocarbon compounds based on volatility.

4.1.5.2 Polar, a column with approximately 50% diphenyl group substituted (50% unsubstituted) PDMS (such as Agilent DB-17, Restek Rxi-17, Supelco SPB-17, Phenomenex ZB-50, etc.) or other similar phase that separates hydrocarbon compounds based on compound polarity or hydrogen bonding.

4.1.5.3 The type of column set, normal versus reverse phase, will dramatically alter the order of compound elution and thus the appearance of the resulting chromatogram, Figure 1, and the subsequent integration templates required. Integration templates are specific to each column configuration.

4.1.6 GC×GC modulator system, thermal or pneumatic/flow

4.1.7 Flame ionization detector (FID)

4.1.8 Mass spectrometer (MS) (optional)

4.1.9 Table 2 lists system configurations used by three laboratories included in the development of this method. The specific hardware and operating parameters can vary provided the instrument can resolve the components of the fuel according to the compound classes (Tier 2) and subclasses (Tier 3) listed in Table 4.

4.1.10 GC×GC processing software, capable of background subtraction, peak (blob) identification and integration, and templating or grouping of peaks (such as GC Image™, ChromaTOF®, Chromspace®, etc.)

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Table 4. Example System Configurations Used to Develop Test Method

Component	System 1	System 2	System 3
Gas Chromatograph	Agilent 7890	Agilent 7890	Agilent 7890
Column 1	Restek Rxi-17	Restek Rxi-17	DB-5MS
Column 2	Restek Rxi-1MS	Restek Rxi-1MS	DB-17MS
GC×GC modulator	LECO Pegasus 4D Thermal Modulator	ZOEX Thermal Modulator (ZX1)	Agilent Reverse Fill/Flush CFT Differential Flow Modulator
Detector	Agilent FID	Agilent FID	Agilent FID and MSD
Column Arrangement	Reverse Phase	Reverse Phase	Normal Phase
Processing Software	GC Image™	GC Image™	GC Image™

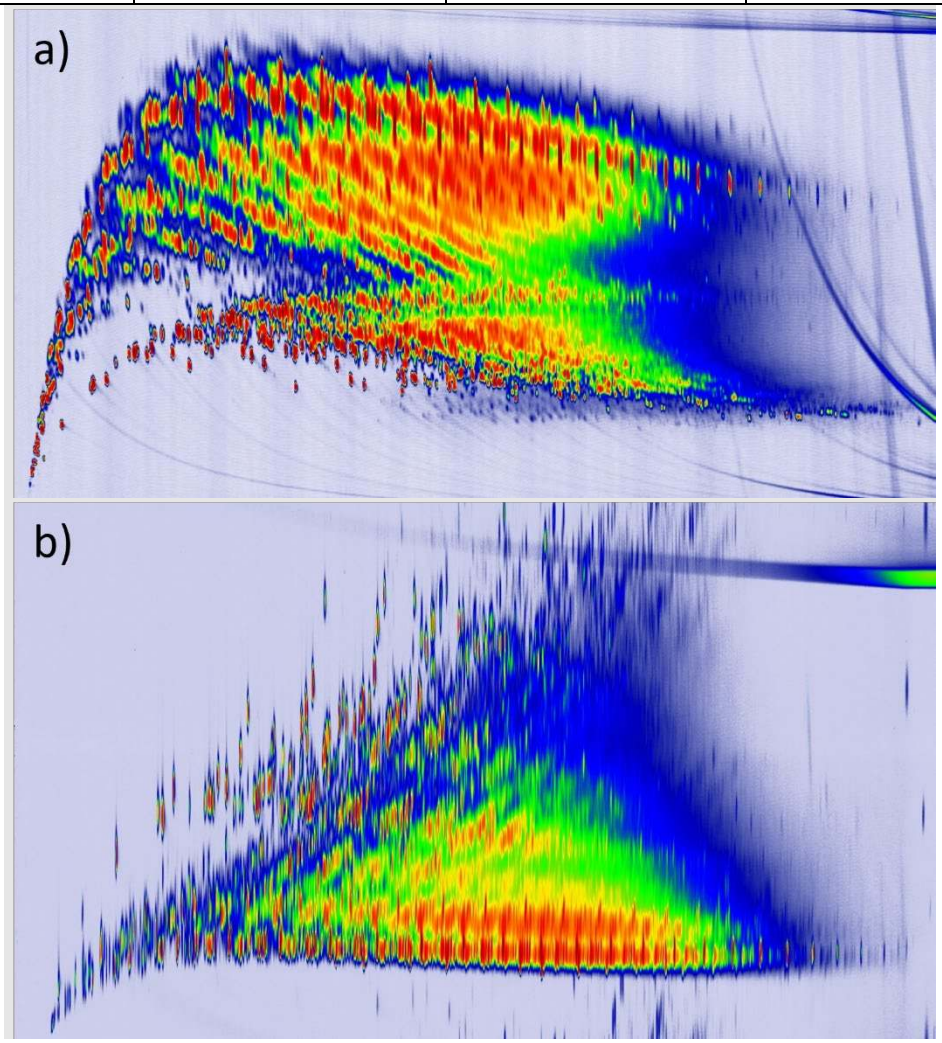


Figure 2. Chromatograms of the reference fuel using a) reverse phase, or b) normal phase column sets

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5. MATERIALS AND REAGENTS

5.1 *Hydrocarbon reference standard*. A gravimetric blend containing a broad range of hydrocarbons that are representative of the reportable compound subclasses, at relevant mass percent concentrations.[1] The hydrocarbon reference standard is used to assess the comprehensiveness of the method during initial instrument setup and/or after major system modifications. The standard can be used to aid development of integration templates for data analysis when mass spectral data is not available. Table 5 lists all 42 compounds present in the gravimetric blend used for the development of this method; included in the table are the compound concentrations, and the Tier 3 subclass elution regions that the operator should expect the compounds to elute. General instrument operation shall be confirmed by using the hydrocarbon standard to conform to the accuracy statement in Section 9.2.

5.2 *Reference fuel sample*. A diesel fuel selected to serve as a known reference fuel sample.[2] The reference fuel sample is used to confirm instrument configuration, template construction, template alignment and data analysis tools more precisely than the hydrocarbon reference standard. Specifically, the integration template should be finalized by confirming the data generated meets the precision statement specified in Sections 9.3.1 and 9.3.2.

5.3 *Quality control (QC) sample.* A stable, representative fuel sample for a given laboratory. The QC sample may be the reference fuel sample or a different fuel in the middle distillate region, e.g., Jet A. The QC sample shall be run regularly and used to assess the integrity of the method and analysis.

5.4 *Solvents*. Wash solvent(s), such as chromatography grade dichloromethane, isopropyl alcohol, acetone, or carbon disulfide, to rinse sample injection syringes between uses. Dilution of the sample should be avoided.

5.5 *Samples*. This method is intended primarily for analysis of aviation turbine fuels such as JP-5, JP-8, Jet A, Jet A-1, and F-24, as well as diesel fuels such as F-76, MGO, or D-2. This method can also be applied to SATF's with additional considerations (Appendix A).

[1]The sole source of supply of the hydrocarbon reference standard (PNA in AVTUR Gravimetric blend, PAC PN 20001.643; available for purchase from Fisher Scientific, PN NC2006242).

[2]The diesel fuel sample used for the development of this method can be obtained by contacting AccuStandard (S-104632N - Ultra-Low Sulfur F76 Diesel Fuel).

6. PROCEDURE

6.1 Standard Operating Conditions

6.1.1 Detailed GCxGC-FID instrument and method parameter are listed in Table 3. Variations from the listed parameters are acceptable provided adequate separation is achieved to meet the specified accuracy (Section 9.3) and precision (Section 9.2) guidelines.

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Table 5. Detailed System Operation Conditions Used to Develop Method

	System 1	System 2	System 3
Injection Volume	0.2 µL	0.05 µL	0.4 µL
Inlet	split ratio 200:1, septum purge 3 mL min^{-1} and temperature of 270 °C	split ratio 500:1, septum purge 3 mL min^{-1} and temperature of 300 °C	split ratio 500:1, septum and temperature of 280 °C
Carrier Gas*	Hydrogen	Hydrogen	Hydrogen
Column 1	DB-17SI, 30 m, 250 µm ID 0.25 µm df	RSI-17SIL MS, 30 m, 250 µm ID, 0.25 µm df	DB-5MS, 40 m, 180 µm ID, 0.18 µm df
Column 2	RXi-1, 2 m, 250 µm ID 0.25 µm df	RXi-1MS, 2.09 m, 100 µm ID 0.10 µm df	DB-17, 5 m, 320 µm ID, 0.25 µm df
Column 1 Flow	1.50 mL min^{-1}	0.5 mL min^{-1}	0.3 mL min^{-1}
Column 2 Flow	Same as Column 1	Same as Column 1	14 mL min^{-1}
Oven Program	40 °C with a 1 min hold, then ramp at 3 °C min^{-1} to 260 °C, with a 1 min hold. Secondary oven offset by +15°C and modulator offset by +30°C	50 °C with a 1 min hold, then ramp at 3 °C min^{-1} to 310 °C. Modulator: 220 °C, with ramp at 3 °C min^{-1} to 380 °C with 35 min hold. No secondary oven present.	35 °C with a 1 min hold, then ramp at 2 °C min^{-1} to 300 °C, hold for 5 min.
Modulator Program	2-stage thermal modulator, 7 s modulation period, 0.6 s hot pulse time, 2.9 s cool time between stages	4.5 s modulation period, 300 ms modulation time and cold flow rate 7.5 slpm.	6.5 s modulation period, 0.22 s inject time, and 0.01 min modulation delay
Detector parameters	Temperature of 280 °C, air flow 450 mL min^{-1}, hydrogen flow 45 mL min^{-1}, helium makeup flow 40 mL min^{-1}	Temperature of 325 °C, air flow 400 mL min^{-1}, hydrogen flow 40 mL min^{-1} (combined), nitrogen makeup flow 25 mL min^{-1}	Temperature of 280 °C, air flow 450 mL min^{-1}, hydrogen flow 45 mL min^{-1} (combined), nitrogen makeup flow 20 mL min^{-1},

* GC systems using helium (He) carrier gas were also evaluated and provided adequate separation. However, no systems using He were included when developing the accuracy and precision statements listed herein.

6.2 Preparation of Apparatus

6.2.1 Prior to analysis of samples, the hydrocarbon reference mixture and reference fuel sample described in Section 5 should be analyzed to verify an appropriate signal-to-noise ratio as follows:

6.2.1.1 Locate the tridecane (n-paraffin C13) peak in the reference fuel sample chromatogram (approx. 1.00 mass %). In the 1D chromatogram from the instrument control software verify that a peak is present with a signal-to-noise ratio of at least 1200 using the 6× standard deviation noise calculation value from the start of the run. Sample injection volume and/or GC inlet split ratio may be adjusted to ensure sufficient signal-to-noise ratio so long as overloading of the GC columns is avoided.

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6.2.2 The integration template shall be drawn to delineate the areas of the 2D chromatogram containing the chromatographic peaks corresponding to each compound subclass as specified in Table 4. Template boundaries are drawn using tandem MS data and/or procedures detailed elsewhere.[3]

[3]Johnson, K.J.; Loegel, T.N.; Metz, A.E.; Wrzesinski, P.J.; Shafer, L.M.; Striebich, R.C.; West, Z.J. *Method for Detailed Hydrocarbon Analysis of Middle Distillate Fuels by Two-Dimensional Gas Chromatography*; NAVAIR Report 2020-361, May 2020.

6.3 Calibration

6.3.1 *Subclass Mass Fraction (Carbon Basis).* The response of the flame ionization detector (FID) is nearly proportional to the number of carbon atoms in a hydrocarbon molecule. The FID area fraction, a_i, can be used directly to represent the carbon mass fraction of each subclass without the necessity of independent detector or compound calibration (see Equation 1). That is to say a_i is the subclass mass fraction on a carbon basis and has the units of: (g carbon *i*)×(g carbon sample)$^{-1}$. Calculation of a_i is shown in Equation 1 where the integrated FID response area for a single subclass *i*, A_i, is divided by the sum of all integrated subclass responses.

Eq 1. $$Mass\ Fraction\ of\ Carbon\ in\ i^{th}\ Subclass\ \approx\ a_i = \frac{A_i}{\sum A_i}\ [=]\ \frac{grams\ c_i}{grams\ c_{total}}$$

6.3.2 *Subclass Mass Percent (Compound Basis).* The subclass mass fraction of carbon, a_i, shall be converted to the subsequent subclass mass percent, M_i, having units of (g compound *i*)×(100%)×(g fuel sample)$^{-1}$ by performing a unit conversion. The unit conversion can be accomplished by multiplying each a_i by a response conversion factor, F_i, which accounts for the C/H ratio of that subclass, and then dividing the product, a_iF_i, by the summation of all corrected subclasses (see Equations 2 and 3). Equation 2 shows the calculation of F_i where MW_i is the molecular weight of the representative compound of the ith subclass, C_i is the number of carbons in this compound subclass, and MW_c is the atomic weight of carbon, 12.01 g/mol. Relative mass percent concentrations for each subclass is calculated as shown Equation 3. Note that for GC Image™ software volume fractions will be used instead of area fractions. Sometimes 2D visualization software refers to the blob integration as an area, because multiple cross sections of a blob are integrated and the areas are summed.

Eq 2. $$F_i = \frac{MW_i}{MW_c x C_i}\ [=]\ \frac{grams\ i}{grams\ C_i}$$

Eq 3. $$M_i = \frac{a_i x F_i}{\sum a_i x F_i}\ x\ 100\%\ [=]\ \frac{grams\ i}{grams\ fuel}\ x\ 100\%$$

6.4 Sample Analysis Procedure

6.4.1 Fuel samples are analyzed according to the following procedure:

6.4.1.1 The GC×GC-FID instrument is configured as described in Section 4 and prepared as described in Section 6.1 to 6.3.

6.4.1.2 The QC sample is analyzed by GC×GC-FID prior to sample analysis to verify instrument operation, i.e., signal-to-noise and subclass group precision is acceptable.

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6.4.1.3 Fuel samples are then analyzed by GC×GC-FID in sequence. A QC sample and a solvent blank (i.e., dichloromethane, isopropyl alcohol, acetone, or carbon disulfide) should be run at an appropriate interval, dependent on instrument configuration. When utilizing thermal modulation, chromatographic shifts tend to occur more frequently due to repeated heating and cooling of the modulator. A recommended practice with this configuration is to run a QC sample and solvent blank every 10 samples. Chromatographic fluctuations when using flow modulation result from configuration changes, which occur less frequently. At a minimum, the QC sample should be run after any alterations in the instrument or method configuration.

7. CALCULATIONS

7.1 Acquired GC×GC-FID data is processed in conjunction with the reference fuel previously acquired. The reference fuel sample chromatogram is processed first to verify consistent instrument performance, normalize peak detection settings, verify integration template alignment, and aid in template adjustment to compensate for changes in retention time.

7.2 *Set the peak detection limits*. Peak detection limits will be set using a signal-to-noise ratio (SNR) limit or specifying detection parameters such as minimum peak area, volume and peak values. Approximately 2900-3400 total peaks should be detected in the reference fuel for proper analysis.

7.3 *Load Template.*

7.4 *Verify Template*. Some of the subclasses contain a single isomer, i.e., all the n-paraffins (from C7 to C30), benzene C6, toluene C7, indan (cycloaromatic C9), and naphthalene (diaromatic C10). Verify that no more than one blob is in each of these single isomer bins. If misalignment is observed, then adjust the template for that fuel sample chromatogram.

7.5 *Apply the Template.*

7.6 *Export the Data.* The integrated FID signal associated with each compound subclass is processed as described in Section 6.3 to obtain relative mass percent concentrations for each compound class and subclass.

8. REPORTING

8.1 *Report* the relative mass percent for all ten compound classes and all 112 subclasses listed in Table 4 to the nearest 0.01%. Compound classes and subclasses with a mass percent of less than 0.01% shall be reported as "<0.01%".

9. BIAS AND PRECISION

9.1 An interlaboratory study involving four laboratories was conducted to develop the accuracy and precision criteria listed below. Each laboratory independently ran and analyzed the hydrocarbon reference standard in triplicate, as well as the reference fuel sample in either duplicate or triplicate sets. The accuracy and precision outlined herein should be verifiable with a 95% confidence interval. Any values that fall outside of the defined acceptable variance should arise no more than once every twenty occurrences, e.g. no more than 5 occurrences per 109 bins for the reference fuel.

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9.2 Method – Accuracy: The hydrocarbon reference standard should be used to assess the accuracy of the method. The measured concentrations should vary no more than ±0.2 absolute mass % from the prescribed gravimetric values (see Table 5, column 3).

9.3 Method – Precision: Precision metrics should be evaluated using a finalized data analysis template in conjunction with the reference fuel sample.

9.3.1 Repeatability of this method is defined at the Tier 3 level. This calculation should be based on three consecutive runs of the reference fuel sample in one laboratory and analyzed by a single operator. The standard deviation among the set of triplicate runs should be no more than 0.04 mass % for each Tier 3 subclass.

9.3.2 Reproducibility data should conform to the Tier 3 mass % concentrations reported for the reference fuel sample (see Table 4, column 3). The reported concentrations are the average values among the laboratories included in the interlaboratory study. Compositional data produced in accordance with this method, for the reference fuel sample, should vary no more than the individual reproducibility metric listed at each respective Tier 3 subclass (see Table 4, column 4).

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Table 6. Detailed Hydrocarbon Class Information for Reference Fuel†

		Reference Fuel Sample (n=4)††	
Tier Class	**Subclass Name**	**Average Concentration (Mass %)**	**Reproducibility, 2×σ (Mass %)**
Tier 3	**Light Hydrocarbons (LHC)**	0.04	0.05
Tier 2	**n-Paraffins ≥ C7**	**13.5**	
Tier 3	n-paraffin-C7	0.03	0.01
	n-paraffin-C8	0.08	0.02
	n-paraffin-C9	0.25	0.03
	n-paraffin-C10	0.5	0.04
	n-paraffin-C11	0.88	0.06
	n-paraffin-C12	0.94	0.04
	n-paraffin-C13	0.97	0.11
	n-paraffin-C14	1.28	0.09
	n-paraffin-C15	1.36	0.14
	n-paraffin-C16	1.37	0.11
	n-paraffin-C17	1.35	0.13
	n-paraffin-C18	1.16	0.13
	n-paraffin-C19	1.00	0.14
	n-paraffin-C20	0.83	0.08
	n-paraffin-C21	0.62	0.09
	n-paraffin-C22	0.43	0.08
	n-paraffin-C23	0.27	0.05
	n-paraffin-C24	0.15	0.03
	n-paraffin-C25	0.08	0.02
	n-paraffin-C26	0.04	0.02
	n-paraffin-C27	0.02	0.01
	n-paraffin-C28	<0.01	-
	n-paraffin-C29	<0.01	-
	n-paraffin-C30	<0.01	-
Tier 2	**iso-Paraffins ≥ C7**	**20.8**	
Tier 3	i-paraffins-C7	0.06	0.07
	i-paraffins-C8	0.11	0.04
	i-paraffins-C9	0.24	0.03
	i-paraffins-C10	0.59	0.05
	i-paraffins-C11	0.84	0.08
	i-paraffins-C12	1.09	0.11
	i-paraffins-C13	1.33	0.10
	i-paraffins-C14	1.61	0.13
	i-paraffins-C15	1.95	0.21
	i-paraffins-C16	2.17	0.14
	i-paraffins-C17	1.80	0.27
	i-paraffins-C18	1.97	0.31
	i-paraffins-C19	2.04	0.32

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	i-paraffins-C20	1.87	0.29
	i-paraffins-C21	1.10	0.26
	i-paraffins-C22	0.87	0.19
	i-paraffins-C23	0.6	0.22
	i-paraffins-C24-31	0.79	0.40
Tier 2	**Monocycloparaffins ≥ C6**	**21.6**	
Tier 3	monocycloparaffins C6-C7	0.10	0.12
	monocycloparaffins C8	0.29	0.13
	monocycloparaffins C9	0.50	0.07
	monocycloparaffins C10	0.84	0.13
	monocycloparaffins C11	1.22	0.09
	monocycloparaffins C12	1.57	0.21
	monocycloparaffins C13	1.76	0.13
	monocycloparaffins C14	1.97	0.12
	monocycloparaffins C15	2.18	0.2
	monocycloparaffins C16	2.16	0.08
	monocycloparaffins C17	2.06	0.21
	monocycloparaffins C18	1.90	0.24
	monocycloparaffins C19	1.63	0.18
	monocycloparaffins C20	1.24	0.30
	monocycloparaffins C21	0.97	0.08
	monocycloparaffins C22-26	1.27	0.48
Tier 2	**Dicycloparaffins**	**13.1**	
Tier 3	dicycloparaffins C8	<0.01	-
	dicycloparaffins C9	0.10	0.05
	dicycloparaffins C10	0.37	0.09
	dicycloparaffins C11	1.02	0.12
	dicycloparaffins C12	1.31	0.17
	dicycloparaffins C13	1.50	0.47
	dicycloparaffins C14	1.55	0.70
	dicycloparaffins C15	1.60	0.39
	dicycloparaffins C16	1.40	0.30
	dicycloparaffins C17	1.15	0.28
	dicycloparaffins C18	0.95	0.38
	dicycloparaffins C19-25	2.11	1.24
Tier 2	**Tricycloparaffins**	**2.96**	
Tier 3	tricycloparaffins-C10	<0.01	-
	tricycloparaffins-C11	0.02	0.05
	tricycloparaffins-C12	0.11	0.18
	tricycloparaffins-C13	0.20	0.17
	tricycloparaffins-C14	0.37	0.05
	tricycloparaffins-C15	0.59	0.14
	tricycloparaffins-C16	0.48	0.04
	tricycloparaffins-C17	0.39	0.08
	tricycloparaffins-C18-20	0.76	0.19
Tier 2	**Alkylbenzenes**	**11.0**	
Tier 3	benzene C6	<0.01	-
	toluene C7	0.05	0.02

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	alkylbenzenes C8	0.25	0.05
	alkylbenzenes C9	0.62	0.05
	alkylbenzenes C10	0.87	0.09
	alkylbenzenes C11	1.13	0.07
	alkylbenzenes C12	1.16	0.22
	alkylbenzenes C13	1.00	0.24
	alkylbenzenes C14	0.87	0.13
	alkylbenzenes C15	0.82	0.17
	alkylbenzenes C16	0.90	0.59
	alkylbenzenes C17	0.74	0.08
	alkylbenzenes C18	0.66	0.16
	alkylbenzenes C19	0.60	0.18
	alkylbenzenes C20-C24	1.04	0.99
Tier 2	**Cycloaromatics**	**11.6**	
Tier 3	cycloaromatics C9	0.06	0.01
	cycloaromatics C10	0.55	0.04
	cycloaromatics C11	1.33	0.26
	cycloaromatics C12	1.46	0.16
	cycloaromatics C13	1.59	0.16
	cycloaromatics C14	1.41	0.23
	cycloaromatics C15	1.37	0.10
	cycloaromatics C16	1.16	0.37
	cycloaromatics C17-21	2.64	1.33
Tier 2	**Diaromatics**	**4.96**	
Tier 3	diaromatics C10	0.03	0.01
	diaromatics C11	0.13	0.02
	diaromatics C12	0.39	0.13
	diaromatics C13	0.71	0.18
	diaromatics C14	0.80	0.09
	diaromatics C15	0.94	0.42
	diaromatics C16	0.86	0.15
	diaromatics C17-20	1.03	0.29
Tier 2	**Triaromatics C14-C18***	**0.50**	**0.34**

† Reference fuel subject to change based on fuel availability. Utilize most recent certificate of analysis associated with the reference standard used.

†† Reference fuel data developed from four different labs using various instrument configurations, i.e., normal or reverse phase column configurations, and flow or thermal modulation.

* Note: Triaromatics were not reported at the Tier 3 level

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Table 7. Detailed Hydrocarbon Reference Standard Class Information‡

Subclass Elution Region	Hydrocarbon Reference Std Compounds	Concentration (Mass %)
Light Hydrocarbons (LHC)	n-Pentane	0.50
Light Hydrocarbons (LHC)	n-Hexane	1.00
n-paraffin-C7	n-Heptane	2.00
n-paraffin-C8	n-Octane	2.50
n-paraffin-C9	n-Nonane	3.50
n-paraffin-C10	n-Decane	4.00
n-paraffin-C11	n-Undecane	4.75
n-paraffin-C12	n-Dodecane	5.25
n-paraffin-C13	n-Tridecane	5.50
n-paraffin-C14	n-Tetradecane	5.00
n-paraffin-C15	n-Pentadecane	4.25
n-paraffin-C16	N-Hexadecane	3.50
n-paraffin-C17	n-Heptadecane	2.75
n-paraffin-C18	n-Octadecane	2.00
n-paraffin-C19	n-Nonadecane	1.00
n-paraffin-C20	n-Eicosane	0.50
monocycloparaffins C6-C7	Cyclohexane	1.25
monocycloparaffins C6-C7	Methylcyclohexane	1.25
monocycloparaffins C8	Ethylcyclohexane	2.00
monocycloparaffins C8	Cycloheptane [C]	2.00
monocycloparaffins C9	Propylcyclohexane	2.75
monocycloparaffins C10	Butylcyclohexane	3.50
monocycloparaffins C11	Pentylcyclohexane	4.00
monocycloparaffins C12	Hexylcyclohexane	1.50
dicycloparaffins C9	Cyclooctane [C]	2.75
dicycloparaffins C10	trans-Decahydronaphthalene	2.75
dicycloparaffins C11	Cyclopentylcyclohexane	0.75
dicycloparaffins C12	Bicyclohexyl	2.35
tricycloparaffins-C13 [A,B] dicycloparaffins C12 [B]	Cyclododecane [C]	1.25
benzene C6	Benzene	0.50
toluene C7	Toluene	1.05
alkylbenzenes C8	Ethylbenzene	2.00
alkylbenzenes C8	o-Xylene	2.75
alkylbenzenes C9	2-Ethyltoluene	3.00
alkylbenzenes C9	n-Propyl-benzene	4.00
alkylbenzenes C9	1,2,4-Trimethyl-benzene	3.50
alkylbenzenes C10	1,2,4,5-Tetramethylbenzene	2.75
alkylbenzenes C16 [A] tricycloparaffins-C16 [A,B] dicycloparaffins C15 [B]	Cyclopentadecane [C]	0.75
cycloaromatics C12	Pentamethylbenzene	2.00
cycloaromatics C13	Hexamethylbenzene	0.50
diaromatics C10	Naphthalene	0.55
diaromatics C12	2-Ethylnaphthalene	0.55

‡ Certified analysis of PNA in AVTUR Gravimetric blend standard from PAC used for the development of this method

[A] Observed elution regions when system is configured in normal phase

[B] Observed elution regions when system is configured in reverse phase

[C] See section A1.4.1. for further explanation on elution region

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APPENDIX A

A1. Special Considerations for Synthetic Aviation Turbine Fuels (SATF)

A1.1 This appendix outlines special considerations regarding quantifying hydrocarbon types of *synthetic* aviation turbine fuels (SATF) that have unique and/or non-typical chemical compositions compared to traditional *petroleum-derived* aviation turbine fuels.

A1.2 Federal Test Method 7508 details the procedure for quantifying hydrocarbon types in *petroleum-derived* aviation turbine fuels. In some cases, these methods and templates can be used directly for the analysis of *synthetic* aviation turbine fuels (SATF)—sometimes referred to as sustainable aviation fuel (SAF)—to determine the hydrocarbon type composition. However, there are cases where special consideration must be given to the analysis of SATF samples, such that templates and procedures must be modified to properly characterize and quantify the hydrocarbon types.

A1.3 *Non-Typical/Skewed Hydrocarbon Distributions*—Petroleum-derived aviation turbine fuels are comprised of about five to seven hydrocarbon types; however, SATF can be comprised of only one or two hydrocarbon types (e.g., iso-paraffins and normal paraffins). When fewer hydrocarbon types are present in a fuel, the concentration of compounds within the types increases and/or the ratio of compound concentrations among differing types is altered disproportionately, which results in a non-typical or skewed hydrocarbon distribution. These non-typical hydrocarbon distributions can negatively impact the chromatographic separation (e.g., cause co-elution of compounds and skewed elution times) and interfere with the correct classification of material. For example: large increases in iso-paraffinic compounds can cause overloaded peaks (or blobs). These overloaded peaks can “bleed” into nearby regions of the chromatogram and can mask or inhibit the detection of smaller peaks in the nearby region (causing an under-reporting of the smaller peak types). If overloading is significant, peaks can “streak” across an extended portion of the chromatogram (on the secondary axis) and lead to false positive detection of compound types. Figure 2 shows a hydroprocessed ester and fatty acids (HEFA) derived fuel that typically contains high concentrations of isoparaffins, which “bleed” into the monocycloparaffin region.

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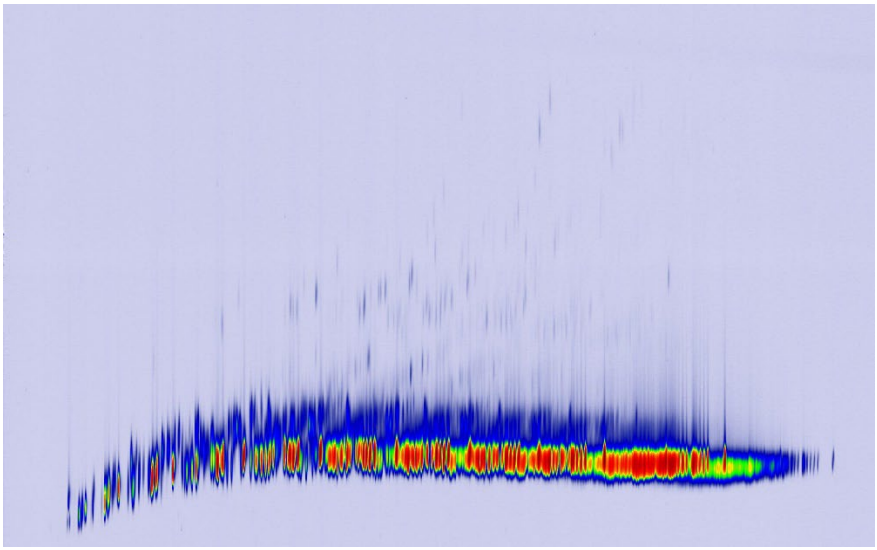


Figure 2. HEFA fuel analyzed with a normal phase configuration instrument

A1.4 *Highly Cycloparaffinic SATF*—Typical petroleum-derived fuels contain <2 mass% tricycloparaffins, which elute near the alkylbenzenes (10-20 mass %) by normal-phase two-dimensional gas chromatography (GC×GC). SATF that contains high concentrations of cycloparaffins also tend to exhibit increased concentrations of the tricycloparaffins, which in turn expand into the alkylbenzenes regions and elute amongst them. This may necessitate template modification to accurately quantify the hydrocarbon types in these types of fuels.

A1.4.1 The hydrocarbon reference standard used in this method contains monocycloparaffins that are not typically present in jet fuel. Specifically, cycloheptane, cyclooctane, cyclododecane and cyclopentadecane are large monocycloparaffins that have unique retention behavior which causes them to elute in template regions that do not accurately represent their compound class and structure. The shifting of these compounds is dependent on whether the GCxGC column configuration is in normal or reverse phase. Table 5 denotes the template region a user can expect all of the compounds present in the hydrocarbon reference standard to elute. The atypical monocycloparaffins mentioned above have more than one elution region listed based on the column configuration.

A1.5 *Highly-Branched Compounds*—Most of the branched compounds in petroleum-derived fuels contain one or two branches (e.g., methyl or dimethyl) with some trimethyl branches. In contrast, SATF fuels can contain compounds with four or more carbon branches. These highly-branched compounds have lower boiling points than compounds of the same carbon number and hydrocarbon type that have fewer branches. As a result, the highly-branched compounds can elute in the same GC×GC chromatographic region as lower carbon number compounds with less branching, especially apparent when using normal phase GC×GC. To accurately quantify the hydrocarbon types of these highly-branched fuels by carbon number, the data analysis template used for petroleum-derived

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jet fuels will likely require modifications. Figure 3 shows an example of an alcohol to jet (ATJ) fuel, which notoriously contains highly branched isoparaffins.

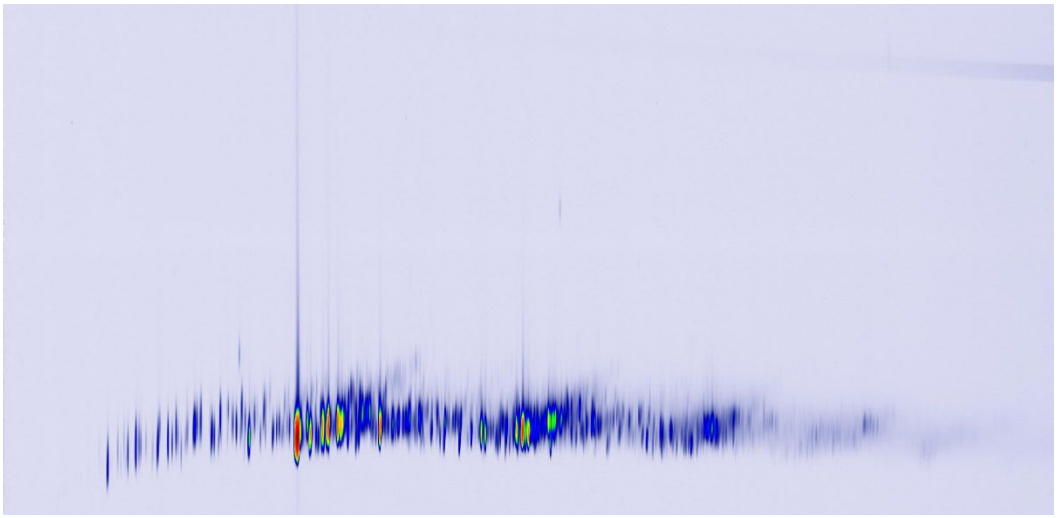


Figure 3. ATJ fuel analyzed with a normal phase configuration instrument

A1.6 Mass spectral (MS) analysis can be used to help determine if compounds are eluting in a chromatographic region other than that designated in the data analysis template, or if they are co-eluting with other more abundant compounds (i.e., MS analysis can provide enhanced characterization of sample compounds). If the GC×GC system has simultaneous flame ionization detection (FID) and MS detection, the FID data analysis template can be applied to the mass spectral chromatogram. The selected or extracted ion chromatogram (SIC or EIC, respectively) can be used to detect where groups of compounds are eluting if the compounds have molecular ions that are detectable in the mass spectra. Many of the highly branched compounds do not have a detectable molecular ion but can be identified by the fragment ion of the next lower molecular weight. Some “coarse” adjustments to the template can be made on the MS chromatogram, which can then be applied to the FID chromatogram for more “fine” adjustments. For SATF where low-level compounds are suspected to be co-eluting with high-level compounds (such as monocycloparaffins in a highly iso-paraffinic fuel), the presence of co-elution can be established by examining the mass spectra in the region of overlap. Then blob-splitting of the co-eluting compounds in the FID chromatogram can be used to rectify this type of discrepancy.

A1.7 If simultaneous mass spectral analysis is not available, or does not have sufficient sensitivity for certain compounds, separate 1D or 2D analysis (GC-MS or GC×GC-MS, respectively) may be warranted to aid in adjusting the data analysis template for various types of SATF.

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A1.8 "Streaking" of high-concentration compounds into other hydrocarbon regions can be addressed by selecting the blobs that are part of the streak, and excluding them from volumes in the affected regions. The volume that is excluded is often an insignificant percentage of the total volume of the peak from which it originated. In the case of fuels where the majority of the composition is only one or two compounds, the data analysis template may need to be modified to include the streaking blobs with those compounds.

DATE	DESCRIPTION OF CHANGE	REASON FOR CHANGE	VERSION	REVISED BY
25 September 2025	Footnote added to Table 4.	Reference standard CofA subject to change when a new fuel is needed for the reference fluid due to availability	1	Navy

Method Prepared by:
US Navy and US Air Force - 2023

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Method 7509.0
January 8, 2024

7509.0 - METHOD FOR DETERMINATION OF FILTRATION TIME AND TOTAL SOLIDS

1. SCOPE

1.1 This method describes the process for determining singularly or simultaneously the filterability characteristics and solids contamination of jet fuel. The purpose is to detect and prevent contaminants in jet fuel that can plug and cause rupture of ground filtration equipment, thereby affecting flight reliability of aircraft. This Appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

2. SUMMARY

2.1 A 3.785 liter (1 gallon) volume of jet fuel is filtered through a membrane filter in the laboratory. The time required to filter this volume is measured in minutes and solids content is determined gravimetrically.

3. APPARATUS

3.1 Membrane filter: White, plain, 47 mm diameter, nominal pore size 0.8 µm. The membrane filter must conform to ASTM D5452 or IP 423 requirements.

3.2 Filtration apparatus: The apparatus, constructed of stainless steel, consists of a funnel and a funnel base with a filter support such that a membrane filter and a flow reducing washer can be securely held between the sealing surface of the funnel and funnel base (see Figure 1 "Apparatus for Determining Total Contaminant" in ASTM D5452 or Figure 2 "Apparatus for Filtering and Dispensing Flushing Fluid).

3.3 Flow reducing washer: A 47-mm diameter flow reducer washer with an effective filtration area of 4.8 cm^2 (NSN 6640-01-018-2795).

3.4 Vacuum flask: A minimum of 4 liters.

3.5 Vacuum system: That develops a vacuum between 67.5 kPa (20 inches of mercury) and 84.5 kPa (25 inches of mercury).

3.6 Oven: Of the static type (without fan assisted circulation) controlling to 90 °C ± 5 °C (194 °F ± 9 °F).

3.7 Forceps: Flat-bladed with non-serrated non-pointed tips.

3.8 Dispenser, rinsing solvent (petroleum ether): Containing a 0.45 µm membrane filter in the delivery line. If solvent has been pre-filtered using a 0.45 µm filter then an inline filter is not required.

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3.9 Glass petri dish: Approximately 125 mm in diameter with removable cover.

3.10 Analytical balance: Single or double pan, the precision standard deviation of which must be 0.07 mg or better.

4. PREPARATION

4.1 Preparation of apparatus and sample containers. All components of the filtration apparatus (except the vacuum flask), sample containers and caps must be cleaned as described in section 9 (Preparation of Apparatus and Sample Containers) of ASTM D5452 or in section 6 (Apparatus cleaning) of IP 423. All metal parts of the filtration apparatus are to be electrically bonded and grounded, including the fuel sample container. See ASTM D5452 or IP 423 for other safety precautions.

5. SAMPLING

5.1 Sampling. Obtain a representative 3.785 liter (1 gallon) sample as directed in section 8 (Sampling) of ASTM D5452 or in section 7 (Sampling) of IP 423. When sampling from a flowing stream is not possible, an all-level sample or an average sample, in accordance with ASTM D4057 and/or ASTM D4177 shall be permitted. The 3.785 liter (1 gallon) sample container shall be an interior epoxy-coated metal can, a brown glass bottle, or a clear glass bottle protected by suitable means from exposure to light.

6. PROCEDURE

6.1 Test procedure.

6.1.1 Using forceps, place a new membrane (test) filter in a clean petri dish. Place the petri dish with the lid slightly ajar in a 90 °C ± 5 °C (194 °F ± 9 °F) oven for 30 minutes. Remove the petri dish from the oven and place it near the balance with the lid slightly ajar, but still protecting the filter from airborne contamination, for 30 minutes.

6.1.2 Weigh the test filter. A filter weighing more than 90 mg will not be used for time filtration testing.

6.1.3 Place a flow reducing washer (required only for time filtration testing) on top of funnel base then place a test filter on top of the reducing washer and secure the funnel to the funnel base.

6.1.4 Immediately prior to filtering the fuel, shake the sample to obtain a homogeneous mix and assure that fuel temperature does not exceed 30 °C (86 °F). Clean the exterior or top portion of the sample container to ensure that no contaminants are introduced. Any free water present in the fuel sample will invalidate the filtration time results by giving an excessive filtration time rating.

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6.1.5 With the vacuum off, pour approximately 200 mL of fuel into the funnel.

6.1.6 Turn vacuum on and record starting time. Continue filtration of the 3.785 liters (1 gallon) sample, periodically shaking the sample container to maintain a homogenous mix. Record the vacuum (kPa or inches of mercury) 1 minute after start and again immediately prior to completion of filtration. Throughout filtration, maintain enough fuel in the funnel so that the membrane filter is always covered.

6.1.7 Report the filtration time in minutes expressed to the nearest whole number. If filtration of the 3.785 liters (1 gallon) is not completed within 30 minutes, the test will be stopped, and the volume of the fuel filtered will be measured. In these cases, report filtration time as ">30 minutes" and the total volume of fuel filtered.

6.1.8 Report the vacuum (kPa or inches of mercury) as determined from the average of the two readings taken in 6.6.

6.1.9 After recording the filtration time, shut off the vacuum and rinse the sample container with approximately 100 mL of filtered petroleum ether and dispense into the filtration funnel. Turn vacuum on and filter the 100 mL rinse. Turn vacuum off and wash the inside of the funnel with approximately 50 mL of filtered petroleum ether. Turn vacuum on and filter. Repeat the funnel rinse with another 50 mL of petroleum ether but allow the rinse to soak the filter for approximately 30 seconds before turning the vacuum on to filter the rinse. With vacuum on, carefully remove the top funnel and rinse the periphery of the filter by directing a gentle stream of petroleum ether from the solvent dispenser from the edge of the filter toward the center, taking care not to wash contaminants off the filter. Maintain vacuum after final rinse for a few seconds to remove the excess petroleum ether from the filter.

6.1.10 Using forceps, carefully remove test filter (from the funnel base and flow reducing washer if present) and place in a clean petri dish. Dry in the oven at 90 °C ± 5 °C (194 °F ± 9 °F) for 30 minutes with the cover on the petri dish slightly ajar. Remove the petri dish from the oven and place it near the balance with the lid slightly ajar, but still protecting the filter from airborne contamination, for 30 minutes. Reweigh the filter.

6.1.11 Report the total solids content in mg/L by using the following formula:

$$\frac{\text{weight gain of filter in mg}}{3.785\ \text{L}} = \text{mg/L}$$

6.1.12 Should the sample exceed the 30-minute filtration time and a portion of the fuel is not filtered, the solids content in mg/liter will be figured as follows: Determine the volume of fuel filtered by subtracting the mL of fuel remaining from 3785 mL.

$$\frac{\text{weight gain of filter in mg}}{\text{mL of fuel filtered} \times 0.001} = \text{mg/L}$$

6.2 Test conditions for filtration time.

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6.2.1 The vacuum should exceed 67.5 kPa (20 inches of mercury) throughout the test. The differential pressure across the filter should exceed 67.5 kPa (20 inches of mercury).

6.2.2 The fuel temperature shall be between 18 °C and 30 °C (64 °F and 86 °F). If artificial heat (such as a hot water bath) is used to heat the sample, erroneously high filtration times may occur, but this approach is allowed.

7. NOTES

7.1 Filtration time. If it is desired to determine the filtration time and not the total solids content, perform the test by omitting steps 6.1.9, 6.1.10, 6.1.11, and 6.1.12.

7.2 Total solids. If it is desired to determine the total solids content and not the filtration time, use of the flow reducing washer may be omitted. When a reducing ring is not used, then total solids shall be determined per ASTM D5452 or IP 423 and the use of a control filter shall be required.

Method prepared by:
Air Force - AFPET/PTP - 2024

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APPENDIX A

TABLE A-I. List of materials cited in each test method.

METHOD	NAME OF THE TEST	MATERIALS
313.4	Penetration of lubricating greases after prolonged working	1. Grease to be tested
350.3	Evaporation loss of lubricating greases and oils (high temperature)	1. VM&P Naphtha 2. Petroleum Ether 3. Tissue, Facial
361.5	Stroking properties of hydraulic brake fluids	1. Ethyl alcohol (ACS) 2. Crocus cloth 3. Wheel cylinder assemblies 4. Master cylinder assembly 5. Rubber cups 6. Wheel cylinder pistons 7. Master cylinder pistons
1151.3	Explosive vapor in boiler fuel oil	1. Hydrocarbon gas
1203.0	Determination of Particulate Matter in Aviation and Distillate Fuels by Automatic Particle Counting	1. Heptane
3005.5	Dirt content of grease	1. None
3007.2	Water displacement and water stability	1. Steel test panels, FS 1009, (ASTM A109/A109M) 2. Naphtha, aliphatic (TT-N-95) 3. Methyl alcohol (O-M-232) 4. Distilled water 5. Silica sand, white, dry, sharp 6. Surgical gauze 7. Absorbent tissue 8. Emulsion, water and oil 9. Isopropyl alcohol 10. Heptane
3010.2	Solid particle contamination in aircraft turbine engine lubricants (gravimetric procedure)	1. Isopropyl alcohol, reagent grade 2. Petroleum ether, reagent grade boiling point range 30°C to 60° C 3. Detergent, liquid, water soluble
3012.1	Determination of particulate matter in aerospace hydraulic fluids	1. ISO 11171 calibration standard 2. Detergent, Micro-90 cleaning solution 3. Hexanes (Certified ACS)

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TABLE A-I. List of materials cited in each test method – Continued.

METHOD	NAME OF THE TEST	MATERIALS
3013.2	Determination of particulate contamination in synthetic turbine engine lubricants	1. Isopropyl alcohol, reagent grade 2. Petroleum ether, 30°C to 60°C boiling range, reagent grade
3213.2	Foaming characteristics of air craft turbine lubricating oils (static foam test)	1. Solvents, dry cleaning; such as Stoddard Solvent, petroleum distillate, VM&P naptha, or similar 2. Toluene conforming to ASTM D892 specifications 3. Petroleum ether (Boiling Range 30 °C to 60°C), ACS 4. Detergent, liquid, water soluble
3403.3	Compatibility of turbine lubricating oils	1. Referee lubricant 2. n-heptane 3. Cleaning solution. Any suitable laboratory glass cleaning fluid may be used providing it removes all evidence of staining and deposits from previous tests.
3410.2	High temperature deposit and oil degradation characteristics of aviation turbine oils	1. Lubricating oil, qualified under MIL-PRF-23699 2. Dry cleaning solvent, (MIL-PRF-680, Type 1) 3. Oakite LRS (REL-4576) 4. No. 400 emery paper 5. No. 30 grit aluminum oxide 6. Bi solvent; Isopropyl alcohol: Acetone 7. Heptane
3411.2	Thermal stability and corrosivity of air craft turbine engine lubricants	1. Toluene, reagent grade 2. Acetone, reagent grade 3. Nitric-sulfuric acid mixture, approximately 1:10 4. Abrasive paper, silicon carbide, 240 grit 5. Steel specimens of approximately 0.15, 0.7, and 7.5 cm made from C1008/C1010 cold rolled half hard with #2 finish and #4 round edge 6. Detergent, liquid, water soluble

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TABLE A-I. List of materials cited in each test method – Continued.

METHOD	NAME OF THE TEST	MATERIALS
3432.2	Compatibility of elastomers with aircraft turbine lubricants	1. Standard test rubber sheets, type F-A, conforming to SAE AMS 3217/4 2. Standard test rubber sheets type FS conforming to SAE AMS 3217/5 3. Cleaning solution, Any suitable laboratory glass cleaning fluid may be used providing it removes all evidence of staining and deposits from previous tests. 4. Dry cleaning solvent (such as Stoddard Solvent or similar) 5. Ethyl alcohol, ACS grade 6. Acetone, ACS grade 7. Distilled water 8. Filter paper 9. Detergent, liquid, water soluble
3433.1	Compatibility of synthetic aircraft turbine lubricants with silicone rubber	1. Standard test rubber sheets, type F-A, conforming to US Air Force Specification bulletin 539, standard elastomer stocks 2. Cleaning solution, glass (concentrated sulfuric acid saturated with potassium or sodium dichromate) 3. Methyl alcohol, ACS, particle free 4. Solvent-Naphtha or toluene, ACS 5. Acetone, ACS, particle free
3457.3	Hydrolytic stability	1. Emery paper, 00 or finer 2. Litmus paper 3. Petroleum ether (O-E-751) 4. VM&P Naphtha 5. n-Hexane, technical grade 6. Sodium sulfate, anhydrous (ACS) 7. Copper strip (ASTM B187/B187M, ASTM B272)
3458.2	Low temperature stability test for oil	None
3463.3	Stability of grease in hot water (water immersion)	1. Distilled water
3467.2	Storage stability of lubricating grease	1. None
3480.3	Volatility of blended oils	1. Test oils

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TABLE A-I. List of materials cited in each test method – Continued.

METHOD	NAME OF THE TEST	MATERIALS
3500.2	Monobasic acid components of synthetic ester lubricants by gas chromatography	1. Potassium hydroxide, reagent grade 2. Ethyl alcohol, 95 percent 3. Distilled water 4. Hydrochloric acid, dilute 5. Ethyl ether, reagent grade 6. Sodium sulfate, anhydrous, reagent grade 7. Diethylene glycol adipate polyester 8. Gas chrom p, 60 to 80 mesh, acid based washed or equivalent 9. Chloroform, reagent grade 10. Phosphoric acid, reagent grade, 85 percent 11. Helium, oil free 12. Butyric acid, reagent grade 13. Isovaleric acid, reagent grade 14. Valeric acid, reagent grade 15. Caproic acid, reagent grade 16. Heptylic acid, reagent grade 17. Caprylic acid, reagent grade 18. Pelargonic acid, reagent grade 19. Capric acid, reagent grade
3603.6	Swelling of synthetic rubbers	1. Standard rubber test sheets, as required by the specification. 2. Alcohol, ethyl or methyl, anhydrous. 3. Filter paper 4. Distilled water
3604.3	Swelling of synthetic rubber by aircraft	1. Standard test rubber sheets, type H or FA 2. Cleaning solution, glass (concentrated sulfuric acid saturated with potassium or sodium dichromate, technical grade) 3. Naphtha, technical grade 4. Toluene, technical grade 5. Alcohol, ethyl 6. Emery cloth No. 60 7. Naphtha

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TABLE A-I. List of materials cited in each test method – Continued.

METHOD	NAME OF THE TEST	MATERIALS
3720.3	Molybdenum disulfide content of lubricating grease	1. n-Hexane, ACS grade 2. Oleic acid, USP grade 3. Petroleum ether (O-E-751)
3722.3	Molybdenum disulfide content of non-soap thickened lubricating grease	1. Hexane, normal ACS 2. Extraction thimble (double thickness 3.3 x 8.0 cm) 3. Filter paper (Whatman No. 1 or equivalent) 4. Filter paper (Whatman No. 40 or equivalent 5. Eschka's mixture-2 parts calcined (light) magnesia thoroughly mixed with one part of sodium carbonate 6. Bromine water, saturated 7. Barium chloride 10 percent 8. Sodium carbonate CP 9. Potassium carbonate, CP 10. Hydrochloric acid
5003.3	Deposit- forming tendencies of air craft turbine lubricants	1. Silicone-carbide paper. No. 400 2. Solvent, dry-cleaning (such as Stoddard Solvent or similar). 3. Cleaning solution, glass, Any suitable laboratory glass cleaning fluid may be used providing it removes all evidence of staining and deposits from previous tests. 4. n-Hexane, thiophene-free (ACS) 5. Isopropyl alcohol (ACS) 6. Petroleum ether (ACS) 7. Distilled water 8. Scouring powder, commercial grade 9. Steel wool 10. Pipe cleaners 11. Cleaning compound, Any suitable water based non corrosive cleaner

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TABLE A-I. List of materials cited in each test method – Continued.

METHOD	NAME OF THE TEST	MATERIALS
5305.2	Corrosiveness of lubricants at 232 ° C or (450 °F)	l. Solvent, dry-cleaning (MIL-PRF-680, type I) 2. Acetone (ACS) 3. Nitric acid solution (3 parts concentrated nitric acid and 7 parts distilled water 4. Pumice, FFF grade 5. Eraser (Eberhard Faber “pink pearl" No. 100 or equivalent) 6. Cleaning solution, glass (concentrated sulfuric acid saturated with ALNOCHROMIX™) 7. Distilled water
5309.6	Corrosiveness of greases (copper strip, 100 ° C)	1. Copper strip, soft tempered, annealed 2. n-Hexane (ACS grade) 3. Acetone (ACS grade) 4. Fine silicon carbide or aluminum grit paper 5. 240 grit silicon- carbide paper 6. 150 mesh silicon - carbide paper
5321.3	Corrosion of lead by lubricating oils	1. Test panel, Chemical-lead (ASTM B29-55 or equal) 2. Test panel, copper, Electrolytic (ASTM B152/l52M) 3. Naphtha, petroleum (TT-N-97) 4. Acetone (ASTM D329) 5. Steel wool 6. Pads, Absorbent- cotton
5329.3	Corrosion Protection (Humidity Cabinet)	1. Naphtha, aliphatic (TT-N-95) 2. Methyl alcohol (O-M-232) 3. Silica sand, white, dry, sharp, (size IAW ASTM E11) 4. Test panels FS 1009 steel, fabricated from open-hearth, "killed" type low carbon, No. 4, soft-temper, cold rolled sheet or strip conforming to ASTM A109/A109M 5. Surgical gauze

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TABLE A-I. List of materials cited in each test method – Continued.

METHOD	NAME OF THE TEST	MATERIALS
5331.2	Corrosion protection of steel against sulfurous acid - salt spray by solid film lubricants	1. Steel specimen, carbon steel conforming to FS 1010 2. Synthetic sea water sulfurous acid test solution (formula ASTM D1141) 3. Aluminum oxide cloth 240 grit 4. VM&P Naphtha 5. Distilled water (ASTM D1193. type 111) 6. Sulfurous acid, ACS
5414.5	Resistance of grease to fuel	1. Standard test fluid (ASTM D471, type 111) 2. Aluminum strip (ASTM B209/B209M, tempers T-3 or T04)
5415.2	Resistance of grease to water and a 1:1 water –ethanol solution	1. Distilled water
6052.2	High temperature, high pressure spray ignition	1. Nitrogen gas
6053.2	Manifold ignition test	1. Steel wool
6508.3	Load carrying capacity of lubricating oils (Ryder gear machine)	1. Lubricating oil, DOD-PRF-85734. 2. Dry cleaning (A-A-59601D). 3. Trimethylpentane (540-84-1) 4. Oakite LRS (REL-4576) 5. Test gear retaining nut lock washer, Fafnir WO-6 (ERDCO® Engineering Corp- oration, Addison, ILL, or any commercial supplier of Fafnir parts).
7501.1	Low temperature torque test method for lubricating greases	None
7502.1	Solvent cleaning power by soil test method	1. Toluene CP at least 99%
7503	High-Temperature Rolling Contact Fatigue Life Testing (RCF) of Lubricating Oils and Bearing Materials (Ball-on-Rod RCF Tester)	1. Isopropyl Alcohol
7504	Efficiency of Axle Gear Lubricants	None
7505	Load Carrying Capacity of Lubricating Oils Using a Modified FZG Gear Test Rig (FZG-Ryder Rig)	1. Dry cleaning solvent such as Stoddard Solvent or similar
7506	Gear Lubricant Compatibility for Stryker Vehicle Limited Slip Differential Clutches	1. Heptane

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TABLE A-I. List of materials cited in each test method – Continued.

METHOD	NAME OF THE TEST	MATERIALS
7507.0	Isentropic Bulk Modulus Of Aviation Turbine Fuels And Diesel Fuels	1. Heptanes 2. n-pentane 3. Isopropyl Alcohol 4. De-ionized water 5. Nitrogen, dry 6. Diesel Fuel
7508		1. dichloromethane, isopropyl alcohol, acetone, or carbon disulfide 2. hydrocarbon reference standard: see 7508 TABLE I.
7509		

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September 30, 2019

APPENDIX B

TABLE B-I. Cross reference list between material name and test method numbers where the material is cited.

Compound's Name	Test Method Where the Material is Cited
Acetone	3410, 3411, 3432, 3433, 5305, 5309, 5321
Barium chloride	3722
Bromine	3722
Butyric acid	3500
Capric acid	3500
Caproic acid	3500
Caprylic acid	3500
Chloroform	3500
Cleaning solution, glass	5003
Detergent, liquid, water soluble, commercial grade	3010, 3012, 3213, 3411, 3432
Diesel fuel	7507
Diethylene glycol adipate	3500
Dry cleaning solvent	3213, 3410, 3432, 6508, 7505
Ethyl alcohol	361
Ethyl ether	3500
n-Heptane	1203, 3007, 3403, 3410, 7506, 7507
Heptylic acid	3500
n-Hexane	3012, 3457, 3720, 3722, 5003, 5309
Hydrochloric acid	3500, 3722
Isopropyl alcohol	3007, 3010, 3013, 3410, 5003, 7503, 7507
Methyl alcohol	3007, 3433, 3603, 5329
Naphtha	350, 3007, 3604, 5321, 5329
Nitric acid, (Nitric-sulfuric acid mixture)	3411, 5305
Nochromix cleaning solution	5305
Oakite LRS	3410, 6508
Oleic acid	3720
n-Pentane	7507
Petroleum Ether	350, 3010, 3013, 3213, 3457, 3720, 5003
Phosphoric acid	3500
Potassium carbonate	3722
Potassium hydroxide	3500
Sodium carbonate	3722
Sodium dichromate	3433, 3604
Sodium sulfate	3457, 3500

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TABLE B-I. Cross reference list between material name and test method numbers where the material is cited – Continued.

Compound's Name	Test Method Where the Material is Cited
Sulfuric acid	3411, 3433, 3604, 5305,
Sulfurous acid	5331
Toluene	500, 3411, 3433, 3604
VM&P naphtha	5331, 3457
Valeric acid	3500

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy – SH
Air Force – 68
DLA – PS

Preparing Activity:
Army - AT
Project number 91GP-2026-002

Reviewing Activity:
Air Force – 11, 20
Army – AR, ME

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